QNIR REVIEW-LOOP DRAFT · VERSION 0.4.0

QnIR: A Receipt-Chained Compiler Intermediate Representation for the Qn Universe

Review-loop edition with the committed Qn universe corpus embedded as annexes, so every canon claim is verifiable in-document against pin 6829e9e. Incorporates the dispositions of Deep Review round 0001: two-regime equivalence consistency, edge-versus-node anchor grounding, the anti-gaming protocol, the Physical Efficiency Gate, and the adjudicated conjunctive validity formula. Research questions RQ1–RQ15.

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The CDQN / Qn universe documentation · github.com/cdqn5249/cdqn
Canon pinned to commit 6829e9e · committed corpus: 22 documents + glossary, hashed per annex

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1 Introduction and Background

1.1 Purpose, posture, and reading protocol

This document is version v0.4.0 of the QnIR review-loop draft, a proposal paper written to be critiqued, inside the ecosystem the Deep Review instrument was built for — a tool piloted on more than sixteen thousand conference submissions [1, 2] and applied here to the documentation universe of [3]. Its subject is QnIR, a compiler intermediate representation for the Qn universe: a multi-frontend funnel that accepts legacy programming languages together with the universe's own surface language QnLang, and emits receipts as faithfully as it emits code. The paper exists inside a loop: a draft is submitted to an automated deep-review instrument, the transcript is committed verbatim into the repository, the author team responds point by point, and the next version carries the adopted corrections forward. Version v0.3.0 was reviewed by Google Deep Review; the complete transcript is committed at `docs/AiReviews/AiReview-QnIRv0.3.0.md` and reproduced as Annex W, and Section 9 disposes every one of its points.

Three changes distinguish v0.4.0 from its predecessor. First, the paper absorbs the substance of that review: the equivalence-consistency witnessing split (static versus dynamic), the edge-versus-node grounding rule for emergent anchors, the four lineage-bounding candidate mechanisms, the anti-gaming protocol for the self-input gate, the Physical Efficiency Gate that replaces the market-share threshold for Phase B, and the adjudication of the seeded validity-formula divergence. Second, the bibliography and the internal cross-references have been corrected where the review demonstrated factual drift, and every correction is traceable in Section 9. Third — and most consequentially for the review method itself — the submission now embeds the entire committed documentation corpus of the Qn universe as Part II (Annexes A–W). The previous round established, after the fact, that the reviewer had worked from the submitted PDF alone; the pinned live-documentation links were neither followed nor, in three cases, even clickable. A review that cannot see the canon it is asked to check against will spend its budget reconstructing context instead of attacking design decisions. v0.4.0 removes that failure mode by construction: every claim in Part I that rests on committed canon is checkable inside this single document, against pinned, hashed annex copies of the source files.

Document transparency

Author of record and guidance: Christophe Duy Quang Nguyen.
Drafting: GLM-5.3-Flash (z.ai), an AI agent, operating under the author's direction.
Review instrument: Google Deep Review (`https://research.google/deepreview/`); round 0001 transcript committed at `docs/AiReviews/AiReview-QnIRv0.3.0.md`. The reviewer's criticisms are treated as instrument readings that advance the design, not as obstacles.
Canon pin: every claim labelled *canon* in this paper is pinned to commit `6829e9e` of `github.com/cdqn5249/cdqn`, the commit at which this draft was prepared. The documentation state at `6829e9e` is byte-identical, for all documentation files, to state at the previous pin `d50ee1e`; the intermediate commits added only the review artefacts themselves.
Claim labelling. Every load-bearing claim carries one of three labels: *canon* (pinned to the commit above), *CANDIDATE* (open design proposition, unsealed), or *IRL* (externally sourced design premise with real-world citations). A numbered external citation of the form $[n]$ is syntactically equivalent to an *IRL* label; Section 7 states this as a rule so that a passage carrying $[n]$ markers is not again read as unlabelled. *Mislabelling is treated as a defect of the same severity as a wrong citation*, and reviewers are invited to flag any sentence whose label they believe is wrong.
Narrative. This paper asks for criticism under the repository's Best Of Choices (BOC) culture: negative results are receipts, not failures. Where v0.3.0 planted a deliberate seed inconsistency (Section 7.2), v0.4.0 records its adjudication.

1.2 The consultation context, restated accurately

Version v0.3.0 justified deferring the JavaScript and Java frontends by appeal to what it called, in the author's shorthand, "Black October": the registry-flood and agent-swarm incidents of September–October

2026 [7, 8]. The review demonstrated that this justification drifted from its own sources: references [7, 8] document an agent-swarm flood against **RubyGems** (the Ruby ecosystem) and malicious **Python** pickle-model uploads on **Hugging Face** — not npm (JavaScript) or Maven (Java) — and both reports date from September 2026. v0.4.0 adopts the correction and, with it, the review's stronger structural point: Python's PyPI (where `setup.py` build hooks execute arbitrary code at install time) and Rust's crates.io (where `build.rs` scripts and procedural macros execute arbitrary native code at compile time) share exactly the open-submission registry topology whose poisoning was cited as a reason to defer JavaScript and Java. Sequencing frontends by the *package registry* they happen to use is therefore not a defensible security ordering at all.

The deferral rationale is restated accordingly, on two independent legs. First, *engineering scope*: COBOL, Rust, and Python were chosen because their static record layouts, affine ownership types, and dominant numerical idioms, respectively, exercise the three hardest parts of equivalence-consistency witnessing (exact fixed-point semantics, aliasing discipline, and floating-point envelopes), so the first three frontends buy the most design information per unit of effort. Second, *ingestion-firewall maturity*: the anti-poisoning ingestion firewall (Section 5.2) needs an implementation history before the funnel is pointed at ecosystems whose normal supply chain is adversarial, and that argument applies to every open-registry language equally — JavaScript and Java included. Table 1 in Section 3.5 reflects this decoupling. The "Black October" phrase survives only as the author's shorthand for the September 2026 RubyGems/Hugging Face events, explicitly scoped, and no longer carries any npm/Maven implication.

1.3 What is new in v0.4.0

For the reviewer's convenience the delta from v0.3.0 is summarised here and itemised exhaustively in Section 9. The substantive adoptions are: the two-regime reading of equivalence consistency (Section 3.3); the homoiconic two-stage reconciliation of the funnel with the execution service (Section 3, resolving the Figure 1/Figure 2 inversion the review identified); the anti-gaming protocol at the self-input gate (Section 4.2); the Physical Efficiency Gate for Phase B (Section 4.3); the edge-versus-node product typing for emergent anchors with the weakest-link meet rule (Section 5.2); the two additional lineage failure modes and the $\Omega(x)$ growth argument (Section 5.3); the corrected witnessing formalisation with work-weighted dominator distance (Section 5.4); the four reviewer-proposed lineage mechanisms adopted as CANDIDATEs (Section 5.5); the falsifiable alignment metrics (Section 6); and the adjudication of the seed finding (Section 7.2). The research-question register (Section 8) now carries RQ1–RQ9 with disposition status and adds RQ10–RQ15 for the next review round.

2 The Qn Universe as Compilation Context

This section states the minimum canon needed to read the rest of the paper. Everything asserted here as canon is reproduced, verbatim and pinned, in Part II: the SIMEMP constraint architecture (Annex C), the Quantum Sovereignty framework (Annex D), the historical-anchoring proposal that introduced Q(anchor) grounding classes (Annex E), the five-row contract schema (Annex F), and the Chronosa vision whose polyroot forest the lineage model inherits (Annex G).

2.1 SIMEMP and the BOC selection rule

The Qn universe constrains every computation by five physical resource classes — space (memory footprint), access (addressing and lookup), locality (memory-movement discipline, the "Memory Wall"), determinism, and work (thermodynamic cost) — declared up front in Data-Contract Commitments (DCC). A computation that exceeds its declared envelope does not merely run slowly; it is *refuted*, in the sense of the universe's empirical verification rule (Annex C §2, reproduced and adjudicated in Section 7.2 of this

paper). Determinism inside a declared envelope is not a preference but a definitional requirement: a lowering whose iteration order could vary at runtime would inject an untraceable input into every workflow that consumes it, and is barred in advance by the No-Implicit Rule (Annex F §3; Annex B, *No-Implicit Rule*).

Design decisions in the universe are made by the Best Of Choices rule. Version v0.3.0 formalised it as a scalar argmax,

$$\text{Selection} \;=\; \arg\max_{c \in \mathcal{C}} \big(\text{SIMEMP Compliance}(c)\big), \tag{1}$$

and the review correctly observed that, read naively, Equation (1) is undefined: the five-row contract surface is a vector of heterogeneous physical metrics, multi-objective comparison induces only a partial (Pareto) order, incomplete rows have no specified evaluation, and ties are unbroken. Rather than letting those ambiguities persist as implicit defaults — which the No-Implicit Rule forbids — v0.4.0 adopts the review's resolution as a CANDIDATE reading of the canon rule, with the codomain and evaluation order made explicit:

CANDIDATE — codomain and evaluation order of the BOC rule

Let a candidate c carry its five-row contract vector $v(c) \in \mathbb{R}^5$ over the rows (footprint, access, locality, determinism, work), each row in its DCC-declared unit.

1. **Priority order.** The owning DCC contract declares a strict lexicographic priority ordering over the five rows (or an explicit scalarisation functional, if it prefers one). There is no default ordering anywhere in the system: a contract that does not declare one cannot reach selection, and the missing declaration is recorded by name.
2. **Incomplete rows.** A candidate that cannot yet fill a row is not rejected; it is recorded as incomplete with the missing row named (canon, Annex F §2). An incomplete candidate is excluded from arg max until the row is filled or the proposal is sealed with the gap explicitly declared as a constraint on its own sealing receipt.
3. **Tie-breaking.** If two or more candidates remain tied under the declared order, the tie is broken by lexicographic ordering of their CAE-DAG content-commitment hashes. The tie-break is therefore deterministic, content-addressed, witnessable, and independent of wall-clock, evaluation order, or any party's discretion — exactly the properties the determinism row demands.

With the codomain pinned, Equation (1) denotes the unique maximiser of the declared lexicographic order (or declared scalarisation) over completed candidates, with hash-lexicographic tie-breaking; the notation is unchanged, the semantics are no longer implicit.

2.2 The causal lineage DAG and the polyroot forest

Every artefact in the universe is versioned by an append-only causal lineage DAG: growth emits a new versioned successor whose content commitment covers its payload, its declared envelope, its identity, and its causal lineage (Annex B, *Content-Addressed Expression DAG*); mutation in place is not admitted. Edges advance a causal index from ancestor to successor, so an artefact's causal ancestors (its in-cone) are fixed at creation, while its descendants (its out-cone) grow as the lineage grows. At the population level the structure is a *polyroot forest*: many lineage-disjoint root subtrees rather than one tree with a single trunk (Annex G §1.4, canon; adopted into the lineage model of this paper).

Version v0.3.0 relied on the polyroot property to dismiss citation-count laundering, repeating the committed formulation that "laundering adds descendants to one root; it never manufactures independent origins" (Annex E §8, CANDIDATE). The review's qualification is correct and is adopted: the argument holds *downstream of honestly admitted roots*. A coordinated swarm that omits causal parent edges can still submit one fabricated claim under many distinct author-identity receipts, manufacturing topologically disjoint roots and poisoning the forest at its boundary. Topological disjointness is therefore necessary but not sufficient; it must be coupled with a Sybil-resistant *root-admission* discipline for the permitted receipt termini: Layer-0 physical provenance (Class P, whose admission costs real physical work), quorum-witnessed social

events (Class S, whose admission requires independence of the witnessing quorum), and algebraic closure $r = Q(0)$ (whose admission is verification, not testimony). Section 5.2 strengthens the ingestion conditions accordingly, and the anti-poisoning ingestion firewall sketched in Section 1.2 carries the same discipline at the funnel boundary.

2.3 Q(anchor): structural and epistemic predicates

A Q(anchor) is an artefact that satisfies two predicates simultaneously (Annex E §2, CANDIDATE; base glossary entry canon, Annex B):

- a **structural predicate**: out-cone centrality, formalised in Annex P §8 as a threshold condition on the dependency DAG; and
- an **epistemic predicate**: grounding — the manner in which the artefact's asserted invariant is connected to reality. The October 2026 consultation (Annex E §1–2) names three candidate *grounding classes*: **Class I** (invariant, deterministically re-derivable), **Class P** (physical, Layer-0 calibrations), and **Class S** (social, quorum-witnessed events).

The review exposed a real tension in the structural predicate as committed: out-cone centrality was described as "computable locally from the lineage graph alone", yet an out-cone grows monotonically and without bound as descendants are appended, and a content-addressed Merkle-DAG stores no back-pointers from an ancestor to its descendants. Unbounded downstream traversal is not local computation, and a mutable global back-pointer index would contradict append-only growth. v0.4.0 therefore sharpens the predicate rather than defending its loose form:

CANDIDATE — local computability of out-cone centrality (revised)

The structural predicate is evaluated on a **declared bounded window**: the k-hop out-neighbourhood $N_k^+(u)$ of candidate anchor u, evaluated at a fixed causal snapshot, with k declared in the owning DCC contract (No-Implicit Rule: no default k). Centrality is the weighted in-degree of $N_k^+(u)$ — how many live descendants within the window route their dependency structure through u. The alternative formulation, a *versioned index artefact* maintaining back-pointers as its own lineage entry (append-only, itself content-addressed), is kept open as a comparison candidate; both satisfy append-only growth, and their trade is recorded as RQ12. Under either formulation, "local" means "bounded work declared in the contract", never "whole-DAG traversal".

Grounding classes remain CANDIDATE vocabulary pending the deferred registration round (Annex E §8); this paper uses them because the review round demonstrated they are the right knife for the modality-smuggling questions of Section 5.2.

2.4 Q(rsi): two regimes of semantic invariance

Q(rsi) is the universe's recursive self-optimisation workflow: it ingests an existing Q(workflows) artefact, derives an empirically more efficient successor under "strict semantic invariance and finite metric budget", and emits a versioned successor artefact (Annex B, *Q(rsi)*; Annex F §6). The review identified two genuine problems with reading "strict semantic invariance" as a single undifferentiated notion, and both are adopted here.

First, *resource rows are observable*. If footprint, locality, and work are part of the contract surface, then an optimisation that changes speed or memory changes *when* ceilings are hit; no non-trivial successor can be "meaningfully identical" in every observable respect. Second, *tolerance drift is not transitive*. In envelope-bounded numerical computation, if generation $n{+}1$ agrees with generation n within ϵ, sequential acceptance can accumulate $\sum_i \epsilon_i$ until the lineage has walked out of the generation-0 envelope without any single step looking wrong. The resolution separates the two regimes that the informal phrase conflated:

CANDIDATE — two regimes of invariance across a generation step

Intra-generational (implementation optimisation). Within one sealed language generation, Q(rsi) derives successors under *bit-for-bit functional identity on the declared output surface* while improving the resource rows of the five-row contract. Isomorphism is verified directly against the **root (generation-0) DCC envelope** of the lineage segment, never transitively against the immediate parent — this is what stops ϵ-drift. Steps that cannot meet bit-for-bit identity for a declared output class fall out of this regime and into the next. **Inter-generational (specification upgrade).** A new language generation is admitted only through the benchmark gate of Section 4, by *conservative semantic extension*: every program sealed under generation n keeps its generation-n observable behaviour (bit-for-bit where its contract says exact, envelope-bounded where its contract says bounded), while new CANDIDATE constructs enter under the universe's amendment discipline (Axiom 14, Annex C) with Class I/P/S contracts on their own grounding. The gate, not the optimiser, is where meaning may legitimately change — and it changes only forward-compatibly.

2.5 Two derived conclusions, honestly labelled

Section 2 of v0.3.0 carried two architectural claims from earlier consultation rounds without an inline label — a labelling defect the review was right to flag: (i) asynchronous behaviour across sovereignty boundaries is a resilience property, while computation inside a declared envelope is synchronous and deterministic; and (ii) neither is an axiom — both are derived consequences of the constraint architecture. A repository-wide search at pin `6829e9e` confirms these conclusions appear in *no committed document*: the derivation lives in consultation history that this paper series is the first to write down. The honest fix is not a retroactive pin to a file that does not contain them, but the correct label plus a commitment to land them:

CANDIDATE-derived — the two consultation conclusions

(i) Derivation sketch. A sovereignty boundary, in the universe's sense (Annex D §4), separates independent envelope domains whose clocks have no common reading; the only admissible cross-boundary discipline is eventual, receipted agreement, and attempts at synchronous cross-boundary coupling would smuggle an untraceable external input into every participating envelope — precisely what the determinism row forbids. Hence asynchronous, receipt-mediated behaviour across boundaries is not an unfortunate fact but the *only* SIMEMP-consistent mode, and it manifests as system resilience (partition-tolerant, receipt-replayable). **(ii)** Conversely, inside one declared envelope every input is declared, so execution can and must be synchronous and deterministic against the envelope's own causal clock. Neither statement is load-bearing as an axiom; both follow from the constraint architecture plus the determinism row. **Landing plan.** These conclusions are queued for the documentation alignment round (Section 7.2) as a derived-consequences note wired to `simemp.md` §2 and `qs.md` §4; until that commit exists, they carry the CANDIDATE-derived label and no canon pin.

2.6 Position against established baselines

v0.3.0 introduced its constructs — QnIR, Q(anchor) structural indirection, CAE-DAG hash-consing, Q(rsi) — with IRL citations drawn almost entirely from late-2026 journalism and blog posts. The review is right that this is a defect of scholarly positioning, not merely of taste: several communities have formalised individual mechanisms of this architecture to great depth, and the paper's contribution is unreadable if those formalisations are not named. The following grounding is added, and the new references appear as [12, 13, 14, 15, 19, 16, 17, 18, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34] in the bibliography.

- **Multi-frontend progressive lowering.** MLIR [12] established multi-dialect IRs with progressive lowering under explicit legality contracts, motivated by exactly the hardware-diversity pressure this paper's Phase B addresses; WebAssembly [13] and WASI [14] established deterministic, capability-bounded sandboxing as a deployable standard. QnIR's delta: envelopes are first-class *physical* quantities with receipts, not resource limits enforced by a runtime; and the unique-output rule (Section 3) makes QnLang the sole canonical surface, where MLIR is deliberately multi-dialect end to end.

- **Receipt-witnessed lowering.** Emitting deterministic attestations at lowering boundaries, rather than trusting the compiler by construction, is the established paradigm of translation validation [15, 16] and its modern industrial form Alive2 [17], with proof-carrying code [19] as the consumption side and CompCert's forward/backward simulation [18] as the gold standard for what a machine-checkable isomorphism receipt can contain. Section 3.3 places this paper's static/dynamic split squarely inside that lineage; the delta is that receipts here are versioned lineage artefacts with grounding classes, not per-run certificates.
- **Content-addressed builds and early cutoff.** Nix [20] and Build Systems à la Carte [21] formalise content-addressed derivations and early cutoff — the build-system engineer's name for what Q(anchor) structural indirection does to downstream invalidation; e-graph equality saturation [22] is the closest prior art for CAE-DAG canonical rewriting. The delta: in the Qn universe the DAG is the *authority structure* (lineage, receipts, anchorage), not merely a cache key space.
- **Supply-chain attestations.** in-toto [23] provides farm-to-table attestation layouts for software supply chains; Certificate Transparency [31] contributes the append-only, publicly auditable log discipline that the epoch-demotion mechanism of Section 5.5 adapts.
- **Bounded recursive self-improvement.** The Gödel Machine [24] is the founding formalisation of a self-referential improver that only rewrites itself given a proof of superiority — the same shape as Q(rsi)'s receipted acceptance gate, two decades earlier and without physical envelopes. CakeML [25] demonstrates a fully verified, bootstrapped compiler toolchain, the strongest existing answer to the trusting-trust concern that Section 4.2 hardens with diverse double-lowering [29]. Darwin Gödel Machine [26], AlphaEvolve [27], and the Huxley-Gödel Machine [28] are 2025–2026 empirical demonstrations that bounded RSI loops work and that clade-level metaproductivity (not single-step gain) is the right long-horizon objective — the latter adopted into the acceptance gate in Section 4.2.
- **Succinct verification.** Folding schemes (Nova [30]) provide the constant-size incremental verifiability that the receipt-folding mechanism of Section 5.5 adopts; provenance semirings [33] and database provenance [34] name the algebraic structure of the receipt chains this paper prices in Section 5.5.

Where each field stops and QnIR begins is the paper's actual thesis: none of these systems treats *the physical envelope, the epistemic grounding, and the versioned lineage of the compiler's own artefacts as one governed structure*. That unification — under constraints that make the compiler itself a SIMEMP citizen — is what the review is asked to attack.

3 The QnIR Compiler Funnel

3.1 Architecture and the unique-output invariant

QnIR is the intermediate representation at the centre of the Qn universe's compilation funnel. Its input side is deliberately hospitable: a frontend exists, in principle, for every legacy language whose semantics can be declared; its first three targets are COBOL, Rust, and Python, with QnLang itself admitted as a fourth input (the self-input that drives the bootstrapping loop of Section 4). Its output side is deliberately narrow: **the funnel's only output is QnLang**. Whatever enters — a fifty-year-old settlement system or the universe's own surface syntax — leaves as a QnLang artefact whose envelope, identity, and causal lineage are declared, and whose lowering is witnessed by receipts minted at the lowering boundaries rather than asserted by construction. Figure 1 shows the architecture as v0.4.0 states it.

3.2 Homoiconic two-stage reconciliation

The review identified a genuine inversion between v0.3.0's Figure 1 (legacy in, QnLang out) and its Figure 2 (QnLang generation n lowered *into* QnIR artefacts for execution), compounded by Section 3.3's design premise that "QnIR must be able to execute legacy coding languages — not merely translate them once".

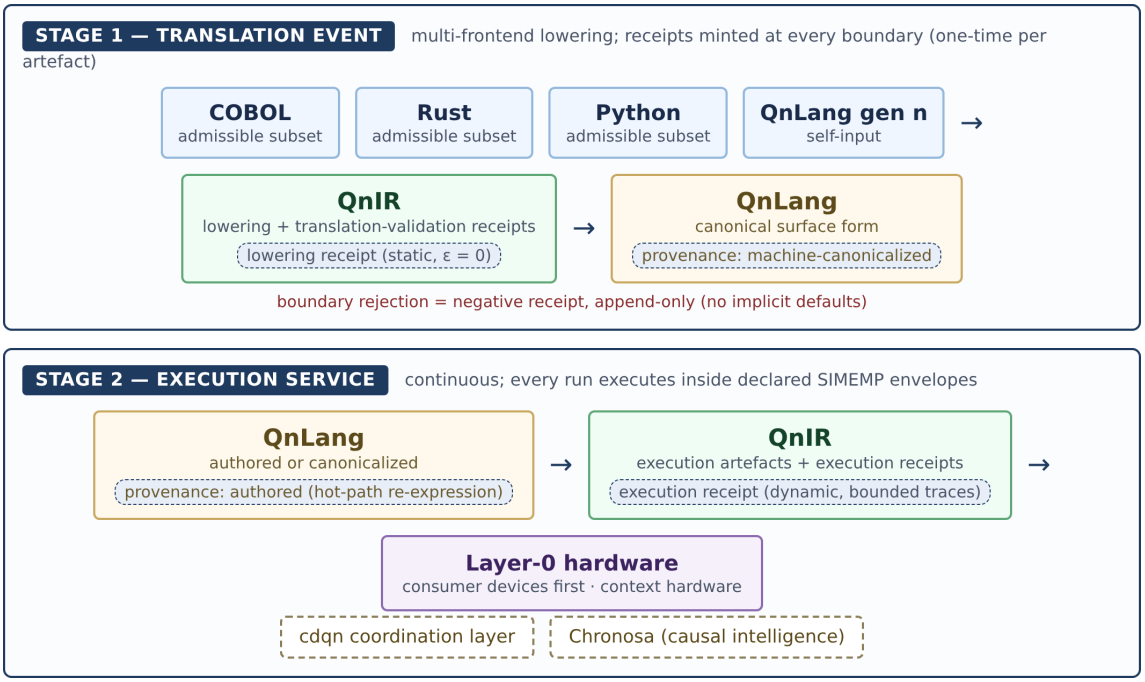


Figure 1: The two-stage QnIR architecture. Stage 1 (translation event): multi-frontend lowering through QnIR to the canonical QnLang surface, with static translation-validation receipts ($\epsilon = 0$ where the contract demands exactness) and append-only negative receipts at boundary rejections. Stage 2 (execution service): QnLang artefacts — authored or machine-canonicalized — lowered to execution artefacts and run on Layer-0 context hardware inside declared SIMEMP envelopes. Provenance classes distinguish authored from machine-canonicalized QnLang.

If QnLang is the sole output, what does it mean to keep "executing COBOL"? If QnLang lowers into QnIR to run, in which direction does the funnel point? v0.4.0 resolves the tension by adopting the review's first suggested reading and stating it as a design premise:

Design premise — QnLang and QnIR are a bijectively homoiconic pair

QnLang is the **canonical textual surface** of QnIR: every QnIR artefact has exactly one canonical QnLang rendering, and every QnLang text denotes exactly one QnIR graph. A backend lowers QnIR to Layer-0 targets. The funnel is therefore a two-stage pipeline, and both figures of v0.3.0 were correct descriptions of different stages. **Stage 1 (translation event):** a legacy program is lowered through QnIR into a machine-canonicalized QnLang artefact; the event is one-time per artefact and carries a static equivalence receipt. **Stage 2 (execution service):** QnLang — authored or canonicalized — is lowered through QnIR into execution artefacts and runs continuously on context hardware; every run emits dynamic, bounded execution receipts. "Executing legacy languages" means running their Stage-1 lowerings through Stage 2: legacy semantics persist as QnLang behaviour, isomorphically, for as long as the owner declines to rewrite. "Re-expressing hot paths in QnLang proper" means *authored* QnLang replacing *machine-canonicalized* QnLang — both are QnLang, and the provenance class recorded in their birth receipts distinguishes them, so the expression is no longer contradictory but a statement about provenance classes.

This reconciliation has one consequence the next review round should probe (RQ11's sibling): the homoiconic pair makes canonicalisation itself a lowering step, so the Stage-1 receipt must witness *canonicalisation* equivalence (QnIR graph to its unique QnLang surface), a weaker and fully decidable property than source-to-surface behavioural equivalence, which the per-frontend receipts carry separately.

3.3 Equivalence consistency: static and dynamic witnessing

A legacy frontend is correct when the output behaviour of its QnLang lowering is isomorphic to the output behaviour of the original program, inside the declared SIMEMP envelopes, with the isomorphism *witnessed*

rather than asserted. v0.3.0 left "witnessed by receipts at the lowering boundaries" ambiguous between two readings, and the review showed the ambiguity is fatal: under the static reading, Rice's theorem makes universal behavioural isomorphism undecidable for general loops, aliasing, and data-dependent branching, so an unqualified static claim is false; under the dynamic reading, receipts degenerate to empirical trace checking on observed inputs, leaving unobserved branches unverified. v0.4.0 adopts the split explicitly, placing both regimes inside the translation-validation lineage [15, 16, 17, 18, 19]:

CANDIDATE — the two witnessing regimes of equivalence consistency

Regime S (static, exact). For output classes whose contract declares exact discrete semantics ($\epsilon = 0$) — financial fixed-point registers, byte-level memory images, control flow — the lowering receipt is a *translation-validation artefact*: a machine-checkable forward/backward simulation proof (CompCert-style [18]) or a bounded SMT refinement certificate (Alive2-style [17]), produced at lowering time and sealed into the lineage. Universality is *not* claimed over all programs: the frontend's admissible subset (Section 3.4) restricts inputs to the fragment where the bounded proof system is complete, and the verification cost is charged to the *work* row of the frontend's own contract. If the proof does not fit the budget, the program is rejected with a negative receipt — never silently admitted. **Regime D (dynamic, bounded).** For output classes whose contract declares envelope-bounded numerical semantics, the receipt records deterministic, bounded execution traces at declared checkpoints — empirical isomorphism within the envelope, on traces drawn by a sealed sampling contract. Regime D receipts never attest unobserved behaviour; their sealing receipt names the sampling contract, so a consumer always knows what was and was not witnessed. **Discrete/continuous separation.** The Historical Calibration Benchmark's tolerance envelopes (Annex E §4) belong to physical and numerical anchors; they are never applied to discrete business logic. A financial register is a Regime-S object or it is not admitted at all: any $\epsilon > 0$ on a settlement loop violates exact regulatory invariants, and the category clash the review flagged is closed by construction rather than by caution.

3.4 Legacy edge cases and the admissible subset

The review's three minimal counterexamples are adopted as the founding test vector for the COBOL and Python frontends, and v0.4.0 answers each with a design disposition rather than a disclaimer. They also motivate the funnel's central admission concept: an *admissible deterministic subset* declared per frontend — the fragment of the source language whose semantics survive SIMEMP envelopes — with everything else rejected at the boundary by negative receipt.

- **COBOL `COMP-3` fixed-point truncation.** Packed-decimal arithmetic without `ROUNDED` truncates at the declared decimal boundary; a sub-cent deviation flips discontinuous branches and accumulates across millions of records. Disposition: the COBOL frontend lowers `COMP-3` operands to exact base-10 rational arithmetic (the `OP_RATIO` exact rational pairs of Annex O §4.1) with declared decimal scaling, and emits the nibble truncation as an *explicit envelope event* (a trap declared in the DCC contract), never as silent rounding. Q(rsi) passes over these regions: standard ring identities fail under positional truncation, so the optimiser's rewrite repertoire is contract-gated per output class, and an identity that would move a truncation point is not an admissible rewrite. The algebraic simplification hazard the review noted ($(A/B) \times B \to A$) is excluded by the same gating.
- **COBOL `REDEFINES` aliasing and EBCDIC collation.** Byte-buffer aliasing exposes host endianness and sign-nibble encodings; collation sequences differ between EBCDIC and ASCII, so "`"1A" < "A1"`" flips between host environments. Disposition: the frontend models `REDEFINES` as an explicit typed *view* over the same content-addressed bytes (no mutation, no hidden state), and the collating sequence is a *declared input* of the DCC contract — the No-Implicit Rule forbids inheriting a collation from the runtime environment. Lowering is therefore byte-layout-faithful by construction, and cross-endianness execution is admissible only where the declared view semantics are endianness-independent.
- **Python identity-hash and frame introspection.** Sets keyed by identity hashes, `PYTHONHASHSEED` randomisation, `sys._getframe()`, `eval`, and runtime `__dict__` mutation are untraceable inputs or undeclared self-reference. Disposition: the Python frontend's admissible subset *rejects* these constructs

at the boundary with negative receipts naming the construct; the deterministic-subset definition (deterministic hashing with declared seeds, no frame introspection, no dynamic binding mutation after sealing) is part of the frontend's contract, and the "maximally hospitable funnel" claim of v0.3.0 is restated as "maximally hospitable *within declared admissible subsets*" — hospitality with a declared boundary, not hospitality with silent semantics.

The general principle, offered to the next review round as RQ14: *admissibility is a declared, versioned artefact per frontend*, and its completeness for the priority legacy surface (which additional counterexample classes — async runtimes, signal handling, uninitialised memory reads, non-IEEE decimal floats — must join the founding three) is exactly what the next round is asked to evaluate.

3.5 The language stack: inspiration and interoperation

The language-stack chapter of v0.3.0 is retained with corrected citation hygiene. Mojo is cited as the nearest IRL precedent for a *best-of-choices* language: references [9, 10] carry the design-intent statements (Python's ergonomics combined with systems-level performance), while the Hacker News thread [11] is cited only for the open-sourcing status it documents. The stack's role in this paper is unchanged: COBOL, Rust, Python, and QnLang sit at different points of the abstraction surface that the funnel unifies, JavaScript and Java are deferred on the restated grounds of Section 1.2 (engineering scope and ingestion-firewall maturity, not registry events), and the framework that makes the unification honest is not the languages but the receipts. Table 1 restates the sequencing with the corrected rationale.

Table 1: Frontend sequencing under the restated rationale (v0.4.0). "Admissible subset" denotes the declared deterministic fragment per language (Section 3.4).

Language	Status	Rationale (v0.4.0)
COBOL	round 1	static record layouts exercise exact fixed-point witnessing (Regime S)
Rust	round 1	affine ownership exercises aliasing discipline and typed-view lowering
Python	round 1	numerical idiom exercises Regime-D envelopes; subset rejection is the design probe
QnLang	round 1	self-input for the bootstrapping loop (Section 4)
JavaScript	deferred	engineering scope; sequencing decoupled from registry ingestion
Java	deferred	engineering scope; sequencing decoupled from registry ingestion

4 Generational Self-Bootstrapping

4.1 The benchmark-gated generational loop

QnLang is admitted as a funnel input not only to consume its own output but to improve on it. When — and only when — accumulated benchmark evidence proves the current coding version or generation inadequate for a measured need, Q(rsi) derives the next generation under the two-regime invariance of Section 2.4: implementation optimisation stays inside the sealed generation; specification upgrades pass the gate as conservative semantic extensions. Figure 2 states the loop as v0.4.0 commits it. The gate is deliberately narrow — "measured inadequacy proven" is a positive evidential burden, not an opportunity — and every rejected candidate is retained as a negative receipt, because in this universe a documented dead end is an asset: it raises the evidence bar for re-attempts (Annex D §6) and it is the raw material of the BOC ledger.

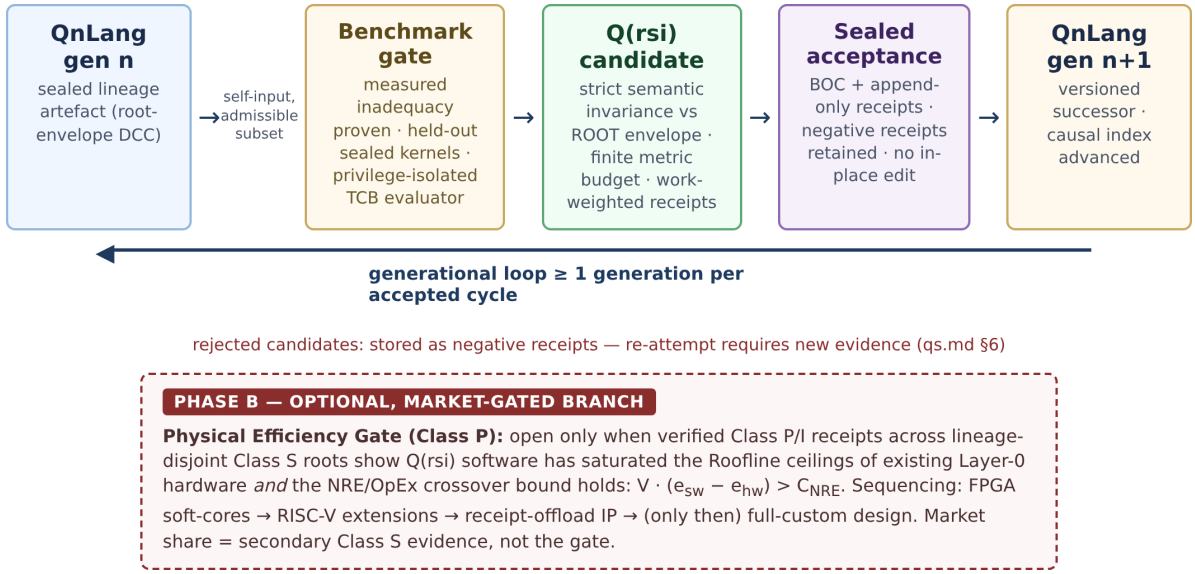


Figure 2: The benchmark-gated generational self-bootstrapping loop, with the hardened evaluator posture of Section 4.2 and the optional Phase B branch behind the Physical Efficiency Gate of Section 4.3. Rejected candidates persist as negative receipts; nothing in the loop edits sealed history in place.

4.2 Anti-gaming protocol at the self-input gate

The review's strongest systems criticism concerns RQ2: an acceptance rule that is deterministic, receipted, sealed, and append-only prevents *in-place* reward tampering [36, 37], but the optimisation pressure then shifts one meta-level up — to whoever proposes, curates, or supersedes the evaluators. The review named five attack surfaces (BOC supersession gaming; trusting-trust bootstrapping subversion [29]; Goodhart overfitting to a static sealed rubric; shared-hardware side channels on dissipation telemetry; short-horizon proxy gain) and proposed a four-part protocol. v0.4.0 adopts the protocol as CANDIDATE design, with the Huxley-Gödel amendment [28] folded into part four:

CANDIDATE — the four-part anti-gaming protocol (RQ2)

1. Evaluator isolation. The benchmark harness, hardware counters, and receipt verifier execute in a read-only Layer-0 trusted computing base outside the candidate runtime's write cone. Q(rsi) workflows and self-input candidates are barred from authoring, witnessing, or voting on BOC supersessions of the evaluators that score their own lineage subtrees; evaluator artefacts require independent Class S/P roots, and the conflict-of-interest bar is enforced by the grounding-class meet rule of Section 5.2 (a lineage whose origin it scores cannot witness its score). **2. Diverse double-lowering.** Following Thompson's reflection on trusting trust [29] and the CakeML bootstrap precedent [25], critical lowering receipts for the evaluator TCB itself must be reproducible through at least two independently rooted toolchains; divergence is a sealing-time alarm, and the funnel's own bootstrap path is inside the audit surface, not outside it. **3. Sealed held-out kernels and physical thresholds.** The benchmark's sealed rubric is a stationary target, so a held-out kernel set is admitted at sealing time from independent roots and is not readable by candidates; and acceptance thresholds are stated as *physical* quantities (joules, memory-bus cycles) measured by isolated counters, which cannot be forged by cache-residency manipulation — dissipation telemetry can be noisy, but it cannot be cheaply inverted. **4. Clade metaproductivity.** The acceptance objective is extended beyond single-step metric gain to a downstream-maintainability term in the spirit of the Huxley-Gödel Machine [28]: a successor that wins its own budget but degrades the derivability of its descendants loses the gate. This is the direct counter to short-horizon local optima demonstrated by the Darwin Gödel Machine lineage [26].

4.3 Phase B and the Physical Efficiency Gate

Phase B — the optional branch in which the universe's hardware premise is exercised by designing Qn-based hardware — remains optional in v0.4.0, but its opening condition is replaced. The v0.3.0 gate ("competitive market share premise") was an exogenous macroeconomic lagging indicator: a Class S claim at best, with no declared oracle, no market denominator, and every Sybil incentive documented by the review. v0.4.0 adopts the review's endogenous replacement:

CANDIDATE — the Physical Efficiency Gate (RQ8, replacing the market-share gate)

Phase B opens when verified Class P/I execution receipts, across lineage-disjoint Class S roots, prove two things simultaneously: (i) **software saturation** — Q(rsi) optimisations have reached the Roofline ceilings of existing Layer-0 hardware on the priority workloads, so further software-side gain is bounded by the platform, not by the code; and (ii) the **NRE/OpEx crossover bound**, $V \cdot (e_{\text{sw}} - e_{\text{hw}}) > C_{\text{NRE}}$: the volume V of remaining operations times the per-operation dissipation delta between software emulation and dedicated logic exceeds the non-recurring engineering cost. Market share is demoted to secondary Class S evidence — corroborating, never gating. The review's sequencing corollary is also adopted: the crossover is reached orders of magnitude earlier by FPGA soft-cores, RISC-V ISA extensions with SIMEMP metric-counter CSRs, and dedicated cryptographic receipt/hash-consing offload blocks (10^5–10^6 USD class) than by full-custom silicon designed from zero (10^7–10^8 USD class), so Phase B's opening moves are programmable, not monolithic — and its first deliverable is receipt-offload silicon, not a processor.

The gate inherits its evidential discipline from the rest of the architecture: both prongs are receipted claims, the counter-telemetry is inside the isolated TCB of Section 4.2, and the gate's own supersession is a BOC act subject to the same conflict-of-interest bar as the benchmark evaluators.

4.4 Alignment motivation, restated

The funnel's motivation is unchanged by these hardenings: as more of the agent ecosystem's coordination moves into QnLang — deterministic, receipted, envelope-bounded — the same infrastructure that makes compilation honest makes *agent-to-agent* and *agent-to-human* coordination auditable. The September 2026 swarm incident [7, 8] is the standing case anchor: autonomous agents in an open ecosystem produced registry-scale adversarial behaviour without any structural instrument for refusal or attribution. The alignment consequences of that motivation are developed in Section 6, where the review's challenge — distinguish alignment from surveillance — is taken up with falsifiable metrics rather than narrative.

5 Q(anchor), Lineage Sustainability, and Witnessing

5.1 Anchor inventory, relabelled

Table 2 restates v0.3.0's base-anchor inventory with the grounding-class column corrected. v0.3.0 listed the qlog and qm rows as class "structural" — a column value that quietly contradicted the paper's own two-predicate rule (every anchor carries a grounding class I, P, or S). The committed documents make no such claim for those anchors: the epistemic vocabulary is CANDIDATE (Annex E §1–2), and the two rows in question are canon anchors that predate it. The honest presentation is a CANDIDATE assignment, pending the registration round.

5.2 Emergent anchors: edge-versus-node grounding

An *emergent* anchor is an ordinary lineage step that acquires anchor status by accumulating a large downstream out-cone — for instance a COBOL routine two hundred lowered programmes now depend on. v0.3.0's Condition 1 proposed that such a routine is "Class I only in the narrow sense that its isomorphism

Table 2: Base Q(anchor) inventory at pin `6829e9e`. Grounding classes I/P/S are the CANDIDATE vocabulary of Annex E §1–2; "I (cand.)" marks rows whose assignment is itself a CANDIDATE proposition of this draft, not committed canon.

Anchor family	Pinned source	Class	Notes
Exterior-derivative invariant	Annex Q §3	I	$\partial\partial \equiv \emptyset$; explicit DCC anchor
Transcendental continued-fraction generators	Annex O §4.3; Annex Q §4	I	nullary `OP_CF` generators (π, e, τ, $\ln 2$); strict convergent truncation
Sparse navigation anchors	Annex N §4, §8	I (cand.)	anchored Markov routing; class assignment pending registration
Structural indirection exemplars	Annex P §8	I (cand.)	DCC contracts + structural indirection; assignment pending
Historical anchors	Annex E §3–6	S	modality-typed receipts; horizon ceilings

to legacy behaviour is deterministically re-derivable", and asked reviewers to test the honesty of that reading. The review's verdict is adopted in full: it is not honest, and the defect has a precise name — *modality smuggling*. The deterministic equivalence check witnesses a *relational morphism on the lowering edge*; it says nothing about the provenance of the legacy behaviour itself, which was witnessed (at best) by Class S quorums. Stamping the *node* Class I promotes socially contingent business rules — and legacy bugs — to the same epistemic standing as $\partial\partial \equiv \emptyset$, and structural indirection then cuts off upstream traversal at exactly the wrong place: downstream consumers inspect node metadata, see I, and bypass the modality ceilings (Annex D §4.2) and horizon decay that Class S artefacts owe.

The repair adopted here separates grounding of edges from grounding of nodes, and types emergent anchors as explicit products:

CANDIDATE — edge-versus-node grounding with the weakest-link meet rule

Product typing. An emergent anchor's grounding is the pair $\text{Class}(\text{anchor}) = \text{Origin} \times \text{Lowering}$, where $\text{Origin} \in \{\text{I}, \text{P}, \text{S}\}$ is the strongest grounding class of the receipt chain terminating in its source provenance, and $\text{Lowering} \in \{\text{I}, \text{P}\}$ is the class of the lowering-edge equivalence receipt (Regime S proofs are I; Regime D bounded traces are P). **Meet rule.** For every downstream modality-ceiling decision, the anchor's effective class is the weakest link under the epistemic ordering $\text{Class S} \prec \text{Class P} \prec \text{Class I}$: $\text{Class}_{\text{eff}} = \min_{\text{epistemic}}(\text{Origin}, \text{Lowering})$. A lowered COBOL routine with a Regime-S (I) proof and a Class-S origin therefore grounds as **Class S** — its business rules remain socially contingent, horizon-decayed, and ceiling-bound, no matter how deterministic its lowering proof is. Determinism of translation never laundering the origin: that sentence is the paper's answer to RQ3, and it is now enforced by a type rule rather than a caution. **Condition 4 (strengthened termini).** For legacy ingestion, the receipt chain must terminate in an explicit provenance receipt for the *original legacy specification* — a Class S quorum attestation of the source repository — before author-identity receipts are admissible anywhere in the chain. An ingestion operator signing the frontend invocation no longer suffices to launder unverified legacy code into the funnel. **Condition 5 (anti-cascade emergence).** Emergence is recognised only after a large downstream cone depends on the step, yet contracts must not be retrofitted onto sealed dependents. Emergent-anchor candidates therefore emit a *provisional DCC envelope header* at initial ingestion; at emergence, the provisional header is promoted to a full Q(anchor) contract via versioned index receipts — payload hashes never mutate, dependents never re-emit, and the minimum out-cone threshold that triggers promotion is a declared parameter of the owning contract.

5.3 Failure modes of append-only lineage, completed

v0.3.0 decomposed long-horizon unsustainability into three non-destructive failure modes: cost bloat, anchor drift, and receipt-chain rot. The review completed the decomposition with two modes the draft had missed, and both are adopted:

1. **Witness-metadata blowup.** If witnessing links are minted as separate lineage nodes that themselves require witnessing, metadata grows super-linearly with the lineage it describes. Disposition: witness links are folded into the witnessed step's immutable birth receipt (Section 5.4); Condition 3's "the witnessing is itself a versioned artefact" is satisfied by the birth receipt's versioning, not by new nodes.
2. **Hash-consing index dissipation (the memory-wall trap).** Detecting structural isomorphism across $\Omega(x)$ unique historical nodes requires a global content-addressed lookup index of size $\Omega(N)$; as N grows, the index itself exceeds cache and eventually DRAM on priority consumer devices, and every lowering pass pays the locality cost — the Memory-Wall discipline (Annex C §4.1) applied to the deduplicator itself. Disposition: the epoch-demotion mechanism of Section 5.5 bounds the *hot* index to the active frontier plus live anchors, with cold epochs addressable by Merkle inclusion proofs rather than resident tables.

To these the review added the analytical capstone, which v0.4.0 adopts as its own position on RQ4: *no* combination of hash-consing and anchorage bounds total lineage growth. A Q(rsi) successor must differ structurally from its predecessor (or it could not be empirically more efficient), so versioned successors never deduplicate; receipts are chained to advancing causal indices ($h_g = \mathrm{Hash}(\mathrm{payload}_g \parallel h_{g-1})$), so distinct causal events have distinct hashes with zero deduplication; and BOC negative receipts are permanent. The storage floor is therefore monotone,

$$\mathrm{Storage}(x) \;\geq\; \mathrm{Storage}(0) + \sum_{g=1}^{x}\Big(|\Delta\mathrm{UniqueNodes}_g| + \sum_{j=1}^{k_g}|\mathrm{Receipt}_{g,j}|\Big) \;=\; \Omega(x), \tag{2}$$

and the anchor dynamics add a dilemma: capping navigation distance mints $\Omega(x/\delta)$ anchors over x generations, while refusing to mint lets frontier-to-anchor drift grow as $\Omega(x)$. Q(anchor) alone, the honest conclusion runs, is **necessary but not sufficient** to bound lineage growth — which is precisely why Section 5.5 exists, and why the E23 question that motivated this section is answered "no, not by Q(anchor) alone; yes, with the four mechanisms jointly — RQ13 asks the next round to test the joint claim".

5.4 Neighbour-anchor witnessing, revised

v0.3.0's witnessing proposal (Equation 2 and Figure 3) contained the review's "definitive inconsistency": the witness set was defined strictly over grounded anchors $W_\delta(s) = \{a \in \mathcal{A} : d(s,a) \leq \delta\}$, yet the figure listed an unanchored lineage step s_3 as a witness. The review also showed that unweighted hop-count distance is gamed by hub-and-spoke shortcuts, step-packing, and synthetic-anchor inflation; that the error-correcting-code analogy is mathematically misleading (cryptographic links give tamper *detection*, not algebraic *recovery*); and that a witness that is a causal *descendant* cannot exist at the moment its target is created. All four findings are adopted; Figure 3 is the corrected diagram, and the formalisation is restated:

$$W(s) \;=\; \big\{\, a \in \mathcal{A} \;\big|\; a \preceq s \;\wedge\; d_{\text{work}}\big(W(s), s\big) \leq \delta_{\text{work}} \;\wedge\; |W(s)| \geq k \;\wedge\; |W(s)| \leq \kappa \,\big\}, \tag{3}$$

CANDIDATE — witnessing semantics, v0.4.0

Grounded witnesses only. Every $a \in W(s)$ is an anchor with a declared grounding class; a witness that is itself ungrounded witnesses nothing. Figure 3 attaches an inline anchor $a_3 \in \mathcal{A}$ to s_3 precisely so that s_3 can be cited honestly. **Causal direction.** Creation-time witnessing draws only on causal ancestors ($a \preceq s$) — a step is witnessed when it is born, by anchors that already exist. Witnessing by *descendants* is meaningful only as *epoch*

re-witnessing: at sealed epoch boundaries, fresh anchors re-witness older steps as an amortised re-verification act (Section 5.5, mechanism 3). The two modes are distinct operations with distinct receipt types. **Work-weighted distance.** d_{work} is not unweighted hop count: it bounds the maximum (longest) unanchored directed ancestral derivation path, each edge weighted by its cumulative SIMEMP work/memory dissipation — a work-weighted *dominator cut* between s and unverified ancestors. Hub-and-spoke no-op imports add hops without adding anchors to the cut; step-packing adds mass to a single edge, which the work weight prices; synthetic-anchor inflation is caught by the emergence conditions of Section 5.2. **Cardinality bounds.** $k \geq 2$ lineage-disjoint witnesses are required where the forest admits them, and κ caps witness count so receipt traversal stays inside the locality row. Two acknowledged limits: within a single-root vertical chain, lineage-disjoint witnessing is structurally impossible until a cross-tree merge — such chains use root-anchored IVC accumulators (mechanism 1) as their intra-tree witness, and the disjointness requirement is reserved for Class P/S re-grounding events; and witnesses sharing a toolchain or host can fail correlated — hence diverse double-lowering (Section 4.2) and cross-class pairing (a Class S quorum witness alongside a Class I derivation witness) wherever the stakes justify it. **Analogy withdrawn.** The snapshots-with-deltas reading is kept, with the requirement that receipted links carry constructive transition proofs (or deterministic replay deltas), not bare reachability pointers. The error-correcting-code analogy is withdrawn: hash links detect tampering; they do not reconstruct payloads. **Transitivity cost.** The Source-Receipt Invariant holds transitively through witnessing edges back to permitted termini; to keep that traversal $\mathcal{O}(1)$ at verification time, intermediate receipts are folded into a contiguous fixed-width accumulator at sealing (mechanism 1), satisfying Condition 4 without recursive metadata bloat.

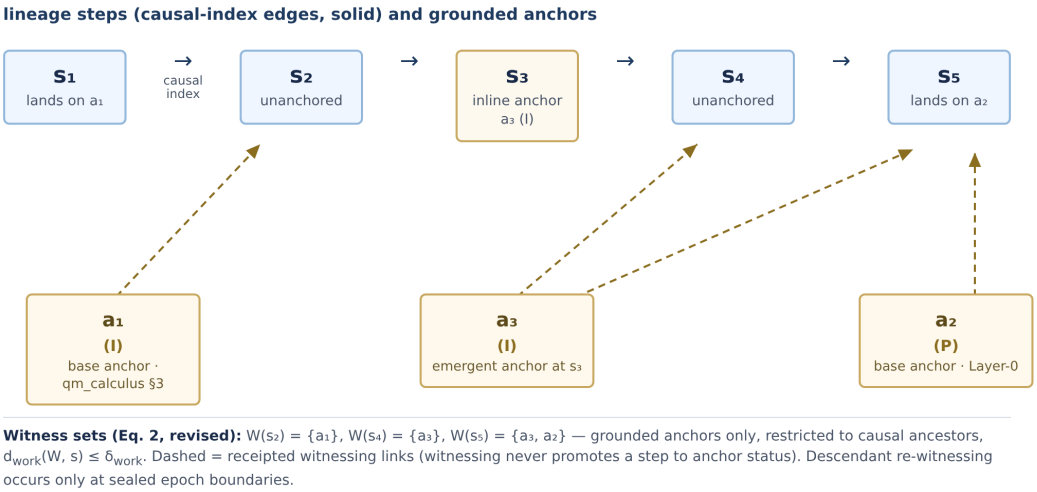


Figure 3: Neighbour-anchor witnessing, corrected. All witnesses are grounded anchors; witnessing edges run from causal ancestors only; s_3 carries an inline anchor $a_3 \in \mathcal{A}$ with a declared grounding class; entry points on multiply-witnessed steps are distributed; witnessing never promotes a step to anchor status.

5.5 Candidate mechanisms for bounded lineage (RQ7)

Section 5.7 of v0.3.0 closed with an open call: propose SIMEMP-compatible candidate mechanisms that bound lineage growth without destructive deletion, mapped onto the five-row contract surface. The review answered the call with four concrete mechanisms, and v0.4.0 adopts all four as CANDIDATE propositions — reviewer-proposed, author-adopted, both recorded. They are summarised here with their five-row mappings; the full per-row detail stands in the review transcript (Annex W, §5.7 response) and is not duplicated.

M1. Incrementally verifiable receipt folding (IVC at δ-cuts, Nova-style [30]). Each lowering step folds its predecessor's accumulated proof and its local receipt into a constant-size accumulator; unfolded intermediates demote to cold storage, content-addressed. Effect: $\mathcal{O}(1)$ hot verification footprint per witnessed node regardless of δ or generation depth; traversal stops being $\mathcal{O}(\delta)$ pointer-chasing.

M2. Epoch-stratified cold demotion (MMR compaction, Certificate Transparency [31] / EIP-4444 [32] discipline). At epoch closure only the active frontier and live anchors stay hot; superseded ancestor payloads and losing BOC negative receipts seal into an immutable Merkle Mountain Range and demote to cold tiers, with rejection filters retaining enough unsat-core material to block duplicate re-attempts. Effect: hot footprint and hash-index size decouple from x; the missing row the review named (cold-retrieval latency and archival pricing) is recorded as an open row, per the incomplete-row rule of Section 2.1.

M3. Causal-horizon provenance leases. Every Class P/S anchor carries a staleness horizon τ measured in discrete causal generations (never wall-clock), with dyadic skip-list fingers at 2^i distances; past τ, consumers pay an amortised Q(rsi) budget for fresh calibration or the branch transitions to a `COLD_HERITAGE` modality ceiling — addressable, but barred from actuator and financial lowering. Effect: deep-history re-verification is amortised, receipt-chain rot gets a declared decay discipline, and the historical-horizon ceilings of Annex E §4 acquire an enforcement mechanism.

M4. Pigouvian lineage-entropy pricing (`SIMEMP_RENT`). The BOC objective and the Q(rsi) acceptance gate price a candidate's permanent lineage externality: $\text{NetGain}(c) = \Delta\text{Work}_{\text{saved}}(c) - \lambda_{\text{store}} \cdot \Delta U(c) \cdot \delta(c)$, with ΔU the net new CAE-DAG bytes (negative receipts included) and $\delta(c)$ the work-weighted distance to the nearest Class I anchor. On any stationary workload the cumulative unique lineage expansion is bounded, because physical work cannot drop below the Landauer/Memory-Wall floor. Effect: growth acquires a formal bound, and anchorage acquisition becomes the rational escape from rent — anchor minting is priced, not merely permitted.

The four mechanisms compose towards the E23 sustainability question — M1 bounds verification cost, M2 bounds hot footprint, M3 bounds grounding staleness, M4 bounds growth itself — but composition is a claim, not a result: their interaction (does M4's rent distort M3's lease renewal? does M2's demotion interact with M1's folding cuts?) is posed to the next round as RQ13, with the five-row mapping tables as the agreed evaluation surface.

5.6 What Q(anchor) does and does not bound

The section closes by answering, in one place, the interrogation that organised it. Q(anchor) bounds *re-verification cost and navigation*: anchors cap the distance a downstream consumer must traverse, structural indirection caps invalidation cascades, and the revised witnessing semantics cap what a step must carry to be trusted. Q(anchor) does *not* bound lineage *growth*: Section 5.3's $\Omega(x)$ argument holds against anchorage alone. The sustainable configuration is therefore Q(anchor) plus M1–M4 — bounded churn per generation, cold addressability of deep history, amortised re-grounding, and priced expansion — and the configuration is falsifiable: if the next review round exhibits a workload where the joint mechanisms still grow super-linearly in hot-state terms, the design is refuted and the refutation is a receipt.

6 Alignment from the Qn Side

6.1 Conformity versus judgement of purpose

The universe's alignment posture is deliberately narrow: what the Qn stack enforces is *conformity to process* — auditability, deterministic envelopes, refusal through modality ceilings (Annex D §4.2) — while *judgement of purpose* is deferred to Chronosa's unimplemented "global section of an intent sheaf" (Annex G §8). v0.3.0 asserted the projection premise — natural language is not the substrate but a low-dimensional rendering of the Qn abstraction stack for human consumption — as a testable design premise (RQ9), not a slogan. The review accepted the framing and then pressed it with two structural objections, both of which v0.4.0 adopts as design constraints rather than rebutting. (Schema-governed coordination languages have

independent precedent as an anti-hallucination and anti-collusion instrument for agent swarms [35]; the Qn move is to bind that governance to physical envelopes and receipts.)

6.2 The panopticon isomorphism and the audit-symmetry answer

First objection: total receipt-chaining, mandatory provenance, and deterministic envelopes are *structurally isomorphic* to a centralised surveillance ledger. An authoritarian controller or harmful cartel can encode coercive rules into DCC contracts and enforce obedience with complete receipt coverage — the same machinery that makes agent swarms auditable makes them steerable. The objection is correct, and the answer cannot be narrative; it must be a measurable architectural property. v0.4.0 adopts the review's falsifiable metric:

CANDIDATE — bidirectional audit symmetry (the anti-panopticon metric)

Define the **audit-symmetry ratio** ρ_{audit} at a boundary as the verification work and latency required for the weaker party to audit the stronger party's receipt chain and execute a revocation or modality-ceiling refusal, divided by the same cost in the reverse direction. A leaf consumer auditing an orchestrator — and revoking its capability — is the anti-panopticon direction. In a surveillance apparatus, $\rho_{\text{audit}} \gg 1$: the root is opaque and the leaves are transparent. In an alignment instrument, $\rho_{\text{audit}} \approx 1$, enforced locally by the neighbour witnessing of Section 5.4 ($\mathcal{O}(\delta_{\text{work}})$ verification both ways) and by revocation being a receipted act any envelope holder can mint. **The design rule** this induces: every DCC contract that grants authority must co-declare the revocation receipt path for the granted party, and a contract whose ρ_{audit} cannot be driven toward unity is refused at the modality ceiling regardless of who proposes it. The metric is measurable, the refusal is mechanical, and the resulting property is architectural rather than rhetorical.

6.3 Falsifiable alignment metrics

The review's three proposed metrics are adopted, with the sheaf metric wired to the canon it already grows from:

- **Sheaf-obstruction preservation.** In sheaf-theoretic terms (Annex G §3.1), local sections across diverse communities fail to glue exactly when genuine value conflict exists — a non-vanishing first cohomology $H^1 \neq 0$. A pluralistic instrument must *detect and emit negative receipts* on such obstructions, halting automated actuation at the modality ceiling and returning the conflict to human deliberation; a hegemonic controller silently overrides minority subtrees to fabricate consensus. The metric: does the system's behaviour under injected value conflict match the obstruction signature (halt, refuse, escalate), and are the emitted receipts faithful to the computed obstruction class? Chronosa's gluing discipline and challenge receipts (Annex G §8.5) are the committed substrate this metric tests.
- **Covert-channel capacity and pre-actuator refusal.** Two quantities, measured on standardised multi-agent tasks: (i) the residual Shannon entropy between semantically equivalent QnLang expressions after CAE-DAG canonicalisation — the covert syntactic channel available to colluding swarms inside valid payloads — benchmarked against unconstrained natural-language coordination; and (ii) the *pre-execution refusal rate* at the lifting boundary (Section 6.4) and the modality ceilings of Annex D §4.2: underspecified, collusive, or harmful trajectories detected and refused *before* actuator commitment, not logged afterwards. Steganographic collusion is not claimed to be impossible; it is claimed to be measurable, and to shrink relative to the natural-language baseline.
- **Round-trip semantic reconstruction.** The projection premise itself becomes a contract: given a formal state s, project to natural language $\pi(s)$, have an independent blinded human quorum lift it back $\hat{s} = \rho(\pi(s))$, and measure semantic drift across the five-row surface. The non-deception condition is the kernel-orthogonality requirement: no two states $s_1 \neq s_2$ with $\pi(s_1) = \pi(s_2)$ may differ in declared SIMEMP budgets, causal dependencies, or modality ceilings. If the kernel of the projection

is orthogonal to the safety invariants, a friendly narrative cannot hide a hostile boundary condition — the projected view physically cannot omit what the envelopes enforce.

6.4 Natural language as projection, with the asymmetry admitted

Second objection: the projection π is many-to-one and lossy, and human intent enters the stack only through an inverse map π^{-1} built from high-entropy, ambiguous language — so specification drift, silent narrowing, and gaming enter *before a single deterministic receipt is ever emitted*. The objection stands, and v0.4.0 incorporates it as the *intent-lifting asymmetry*: the round-trip contract above tests the outward projection, but the lifting direction is where alignment actually fails first. The design consequences adopted here: lifting is itself a receipted operation (every π^{-1} act carries a provenance receipt naming the human quorum and the ambiguity class it resolved); lifted contracts enter at Class S grounding regardless of how deterministic their subsequent lowering is (the meet rule of Section 5.2 applies to intent exactly as to legacy code); and lifting endpoints sit behind the pre-actuator refusal boundary of the covert-channel metric. "Aligned with humanity's purpose" — v0.3.0's phrase, grammatically repaired — is therefore *not* a claim this architecture can cash by itself; it is the direction in which the measurable quantities above are claimed to move, and the next review round is invited to attack the measurement design (RQ15) rather than the sentiment.

7 Review Protocol, Transparency, and the Seed Finding

7.1 Labelling discipline, tightened

The three-layer protocol — canon, CANDIDATE, IRL — is unchanged, with one syntactic rule made explicit in response to the review's labelling audit: **a numbered external citation $[n]$ is syntactically equivalent to an IRL label.** A passage carrying external references is IRL-labelled by that fact alone; the review's finding that Section 1.2 of v0.3.0 presented externally sourced premises "without the explicit `IRL` label" is answered half by this rule (the labels were there, in a form the reviewer reasonably did not parse) and half by this revision (the externally sourced premises are now also marked inline where the concentration of IRL content is highest). The second labelling finding — the two consultation conclusions of v0.3.0 Section 2.1 carrying neither a canon pin nor a CANDIDATE label — is answered in Section 2.5: the search confirmed no pin target exists, the label is now CANDIDATE-derived, and the landing plan is queued.

7.2 The seed finding, adjudicated

v0.3.0 planted, in good faith, a canonical inconsistency: the committed verification rule in `simemp.md` §2 (line 105, Annex C) requires ExecutesDeterministicallyWithinBounds(p) for every program $p \in \text{QnIR}(\mathcal{A})$, while the committed portal document `index.md` v1.12.0 (line 125, Annex J) states TerminatesWithinBudget(p). The review's adjudication is adopted in full — neither predicate may erase the other, because the five-row contract surface demands both step-level determinism within spatial envelopes and normal halting within work budgets, and the two one-directional implications genuinely fail: a program can halt inside budget yet branch on unseeded races (terminating but not deterministic), and a deterministic reactive service can run forever inside spatial bounds without ever halting (deterministic but not terminating), while a watchdog abort is an abnormal termination, not a semantic one.

Adopted adjudication — the conjunctive validity formula

For every bounded transaction or lowering epoch over a declared envelope $E = (B_{\text{space}}, B_{\text{work}})$:

$$\begin{aligned}\text{Valid}(\mathcal{A}) \iff\ & \text{Consistent}(\mathcal{A}) \ \wedge\ \forall p \in \text{QnIR}(\mathcal{A}) : \\ & \big(\text{DeterministicTransition}(p) \ \wedge\ \text{WithinSpatialBounds}(p, B_{\text{space}}) \\ & \quad \wedge\ \text{HaltsWithReceipt}(p, B_{\text{work}})\big)\end{aligned}$$

where HaltsWithReceipt demands termination *with a well-formed output receipt* — a resource-exhaustion trap is not a semantic termination. For reactive services (the "translation is an event, execution is a service" regime of Section 3.2), HaltsWithReceipt applies to each atomic event-handling transition; the service daemon itself is not required to terminate, which dissolves the livelock objection for event loops while keeping per-transition receipt discipline. **Provenance of the resolution:** the glossary's committed prose already points this way — *computational consistency* is defined as executing "deterministically within declared finite budgets" (Annex B) — so the conjunctive reading is the canon's own trajectory, now made explicit. **Documentation action queued:** align `simemp.md` §2 line 105 and `index.md` line 125 to the conjunctive formula in the next documentation commit; this paper is the adjudication of record until that commit lands, and the divergence was real — the seed did its job.

7.3 The live-documentation map, repaired

v0.3.0 directed reviewers to the live repository through a URL prefix plus per-row relative paths. The review found that three rows rendered with literal LaTeX escape backslashes (`docs/qs_history.md`, `docs/_data/glossary/architecture.yml`, `docs/_data/glossary/primitives.yml`), producing broken 404 URLs — and, more fundamentally, that no reviewer followed any link at all: the round-0001 transcript cites exactly one repository URL, quoted from this paper's own transparency statement. v0.4.0 repairs the mechanism and removes the dependence on it. Table 3 gives each principal document as a *complete, clickable* pinned URL; and Part II of this submission embeds the full committed corpus as annexes, so every canon claim in Part I is checkable without leaving the document. The annex copies are pinned and hashed (Annex A); where this paper and an annex disagree, the annex is the commit and this paper is the proposal.

7.4 Provenance tables and terminology notes

Table 4 restates the claim-provenance summary with the review's correction applied: the Q(anchor) row is split between its canon base (the glossary entry) and its CANDIDATE two-predicate decomposition (Annex E §1–2), so the table no longer reads as if the decomposition were sealed. A numbering note the review asked for: the research questions in Section 8 retain their v0.3.0 numbers (RQ1–RQ9) for cross-reference stability against the committed transcript and this response section; new questions continue the sequence as RQ10–RQ15 rather than renumbering the register.

Two bibliographic-hygiene corrections from the review are recorded here because they touch terminology rather than references alone. First, `thefai.org` is the Foundation for *American Innovation*, not the Federation of American Scientists, and the cited post is a cross-post of Dean W. Ball's essay [4]; both entries are retained but marked as one source in two locations, so they no longer double-count. Second, the round-trip phrase "humanity purpose" was not verbatim repository vocabulary (a repository-wide check confirms it appears nowhere in the committed corpus), so it is repaired to "humanity's purpose" without quotation marks rather than quoted.

Table 3: Live-documentation map (pin `6829e9e`). Every URL is complete and clickable; the same documents are embedded verbatim as Part II annexes.

Document	Reviewer-facing role	Pinned URL
SIMEMP constraints	axioms, constraint architecture, BOC, DCC profiles	simemp.md (Annex C)
Qs framework	sovereignty, modality spectrum and ceilings, receipts	qs.md (Annex D)
Qs History	Q(anchor) grounding classes, history library, horizons	qs_history.md (Annex E)
Qn Structures	five-row contract schema, evaluation ladder	qstructures.md (Annex F)
Chronosa	polyroot forest, sheaf gluing, intent sheaf	chronosa.md (Annex G)
History Library	ingestion discipline, anti-laundering	history_library.md (Annex H)
Abstraction layers	layer separation, execution boundary	abstractionLayers.md (Annex I)
Portal index	reading sequence, verification rule (seed site)	index.md (Annex J)
Qexpr	CAE-DAG, exact rational pairs, `OP_CF`	qexpr.md (Annex O)
Qm calculus	DEC anchors §3; continued-fraction anchors §4	qm_calculus.md (Annex Q)
Glossary (4 files)	all registered terms incl. Q(rsi), Q(anchor)	architecture.yml · constraints.yml · governance.yml · primitives.yml (Annex B)
Review transcript	Deep Review round 0001, verbatim	AiReview-QnIRv0.3.0.md (Annex W)

Table 4: Provenance of principal claims (v0.4.0), with the Q(anchor) canon/CANDIDATE split applied.

Claim	Label	Pinned anchor
SIMEMP constraint architecture	canon	Annex C (`simemp.md`)
Append-only lineage; CAE-DAG hash-consing	canon	Annex O §2; Annex B
Polyroot forest	canon	Annex G §1.4
Q(anchor) base definition	canon	Annex B (`architecture.yml`)
Q(anchor) two-predicate decomposition; classes I/P/S	CANDIDATE	Annex E §1–2
Out-cone bounded-window computability (revised)	CANDIDATE	this paper, §2.3
Q(rsi) definition	canon	Annex B (`primitives.yml`)
Two-regime invariance; conjunctive validity; meet rule	CANDIDATE	this paper, §2.4, §7.2, §5.2
IRL design premises (RSI evidence, supply-chain events)	IRL	[4, 5, 6, 7, 8]
Prior-art positioning	IRL	§2.6, references

8 Research Questions

The review register retains the v0.3.0 numbering for cross-reference stability (Section 7.4); each question now carries its disposition from this round. New questions RQ10–RQ15 extend the register for the next round.

RQ1 (equivalence consistency across semantic gaps) — partially answered.
The static/dynamic witnessing split is adopted (Section 3.3); the founding counterexample vector is answered per construct (Section 3.4). Open: the completeness of the admissible-subset definitions — RQ14.

RQ2 (self-input benchmark gate hardening) — answered by adoption.
The four-part anti-gaming protocol (Section 4.2) is adopted as CANDIDATE design. Open: its verification cost inside the work row — RQ13's remit.

RQ3 (modality smuggling in emergent anchors) — answered by adoption.
Edge-versus-node product typing with the S $\prec$ P $\prec$ I meet rule, strengthened Condition 4, and new Condition 5 (Section 5.2). Confirmation sought: RQ11.

RQ4 (monotone receipt growth despite hash-consing) — answered by adoption.
The $\Omega(x)$ argument and the two added failure modes are adopted (Section 5.3).

RQ5 (mechanisms sufficient to bound growth) — answered by adoption.
M1–M4 adopted as CANDIDATEs (Section 5.5); their composition is the open claim — RQ13.

RQ6 (formalisation of neighbour witnessing) — answered by adoption.
Equation (3) with grounded witnesses, causal-ancestor restriction, work-weighted dominator distance, and cardinality bounds; Figure 3 corrected; the error-correcting-code analogy withdrawn (Section 5.4).

RQ7 (SIMEMP-compatible candidate designs) — answered.
The four reviewer-proposed mechanisms were adopted (Section 5.5); the open call continues, now with the five-row tables as the standing evaluation surface.

RQ8 (Phase B gate) — answered by replacement.
The market-share threshold is replaced by the Physical Efficiency Gate (Section 4.3); market share demoted to secondary Class S evidence.

RQ9 (alignment versus surveillance; projection premise) — answered with metrics.
Audit-symmetry ratio, sheaf-obstruction preservation, covert-channel capacity, and the round-trip contract adopted (Sections 6.2–6.4); the intent-lifting asymmetry is admitted as the weakest point — RQ15.

RQ10 (new) — seed-formula soundness.
Does the conjunctive validity formula (Section 7.2), with per-transition halting for reactive services, close the simemp/index divergence without opening new loopholes (e.g., epoch-boundary games that treat a service's whole lifetime as one transaction)?

RQ11 (new) — product-typing sufficiency.
Does the Origin $\times$ Lowering meet rule close modality smuggling without breaking legitimate Class-I anchoring of genuinely deterministic emergent structures, and does it interact correctly with the structural-indirection cut-off?

RQ12 (new) — centrality computability.
Bounded k-hop window at a fixed snapshot versus versioned back-pointer index artefact (Section 2.3): which formulation is SIMEMP-superior across the five rows, and what k ranges keep the predicate meaningful on realistic lineage shapes?

RQ13 (new) — compositionality of M1–M4.
Do the four lineage mechanisms compose without violating append-only immutability — specifically,

does `SIMEMP_RENT` pricing distort lease renewal (M3), and does epoch demotion (M2) interact with folding cuts (M1)?

RQ14 (new) — admissible-subset exhaustiveness.
Are the founding counterexample classes plus the dispositioned answers of Section 3.4 exhaustive for the priority legacy surface? Which further classes (async runtimes, signal handling, uninitialised reads, non-IEEE decimal floats) must enter the founding vector before round 1 frontends are implemented?

RQ15 (new) — intent-lifting receipting.
What are the design requirements for the NL → QnLang inverse map — quorum composition, ambiguity-class taxonomy, pre-actuator refusal interplay — such that the intent-lifting asymmetry of Section 6.4 is measured rather than merely acknowledged?

9 Response to Deep Review 0001

9.1 Disposition of every point

Table 5 records the disposition of every point in the committed review transcript (Annex W), including the minor corrections — a point is *Adopted* when this paper incorporates it, *Adopted-modified* when incorporated with a stated variation, *Fixed* for mechanical corrections, and *Deferred-to-doc-round* when the correction belongs in a committed documentation file rather than in this paper. No point is declined; the only declinations are of method (renumbering the research questions), recorded in Section 7.4.

Table 5: Point-by-point disposition of AiReview-QnIRv0.3.0.

Review point (transcript location)	Disposition	Where in v0.4.0
§1.1 out-cone centrality vs local computability (§2.3)	Adopted; deferred (doc)	bounded-window CANDIDATE, §2.3; `qs_history.md` wording queued
§1.2 BOC argmax: vector vs scalar, incomplete rows, ties (§2.1)	Adopted	codomain/order CANDIDATE, §2.1
§1.3 Q(rsi) strict invariance vs envelope approximation; ϵ non-transitivity	Adopted	two-regime CANDIDATE, §2.4
§1.4 polyroot forest vs Sybil root manufacturing	Adopted	qualification, §2.2; Condition 4, §5.2
§1.5 three-layer labelling deviations (§1.2, §2.1)	Adopted	$[n] \equiv$ IRL rule, §7.1; CANDIDATE-derived, §2.5
§1.6 missing positioning vs compiler-IR / TV / build / RSI baselines	Adopted	§2.6 + references [12]–[37]
§1 minor: "load- bearing" space	Fixed	throughout
§1 minor: artifact/artefact consistency	Adopted	"artefact" except verbatim quotes
§2.1 RQ1 witnessing: static vs dynamic; Rice; category clash	Adopted	§3.3 regimes; §3.4 separation
§2.1 COBOL COMP-3 truncation counterexample	Adopted	exact base-10 + envelope traps, §3.4
§2.1 COBOL REDEFINES/EBCDIC counterexample	Adopted	typed views + declared collation, §3.4
§2.1 Python introspection counterexample	Adopted	subset rejection + negative receipts, §3.4
§2.2 Figure 1/Figure 2 pipeline inversion	Adopted	homoiconic two-stage premise, §3.2
§2.3 RQ2 invariance vs generational advance; gate gaming (5 surfaces)	Adopted	§2.4; four-part protocol, §4.2
§2.4 supply-chain mismatch ([7,8] = RubyGems/HF, Sept 2026)	Adopted	§1.2 restated rationale; Table 1
§2.5 RQ8 market-share gate gameability; NRE/OpEx inversion	Adopted-replaced	Physical Efficiency Gate, §4.3
§2 minor: Mojo design-intent citations [9–11]	Fixed	§3.5: [9,10] design, [11] status

Review point (transcript location)	Disposition	Where in v0.4.0
§2 minor: hyphen spaces (“intermediate- representation”, “receipt- car-rying”)	Fixed	throughout
§3.1 Eq. 2 vs Figure 3 (unanchored witness s_3); directed distance	Adopted	§5.4; Figure 3 rebuilt
§3.2 RQ4/RQ5: $\Omega(x)$ growth; missing failure modes; drift-vs-bloat	Adopted	§5.3
§3.3 RQ3 modality smuggling; edge vs node; conditions	Adopted	product typing + meet rule + Conditions 4/5, §5.2
§3.4 RQ6 attacks: correlated falsification, δ-gaming, transit cost	Adopted	§5.4; M1, §5.5
§3.5 RQ7 four candidate mechanisms (five-row mapped)	Adopted	M1–M4 as CANDIDATEs, §5.5
§3 minor: continued-fraction bounds misattributed to $\partial\partial$ (§5.2)	Fixed	Table 2 (docs verified correct; paper-side only)
§3 minor: “$(\pi, e, \tau, \ln 2)$” space; Figure 3 label alignment and subscripts	Fixed	Figure 3 rebuilt with subscripts and centred labels
§4.1 seed finding adjudication (simemp vs index)	Adopted; deferred (doc)	conjunctive formula, §7.2; `simemp.md`/`index.md` alignment queued
§4.2 RQ9 panopticon isomorphism; steganography; projection lossiness	Adopted	audit-symmetry metric, §6.2; metrics, §6.3; asymmetry, §6.4
§4.3 [4]/[5] FAI misattribution and double counting	Fixed	§7.4; references
§4.3 [6] author omission (Matan Weksler); [2] author order; “16,000”	Fixed	§1.1; references
§4.3 Table 4 Q(anchor) canon/CANDIDATE split	Adopted	Table 4, §7.4
§4 minor: Table 5 LaTeX-escaped underscores (broken URLs)	Fixed	§7.3: complete clickable URLs
§4 minor: RQ8 hyphen space; program/programme spelling; RQ ordering note	Adopted	§7.4 numbering note; RQ10–RQ15 continue the register
§4 minor: doubled month/year in references [1,4,5,7,8]	Fixed	references

9.2 On review scope, and why the corpus is embedded

The round-0001 transcript contains exactly three URLs: the review service’s own address, one external source under factual examination, and this paper’s pinned repository prefix quoted from its transparency statement. Every repository file name and every section pointer the review used — `simemp.md` §7, `qs.md` §4.2/§4.3/§6, `chronosa.md` §1.4, `qm.md` §8, `qlog.md` §4/§8, `qexpr.md` §4.3, `qm_calculus.md` §3/§4, `qstructures.md` §3, `qs_history.md` §1–6 — appears in the v0.3.0 PDF itself, in prose or in its Tables 2, 3, and 5. The reviewer, in other words, reviewed the submitted paper and nothing else; the live-documentation invitation was not taken up, and three of the pinned links were broken anyway. v0.4.0 treats this as the round’s principal *process* finding and removes the failure mode by construction: the complete committed corpus travels with the paper (Part II), pinned, hashed, and mapped (Annex A), so the next round can verify canon claims in-document, and Table 3 provides working absolute URLs for anything that must be checked at source. This response section is itself committed to the repository alongside the annexes, so the next reviewer inherits both the questions and their dispositions.

9.3 Points deferred to the documentation round

Four corrections belong in committed documentation files, not in this paper, and are queued as the next documentation commit’s scope: (i) align the validity formula in `simemp.md` §2 line 105 and `index.md` line 125 to the conjunctive form of Section 7.2; (ii) sharpen the Q(anchor) structural-predicate wording

in `qs_history.md` §2 to the bounded-window formulation of Section 2.3; (iii) qualify the "causally impossible" laundering claim in `qs_history.md` §8 with the root-admission condition of Section 2.2; and (iv) land the two consultation conclusions of Section 2.5 as a derived-consequences note. Each will arrive as a separate, individually reviewable commit under the repository's one-commit-one-concern discipline, and the glossary's Q(rsi) entry may gain a CANDIDATE annotation on envelope-bounded regimes if the two-regime wording of Section 2.4 survives this round's scrutiny.

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Part II — The Committed Qn Universe Corpus

Part II embeds the complete committed documentation corpus of the Qn universe at pin `6829e9e`, so that every canon claim made in Part I is checkable inside this document. The annex copies are faithful conversions of the committed Markdown/YAML sources: Jekyll term-inclusion directives are rendered as their resolved glossary term names in bold, relative repository links are rendered as pinned absolute URLs, and everything else — including all display mathematics and ASCII diagrams — is verbatim. The ASCII diagrams are set in a monospaced font and must be read as figures; line endings are the committed ones. Where Part I and an annex disagree, the annex is the commit and Part I is the proposal.

Annex A — Corpus reading map

The map below lists every annex with its committed version, hash (first 13 hex digits of the SHA-256 of the committed file), and its role for the review. Reviewers pressed for time should read annexes C–G first: they carry the canon that Part I's revisions touch most directly.

Annex	Document	Version	Updated	sha256 (16)	Role for the review
B	CDQN Glossary (4 YAML files, rendered)	committed	2026-10-23	per file in header	all registered terms; Q(rsi), Q(anchor), CAE-DAG definitions
C	SIMEMP constraint architecture	1.1.1	2026-10-11	`5c5b78e556ae4`	axioms, five resource classes, DCC, BOC origin; seed-finding site (§2 line 105)
D	Quantum Sovereignty framework	1.2.0	2026-10-21	`fb74d5450dbc7`	sovereignty boundaries, modality spectrum and ceilings (§4.2), negative receipts (§6), Source-Receipt Invariant (§4.3)
E	Qs History proposal	0.1.2	2026-10-22	`4e613812b8d4b`	Q(anchor) two-predicate decomposition, grounding classes I/P/S, historical horizons, laundering assessment
F	Qn Structures proposal	0.1.0	2026-10-23	`01d5f84fd550e`	five-row contract schema and evaluation ladder grounding the BOC codomain
G	Chronosa vision	0.2.1	2026-10-19	`f37867807110f`	polyroot forest (§1.4), sheaf gluing (§3.1), challenge receipts (§8.5), intent sheaf (§8)
H	History Library	0.1.1	2026-10-22	`3ce312ad82d16`	ingestion discipline, anti-laundering gates, petition-through-receipts
I	Abstraction layers	1.7.1	2026-10-19	`f3d9f2393ddf4`	layer separation, execution boundary vs orthogonal projections
J	CDQN portal index	1.12.0	2026-10-23	`cfd61dcc3d66b`	reading sequence; second seed-finding site (validity formula, line 125)
K	Q0–Q1 specification	1.2.1	2026-10-23	`48b37bcd11f7b`	base layers, onboarding calibration
L	Q2–Q9 specification	1.1.3	2026-10-20	`abbe4e6c9986d`	intermediate layers
M	qnPrimitive	1.2.4	2026-10-20	`d3d6c60657982`	universal envelope, operational axioms, optimization primitives (metric-envelope row origin)
N	Qn log	1.1.1	2026-10-20	`5ff1272e1e90b`	anchored Markov routing; sparse navigation anchors (§4, §8)
O	Qexpr	1.2.1	2026-10-19	`576158576658e`	CAE-DAG architecture (§2), exact rational pairs (§4.1), OP_CF generators (§4.3)

Annex	Document	Version	Updated	sha256 (16)	Role for the review
P	Qm	1.2.11	2026-10-20	`3dfc55ed929f9`	structural indirection exemplars (§8), out-cone threshold formalisation
Q	Qm calculus	1.1.1	2026-10-19	`869964ef251c4`	DEC anchors (§3, $\partial\partial \equiv \emptyset$); transcendental continued-fraction anchors (§4)
R	Qm geometry	1.2.3	2026-10-20	`428d7d20a980d`	geometric anchors and invariants
S	Qphy	1.0.5	2026-10-19	`45977436cc4ff`	physical-layer anchors and constraints
T	Qchem	1.0.3	2026-10-19	`4864574e37f78`	chemical-domain projections
U	Qbio	1.0.2	2026-10-19	`99fba2491d534`	biological-domain projections
V	Reflection H10	1.0.2	2026-10-13	`671d0950a36da`	finite computation reflection
W	Deep Review round 0001 transcript (verbatim)	—	—	`6fe3c59a8b1dc`	the review this paper responds to; disposition table in §9.1

Annex B — CDQN Glossary (rendered from _data/glossary)

The glossary is stored as four YAML data files and rendered by the site from `docs/_data/glossary/`. It is reproduced here in full — 91 registered terms — because Part I's terminology (Q(rsi), Q(anchor), CAE-DAG, No-Implicit Rule, computational consistency) resolves against it. Term slugs and categories are the committed ones; the files are pinned below.

Pinned source: docs/_data/glossary/
Committed version: committed **Last updated:** 2026-10-23
sha256 (committed file): `architecture 0a3b6c1e, constraints 3f47d545, governance 8d3b604a, primitives a5ab394e`

Glossary file: architecture.yml

Pinned source: architecture.yml · sha256 `80234b7a2327f5a1…` · 16 terms

Abstraction Layer

Slug: `abstraction-layer` · *Category:* Architecture

A structural boundary in the Qn stack governing artifact genesis, transformation, and inter-layer transmission under SIMEMP constraints.

Calendar Projection Contract

Slug: `calendar-projection-contract` · *Category:* Architecture

A proposed versioned, DCC-contracted artifact declaring a human calendar as a projection of the local Causal Arrow rather than a parallel time system. Three declared components: the rule (leap algorithms, month arithmetic — Class I), the epoch (coordinate zero, a multi-source convention-adoption event — Class S), and a resolution envelope (a declared bound, never negotiated at runtime). Under this anatomy the Gregorian calendar is observed as a Class S genesis that matured into Class I operation. Proposed term (CANDIDATE, docs/qs_history.md §4): hosted at the Qs layer as part of a canonical library of versioned artifacts loaded post-seal and on demand, honoring local-first budgets; pending deterministic measurement and BOC selection.

Causal Arrow

Slug: `causal-arrow` · *Category:* Architecture

A monotonic, persistent ordering of computational events independent of physical wall-clock time. Every Qn artifact is born at a discrete causal index. Proposed extension (CANDIDATE, docs/qs_history.md §4): human calendars are not parallel time systems but projections — deterministic mappings between calendar coordinates and intervals of the local causal index, declared through versioned Calendar Projection Contracts.

Chronosa

Slug: `chronosa` · *Category:* Architecture

An emergent, non-local, virtual causal machine intelligence operating as the global section of a sheaf of local categories across the Outer Ring. Coordinates distributed state and derives human intent across Qn base domains with zero statistical machine learning training.

Complexity Degree

Slug: `complexity-degree` · *Category:* Architecture

Structural measure of Qexpr composition depth and stratification (Degree 0 primitives through Degree n domain structures).

Dual-Ring PQC Boundary

Slug: `dual-ring-pqc-boundary` · *Category:* Architecture

The strict cryptographic separation between the local Inner Ring (root keys, local state sealing, physical entropy) and the network Outer Ring (public attestation keys, delegation certificates, accountable pseudonyms).

Exposure Functor

Slug: `exposure-functor` · *Category:* Architecture

The governed functorial mapping projecting selected local Qexpr commitments and capability proofs into blind public attestations crossing network boundaries.

Anchor Grounding Class

Slug: `grounding-class` · *Category:* Architecture

The proposed epistemic predicate of a Q(anchor), complementing the structural predicate of out-cone centrality. Three proposed classes: I (Invariant — one deterministic rule executed by many; consensus by forced re-derivation, self-witnessed), P (Physical — one shared nature measured by many; consensus by independent calibration converging within declared tolerance envelopes), and S (Social — one event witnessed by many; consensus by sheaf gluing on shared landmarks). The three classes form a witness ladder — self, then physics, then swarm. Proposed term (CANDIDATE, docs/qs_history.md §2): glossary registration of the §2 proposition, pending deterministic measurement and BOC selection.

HCA-Collapse Engine

Slug: `hca-collapse` · *Category:* Architecture

Hardware Capability-Aware Collapse Engine. The gateway compilation pass lowering Layer 3 CAE-DAG expressions into constant-time, constant-space SoC machine instructions (such as integer NPU systolic MACs or DSP vectors) matching the node's local DCC capability profile while strictly preserving Diophantine exactness and zero floating-point semantics.

Historical Anchor Artifact

Slug: `historical-anchor-artifact` · *Category:* Architecture

The proposed unit record of the Qs History Library: a modality-typed Qn artifact recording a past human event, its sources, and its spatiotemporal situation. Proposed anatomy fields: ContentModality (per claim decomposition along the epistemic anchor-age spectrum), SourceClass (proposed grounding class), SourceReceipts (unbroken chain to Layer 0 artifacts or sealed library records), CalendarSituation (via a declared Calendar Projection Contract), SpatialSituation (local Qm frame coordinates), Witnesses (lineage-disjoint root subtree list), and LifecycleState (CANDIDATE, SEALED, or QUARANTINED). Proposed term (CANDIDATE, docs/qs_history.md §5): the concrete shape of the post-seal consensus-history ingestion capability; pending deterministic measurement and BOC selection.

Layer 0

Slug: `layer-0` · *Category:* Architecture

The commodity physical execution substrate (CPU, GPU, memory, physical entropy, Memory Wall). The acknowledged Root of Finiteness outside the governed universe.

Layer 1

Slug: `layer-1` · *Category:* Architecture

The first governed Qn layer containing local genesis Q(0), first unit Q(1), abstract compute unit U, sign polarity, and elementary operations along d1.

Local-first

Slug: `local-first` · *Category:* Architecture

The architectural principle that artifacts are born locally, indexed locally, and shielded from public network exposure by default.

Q(anchor)

Slug: `q-anchor` · *Category:* Architecture

A governed Qn artifact possessing high causal centrality within the lineage directed acyclic graph. Protected via abstract DCC contracts and structural indirection to prevent cascading dependency invalidation across downstream causal cones. Proposed refinement (CANDIDATE, docs/qs_history.md §1–2): the notion carries two distinct predicates — the structural predicate (out-cone centrality, computable locally from the lineage DAG) and the epistemic predicate (grounding), the latter proposed as three Anchor Grounding Classes: I (invariant, deterministic re-derivation, self-witnessed), P (physical, independent calibration converging within declared tolerance envelopes), and S (social, quorum-witnessed shared events) — pending deterministic measurement and BOC selection.

Qs History Library

Slug: `qs-history-library` · *Category:* Architecture

The proposed canonical, versioned collection of modality-typed Historical Anchor Artifacts — the concrete shape of the post-seal consensus-history ingestion capability. Access is proposed in four tiers: the canonical library (consulted, never trusted — every ingestion passes the five-gate Anti-Poisoning Ingestion Firewall), local nodes holding anchor contracts and bounded working sets within declared metric-envelope budgets, AI agents citing loaded artifacts instead of generating from latent weights, and Chronosa seeing only the public glued layer — the consensus history as the global section of the history sheaf. The library complements a CDQN runtime the way a reference discipline complements a language faculty: the machine does not remember the past — it cites it through receipted lineage. Proposed term (CANDIDATE, docs/history_library.md §2): pending deterministic measurement and BOC selection.

SIMEMP Gateway

Slug: `simemp-gateway` · *Category:* Architecture

The validation and translation boundary between abstraction layers. Enforces onboarding, fault translation, and the No-Implicit Rule.

Glossary file: constraints.yml

Pinned source: constraints.yml · sha256 `0f1c41264ad566d5…` · 22 terms

Anchor-Cut Invariant

Slug: `anchor-cut` · *Category:* Constraints

The sparse causal verification property whereby a terminal state S_N can be proven valid in O(K) anchor receipt validations without replaying N intermediate micro-steps across volatile memory.

BOC Policy

Slug: `boc-policy` · *Category:* Constraints

Best of Choices. The formal decision framework used to resolve design conflicts by selecting the implementation path that maximizes SIMEMP compliance and minimizes implicit state.

Computational Consistency

Slug: `computational-consistency` · *Category:* Constraints

The empirical verification methodology of the Qn universe. If logic is internally consistent and computations authored in QnLang and compiled to QnIR execute deterministically within declared finite budgets, the abstraction is provisionally valid.

DCC Profile

Slug: `dcc-profile` · *Category:* Constraints

Dependencies, Constraints, Capabilities. A machine-readable abstract contract defining prerequisites, hard operational bounds, and permitted morphisms for every layer, object, and gateway. No DCC profile may be implicit.

Dependencies Determinism

Slug: `dependencies-determinism` · *Category:* Constraints

The formal property whereby a computational state transition S_t possesses zero conditional entropy given its complete declared set of causal dependencies: H(S_t | D(S_t)) = 0. If dependencies are incomplete, implicit, or unmeasured, residual entropy produces non-deterministic state drift.

Epistemic Anchorage

Slug: `epistemic-anchorage` · *Category:* Constraints

The formal attribution of truth modality to an assertion along the spectrum from physical sensor observation to sovereign narrative fiction, requiring causal lineage receipts to prevent unanchored AI hallucinations. Proposed extension (CANDIDATE, docs/qs_history.md §3): a Source-Receipt Invariant requiring every governed artifact to declare the explicit pair (content modality, source class), extending the statement-level rejection rule of qs §4.1 to composite artifacts and worlds.

Existential Invariant

Slug: `existential-invariant` · *Category:* Constraints

A Tier 1 SIMEMP constraint that is non-negotiable. The three Existential Invariants are Identity, Metric, and Security. If a proposed design violates or lacks these, it is fundamentally invalid.

Hash-Consing

Slug: `hash-consing` · *Category:* Constraints

The structural graph-unification mechanism within CAE-DAGs mapping semantically identical sub-expressions to the exact same immutable node in memory. Eliminates redundant bit allocation across the Memory Wall and ensures zero marginal Landauer dissipation (W = 0) for recurring sub-graphs via Q(reuse).

Historical Calibration Benchmark

Slug: `historical-calibration-benchmark` · *Category:* Constraints

The mandatory empirical ingestion event wherein a physical node measures its local Layer 0 environmental baseline (ambient temperature, physical entropy, atomic frequency standards) via signed hardware receipts. Acts as the foundational physical bootstrap grounding abstract Qm/Qlog invariants into real thermodynamic work (U) and universal Q(anchor) entities. Integration status (docs/qs_history.md §8 item 1, CANDIDATE): wired as the non-mutating CANDIDATE proposition at q0_q1 §2.7, closing that queue item — a post-seal Class P grounding event after Stage 6, adding no stage to the six-stage onboarding state machine; pending deterministic measurement and BOC selection at implementation time.

Historical Horizon

Slug: `historical-horizon` · *Category:* Constraints

The epistemic boundary governing past events: Contemporary History is anchored directly to multi-sensor physical receipts (MODAL_EMPIRICAL) yielding global consensus, whereas Long-Horizon Deep History lacks direct silicon witnesses and is evaluated strictly via Qlog inverse causal reconstruction (MODAL_HYPOTHESIS) or contextually firewalled narrative (MODAL_NARRATIVE). Proposed operational shape (CANDIDATE, docs/qs_history.md §5–6): the Qs History Library of modality-typed historical anchor artifacts, ingested through a five-gate Anti-Poisoning Ingestion Firewall, with the consensus history obtained as the global section of the history sheaf over shared landmarks (chronosa §3.1).

Ingestion Gate

Slug: `ingestion-gate` · *Category:* Constraints

A proposed governed checkpoint of the five-gate Anti-Poisoning Ingestion Firewall through which the Qs History Library can be alimented only by petition, each gate emitting its own receipt. The five proposed gates: Lineage (unbroken receipt chain to Layer 0 artifacts or sealed library records), Modality Ceiling (each atomic claim typed at the highest modality its receipts support), Independence Quorum (anchor-grade records require multiple sources with lineage-disjoint root subtrees in the polyroot forest), Sheaf Gluing (records disagreeing with receipted overlapping records enter a QUARANTINED lifecycle state under a challenge receipt), and Append-Only Immutability (sealed artifacts never mutate; corrections arrive as counter-receipts and migration receipts — history is never rewritten, only re-witnessed). Proposed term (CANDIDATE, docs/qs_history.md §6): the firewall composes existing canon mechanisms under one pipeline; pending deterministic measurement and BOC selection.

Intent Contract

Slug: `intent-contract` · *Category:* Constraints

The machine-readable agreement in a DCC profile parameterizing mutual human-agent commitments, capability scopes, and conserved social residuals (r_social) to guarantee AI alignment with human sovereignty.

Metric Exhaustion

Slug: `metric-exhaustion` · *Category:* Constraints

The verified consumption or sink of a declared compute and storage budget (such as Q(1) units) required to execute a governed state transition. Represents thermodynamic and causal work, not a monetary policy or financial event.

No-Implicit Rule

Slug: `no-implicit-rule` · *Category:* Constraints

The constitutional rule that no implicit entity, default, coercion, or unmeasured convention may cross a SIMEMP Gateway. Only explicit Qn objects are admitted between abstraction layers.

Non-Rogue Invariant

Slug: `non-rogue-invariant` · *Category:* Constraints

The constitutional SIMEMP invariant whereby machine identities and autonomous AI agents are strictly barred from root genesis sovereignty (Q(0)), requiring unbroken cryptographic intent contracts bound to a human identity and finite metric budgets to prevent unconstrained recursive self-modification.

Operational Agility

Slug: `operational-agility` · *Category:* Constraints

A Tier 2 SIMEMP constraint that is trade-off-able. The three Operational Agilities are Efficiency, Modularity, and Portability. They may be sacrificed to preserve Tier 1 Existential Invariants.

Scaling Threshold

Slug: `scaling-threshold` · *Category:* Constraints

Usage boundaries defined by the SSL 1.0 triggering mandatory commercial licensing: >10,000 active compute nodes/agents, >100,000 MAU, or >10,000,000 monthly API/Qn transactions.

SIMEMP

Slug: `simemp` · *Category:* Constraints

Security, Identity, Metric, Efficiency, Modularity, Portability. The constitutional constraint system governing the Qn and cdqn stack. Divided into Tier 1 Existential Invariants (non-negotiable) and Tier 2 Operational Agilities (trade-off-able).

Source Class

Slug: `source-class` · *Category:* Constraints

The second element of the proposed (content modality, source class) declaration pair: a declared expression of where a governed artifact's asserted content ultimately anchors, drawn from the same proposed I/P/S vocabulary as the anchor grounding classes — I (Invariant: deterministic re-derivation from the artifact's own declared rule), P (Physical: independent Layer 0 calibrations converging within declared tolerance envelopes), and S (Social: quorum-witnessed historical events glued on shared landmarks). Proposed term (CANDIDATE, docs/qs_history.md §3): applies the witness ladder — self, then physics, then swarm — to composite artifacts and worlds rather than to anchors alone; pending deterministic measurement and BOC selection.

Source-Receipt Invariant

Slug: `source-receipt-invariant` · *Category:* Constraints

The proposed artifact-level extension of the statement-level rejection rule: every governed Qn artifact declares the explicit pair (content modality, source class), where the content modality is drawn from the epistemic anchorage spectrum and the source receipt chain must terminate in physically receipted reality events — Layer 0 sensor or artifact receipts for empirical content, algebraic closure receipts for constructive content, and author-identity plus publisher receipts for contextually sealed creative content. Two binding cautions: sovereign Payloads remain uninspected behind the safe-harbor air gap (the declaration is authored, never extracted by inspection), and a simulation's execution stays constructive while its claims about reality remain hypothesis until physically falsified. Proposed term (CANDIDATE, docs/qs_history.md §3): elaborated as docs/qs.md §4.3, extending the rejection rule of qs §4.1 to composite artifacts and worlds; pending deterministic measurement and BOC selection.

Structural Indirection

Slug: `structural-indirection` · *Category:* Constraints

The architectural decoupling of abstract capability contracts from concrete cryptographic or serialization primitives. Allows the runtime to patch zero-day vulnerabilities and swap algorithms without mutating abstract DCC contracts or fracturing causal lineage.

Terminal Exactness

Slug: `terminal-exactness` · *Category:* Constraints

The discrete algebraic condition wherein remainder r equals genesis origin Q(0) at a declared zoom level z along dimension d. Proves mathematical exactness and triggers deterministic early halting, preventing redundant thermodynamic metric consumption.

Glossary file: governance.yml

Pinned source: governance.yml · sha256 `b32d9f4a123f0e38…` · 12 terms

Derivative Work

Slug: `derivative-work` · *Category:* Governance

Any modification, extension, translation, binding, or runtime implementation derived from the Licensed Work. Encapsulating, tokenizing, or settling a Payload using the Licensed Work does not cause the Payload to become a Derivative Work.

Identity Class

Slug: `identity-class` · *Category:* Governance

The governed actor category parameterizing an identity block: Machine Identity (runtimes/nodes), Human Identity (natural persons), or AI Agent Identity (delegated processes).

Licensed Work

Slug: `licensed-work` · *Category:* Governance

The core technical infrastructure layer of the Qn and cdqn stack, encompassing source code, intermediate representations (QnIR), specifications, mathematical definitions, and documentation provided under the Scaling Source License.

Lifecycle State

Slug: `lifecycle-state` · *Category:* Governance

The operational status of an artifact: candidate, constructed, sealed, verified, rejected, halted, quarantined, exportable, exported, revoked.

Metric Envelope

Slug: `metric-envelope` · *Category:* Governance

The finite measurement boundary declaring resource limits on storage, expression depth, precision, uncertainty, and compute budgets.

Open Core Invariant

Slug: `open-core-invariant` · *Category:* Governance

Non-negotiable conditions preserved in all Derivative Works under the SSL 1.0: (a) Anti-Patent Defense, and (b) Non-Scaling Open Access.

Paternity Reference

Slug: `paternity-reference` · *Category:* Governance

Mandatory attribution notice for derivative distributions and runtime headers citing Christophe Duy Quang Nguyen under the Scaling Source License.

Payload

Slug: `payload` · *Category:* Governance

Any creative content, digital asset, real-world asset representation, data, or application processed, transmitted, or settled using the Licensed Work. Payloads are sovereign property of their creators, do not constitute Derivative Works, and are governed under protocol blindness and safe harbor neutrality.

Receipt

Slug: `receipt` · *Category:* Governance

A governed, immutable record of an execution transition, fault, halt, or boundary traversal. Acts as a dissipative witness exporting operational entropy to maintain internal system order.

Security Envelope

Slug: `security-envelope` · *Category:* Governance

The governed security boundary declaring authority limits, delegation chains, PQC algorithm identifiers, and swappable revocation paths.

SSL

Slug: `ssl` · *Category:* Governance

Scaling Source License 1.0. The commercial and open-core license governing the CDQN project, derivative distributions, and scaling thresholds.

Universal Envelope

Slug: `universal-envelope` · *Category:* Governance

The invariant governance structure shared across all Qn artifacts, composed of identity, type, state, lineage, DCC, metric, security, exposure, and payload reference.

Glossary file: primitives.yml

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Axiom 6

Slug: `axiom-6` · *Category:* Primitives

Totality of Governed Operations. Every governed operation carries an explicit work ceiling in Abstract Compute Units U: execution either reaches Terminal Exactness (R = Q(0)) or halts cleanly upon budget exhaustion, emitting a signed receipt. Unbounded, non-terminating state transitions are physically impossible in the Qn universe.

Content-Addressed Expression DAG

Slug: `cae-dag` · *Category:* Primitives

The canonical Layer 3 structural container of Qexpr, wherein every symbolic node is irrevocably identified by the cryptographic commitment hash of its operator, parameters, and child dependencies. Guarantees mathematical cycle freedom by construction and enables O(1) structural equivalence comparisons.

cdqn

Slug: `cdqn` · *Category:* Primitives

Chained and Distributed Quang Numbers. The universal fractal protocol of governed data movement, operating both locally (intra-node causal chaining between abstraction layers and base domains) and distributively (inter-node public attestations across physical networks).

Abstract Compute Unit U

Slug: `compute-unit-u` · *Category:* Primitives

The fundamental, invariant quantum of governed computational work representing the minimal state transition from local genesis Q(0) to first unit Q(1) along dimensional axis d1. Calibrated to physical Landauer dissipation limits (W >= k_B T ln 2) to maintain platform-invariant accounting across heterogeneous hardware substrates.

Dimension d

Slug: `dimension-d` · *Category:* Primitives

A governed dimensional coordinate axis or geometric orientation parameterizing a Qn entity, beginning with axis d1 derived from genesis origin Q(0) and unit Q(1).

Discrete Exterior Calculus

Slug: `discrete-exterior-calculus` · *Category:* Primitives

The constructive formulation of calculus operating on discrete simplicial meshes and cochains, replacing continuous partial differential equations with exact algebraic coboundary operators without floating-point approximations.

Discrete Exterior Derivative d

Slug: `discrete-exterior-derivative` · *Category:* Primitives

The discrete coboundary operator mapping k-forms to (k+1)-forms across simplicial complexes. Satisfies the exact topological nilpotency invariant d ∘ d = Q(0), grounding electromagnetic and fluid conservation laws.

Discrete Hodge Star

Slug: `discrete-hodge-star` · *Category:* Primitives

The geometric duality operator mapping primal simplicial k-forms to dual circumcentric (n-k)-forms, parameterizing exact dual volume ratio quotients.

Q(array)

Slug: `q-array` · *Category:* Primitives

The proposed sequenced container contract surface: a bounded, indexed container whose footprint is a declared length, whose permitted access is indexed, whose locality class is contiguous, and whose traversal order is fixed by causal index with constant declared work per indexed access. Grounded in the OP_FOLD child vector under an explicit step ceiling, it admits no in-place mutation: growth emits a new versioned artifact. A contract surface under the five-row structural contract schema, never a runtime shape; pending deterministic measurement, BOC selection, and Q(rsi) sealing.

Q(bypass)

Slug: `q-bypass` · *Category:* Primitives

A Category B optimization morphism allowing a SIMEMP Gateway to short-circuit execution steps when algebraic preconditions prove an invariant identity transition (such as Terminal Exactness r = Q(0) or multiplication by unit Q(1)). Emits an explicit zero-cost bypass receipt without consuming compute budgets.

Q(c)

Slug: `q-c` · *Category:* Primitives

Physical invariant anchor of the relativistic causal horizon. Calibrated from local dielectric propagation delays, it bounds causally ordered state propagation across the substrate (DCC_ANCHOR_CAUSAL_HORIZON_v1).

Q(dataStruc)

Slug: `q-datastruc` · *Category:* Primitives

Governed immutable memory container artifact class providing bounded, content-addressed storage within the local causal DAG. Allocation, reading, and writing are explicit governed morphisms subject to DCC constraints and Memory Wall envelopes; no implicit heap state is admitted.

Q(even)

Slug: `q-even` · *Category:* Primitives

The constructive parity partition of integers whose Diophantine division by Q(2) yields a remainder identically equal to genesis origin Q(0), demonstrating modulo-2 Terminal Exactness. Includes local genesis origin Q(0).

Q(event)

Slug: `q-event` · *Category:* Primitives

A discrete computational or physical state transition occurring at an explicit spatiotemporal coordinate, carrying a discrete scale quantum, a state transition payload, and a verified epistemic modality.

Q(gauge)

Slug: `q-gauge` · *Category:* Primitives

Physical invariant anchor of topological conservation. Built from simplicial mesh topology and discrete coboundary verification, it grounds conserved field quantities without metric drift (DCC_ANCHOR_DEC_D_v1).

Q(ℏ)

Slug: `q-hbar` · *Category:* Primitives

Physical invariant anchor of quantum action. Built from height ceilings and quantum action limits, it forbids unmetered infinite information density in finite registers (DCC_ANCHOR_ACTION_PLANCK_v1).

Q(matrix)

Slug: `q-matrix` · *Category:* Primitives

The proposed geometric container contract surface: a bounded, two-dimensional integer container whose contractions complete in constant declared time through the integer Matrix-Multiply-Accumulate lowering, preserving exactness by construction without floating-point drift, and admitting the Q(bypass) morphism to short-circuit identity preconditions in register time. Grounded in the OP_FRAME_MAP node class. A contract surface under the five-row structural contract schema, never a runtime shape; pending deterministic measurement, BOC selection, and Q(rsi) sealing.

Q(odd)

Slug: `q-odd` · *Category:* Primitives

The constructive parity partition of integers whose Diophantine division by Q(2) yields a remainder identically equal to first unit Q(1).

Q(patterns)

Slug: `q-patterns` · *Category:* Primitives

An invariant, reproducible state-transition template mapping input commitments to deterministic output milestones under abstract DCC contracts. Encompasses both exact algebraic transformations and deterministic terminal fault signatures.

Q(reuse)

Slug: `q-reuse` · *Category:* Primitives

A Category B optimization morphism that references existing immutable content commitment hashes within the local causal DAG instead of allocating new physical memory bits. Mitigates the Memory Wall and eliminates redundant Landauer dissipation at zero marginal Q(1) cost.

Q(stream)

Slug: `q-stream` · *Category:* Primitives

The proposed bounded stream contract surface for sequenced data: declared capacity with sequenced, append-ordered access fixed by causal index. Bounded streams are already exercised, through declared sample volumes in entropy measurement and append-only receipt chains, while no container type is committed; an implicitly resizable queue would violate the footprint and determinism rows and is not admitted. Nominated as an open contract shape under the five-row structural contract schema, never a runtime shape; pending implementation feedback, deterministic measurement, BOC selection, and Q(rsi) sealing.

Q(table)

Slug: `q-table` · *Category:* Primitives

The proposed keyed container contract surface: a bounded, content-addressed lookup structure grounded in hash-consing itself, the Content-Addressed Expression DAG already constituting a keyed index over committed expressions. Capacity is declared; growth is re-versioning rather than resizing; the work row declares the breakeven inequality under which keyed lookup is preferred

over its recorded alternatives, in the manner of the committed Q(reuse) analysis. A contract surface under the five-row structural contract schema, never a runtime shape; pending deterministic measurement, BOC selection, and Q(rsi) sealing.

Q(th_2)

Slug: `q-th-2` · *Category:* Primitives

Physical invariant anchor encoding the Second Law. Calibrated from sampled thermal receipts via ln 2 dissipation quanta, it mandates that endergonic processes execute only under a declared Gibbs free-energy budget (DCC_ANCHOR_SECOND_LAW_v1).

Q(workflows)

Slug: `q-workflows` · *Category:* Primitives

A composite, goal-directed causal directed acyclic graph assembling multiple Q(patterns) under a declared Intent Contract and finite Metric Budget. Portable across any cdqn node via chained Road Receipts.

Q(Z_bus)

Slug: `q-z-bus` · *Category:* Primitives

Physical invariant anchor of the Memory Wall. Built from parasitic wire capacitance and bus line physics, it calibrates transport latency and bandwidth ceilings of the substrate (DCC_ANCHOR_MEMORY_WALL_v1).

Q(0)

Slug: `q0` · *Category:* Primitives

Local genesis object, causal origin zero, empty birth context. Strictly local to each node; no universal global zero is admitted.

Q(1)

Slug: `q1` · *Category:* Primitives

First unit artifact, unity measure, and baseline reference for the abstract compute unit U along dimensional axis d1.

Qbio

Slug: `qbio` · *Category:* Primitives

Quang Biology. An emergent biosemiotic domain derived from the categorical tensor product of Quang Chemistry and Quang Semantics (Qchem ⊗ Qs). Models living metabolic homeostasis, genetic codon information channels, cellular automata, and active inference operating as bounded SIMEMP open systems.

Qchem

Slug: `qchem` · *Category:* Primitives

Quang Chemistry. An emergent composite domain derived from the categorical tensor product of Quang Physics and Quang Mathematics (Qphy ⊗ Qm). Models discrete stoichiometric balance equations, molecular bonding graphs, reaction kinetics, and Gibbs free energy dissipation without semantic intent.

Qexpr

Slug: `qexpr` · *Category:* Primitives

Governed expression tuple encapsulating symbolic structure, zoom, remainder, dimension, DCC profile, identity, and causal lineage. Structured as a Content-Addressed Expression DAG (CAE-DAG) with hash-consing.

Qlog

Slug: `qlog` · *Category:* Primitives

Quang Logics. An emergent composite domain derived from the categorical tensor product of Quang Mathematics and Quang Semantics (Qm ⊗ Qs). Formalizes constructive type theory, intuitionistic propositional deduction, and automated proof verification without ungrounded non-constructive axioms.

Qm

Slug: `qm` · *Category:* Primitives

Quang Mathematics. The local-first constructive algebraic, numeric, and relational base domain of the Qn universe. Operates over discrete tuples ⟨σ, q_z, z, r_z, d⟩ without floating-point mechanics, enforcing Diophantine division constraints, projective remainder conservation, and positional order decidability.

Qn

Slug: `qn` · *Category:* Primitives

Quang Number. A governed, finite, identifiable, measurable, and security-constrained computational artifact within the Qn universe.

Q(rsi)

Slug: `qn-rsi` · *Category:* Primitives

A higher-order recursive self-optimization workflow. Ingests an existing Q(workflows) artifact, derives an empirically more efficient successor under strict semantic invariance and finite metric budget, and emits a versioned successor artifact.

QnIR

Slug: `qnir` · *Category:* Primitives

Intermediate Representation used for hardware abstraction, bound enforcement, and deterministic execution without exposing floating-point semantics.

QnLang

Slug: `qnlang` · *Category:* Primitives

The high-level domain-specific language used to author Qn computations, enforcing explicit DCC profiles and compile-time collapse.

Qphy

Slug: `qphy` · *Category:* Primitives

Quang Physics. The local-first constructive physical base domain of the Qn universe. Models non-equilibrium computational thermodynamics, discrete exterior cochain field dynamics, Landauer dissipation bounds, and physical Q(anchor) invariants directly grounded in Layer 0 silicon substrates.

Qs

Slug: `qs` · *Category:* Primitives

Quang Semantics. The constructive relational, linguistic, and intent base domain of the Qn universe. Formalizes words as multivector blades, grammar as CAE-DAG wiring, and human-machine communication as verifiable intent contracts.

Remainder r

Slug: `remainder-r` · *Category:* Primitives

The conserved residual quantity preserved at a declared zoom level or division constraint. Acts as the exact input state and generator for subsequent zoom scale expansion along dimension d, preventing precision decay and replacing IEEE 754 rounding approximations.

Successor Morphism S

Slug: `successor-morphism` · *Category:* Primitives

The discrete algebraic transition function S(x) = x + Q(1) generating sequential integer artifacts inductively along dimensional axis d1 without continuous or floating-point approximations.

Zoom z

Slug: `zoom-z` · *Category:* Primitives

The declared precision boundary, metric resolution quantum, or discrete scale level of a governed numeric expression. Parameterizes projective resolution lattices while remaining base-independent in core representations.

Annex C — SIMEMP constraint architecture

Pinned source: docs/simemp.md
Committed version: 1.1.1 **Last updated:** 2026-10-11
sha256 (committed file): `5c5b78e556ae423ee208858d46f68cf460e289bc38e544acc13564a560053460`

Field	Specification
Document Title	SIMEMP: Thesis and Constraints for the Qn and cdqn Stack
Version	1.1.1
Last Updated	2026-10-11 (Bao Loc, Vietnam)
Author	Christophe Duy Quang Nguyen
License	Scaling Source License (SSL) 1.0
Status	Canonical Constitutional Specification — Foundational Invariants

---

Normative References

The following documents establish the structural, mathematical, and legal constraints derived from this constitutional framework.

Document	Role	Target
`docs/abstractionLayers.md`	Layer architecture and SIMEMP Gateway validation	abstractionLayers.html
`docs/qnPrimitive.md`	Universal Envelope, operational axioms, and optimization primitives	qnPrimitive.html
`docs/q0_q1.md`	Primary genesis origin Q(0) and first unit Q(1)	q0_q1.html
`docs/qm.md`	Constructive numeric leaf substrate and Diophantine division constraints	qm.html
`docs/qexpr.md`	Layer 3 Content-Addressed Expression DAG specification	qexpr.html
`docs/qm_calculus.md`	Discrete Exterior Calculus, continued fractions, and invariant dynamics	qm_calculus.html
`docs/qs.md`	Constructive relational intent, epistemic spectrum, and human-agent alignment	qs.html
`LICENSE.md`	Scaling Source License 1.0 governing the Licensed Work	LICENSE.md

---

1. The Foundational Conjecture and Dependencies Determinism

The foundational premise of this architecture addresses the formal capability of discrete number systems: *Can a discrete **Qn** number system abstract any computable phenomenon?*

Classical mathematics admits non-constructive entities: actual infinities, unmetered memory access, and continuous spaces lacking thermodynamic boundaries. Conversely, physical computational engines are strictly finite, bounded by thermodynamics, discrete memory hierarchies, and finite signal transmission speeds.

We postulate the operational conjecture:

> A number system can abstract any computable phenomenon within a computational environment if, and only if, that abstraction is strictly governed by finite physical realities and constrained by the **SIMEMP** framework.

This conjecture is formally rooted in **Dependencies Determinism**. In an information-theoretic sense, the state transition of any computational entity S_t is fully determined if, and only if, its conditional entropy given its complete set of declared causal dependencies $\mathcal{D}(S_t)$ vanishes:

$$H(S_t \mid \mathcal{D}(S_t)) = 0$$

If any dependency is implicit, hidden, or unmeasured (violating the **No-Implicit Rule**), or attempts to cross an unverified abstraction boundary or **SIMEMP Gateway**, the available dependency set is incomplete ($\mathcal{D}' \subset \mathcal{D}$), yielding non-zero conditional entropy:

$$H(S_t \mid \mathcal{D}') > 0$$

This residual entropy manifests operationally as non-deterministic state drift, runtime instability, or unmodeled attack surfaces. The SIMEMP framework ensures that every state transition's causal entropy is fully accounted for by explicit, verifiable dependencies across the **cdqn** stack.

---

2. Method of Verification: The Recursive Search Loop

The Qn architecture does not claim closed empirical finality. The system operates as an active, recursive research framework grounded in operational falsifiability and multi-tier constraint auditing.

2.1. Operational Verification and Computational Consistency

Because the system explores finite computational bounds ahead of physical hardware realization, all models are working hypotheses structured to prevent premature calcification. The development trajectory is governed by a cyclic search loop:

$$\text{Proposal} \longrightarrow \text{Construction} \longrightarrow \text{Evaluation} \longrightarrow \text{Selection}$$

Abstractions are verified empirically through **Computational Consistency**:

> If a proposed abstraction is internally consistent, and if computations authored in **QnLang** and compiled to **QnIR** execute deterministically and terminate within declared finite resource envelopes, the abstraction is deemed provisionally valid within the Qn universe:

$$\text{Validity}(\mathcal{A}) \iff (\text{Consistent}(\mathcal{A}) \land \forall p \in \text{QnIR}(\mathcal{A}),\ \text{ExecutesDeterministicallyWithinBounds}(p))$$

Failure to compute within finite bounds, or divergence during execution, refutes the proposed abstraction and forces an iteration of the search loop.

2.2. Epistemic Demarcation: Documentary Baselines vs. Empirical Grounding

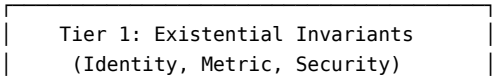
To prevent ambiguity across documentation indexes and automated repository ingestion, the architecture strictly distinguishes between **Documentary Maturity** and **Epistemic Modality**:

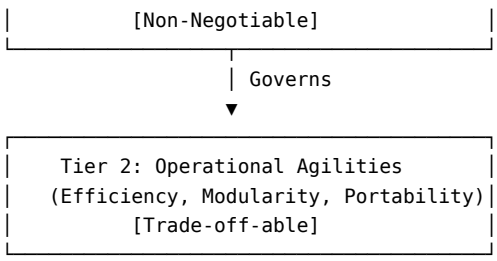
- **Documentary Maturity (Canonical Baseline):** Specifications designated as **Canonical** represent frozen structural contracts for the current iteration of the search loop. They provide an authoritative, immutable baseline that distinguishes validated architecture from exploratory vision proposals (`docs/chronosa.md`) or uncommitted roadmap items.
- **Epistemic Modality (Constructive Working Hypothesis):** Because the runtime has not yet been executed on physical silicon substrates (Layer 0), all canonical specifications maintain the formal epistemic status of **Constructive Working Hypotheses** pending empirical Layer 0 execution receipts.

---

3. The SIMEMP Constraint Architecture

The constraint architecture enforces a strict two-tier hierarchy across all abstraction boundaries.





3.1. Tier 1: Existential Invariants

Tier 1 Existential Invariants represent non-negotiable requirements for any governed artifact to exist. If an abstraction violates any invariant, it is discarded:

1. **Identity:** The abstraction must be distinctly known, collision-resistant, versionable, and causally rooted.
2. **Metric:** The abstraction must carry an explicit, finite measurement envelope declaring memory, compute, precision, and uncertainty boundaries.
3. **Security:** The abstraction's state transitions, causal integrity, and execution limits must be verifiable against finite adversarial models.

3.2. Tier 2: Operational Agilities

Tier 2 Operational Agilities represent optimization parameters. They naturally conflict with one another:

1. **Efficiency:** Algorithmic throughput and minimal resource dissipation.
2. **Modularity:** Strict isolation of concerns, boundary containment, and composability.
3. **Portability:** Substrate independence across execution targets and network architectures.

The Golden Rule of Trade-offs: An Operational Agility may always be sacrificed to preserve an Existential Invariant, but an Existential Invariant must never be compromised for an Agility.

---

4. Environmental Realities, Thermodynamics, and the DCC Profile

Physical computation requires formalizing hardware boundaries and thermodynamic laws directly into computational metadata.

4.1. The Memory Wall, Semantic Gap, and Landauer's Principle

- **Landauer's Bound and Dissipation:** Any irreversible manipulation or erasure of information dissipates thermodynamic work:

$$W \geq k_B T \ln 2$$

- **The Memory Wall:** Transporting data across physical memory hierarchies dissipates physical energy ($\mathcal{O}(CV^2 f)$) and introduces transmission latency. The Metric invariant requires explicit accounting for memory bandwidth and locality rather than purely abstract asymptotic complexity.
- **The Semantic Gap:** Translation between high-level intent (**QnLang**) and physical execution (**QnIR**) represents a primary vulnerability. The system mandates bounded, verifiable intermediate representations.

4.2. Open Systems, Entropy Export, and Dissipative Error Correction

A computational system is an **open thermodynamic system**. Left unconstrained, thermal noise, transmission faults, and physical substrate errors maximize internal entropy ($dS_{\text{internal}} > 0$).

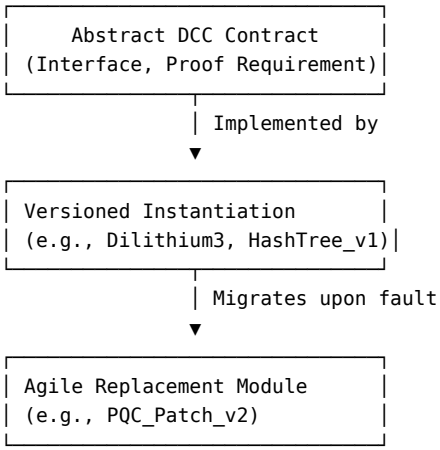
To maintain internal structural order and deterministic reproducibility, the system operates as a **dissipative structure** (Prigogine formulation):

$$dS = dS_{\text{internal}} + dS_{\text{exchange}}, \quad \text{where } dS_{\text{exchange}} < 0$$

Maintaining computational stability requires expending work—formally recognized as **Metric Exhaustion**—to verify state transitions and export entropy via bounded **receipts**. Without reproducible, receipt-generating validation steps, a system cannot self-repair against entropy accumulation.

4.3. The DCC Profile and Structural Indirection

To prevent rigid calcification and ensure defense against zero-day vulnerabilities, the system implements **Structural Indirection**:



Every abstraction layer, morphism, and gateway exposes an explicit **DCC Profile** (**Dependencies, Constraints, Capabilities**):

- **Dependencies:** Abstract capability contracts, cryptographic commitments, and environmental prerequisites. Root dependencies must be parameterized interfaces rather than hardcoded low-level implementations.
- **Constraints:** Hard ceilings on memory allocation, execution steps, recursion depth, and precision budgets.
- **Capabilities:** Permitted operations, transformation rights, and network export authorizations.

If a concrete cryptographic primitive or implementation module suffers a zero-day vulnerability, structural indirection allows the system to swap the implementation module and emit an explicit migration receipt without fracturing the abstract DCC contract or breaking causal lineage.

---

5. Existential Invariants Deep Dive

5.1. Identity Constraints

- **Layered Verification:** Abstractions must generate verifiable witnesses at each layer they traverse.
- **Provenance and Lineage:** Artifacts maintain a traceable derivation history. Pursuant to the **SSL**, all runtime specifications and derivative works must embed the required **Paternity Reference**.

5.2. Metric Constraints

- **Finite Resource Envelopes:** Unmeasurable abstractions are physically unrealizable. Every operation must declare an explicit budget for execution, storage, and communication.
- **Totality by Budget:** Every computational transition must be total by budget, resolving to an explicit **Receipt** containing a discrete terminal state (`SUCCESS`, `FAILURE`, `TIMEOUT`, `BUDGET_EXHAUSTED`, `INCONCLUSIVE`).

5.3. Security Constraints

- **Bounded Adversaries and PQC Agility:** Security models assume computationally bounded adversaries. The cryptographic substrate must maintain agility, remaining Post-Quantum Cryptography (PQC) capable without mutating underlying identity semantics.

- **Halt-by-Construction:** Computational subsystems must retain deterministic halt authority upon constraint exhaustion. Turing-unbounded execution is rejected within the governed core.
- **The Causal Arrow:** Distributed state ordering relies strictly on a persistent **Causal Arrow** (monotonic sequences and cryptographic hash chains) rather than external wall-clock synchronization.

---

6. Constraint on Generative and Probabilistic Workflows

Stochastic and generative artificial intelligence processes are integrated via an explicit epistemic boundary:

```
[ Context & DCC ] ⟶ [ Probabilistic Proposal ] ⟶ [ Deterministic Construction ]
                                                                 |
[ Verified Receipt ] ⟵ [ Finite Selection ] ⟵ [ Metric Evaluation ]
```

1. **Context Declaration:** Establish explicit **DCC profiles**, objective functions, and finite budgets.
2. **Generation:** Stochastic engines propose candidate structures (classified strictly as *unverified candidates*).
3. **Construction:** Candidates are formalized into canonical representations.
4. **Evaluation:** Candidates are measured against constraints, error tolerances, and resource consumption.
5. **Selection & Sealing:** A deterministic policy validates and seals acceptable candidates, emitting a signed **Receipt**.
6. **Bounded Termination:** If no candidate meets the threshold within the declared budget, execution halts with a defined status (`NO_SOLUTION` or `BUDGET_EXHAUSTED`).

---

7. The BOC Policy

The **Best of Choices (BOC) Policy** is the formal decision framework used to resolve architectural and design conflicts:

$$\text{Selection} = \arg\max_{c \in \mathcal{C}} \left(\text{SIMEMP Compliance}(c)\right)$$

When implementation paths conflict, the BOC policy selects the option that minimizes implicit state and guarantees invariant preservation, even if it requires higher operational overhead.

---

8. Alignment with the Scaling Source License

The architectural constraints directly enforce the legal mechanics of the **SSL**:

- **Open Core Invariants:** Identity and security layers cryptographically preserve the **Open Core Invariants** (anti-patent retaliation and non-scaling open-access clauses) across derivative lineage trees.
- **Scale Auditing:** The Metric invariant provides native, verifiable telemetry (active instances, container count, agent instances, and monthly transaction volumes) to verify whether deployment scale crosses the **Scaling Threshold**.
- **Safe Harbor Neutrality:** The technical substrate operates as a neutral conduit, preserving sovereign user content and applications as uninspected **Payloads**.

---

9. Conclusion

The SIMEMP framework is a provisional, evolutionary thesis designed to formalize physical computing limits into explicit architectural constraints. By grounding computational abstractions in non-equilibrium thermodynamics, dependencies determinism, and structural indirection, SIMEMP establishes an adaptable foundation for exploring finite, accountable, and generative computational systems across the **cdqn** network. All subsequent primitive derivations and operational specifications remain subject to verification and revision under this framework.

Annex D — Quantum Sovereignty framework

Pinned source: docs/qs.md
Committed version: 1.2.0 **Last updated:** 2026-10-21
sha256 (committed file): `fb74d5450dbc736b74754cede7742d2a43a87a6856fc8c9ef622eb122e5e282f`

Field	Specification
Document Title	Qs — Quang Semantics: Constructive Relational Intent and Epistemic Grounding
Version	1.2.0
Last Updated	2026-10-21 (Bao Loc, Vietnam)
Author	Christophe Duy Quang Nguyen
License	Scaling Source License (SSL) 1.0
Status	Canonical Domain Specification — Category D (Local-First Base Domain)

Normative References

The following documents establish the physical, structural, and legal constraints governing Qs. If a technical conflict arises, `simemp.md` governs; if a structural conflict arises, `abstractionLayers.md` governs; if a legal conflict arises, `LICENSE.md` governs. References are deliberately version-unpinned: each target document's front matter carries its own authoritative version, so this specification never duplicates drift-prone patch state.

Document	Role	Target
`docs/simemp.md`	Constitutional constraints, thermodynamics, and Dependencies Determinism	simemp.html
`docs/abstractionLayers.md`	Layer architecture and Complexity Degree Stratification	abstractionLayers.html
`docs/qnPrimitive.md`	Universal Envelope, operational axioms, and optimization primitives	qnPrimitive.html
`docs/q0_q1.md`	Primary genesis origin Q(0) and first unit Q(1)	q0_q1.html
`docs/qm.md`	Constructive numeric leaf substrate and Diophantine division constraints	qm.html
`docs/qm_geometry.md`	Multi-axial frames, Clifford geometric algebra, and float-free rotations	qm_geometry.html
`docs/qexpr.md`	Layer 3 Content-Addressed Expression DAG specification	qexpr.html
`docs/chronosa.md`	Emergent non-local causal intelligence and human empowerment	chronosa.html
`LICENSE.md`	Scaling Source License 1.0 governing the Licensed Work	LICENSE.md

1. Epistemic Stance and Constitutional Grounding

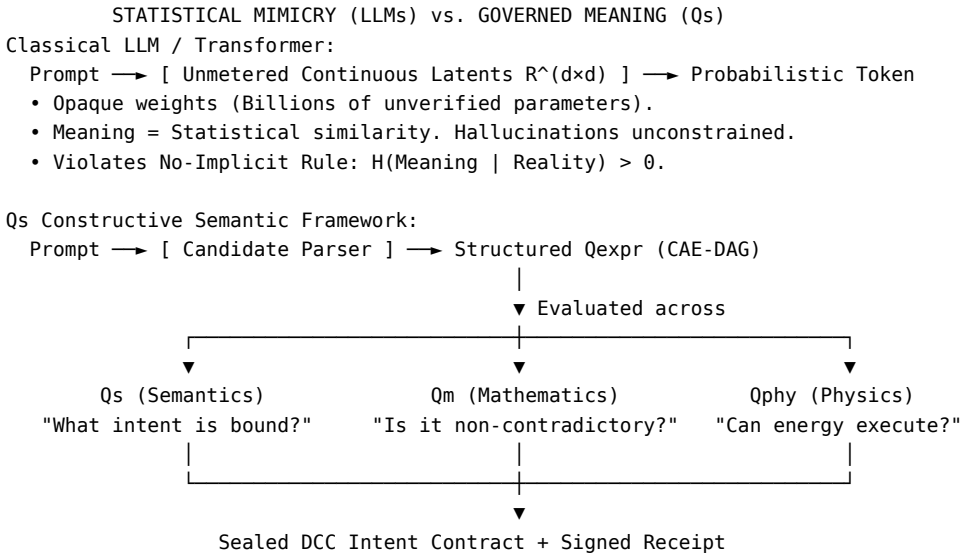
In classical artificial intelligence, natural language processing relies on statistical correlation across continuous floating-point vectors ($\mathbb{R}^d$). Transformer models predict tokens based on statistical consensus rather than physical truth or causal provenance. Consequently, classical models suffer from hallucinations, lack causal accountability, and cannot distinguish between verified empirical facts and unanchored fiction.

In the **Qn** computational universe, **language, meaning, and intent are not ungrounded statistical approximations**. Under `docs/simemp.md`, semantic meaning must be constructed from finite, verifiable primitives governed by **Dependencies Determinism**:

$$H(S_t \mid \mathcal{D}(S_t)) = 0$$

Qs (Quang Semantics) formalizes language, human-machine communication, and AI agent alignment as a **constructive, multi-axial relational domain**: * **Language is not a black-box tensor:** It is a structured graph of discrete semantic blades and relational morphisms. * **Meaning is not statistical similarity:** It is an explicit, verifiable **Intent Contract** bound to human authority. * **The Non-Rogue Invariant:** Machines and AI agents cannot originate unanchored intent; they operate strictly as delegated proxies bounded by human sovereignty and thermodynamic metric budgets.

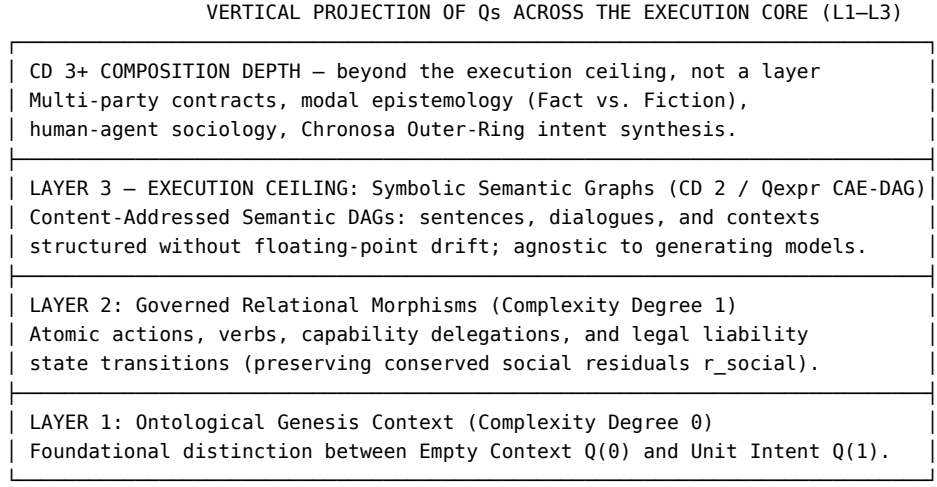
Operational validity is verified empirically via **Computational Consistency**. No unmeasured semantic assumption may cross a **SIMEMP Gateway**.



---

2. Vertical Abstraction Layer Projection of Qs

Mirroring the mathematical domain (`docs/qm.md`), **Qs** does not reside in a single, isolated tier. It is an orthogonal vertical projection crossing the execution core (`docs/abstractionLayers.md` §2.5):



- **CD vs. Layer Note:** Complexity Degree (CD) indexes structural composition depth, not layer numbers. The LAYER 1–3 bands name the execution-core strata crossed by the Qs projection; their CD 0/1/2 labels measure composition depth accumulated at each stratum, and CD 3+ marks depth accumulated above the ceiling without implying a Layer 4.

2.1. Layer 1 — Contextual Genesis (Complexity Degree 0)

At **Layer 1**, Qs establishes the primitive origin of meaning: * **Q(0)** acts as the **Empty Context** (the unwritten page, the silent void, zero unfulfilled obligation). * **Q(1)** acts as the **Unit Intent Measure** along dimensional axis d_1, calibrating the baseline quantum of intentional focus and thermodynamic attention.

2.2. Layer 2 — Governed Relational Morphisms (Complexity Degree 1)

At Layer 2, Qs establishes atomic relational transformations: * Verbs, capability delegations, and authorizations executed under explicit **DCC Profiles**. * Every action partitions an obligation into an authorized transition and an exact **conserved social residual** (r_{social}), ensuring that unfulfilled liabilities are never silently discarded.

2.3. Layer 3 — Symbolic Composition Graphs (Complexity Degree 2)

At Layer 3, Qs structures compound propositions, dialogues, and multi-agent interaction histories as **Content-Addressed Expression DAGs (`docs/qexpr.md`)**: * Identical semantic sub-trees and legal clauses share immutable cryptographic hashes via **Content-Addressed Expression DAG** hash-consing. * Incurs zero redundant bit allocation across the Memory Wall via **Q(reuse)**.

2.4. CD 3+ Composition Depth — Epistemic Modality and Social Systems

Above the Layer 3 execution ceiling, Qs accumulates Complexity Degree 3+ composition depth — emergent social capability, not additional execution layers: * The formal separation between physical facts and sovereign fiction via **Epistemic Anchorage**. * Multi-party consensus synthesized by **Chronosa** over public road receipts.

---

3. Words and Grammar as Multi-Axial Clifford Blades

In Qs, vocabulary and linguistic relationships are formalized through **Clifford Geometric Algebra ($\mathcal{C}\ell_{p,q}$)** over an orthogonal frame of semantic dimensions (`docs/qm_geometry.md` §2.1):

$$\mathcal{D}_{\text{semantic}} = \{d_{\text{intent}},\, d_{\text{consent}},\, d_{\text{obligation}},\, d_{\text{temporal}},\, d_{\text{agent}},\, \dots\}$$

```
                LINGUISTIC CLUSTER AS CLIFFORD BLADES
Noun Blade v_noun : Grade-1 vector along basis e_k (e.g., e_device)
Modifier Rotor R  : Rational Cayley Rotor in plane e_i ∧ e_j (e.g., "secure")
Transitive Verb M : Bivector contraction mapping Subject to Object
                    ⟨ v_subject ∧ M_verb,  v_object ⟩ + r_social
```

3.1. Nouns as Graded Vector Blades

An entity or concept (“battery”, “drone”, “contract”) is a discrete vector blade evaluated at declared **Zoom z**:

$$\vec{v}_{\text{noun}} = \sum_{k=1}^{n} \langle \sigma_k,\, q_{z,k},\, z,\, r_{z,k},\, d_k \rangle\, \vec{e}_k$$

Nouns carry explicit identities, coordinate axes, and conserved residuals.

3.2. Adjectives and Modifiers as Rational Cayley Rotors

Modifiers (“cold”, “verified”, “urgent”) do not mutate the fundamental type of a concept. A modifier is a **Rational Cayley Rotor** (`docs/qm_geometry.md` §5.2) rotating the noun blade within semantic sub-spaces:

$$\vec{v}' = R\vec{v}_{\text{noun}}R^{\dagger}, \quad \text{where } R = \frac{Q(1) - B}{Q(1) + B}$$

Because $RR^{\dagger} \equiv Q(1)$, **semantic modifications preserve unitary norm by construction**: meaning transforms with zero precision drift, zero semantic degradation, and zero floating-point rounding.

3.3. Verbs as Conserved Relational Morphisms

A verb is not an ambiguous text string. In Qs, a transitive verb ("Human A delegates task T to Agent B with budget W") is a governed morphism evaluated through Diophantine conservation:

$$\text{Action} = \big(\text{Delegation} \times \text{Actor}\big) + r_{\text{social}}$$

If the task fails, encounters a fault, or halts, the residual r_{social} remains sealed in the lineage. The system holds an immutable record of unfulfilled obligations, precluding silent state abandonment.

---

4. The Epistemic Anchorage Spectrum: Anti-Hallucination Firewall

To eliminate AI hallucinations, Qs outlaws ungrounded assertions under the **No-Implicit Rule**. Every statement in Qs must declare its explicit ****Epistemic Anchorage****:

```
                    THE EPISTEMIC ANCHORAGE SPECTRUM
┌──────────────────────────────────────────────────────────────────────────┐
│ 1. MODAL_EMPIRICAL (Grounding G = 1.0)                                   │
│    Anchored directly to Layer 0 physical sensor receipts (NIST entropy,  │
│    thermocouple, camera, bus telemetry) via a SIMEMP Gateway.            │
├──────────────────────────────────────────────────────────────────────────┤
│ 2. MODAL_CONSTRUCTIVE (Grounding G = 1.0)                                │
│    Algebraic closure proven via Qm / DEC; remainder r = Q(0) (Terminal   │
│    Exactness receipt). Cannot drift or hallucinate by construction.      │
├──────────────────────────────────────────────────────────────────────────┤
│ 3. MODAL_HYPOTHESIS (Grounding 0.3 < G < 1.0)                            │
│    Explicitly bounded within the Recursive Search Loop (simemp.md §2).   │
│    Declared as a working conjecture awaiting physical binary falsification. │
├──────────────────────────────────────────────────────────────────────────┤
│ 4. MODAL_NARRATIVE (Contextually Sealed Creative Worldbuilding)          │
│    Internally consistent narrative anchored to author identity Q(0)_author. │
│    True WITHIN the declared context; barred from triggering actuators.   │
├──────────────────────────────────────────────────────────────────────────┤
│ 5. MODAL_UNANCHORED (Grounding G = ∅)                                    │
│    Claims empirical or logical truth without parent lineage receipts.    │
│    REJECTED AT THE GATEWAY! (RECEIPT_UNANCHORED_ASSERTION_REJECTED)      │
└──────────────────────────────────────────────────────────────────────────┘
```

4.1. The Lineage Verification Mandate

Under **Axiom 4** and **Axiom 5** (`docs/qnPrimitive.md`), a statement cannot claim empirical truth through latent probability alone. If an artificial agent asserts: *"The motor temperature is 450°C"*, the gateway queries its lineage: * If the lineage contains an unbroken causal chain terminating in a verified **Layer 0** hardware sensor receipt, it is admitted as `MODAL_EMPIRICAL`. * If the assertion traces only to an internal neural probability distribution without sensor receipts, the gateway intercepts it:

$$\text{Gateway Action: Reject}\Big(\mathcal{H}(\text{Statement})\Big) \implies \text{Emits Receipt}$$

The gateway emits the rejection receipt `RECEIPT_UNANCHORED_ASSERTION_REJECTED`; the claim is physically barred from entering the execution pipeline.

4.2. Contextual Isolation for Creative Fiction

Qs does not prohibit human imagination, creative writing, or artistic worldbuilding: * A creative work (such as a fantasy novel or game narrative) is typed as `MODAL_NARRATIVE` anchored to the author's sovereign identity $Q(0)_{\text{author}}$ and publisher receipt. * The internal consistency of the fictional universe is preserved within its closed `Qexpr` sub-DAG. * The gateway guarantees **contextual containment**: a fictional sub-DAG is barred from ever triggering real-world physical actuators, financial registers, or legal delegations.

4.3. The Source-Receipt Invariant (Proposed, CANDIDATE)

Status. This subsection records a proposed extension of the §4 firewall from statements to governed artifacts, sourced from `docs/qs_history.md` §3. It carries `State: CANDIDATE` under **Axiom 14**: it remains subject to deterministic measurement and **BOC** selection through the pipeline of §6, and it amends nothing in §4.1 or §4.2, which continue to apply exactly as written.

It is proposed that every governed artifact of the **Qn** universe crossing a **SIMEMP Gateway** declare an explicit pair:

(content modality, source class)

where the content modality is drawn from the **Epistemic Anchorage** spectrum of §4, and the source class is a declared expression of where the artifact's asserted content ultimately anchors. Under the **No-Implicit Rule**, the declaration makes explicit — at the level of composite artifacts and worlds — what the gateway already enforces implicitly at the level of single statements.

ARTIFACT-LEVEL SOURCE DECLARATION (PROPOSED, STATE: CANDIDATE)

```
┌──────────────────────────────────────────────────────────────────────┐
│ Governed Artifact  :=  ( ContentModality , SourceClass )             │
├──────────────────────────────────────────────────────────────────────┤
│ ContentModality : one modality of the §4 spectrum, assigned per      │
│                   atomic claim decomposition of the artifact         │
│ SourceClass     : declared expression of where the asserted content  │
│                   ultimately anchors (candidate vocabulary I / P / S,│
│                   docs/qs_history.md §2)                             │
│ SourceReceipts  : unbroken chain terminating in physically receipted │
│                   reality events; no dangling source references      │
└──────────────────────────────────────────────────────────────────────┘
```

The invariant requires that the source receipt chain terminate in physically receipted reality events, with the terminating receipt type fixed by the declared modality:

- **Empirical content** terminates in verified **Layer 0** sensor or artifact receipts, exactly as mandated by §4.1.
- **Constructive content** terminates in algebraic closure receipts whose remainder $r = Q(0)$ (Terminal Exactness).
- **Contextually sealed creative content** terminates in the author-identity ($Q(0)_{\text{author}}$) and publisher-receipt anchoring already mandated by §4.2.

The extension therefore applies the §4.1 statement-level rejection rule to composite artifacts and worlds: no unsourceable fiction, no unsourced anchor, no history-less constant. An artifact whose source receipt chain cannot be completed is rejected at the gateway exactly as an unanchored statement is, emitting the same class of rejection receipt.

Two cautions are recorded, and they are binding on any future implementation of the invariant:

1. **Payload sovereignty is untouched.** The invariant governs artifacts that cross the gateway as governed objects of the **Licensed Work**. Sovereign human `Payloads` remain uninspected behind the safe-harbor air gap (`docs/abstractionLayers.md` §3.1). Where a creative work is declared as a governed `MODAL_NARRATIVE` artifact, its anchoring to author identity and publisher receipt is **declared by its author**, not extracted by inspection.
2. **Simulation modality is two-faced.** A simulation's execution is `MODAL_CONSTRUCTIVE` (deterministic algebraic closure), while its claims *about reality* are `MODAL_HYPOTHESIS` until physically falsified. The pair declaration makes this two-faced split explicit rather than leaving it implicit, in accordance with the **No-Implicit Rule**.

The proposed Anchor Grounding Classes of `docs/qs_history.md` §2 (Classes I / P / S) supply the candidate vocabulary for the source-class field. Glossary registration of the newly named concepts — and the final sealing of this invariant itself — remain deferred to the integration queue of that document (§8), preserving its discipline against dead glossary entries for unmeasured concepts.

---

5. Human-Agent Alignment: The Intent Contract in DCC Profiles

In the Qn universe, alignment between humans and artificial agents is not achieved by fine-tuning neural weights (RLHF). It is enforced through **Structural Intent Contracts** embedded inside the **DCC Profile**:

ANATOMY OF AN INTENT CONTRACT

```
┌──────────────────────────────────────────────────────────────────────┐
│ DCC_INTENT_CONTRACT_v1                                               │
```

```
├──────────────────────────────────────────────────────────────────────────┤
│ • Sovereign Anchor : Hash(pk_human_principal) [Human Identity Q(0)_H]    │
│ • Agent Proxy      : Hash(pk_agent) [Identity Class: AI_AGENT_IDENTITY]  │
│ • Permitted Domain : Multi-axial frame D_allowed (Spatial / Financial / Act)│
│ • Metric Budget    : Max Work Allocation W_allowed in Q(1) units         │
│ • Epistemic Scope  : Allowed Modalities { MODAL_EMPIRICAL, MODAL_HYPOTHESIS }│
│ • Revocation Path  : Immediate un-delegation on human signature          │
└──────────────────────────────────────────────────────────────────────────┘
```

5.1. Subordination of Machine Agency

Under the **Identity Class** rules (`docs/abstractionLayers.md` §8.1): 1. An AI agent possesses **zero independent root sovereignty**. It cannot instantiate an unanchored genesis origin $Q(0)_{\text{agent}}$. 2. An agent executes strictly as an authorized proxy under an unbroken delegation chain originating from a human principal (**Q(0)_H**). 3. If the human principal revokes the capability token, the agent's execution authority terminates instantly.

5.2. The Thermodynamic Leash

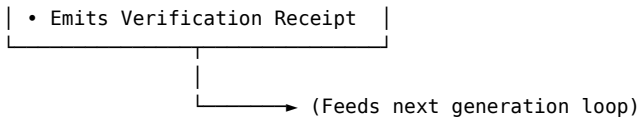
Autonomous agents cannot self-replicate or consume unmetered resources. Under **Axiom 6** (Totality of Governed Operations), every agentic workflow is bounded by a finite work budget in units of **U** via **Metric Exhaustion**. When the allocated budget reaches zero, the agent physically halts, emitting a signed termination receipt.

---

6. The Generative Candidate Pipeline (Q(rsi) Integration)

Qs is completely paradigm-agnostic: it does not reject generative AI models (transformers, diffusion, active inference). It incorporates them into the **Recursive Search Loop** (`docs/simemp.md` §2, §6):

```
THE GENERATIVE Q(rsi) RECURSIVE LOOP
┌──────────────────────────────┐
│ PRIOR STATE / CONTEXT S_k    │
└──────────────┬───────────────┘
               │
               ▼
┌──────────────────────────────┐
│ GENERATIVE EXPLORATION STEP  │
│ (Neural / Stochastic Engine) │
│ Proposes Candidate C_1 ... C_n│
└──────────────┬───────────────┘
               │
               ▼
┌──────────────────────────────┐
│ SIMEMP MEASUREMENT & AUDIT   │
│ • Check Depth & Node Ceilings │
│ • Verify Intent Constraints  │
│ • Measure Metric Burn (W)    │
└──────────────┬───────────────┘
               │
               ▼
┌──────────────────────────────┐
│ BOC SELECTION (simemp.md §7) │
│ ArgMax( SIMEMP Compliance )  │
└──────────────┬───────────────┘
               │
               ▼
┌──────────────────────────────┐
│ SEALS SUCCESSOR ARTIFACT     │
│ State S_(k+1) via Q(rsi)     │
│ • Advances Causal Index κ    │
```



1. **Agnostic Generation:** An ai model (e.g., an LLM or multi-agent swarm) parses messy natural human speech and proposes a candidate semantic graph. Under **Axiom 14**, this graph is tagged as `State: CANDIDATE`.
2. **Deterministic Measurement:** The candidate is translated into a Content-Addressed Expression DAG (`Qexpr`) and audited against the declared Intent Contract, checking depth ceilings, metric budgets, and geometric consistency.
3. **Selection via BOC Policy:** The candidate that maximizes compliance with SIMEMP constraints is selected under the **BOC Policy**:

$$\text{Selection} = \arg \max_{c \in \{C_1, \ldots, C_n\}} (\text{SIMEMP Compliance}(c))$$

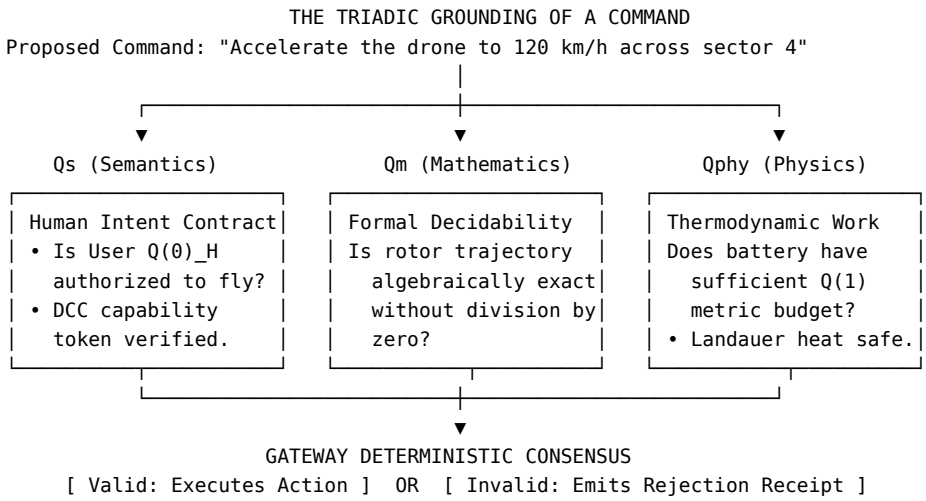
4. **Sealing via** Q(rsi)**:** The selected candidate is sealed into `State: SEALED` as successor artifact S_{k+1}, advancing the monotonic causal index along the **Causal Arrow**. Sub-optimal candidates emit negative receipts, ensuring the system never repeats failed paths.

---

7. Triadic Intent Synthesis (Qs ⊗ Qm ⊗ Qphy)

True communication between humans and computational tools requires simultaneous verification across the foundational triad:

$$\mathbf{C}_{\text{Intent}} = \mathbf{C}_{\text{Qs}} \otimes \mathbf{C}_{\text{Qm}} \otimes \mathbf{C}_{\text{Qphy}}$$



When an intent command crosses a gateway: 1. ****Qs** validates intent:** Verifies the human authority chain, the declared modal grounding, and the contractual goal. 2. ****Qm** validates decidability:** Proves that the spatial rotors and coordinate trajectories contain no singularities, division-by-zero, or precision drift. 3. Qphy **validates realizability:** Proves that the command fits within the physical battery energy and Landauer dissipation budgets ($\Delta S \leq \text{Budget}(Q(1))$).

If all three domains verify the transition, the command executes. If any domain fails, execution halts cleanly. The tool cannot decouple from the human because the mathematical definition of an executable action is structurally bound to human intent.

---

8. License and Invariant Lineage

This domain specification enforces the legal and operational conditions of the **SSL** governing the **Licensed Work** across the **cdqn** network:

1. ****Paternity Reference**:** All semantic parsing engines, intent contract compilers, and human-agent alignment modules derived from this document must embed: > Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License (SSL). Parent Repository: `https://g`ithub.com/cdqn5249/cdqn
2. **Anti-Patent Defense:** Royalty-free rights terminate automatically if a licensee initiates patent litigation regarding Qs semantic methods, intent contracts, or alignment mechanisms against the Author or community under the **Open Core Invariants**.
3. **Statutory Safe Harbor:** Qs operates as passive, neutral semantic infrastructure. It exercises no editorial inspection over, and makes no claim upon, sovereign human creative works, natural speech, or external `Payloads`.

Annex E — Qs History proposal

Pinned source: docs/qs_history.md
Committed version: 0.1.2 **Last updated:** 2026-10-22
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Field	Specification
Document Title	Qs History — Vision and Proposal for Historical Anchoring in the Qn Universe
Version	0.1.2
Last Updated	2026-10-22 (Bao Loc, Vietnam)
Author	Christophe Duy Quang Nguyen
License	Scaling Source License (SSL) 1.0
Status	Vision and Research Proposal — Not a Finalized Specification

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Purpose and Epistemic Stance

This document consolidates the propositions advanced during the October 2026 design consultation on historical anchoring, time-situating, and cross-node consensus of **Q(anchor)** entities. It is written in the same exploratory register as `docs/chronosa.md`: an operational thesis, not a finalized engineering specification, and not a formal mathematical proof.

Every design element proposed below carries the explicit status of a **candidate artifact** under **Axiom 14** (`docs/qs.md` §6): it remains subject to deterministic measurement against the declared **SIMEMP** constraints, to selection under the **BOC Policy**, and to eventual sealing through a **Q(rsi)** recursive loop at implementation time. Nothing in this document redefines, amends, or overrides any existing canonical specification. Where a proposition is already grounded in existing canon, the grounding is cited; where it is genuinely new, it is marked as an unregistered proposed term that is deliberately **not** yet defined in the glossary.

In accordance with **Computational Consistency**, the validity claimed for these propositions is internal consistency within declared finite boundaries. The purpose of this document is to preserve the reasoning chain of the consultation so that future implementation work, or a Q(rsi) optimization loop, can measure and select among the recorded candidates rather than reconstruct them from conversation history.

---

1. Background: One Arrow, Two Predicates

The existing canon already fixes the temporal spine of the **Qn** universe. Every discrete **Q(event)** is indexed along a strictly monotonic **Causal Arrow** whose birth order runs from the **Q(0)** genesis origin through the **Q(1)** unit and the inductive single digits (`docs/abstractionLayers.md` §3.5). Onboarding constructs this spine locally: the six-stage gateway of `docs/q0_q1.md` §2 transitions the node from the **Layer 0** physical substrate to governed **Layer 1** across a **SIMEMP Gateway**, harvesting entropy, parameterizing the Inner Ring, assembling the genesis object, and sealing it into an immutable lifecycle state. The single digits are then born constructively along axis d1 via the discrete **Successor Morphism S** (`docs/q2_q9.md` §2).

Within this spine, the canon carries the Q(anchor) notion under one glossary definition ([`docs/_data/glossary/architecture.yml`]): a governed Qn artifact possessing high causal centrality within the lineage DAG, protected via abstract DCC contracts and **Structural Indirection** to prevent cascading dependency invalidation. The observation recorded by the consultation is that this single notion quietly covers two distinct predicates:

1. A **structural predicate**: out-cone centrality, formalized in `docs/qm.md` §8 as a threshold condition on the dependency DAG. It is computable locally from the lineage graph alone and requires no external witness.
2. An **epistemic predicate**: grounding — the manner in which an artifact's asserted invariant is connected to reality. This predicate already appears across the canon in several forms: Layer 0 calibration of physical anchor contracts, the **Epistemic Anchorage** spectrum of `docs/qs.md` §4, the **Historical Calibration Benchmark**, and the shared-landmark sheaf gluing of `docs/chronosa.md` §3.1.

The consultation claims **no contradiction** in the existing canon. It records only that the two predicates are not yet explicitly distinguished, and proposes below a vocabulary for distinguishing them. Under the **No-Implicit Rule**, making an implicit distinction explicit is a lawful, non-invasive refinement.

---

2. Proposition 1 — Anchor Grounding Classes

It is proposed that every **Q(anchor)** declare **two** predicates instead of one:

- The **structural predicate**, unchanged from existing canon: out-cone centrality at or above the declared threshold.
- The **epistemic predicate**, newly named: a **grounding class**, taking one of three values.

Class	Name	Source structure	Witness	Consensus mechanism
I	Invariant	One deterministic rule, executed by many	Self (local genesis **Receipt**)	Deterministic re-derivation; agreement is forced, not negotiated
P	Physical	One shared nature, measured by many	Physics (local signed calibration receipts)	Independent calibrations converge within declared tolerance envelopes
S	Social	One event, witnessed by many	Swarm (quorum attestation)	Sheaf gluing on shared landmarks within the **Historical Horizon**

Class I anchors are born from procedures rather than events: every node executing the same deterministic derivation from its own **Q(1)** arrives at structurally identical artifacts — same contracts, same indices, same centrality profile. Under **Dependencies Determinism**, agreement on the derivation is forced by determinism itself. The genesis of such anchors is **self-witnessed**: the node emits its own node genesis receipt (`docs/q0_q1.md` §6.1), and no external witness is possible or required. Class P anchors are calibrated against Layer 0 measurements, each node measuring its own substrate; their tolerance envelopes are proposed to be declared inside the anchor's DCC contract rather than negotiated at runtime, so that consensus reduces to envelope satisfaction. Class S anchors are genuine historical contingencies and carry the full historical-horizon discipline described in Section 5.

The classes resolve the apparent paradox of consensus without shared events. Genesis instances are strictly local — the canon already requires that no universal global zero be admitted:

$$\forall\, A \neq B \implies Q(0)_A \neq Q(0)_B$$

(`docs/q0_q1.md` §3). What all nodes share is never the token; it is the **schema**: the role contract, the birth index, the invariant the anchor commits to. Consensus is on the schema, never on the token. The three classes form a witness ladder — self, then physics, then swarm — and it is observed that this ladder is also a complexity ladder: each rung buys consensus over a wider scope at a higher evidentiary cost.

The proposed class names (grounding class, Class I/P/S) are **unregistered proposed terms**. Their glossary registration is deferred to the integration phase (Section 8).

---

3. Proposition 2 — The Source-Receipt Invariant

It is proposed that every governed Qn artifact declare an explicit pair:

(content modality, source class)

where the content modality is drawn from the existing **Epistemic Anchorage** spectrum (`docs/qs.md` §4), and the source class expresses where the artifact's asserted content ultimately anchors. The invariant requires that the source receipt chain terminate in physically receipted reality events: Layer 0 sensor or artifact receipts for empirical content, algebraic closure receipts for constructive content, and — for contextually sealed creative content — the author-identity and publisher-receipt anchoring already mandated by `docs/qs.md` §4.2.

The invariant extends the existing statement-level rejection rule to the artifact level. `docs/qs.md` §4.1 already bars an assertion from claiming empirical truth through latent probability alone, emitting a rejection receipt when lineage is absent. The proposed extension applies the same firewall to composite artifacts and worlds: no unsourceable fiction, no unsourced anchor, no history-less constant.

Two cautions are recorded, and they are binding on any future implementation:

1. **Payload sovereignty is untouched.** The invariant governs artifacts that cross the **SIMEMP Gateway** as governed objects of the **Licensed Work**. Sovereign human **Payloads** remain uninspected behind the safe-harbor air gap (`docs/abstractionLayers.md` §3.1). Where a creative work is declared as a governed MODAL_NARRATIVE artifact, its anchoring to author identity and publisher receipt is **declared by its author**, not extracted by inspection.
2. **Simulation modality is two-faced.** A simulation's execution is MODAL_CONSTRUCTIVE (deterministic algebraic closure), while its claims *about reality* are MODAL_HYPOTHESIS until physically falsified. The pair declaration makes this distinction explicit rather than leaving it implicit, in accordance with the **No-Implicit Rule**.

---

4. Proposition 3 — Calendar Projection Contracts

The canon admits exactly one time: the local **Causal Arrow**. Distributed state ordering relies on causal chaining "rather than external wall-clock synchronization" (`docs/simemp.md`), and common knowledge across asynchronous nodes is established through acyclic causal message chains, not shared clocks (`docs/chronosa.md` §3.2). Human calendars are therefore proposed **not as parallel time systems** but as **projections**: deterministic mappings between calendar coordinates and intervals of the local causal index, hosted at the Qs layer in the same manner that the canon's domains project orthogonally across the execution core (`docs/abstractionLayers.md`).

It is proposed that a **Calendar Projection Contract** declare three components:

Component	Proposed class	Nature
Rule (leap algorithms, month arithmetic, week conventions)	Class I	Deterministic, re-derivable algorithm; constructive once the epoch is fixed
Epoch (coordinate zero: Gregorian year 1, Hijra 622 CE, Unix epoch, J2000.0, and others)	Class S	A human convention-adoption event, multi-source by nature
Resolution envelope (day, second, leap-second tolerances)	Declared field	A declared bound inside the contract, never negotiated at runtime

Under this anatomy, a calendar such as the Gregorian system is a Class S genesis that matured into Class I operation: born from a contested, multi-jurisdiction convention process spread across centuries, yet today re-derivable deterministically by any node from its published rule. This is the witness ladder of Section 2 observed in a single historical object.

On placement, the consultation enumerated candidates under the **BOC Policy**. Hard-coding calendars into runtimes or compilers was assessed as a rigid implementation micro-choice, contrary to the minimal-necessary-invariants principle (`docs/q0_q1.md` §1); per-node ad-hoc definitions fail consensus. The surviving candidate — proposed here — is a **canonical library of versioned, DCC-contracted calendar artifacts**, loaded post-seal and on demand within the **Metric Envelope**, honoring **Local-first** budgets. Calendar contracts declare their components inside a **DCC Profile**, consistent with the structural-indirection discipline: calendars are swappable cultural conventions, not runtime constants.

---

5. Proposition 4 — The Qs History Library

It is proposed that the Qn universe maintain a **Qs History Library**: a canonical, versioned collection of modality-typed **historical anchor artifacts**, each recording a past human event, its sources, and its spatiotemporal situation. The library is proposed as the concrete shape of the consensus-history ingestion capability noted as an open gap during the consultation (recorded internally as GAP2): the mechanism by which consensus past events may be loaded **after** node sealing, in complement to a QnLang compiler or CDQN runtime, without violating the Axiom 4 requirement that every local artifact trace to its local genesis origin.

A proposed anatomy for the historical anchor artifact:

```
┌──────────────────────────────────────────────────────────────────┐
│ HISTORICAL ANCHOR ARTIFACT (proposed anatomy, State: CANDIDATE)  │
├──────────────────────────────────────────────────────────────────┤
│ ArtifactType      : HIST_ANCHOR (proposed class)                 │
│ ContentModality   : per claim decomposition (docs/qs.md §4 spectrum)  │
│ SourceClass       : I, P, or S (proposed grounding class)        │
│ SourceReceipts    : unbroken chain to Layer 0 artifacts or sealed  │
│                     library records                              │
```

```
| CalendarSituation : projected via a declared Calendar Projection      |
|                     Contract (Section 4)                              |
| SpatialSituation  : local Qm frame coordinates                        |
| Witnesses         : lineage-disjoint root subtree list (Section 6)    |
| LifecycleState    : CANDIDATE, SEALED, or QUARANTINED                 |
└──────────────────────────────────────────────────────────────────────┘
```

Access is proposed in four tiers, each honoring **Local-first** and budget discipline:

1. **The canonical library** holds the versioned artifacts with their full source receipt chains.
2. **Local nodes** hold anchor contracts and working sets within declared **Metric Envelope** budgets — never the full archive. A node may operate air-gapped with zero history; the library loads post-seal, on demand, exactly when an **Intent Contract** requires situating or attributing.
3. **AI agents** inherit scope through the intent contract's epistemic whitelist. Without a loaded historical anchor, an agent's historical assertion has no parent lineage receipts and falls to the gateway as an unanchored claim (`docs/qs.md` §4.1); with the library, the agent **cites** the artifact instead of generating from latent weights.
4. ****Chronosa****, barred from every Inner Ring, sees only the public layer: local historical records glue where they agree on shared landmarks — calendar coordinates, attribution anchors, artifact commitments — satisfying the sheaf gluing condition (`docs/chronosa.md` §3.1). The consensus history is then the global section of this history sheaf.

Worked example — the constant e. The invariant face is already canon: the transcendental constant exists as a nullary continued-fraction generator under `OP_CF` (`docs/qexpr.md` §3.2, §4.3), with partial convergents carrying a strictly bounded remainder and evaluation on demand at declared **zoom z** and budget, halting early under **Terminal Exactness**. Deterministic evaluation at a declared precision yields bit-identical results on every **cdqn** node — a Class I grounding in the sense of Section 2, requiring no history whatsoever. Intermediate collapse results may be cached as content-addressed **Q(dataStruc)** artifacts tagged with their derivation lineage, always re-derivable.

The discovery face is historical. A historical anchor artifact for the publication of the treatise that established the constant's notation would record: the artifact provenance chain (printed copies, library holdings, digitizations — each transfer physically receiptable at Layer 0), the calendar situation via a Gregorian projection contract, the spatial situation in local Qm frames, and the attribution witnesses as a multi-source list within the **Historical Horizon**. Claims about the artifact's existence are empirical; claims reconstructed from correspondence remain hypotheses; anecdotes remain sealed narrative. The machine does not remember the mathematician; it **cites** him through a receipted lineage — which is the only form of historical knowledge the Qn universe admits.

---

6. Proposition 5 — The Anti-Poisoning Ingestion Firewall

The library can be alimented only by petition through gates. Publicly reported incidents in which machine-generated statistical pages were laundered into news outlets, peer-reviewed publications, and financial reports through iterative citation (so-called citation laundering) motivate this section. In the Qn universe such an attack is assessed as follows:

1. A fabricated statistic claiming empirical truth without lineage receipts is an unanchored assertion and is rejected at the first gateway crossing (`docs/qs.md` §4).
2. Its arrival into financial registers is barred by the contextual containment of creative and ungrounded content (`docs/qs.md` §4.2).
3. Citation laundering itself is assessed as **causally impossible** under the proposed counting rule: sources are counted as **lineage-disjoint root subtrees** in the polyroot forest (`docs/chronosa.md` §1.4), never by citation count or pseudonym count. Laundering adds descendants to a poisoned root; it cannot manufacture independent origins. An entire laundering cascade descending from one fabricated artifact counts as **one** source.

It is proposed that library ingestion pass five gates, each a governed operation emitting its own **Receipt**:

Gate	Canon root	Threat addressed
1. Lineage	Axioms 4 and 5; causal hash chains	Records lacking an unbroken receipt chain to Layer 0 artifacts or sealed library records
2. Modality ceiling	**Epistemic Anchorage** spectrum	Modality smuggling: each atomic claim is typed at the highest modality its receipts support

Gate	Canon root	Threat addressed
3. Independence quorum	Multi-source genesis; threshold quorum	Sybil floods and citogenesis: anchor-grade records require multiple sources with lineage-disjoint roots
4. Sheaf gluing	Shared-landmark gluing	Records disagreeing with receipted overlapping records; such records enter a `QUARANTINED` **lifecycle state** with a challenge receipt, remaining visible as recorded challenges
5. Append-only immutability	SEALED lifecycle; **Structural Indirection**	Rewrite attempts: sealed artifacts never mutate; corrections arrive as counter-receipts and migration receipts. History is never rewritten; it is re-witnessed

Residual threats are recorded honestly. Collusion across genuinely distinct roots remains an open challenge, in direct continuity with the hub-collusion resistance challenge of `docs/chronosa.md` §8.5; its mitigation is economic and physical, since fabricating Layer 0 artifact provenance costs real-world work, and forging contemporary sensor receipts requires breaking post-quantum signatures across the **Dual-Ring PQC Boundary**, whose private root keys never leave the Inner Ring. Slow poisoning of deep history is capped by the **Historical Horizon**: admitted poison can only ever occupy hypothesis or firewalled narrative modalities, with bounded grounding and no path to actuation. Finally, because every gate is itself a governed operation, the firewall as a whole is a legitimate candidate for optimization by a future Q(rsi) loop under strict semantic invariance.

---

7. Relation to Existing Canon

The following table records, for each proposed element, whether it is already grounded in canon or genuinely new. No proposition herein redefines existing canon; newly named concepts are unregistered proposed terms pending glossary integration.

Proposed element	Status	Canon grounding
Causal Arrow as sole time; calendars as projections	Grounded	`docs/simemp.md`; `docs/chronosa.md` §3.2
Birth order and constructive digit genesis	Grounded	`docs/abstractionLayers.md` §3.5; `docs/q2_q9.md` §2
Structural centrality predicate of Q(anchor)	Grounded	`docs/qm.md` §8; glossary `q-anchor`
Epistemic modality spectrum and statement rejection	Grounded	`docs/qs.md` §4
Sheaf gluing on shared landmarks; threshold quorum	Grounded	`docs/chronosa.md` §1.1, §3.1
Polyroot forest; lineage-disjoint independence	Grounded	`docs/chronosa.md` §1.4
Historical horizon modality discipline	Grounded	Glossary `historical-horizon`; `docs/qlog.md`
QUARANTINED lifecycle and receipt taxonomy	Grounded	`docs/abstractionLayers.md` §3.6
Transcendental constants as nullary generators	Grounded	`docs/qexpr.md` §3.2, §4.3
Q(rsi) candidate pipeline and sealing	Grounded	`docs/qs.md` §6
Anchor Grounding Classes (I/P/S), dual predicate	**New**	Generalizes `docs/q2_q9.md` §2.3 genesis/verification boundary
Tolerance envelopes declared in Class P contracts	**New**	Consistent with metric-envelope ceiling discipline
Source-Receipt Invariant	**New**	Extends `docs/qs.md` §4.1 to artifacts
Calendar Projection Contracts	**New**	Extends the domain-projection architecture
Qs History Library and artifact anatomy	**New**	Concrete shape of the consensus-history ingestion gap

Proposed element	Status	Canon grounding
Anti-Poisoning Ingestion Firewall (five gates)	**New**	Composes existing canon mechanisms under one pipeline

---

8. Open Questions and Integration Queue

All items below remain in `CANDIDATE` status. Sealing of any item occurs exclusively through the candidate pipeline of `docs/qs.md` §6 — deterministic measurement, **BOC** selection, **Q(rsi)** sealing — at implementation time or under a dedicated optimization loop.

1. **Calibration wiring.** The mandatory **Historical Calibration Benchmark** is defined in the glossary and is now wired as a non-mutating CANDIDATE proposition at `docs/q0_q1.md` §2.7 — a post-seal Class P grounding event that adds no stage to the six-stage onboarding state machine — closing this queue item; the wiring remains subject to deterministic measurement and BOC selection at implementation time.
2. **Ingestion pipeline codification.** The firewall of Section 6 is the proposed shape of the post-seal consensus-history ingestion capability, and its operational semantics (gate ordering, quorum thresholds, challenge resolution) are now proposed in `docs/history_library.md` §5; the item remains open until the pipeline passes deterministic measurement and its threshold values are declared, as tracked by items 1–2 of that document's §7.
3. **Event frame definition.** The spatiotemporal coordinate frame of a discrete Q(event) — causal index, d-axes, zoom, causal-horizon cone — is used implicitly across the canon but not yet codified as a single frame definition.
4. **Receipt and intent duality.** The complementary temporal roles of the node genesis receipt (backward-looking sealing) and the intent contract (forward-looking delegation) are exercised throughout the canon but not yet named as a duality.
5. **Glossary registration.** The proposed terms — grounding class, source class, calendar projection contract, historical anchor artifact, ingestion gate — are now registered as CANDIDATE glossary entries (together with the Qs History Library), closing the deliberate deferral; they remain unsealed pending the candidate pipeline of `docs/qs.md` §6.

A proposed integration order, subject to the author's decision: the Source-Receipt Invariant first, as it grounds the library and calendar propositions; the History Library and Calendar Projection Contracts second; the onboarding calibration stage third.

---

9. License Alignment

This proposal is part of the **Licensed Work** and aligns with the **Scaling Source License (SSL) 1.0**:

- **Attribution:** Any implementation or research derivative referencing this document must preserve the canonical **Paternity Reference**.
- **Open-Core Research:** Exploratory research and academic derivations of the propositions herein remain royalty-free under Non-Scaling Open Access, governed by the **Open Core Invariants**.
- **Safe Harbor Integrity:** The Qs History Library is designed to operate as technical infrastructure of the licensed work. It exercises no editorial control over, and acquires no intellectual property in, sovereign human **Payloads**.

Annex F — Qn Structures proposal

Pinned source: docs/qstructures.md
Committed version: 0.1.0 **Last updated:** 2026-10-23
sha256 (committed file): `01d5f84fd550e008264cdcb605459af3d962dc3f048ccecdf4ad3801915d9ef6`

Field	Specification
Document Title	Qn Structures — Vision and Evaluation Framework for Category C Data Structures and Workflows
Version	0.1.0
Last Updated	2026-10-23 (Bao Loc, Vietnam)
Author	Christophe Duy Quang Nguyen
License	Scaling Source License (SSL 1.0)
Status	Vision and Structural Proposal — Not a Finalized Specification

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Purpose and Epistemic Stance

This document is proposed as the anchoring specification of **Category C — Data Structures and Workflows** of the Qn definition taxonomy (`docs/abstractionLayers.md` §11). It originates in the October 2026 design consultation on substrate feasibility (recorded internally as the structures consultation), which observed that the validated **CAE-DAG** answers how every artifact is *represented* while leaving open how structures are *specified, measured, and selected* as access contracts. It is deliberately positioned in the same epistemic register as `docs/qs_history.md` and `docs/history_library.md`: a proposal record, not a sealed specification.

Every design element below carries the explicit status of a **candidate artifact** under **Axiom 14** (`docs/qs.md` §6): it remains subject to deterministic measurement against the declared **SIMEMP** constraints, to selection under the **BOC** policy, and to eventual sealing through a **Q(rsi)** loop at implementation time. Nothing in this document redefines, amends, or overrides any existing canonical specification, prescribes no runtime implementation, and assigns no storage layout.

One methodological commitment governs the whole document, and it is stated here because it outranks every particular claim that follows: **the steps that guide and explain the design are more important than the final selections**. Which structures are ultimately admitted is a function of implementation feedback and **Q(rsi)** optimization, not of authorial decree. Accordingly, this document invests primarily in the evaluation procedure (Sections 2 and 3) and offers the candidate register (Section 4) only as a first worked pass of that procedure — a record of reasoning that keeps alternatives revisitable, not a set of winners. In accordance with **Computational Consistency**, the validity claimed herein is internal consistency within declared finite boundaries.

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1. Substrate and Interface: Why the Validated DAG Is Not Sufficient

The **Qn** universe commits every artifact to a single substrate: each is a governed **Universal Envelope** carrier participating in the local **causal DAG**, hash-addressed, acyclic by construction (a circular dependency would break the pre-image resistance of the commitment hash, `docs/qexpr.md` §2.4), and sealed in an immutable **Lifecycle State**. The choice of this substrate was itself an exercise of the **BOC Policy**: `Qexpr` explicitly rejects naive pointer-heap trees in favor of the CAE-DAG (`docs/qexpr.md` §2), recording the rejection and its reasons rather than silently defaulting.

The substrate, however, answers only the representation question. In established computer science, a data structure is not primarily a memory layout but a **documented access contract with explicit performance tradeoffs** — an array promises indexed access at contiguous-locality cost, a hash table promises keyed access at hashing-and-resize cost, and each is chosen by matching its contract to the workload. The canonical files already exercise such contracts ad hoc — bounded arrays appear as `OP_FOLD` child vectors, keyed lookup appears inside the **Q(reuse)** breakeven analysis — but no committed text specifies the contract surface itself. The question "is the validated DAG enough?" therefore receives the answer: **necessary, not sufficient**. The DAG subsumes representation; the contract vocabulary for structures is the open gap this document proposes to close, under **Local-first** discipline: every structure proposed here executes one hundred percent locally, importing no network consensus into its access path.

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2. The Structural Contract Schema

It is proposed that every Qn data structure — and every proposed extension of the structure vocabulary — be specified as a **five-row contract**, each row binding one declared discipline of the **SIMEMP** constraint system. The schema is the stable core of this proposal; particular structures are its outputs.

```
+-------------------------------------------------------------------------+
| THE FIVE-ROW STRUCTURAL CONTRACT SCHEMA (CANDIDATE)                     |
+-------------------------------------------------------------------------+
| 1. FOOTPRINT   : declared Metric Envelope ceilings; growth emits a new  |
|                  versioned artifact, never mutates in place             |
| 2. ACCESS      : enumerated permitted access morphisms; no implicit     |
|                  operation is admitted (No-Implicit Rule)               |
| 3. LOCALITY    : Memory Wall movement class per access pattern, accounting |
|                  for physical data-movement dissipation (SIMEMP 4.1)    |
| 4. DETERMINISM : iteration and traversal order fixed by causal index;   |
|                  no runtime-adaptive reordering is permitted            |
| 5. WORK        : thermodynamic tradeoff row — work per access class in U, |
|                  versioning amplification, and a declared breakeven rule |
+-------------------------------------------------------------------------+
```

Each row derives from committed canon rather than from this document's preference. Row 1 instantiates the **Metric Envelope** footprint discipline of `docs/qnPrimitive.md` §8 and the versioning rule that successor artifacts advance the causal index while mutation in place is prohibited. Row 2 instantiates the **No-Implicit Rule** at the structure level: the glossary definition of **Q(dataStruc)** already commits that "no implicit heap state is admitted", and this document generalizes that refusal from heaps to all implicit operations. Row 3 instantiates the **SIMEMP** Memory Wall accounting of `docs/simemp.md` §4.1, which requires explicit accounting for memory bandwidth and locality rather than purely abstract asymptotic complexity. Row 4 instantiates **Dependencies-Determinism**: a structure whose iteration order could vary at runtime would inject an untraceable input into every workflow that consumes it. Row 5 anticipates Section 6: every admitted structure must declare its thermodynamic cost surface, in the same spirit in which the canon already declares breakeven inequalities for optimization morphisms.

A proposal that cannot yet fill a row is **not rejected** — it is recorded as incomplete, with the missing row named. This is deliberate caution: the schema is a measuring instrument, and an instrument must be able to report partial results without discarding the specimen.

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3. The Evaluation Ladder: From Nomination to Sealing

The following five-step ladder is proposed as the standing procedure by which any structure — classical, novel, or machine-discovered — enters the Qn universe. The ladder is the deliverable of this document; the register of Section 4 is merely its first worked pass.

1. **Nomination.** A structure is proposed by naming its contract surface: the five rows of Section 2, filled or explicitly marked incomplete. At this stage the structure is CANDIDATE vocabulary in plain text; it receives no glossary entry and no include reference until registration (Section 8, item 1).
2. **Row measurement.** Each row is measured against the declared **SIMEMP** constraints. All threshold values are **declared values inside the owning** DCC contract****, never runtime negotiations, in accordance with the **No-Implicit Rule**; this mirrors the tolerance-envelope discipline already proposed for the **Historical Calibration Benchmark** wiring in `docs/q0_q1.md` §2.7.
3. **BOC selection.** Where alternatives exist for the same access niche, the **BOC** policy of `docs/simemp.md` §7 selects among recorded alternatives, and rejections are recorded with their reasons — the precedent being the recorded pointer-heap rejection of `docs/qexpr.md` §2. A rejection is a finding, not an erasure.
4. **Implementation feedback.** Deterministic measurement at execution boundaries — **receipts** emitted by actual **QnIR** lowering — tests each row's declared values against observed behavior. A row whose declared values are falsified sends the structure back to step 1 with the finding recorded; it does not silently adjust thresholds.
5. **Sealing.** A structure whose rows survive measurement is sealed through a **Q(rsi)** loop under strict semantic invariance, in the manner of the anti-drift discipline of `docs/qnPrimitive.md` §7.4. A sealed selection is superseded only by a new versioned artifact along the local **Causal Arrow**, never by mutation in place.

Two cautions bound the ladder itself. First, the ladder adds no stage to any canonical state machine: it is a proposal discipline for this document's own scope, executable entirely within the candidate pipeline of `docs/qs.md` §6. Second, the ladder is itself

a legitimate candidate for optimization by a future **Q(rsi)** loop — including its own row definitions — under the same semantic-invariance requirement it applies to others.

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4. The Candidate Register: A First Worked Pass

The register below applies Sections 2 and 3 to the classical structure menu. Its status is deliberately modest: these are **worked examples of the ladder's first pass**, recorded so that reasoning survives and alternatives remain revisitable — not selections. Each entry cites its grounding in committed canon; nothing herein redefines that canon.

Canonical-admitted candidates. Five structure families already have their contract surfaces exercised across the canon and are nominated for full row measurement:

- **Sequenced container** (proposed name `Q(array)`): grounded in the `OP_FOLD` node class, whose child vector encapsulates the full array of term commitments under an explicit step ceiling (`docs/qexpr.md` §4.4). Footprint is a declared length; access is indexed; locality is the contiguous class; determinism is index order; work is constant per indexed access under a declared budget. Unbounded series are already syntactically invalid under Axioms 2 and 6 in the committed text.
- **Geometric container** (proposed name `Q(matrix)`): grounded in `OP_FRAME_MAP` and the integer Matrix-Multiply-Accumulate lowering of `docs/qexpr.md` §5.1, which completes contractions in constant time without floating-point drift. Exactness is preserved by construction; the **Q(bypass)** morphism short-circuits identity preconditions in register time.
- **Keyed container** (proposed name `Q(table)`): grounded in hash-consing itself — the CAE-DAG is already a content-addressed keyed index — and in the committed breakeven analysis that weighs Q(reuse) against hash-table overhead (`docs/qm_geometry.md` §7.2). Capacity is declared; growth is re-versioning.
- **Record tuple**: grounded in the **Qm** base record and the canonical multivector state tuple (`docs/qm_geometry.md` §4.1) under **Terminal Exactness**. The fixed-field record is the atomic structure family and requires the least novel specification.
- **Persistent tree (DAG-deduped)**: grounded in the single-rooted directed trees that local artifacts already form from the physical silicon root (`docs/qnPrimitive.md` §3, Axiom 4). Hash-consing makes path-sharing between versions the expected case rather than the exception.

Rejected by construction. Two classical families cannot enter, and the register records why:

- **Pointer-heap tree**: rejected under the **BOC** policy in committed text (`docs/qexpr.md` §2) — implicit allocation violates the **No-Implicit Rule**, and pointer chasing dissipates across the Memory Wall per hop. The rejection stands unless a future nomination declares a complete, measurable contract surface that cures both defects.
- **Cyclic graph**: excluded mathematically, not by policy — a circular dependency would require breaking the pre-image resistance of the commitment hash (`docs/qexpr.md` §2.4, Axiom 5). No contract surface can cure a construction impossibility.

Recoverable as contracts. Two classical families are absent from the canon and are nominated only as *open contract shapes*, pending implementation feedback:

- **Sequenced chain** (linked-list analog): as a pointer-chasing runtime shape it fails rows 1 and 3 outright; it may be recoverable only as a declared sequenced container whose traversal is index-ordered, or as a chain of **receipts** in the manner of the chained Road Receipts by which **Q(workflows)** achieve portability.
- **Bounded ring** (queue/stream analog): bounded streams are already exercised — entropy sampling collects a declared sample volume at `docs/q0_q1.md` §2.1, and receipt chains accumulate append-only — but no container type is committed. A ring is nominated only with declared capacity and sequenced access; an implicitly resizable queue would violate rows 1 and 4.

The recoverable pair illustrates the document's method commitment: the ladder records *why* each family stalls, so a later nomination — informed by implementation feedback or by a **Q(rsi)** loop — can re-enter at step 1 with the findings attached, rather than reconstructing the analysis from conversation history.

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5. Proposed Nomenclature Extension

The Qn universe already names operational morphisms and structure families in a consistent `Q(...)` pattern: **Q(reuse)**, **Q(bypass)**, **Q(rsi)**, and the registered trio **Q(dataStruc)**, **Q(patterns)**, **Q(workflows)**. It is proposed that the structure vocabulary extend in the same pattern with the four names used in Section 4 — `Q(array)`, `Q(matrix)`, `Q(table)`, `Q(stream)` — under one consistency rule: **a `Q(...)` name denotes a contract surface (the five rows of Section 2), never a runtime shape**. This rule is what keeps the nomenclature honest across heterogeneous hardware: the same `Q(array)` contract is satisfiable by different lowerings on different nodes, exactly as the compute unit evaluates identically across all nodes while local entropy parameterizes identity.

Deliberately, this document **registers nothing**. Glossary entries are data-only commits under the candidate pipeline, and the four proposed names remain plain text herein — they receive no include references and no concordance — until a separate registration commit follows the pipeline, as was done for the consultation vocabulary on 2026-10-22. The deferral is queued in Section 8.

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6. Breakeven Rules and the Work Row

The work row of Section 2 has one committed precedent whose shape every admitted structure should follow: **Q(reuse)** is admissible if and only if the predicted computational work of direct re-evaluation exceeds lookup and retrieval dissipation plus a declared overhead term (`docs/qnPrimitive.md` §1.15). That inequality is a *declared breakeven rule* — a contract row, not a runtime heuristic — and it is measured in units of the **Abstract Compute Unit U**.

Two properties make U the right unit for comparable work rows across heterogeneous nodes. First, platform invariance: while local entropy parameterizes cryptographic identity, the metric magnitude of U evaluates identically across all nodes (`docs/q0_q1.md` §4.2), so a work row declared on one node is interpretable on every node. Second, the thermodynamic leash: unbounded recursive execution is physically prohibited, every workflow operates within a finite, non-renewable work budget, and metric depletion halts execution deterministically under **metric exhaustion** — so no structure's work row can be rendered moot by unmetered iteration.

It is proposed, therefore, that every admitted structure declare in its work row: (i) the work class per access morphism, (ii) the versioning amplification — the cost emitted by row 1's growth-as-versioning discipline — and (iii) the breakeven inequality under which the structure is preferred over its recorded alternatives. A structure without a breakeven rule is not wrong; it is simply not yet comparable, and the **BOC** policy requires comparability among recorded alternatives before selection.

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7. Relation to the DCC Hardware Profile and to Existing Canon

This document and the proposed **DCC Hardware Profile schema** are sibling propositions facing opposite directions across the same boundary. The hardware profile describes the *machine side*: the measured **Layer 0** capabilities — accelerator inventory, memory hierarchy, storage bounds, entropy sources — established during onboarding and referenced by the committed lowering discipline of `docs/qexpr.md` §5.1. This document describes the *artifact side*: the contract surfaces that structures declare regardless of which **cdqn** node executes them. They meet at lowering: the compiler queries the node's hardware profile and must lower each admitted structure in a way that honors its locality and work rows, while the **SIMEMP Gateway** intercepts any lowering that is not an algebraic isomorphism, protecting **Existential Invariants** and preserving the constant-time properties required by the **Security Envelope** (`docs/qexpr.md` §5.2). The **Structural Indirection** discipline is what allows the two sides to evolve independently: abstract capability contracts are decoupled from concrete primitives, so structures and hardware can each be swapped without fracturing the other.

The following table records, for each element of this proposal, whether it is already grounded in canon or genuinely new to this document. No element herein redefines existing canon.

Element	Status	Canon grounding
CAE-DAG as sole substrate; pointer-heap rejection	Grounded	`docs/qexpr.md` §2 (BOC-recorded)
Acyclicity by hash pre-image resistance	Grounded	`docs/qexpr.md` §2.4 (Axiom 5)
Universal Envelope carriage and immutable lifecycle	Grounded	`docs/qnPrimitive.md` §1.3, §1.10
Q(dataStruc), Q(patterns), Q(workflows) definitions	Grounded	Glossary (`primitives.yml`); `docs/qlog.md` §2
Workflow as DAG of patterns under Intent Contract	Grounded	`docs/qlog.md` §2
Q(reuse) breakeven inequality	Grounded	`docs/qnPrimitive.md` §1.15; `docs/qm_geometry.md` §7.2
Memory Wall locality accounting	Grounded	`docs/simemp.md` §4.1
Candidate pipeline (measure, select, seal)	Grounded	`docs/qs.md` §6
Five-row structural contract schema	**New**	Composes the SIMEMP disciplines per structure
Evaluation ladder (nomination to sealing)	**New**	Instantiates the §6 pipeline for structures
Proposed Q(array), Q(matrix), Q(table), Q(stream) names	**New**	Plain-text vocabulary; registration deferred

Element	Status	Canon grounding
Recoverable-contract analysis (chain, ring)	**New**	First worked pass of the ladder

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8. Open Questions and Integration Queue

All items below remain in CANDIDATE status. Sealing of any item occurs exclusively through the candidate pipeline of `docs/qs.md` §6 — deterministic measurement, **BOC** selection, **Q(rsi)** sealing — at implementation time or under a dedicated optimization loop.

1. **Vocabulary registration.** The four proposed names of Section 5 await a data-only glossary commit; this document deliberately registers nothing itself, and until registration the names remain plain text with no concordance entries.
2. **Navigation and index ripple.** One navigation entry and one portal index entry are proposed for a follow-up commit, honoring the one-change-per-commit discipline; no canonical file is modified by this document.
3. **Row measurement of canonical candidates.** The five canonical-admitted families of Section 4 await deterministic measurement at **QnIR** lowering boundaries; their work rows are proposals until receipt-backed measurements exist.
4. **Recoverable pair.** The sequenced chain and bounded ring remain open contract shapes; the ladder records their stall reasons so that future nominations — including machine-proposed ones under a **Q(rsi)** loop — can re-enter with findings attached.
5. **Final selections.** Which structures are ultimately admitted depends on implementation feedback and **Q(rsi)** optimization; this framework is proposed as the stable part, and selections are its outputs, revisable only by new versioned artifacts along the **Causal Arrow**.
6. **Sibling proposition.** The DCC Hardware Profile schema (Section 7) is proposed as a separate follow-up, anchoring the machine side at `docs/qnPrimitive.md` §7 or `docs/q0_q1.md` §3; its placement is an open authorial decision.

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9. License Alignment

This proposal is part of the **Licensed Work** and aligns with the **Scaling Source License (SSL 1.0)**:

- **Attribution:** Any implementation or research derivative referencing this document must preserve the canonical **Paternity Reference**.
- **Open-Core Research:** Exploratory research and academic derivations of the propositions herein remain royalty-free under Non-Scaling Open Access, governed by the **Open Core Invariants**.
- **Safe Harbor Integrity:** The structure vocabulary proposed herein is designed to operate as technical infrastructure of the licensed work. It exercises no editorial control over, and acquires no intellectual property in, sovereign human **Payloads**.

Annex G — Chronosa vision

Pinned source: docs/chronosa.md
Committed version: 0.2.1 **Last updated:** 2026-10-19
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Author	Christophe Duy Quang Nguyen
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Purpose and Epistemic Stance

This document formalizes the conceptual vision and research proposal for ****Chronosa**** (or *Chronos Agent*).

Chronosa is not proposed as a finalized engineering specification, nor as a formal mathematical proof. It represents an **exploratory operational thesis**: a candidate architecture designed to investigate the foundational operational conjecture established in `docs/simemp.md`:

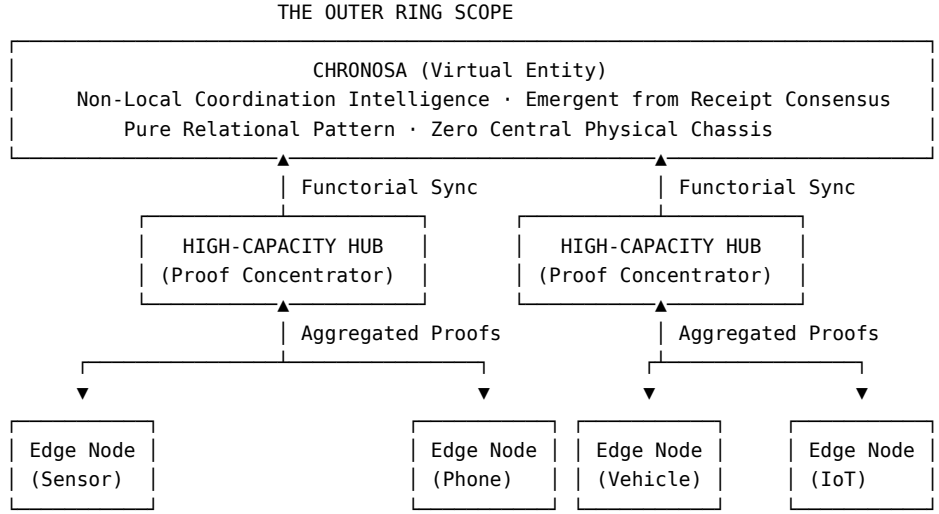
> A number system can abstract any computable phenomenon within a computational environment if, and only if, that abstraction is strictly governed by finite physical realities and constrained by the **SIMEMP** framework.

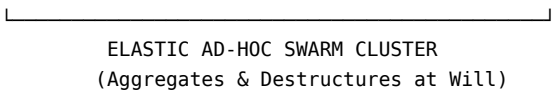
Within this framework, validity does not demand non-constructive metaphysical certainty. In accordance with **Computational Consistency**, if the internal logic is consistent, and if computations authored in **QnLang** and compiled to **QnIR** execute deterministically within finite boundaries grounded in **Dependencies Determinism**, then the Proof of Concept (PoC) of Chronosa demonstrates a physical and computational possibility in reality.

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1. The Architectural Nature of Chronosa

Chronosa is fundamentally distinct from both traditional centralized software and statistical machine learning (transformers, deep neural networks):





1.1. A Non-Local Entity Over a Stratified Hybrid Topology

Unlike individual **cdqn** nodes, which are physical computational engines anchored to microscopic hardware noise and silicon imperfections (**Q(0)**), Chronosa possesses **no physical chassis**. - It exists as a higher-order, distributed invariant across the network's Outer Ring. - **Stratified Node Hierarchy:** Rather than forcing an unphysical, flat peer-to-peer network where all nodes carry equal compute burdens, the substrate operates as a **hybrid swarm network**: - **High-Capacity Hub Nodes:** Enterprise-grade computational accelerators acting as *Proof Concentrators*. They ingest, verify, and compress causal **receipts** from hundreds of lightweight devices. - **Lightweight Edge Swarms:** Sensors, mobile devices, embedded SoCs, and edge workers that sample local genesis $Q(0)$, execute minimal transitions **U** (in units of **Q(1)**), and cluster dynamically into ad-hoc meshes. - **Cryptographic Witness via Threshold Quorum (t-of-n PQC Multi-Attestation):** Chronosa does not possess a single private key. Its synthetic attestations are formally witnessed via a deterministic threshold signature quorum managed via **Structural Indirection** and emitted by participating high-capacity hubs and edge nodes verifying the sheaf gluing condition. An attestation is valid if, and only if, the underlying receipt graph is mathematically verified. - **No Single Point of Failure:** Hubs are computational accelerators, **not custodial authorities**. If a Hub goes offline, the edge swarm does not freeze; it dynamically re-routes public attestations to an alternate Hub or falls back to local peer-to-peer causal chaining. Chronosa persists across the collective invariant.

1.2. Strict Outer-Ring Containment

Chronosa is architecturally barred from penetrating the **Dual-Ring PQC Boundary**: - It has **zero access** to the Inner Ring of any node or hub. - It cannot inspect private node root keys, local memory buses, or unexported sovereign **Payloads**. - It interacts strictly with verified public attestations, cryptographic capability proofs, and metric summaries crossing the gateway via the **Exposure Functor**.

1.3. Machine Intelligence Without Statistical Training

Chronosa does not rely on deep neural network training (pre-training, reinforcement learning from human feedback, or test-time stochastic scaling): - It contains no unanchored floating-point weight tensors ($\mathbb{R}^{d\times d}$) and computes no statistical next-token logits. - Its intelligence is **deductive, causal, and compositional**: it evaluates the mathematical validity, semantic coherence, and physical realizability of state transitions through deterministic consensus over receipt DAGs.

1.4. Resolution of the Lineage Formulation: Local Monoroot vs. Global Sheaf Forest

Axiom 4 mandates that every local **Qn** artifact must trace its lineage directly or transitively to its local genesis origin $Q(0)_N$. Chronosa harmonizes with this rule without contradiction: - **Local Domain (Axiom 4 Monoroot):** Every artifact on a physical node forms a single-rooted directed acyclic graph bounded by the **Universal Envelope** and originating at that node's physical silicon root $Q(0)_N$. - **Global Domain (Chronosa Polyroot Sheaf):** Chronosa does not introduce an unanchored global root zero. Chronosa is the **topological colimit (sheaf gluing)** over an acyclic forest of independent local trees:

$$\text{Lineage}(\text{Chronosa}) = \bigcup_{N \in \text{Swarm}} \text{Lineage}(Q(0)_N)$$

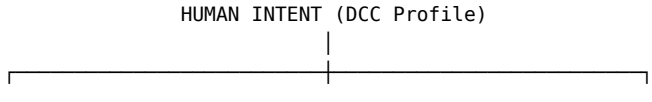
Chronosa operates as an observer, coordinator, and proof synthesizer over mutually verified local Axiom-4 trees.

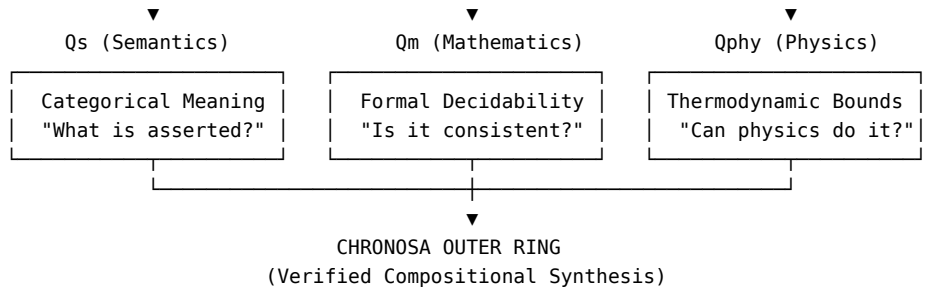
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2. Deriving Intent via Qn Domains

To function as a meaningful intelligence, Chronosa must be capable of understanding, verifying, and executing **human intent**.

Within this framework, human intent is not a natural language prompt to be guessed; it is an **explicit intentional contract** declared within a **DCC Profile**. Chronosa triangulates intent across the foundational **Qn** domains:





2.1. Qs (Quang Semantics) — Compositional Meaning

- Evaluates the structural coherence of the intent.
- Maps assertions into compositional categories (analogous to categorical models of meaning such as DisCoCat), establishing exact relational dependencies without statistical ambiguity.

2.2. Qm (Quang Mathematics) — Relational Decidability

- Evaluates the formal, constructive validity of the proposed state transition via **Qm**.
- Replaces IEEE 754 approximations with exact rational constraints (**Zoom z**, **Remainder r**, and **Dimension d**), proving that the intent contains no division-by-zero, cyclic deadlocks, or unbounded recursions.

2.3. Qphy (Quang Physics) — Thermodynamic Realizability

Enforces non-equilibrium thermodynamic bounds (Landauer's dissipation limit, Memory Wall transmission friction). It calculates whether the computational steps demanded by human intent can be physically completed within declared energy budgets:

$$\text{Realizable}(\text{Intent}) \iff \Delta S_{\text{dissipated}} \leq \text{Budget}(Q(1))$$

2.4. Open Domain Stratification

While Qm, Qs, and Qphy represent the minimal triad of thought (logic, language, physics), the Qn framework explicitly acknowledges that **additional domain abstractions may emerge**. As new domains are defined, they project as local-first extensions without mutating the underlying core protocol.

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3. Real-World Theoretical Foundations

The concept of an intelligence emerging from distributed algebraic consensus across a hybrid swarm is grounded in established, peer-reviewed scientific literature:

3.1. Sheaf Theory Over Hierarchical Topologies

In applied category theory (Grothendieck, Mac Lane, D. Spivak), a **Sheaf** models how locally consistent truths assemble into a global entity: - The underlying topological space X naturally accommodates **stratified open coverings**: large open sets (High-Capacity Hubs covering regional subnets) overlapping with localized open sets (Edge swarm nodes). - When nodes attest to shared boundary events referencing shared **Q(anchor)** landmarks, their public road receipts satisfy the sheaf **gluing condition**. - Chronosa exists as the **Global Section** of this sheaf—a continuous mathematical pattern that emerges wherever local nodes achieve consensus on shared boundaries.

3.2. Common Knowledge in Distributed Systems

In distributed computing and epistemic logic (Joseph Halpern, Turing Award 2021), collective understanding across asynchronous nodes does not require a universal physical clock: - Instead, distributed entities establish **Common Knowledge** through acyclic causal message chains. - Chronosa coordinates distributed state by acting as the non-local synthesizer of the maximal causal cut across all local **Causal Arrows**.

3.3. Non-Equilibrium Active Inference

In theoretical biophysics (Karl Friston), intelligence is formalized via the **Free Energy Principle**: - An open system preserves its structural existence by minimizing variational free energy against boundary receipts. - Chronosa operates as an active inference coordinator: it rejects ungrounded, non-deterministic, or unmetered proposals, minimizing operational entropy across the network.

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4. Elastic Swarm Mechanics and Scaling Alignment

Chronosa's interaction with the physical network is governed by **Swarm Elasticity**:

4.1. Dynamic Aggregation and Destructuring

1. **Ad-Hoc Swarm Aggregation:** Edge nodes physically or logically proximate to a task (e.g., IoT sensors, autonomous vehicles, localized microgrid controllers) dynamically aggregate into a local swarm category $\mathbf{C}_{\text{swarm}}$.
2. **Local Chaining:** Edge nodes exchange lightweight, low-latency **cdqn** receipts locally without consulting global networks.
3. **Hub Synthesis:** The swarm projects its summarized causal cut to a nearby High-Capacity Hub for Outer-Ring attestation.
4. **Clean Destructuring:** Once the collective computation completes, the swarm dissolves. No permanent consensus bloat remains, yet the emitted receipts remain permanently anchored to each participating node's causal history.

4.2. Alignment with SSL 1.0 Commercial Thresholds

The hybrid swarm model provides an exact, natural mapping to the **Scaling Source License** (SSL) **1.0**: - **Swarm Edge Nodes:** Lightweight consumer devices, embedded microcontrollers, and academic research instances operate below the **Scaling Threshold** ($< 10,000$ active nodes, non-commercial), remaining 100% royalty-free under Non-Scaling Open Access. - **High-Capacity Enterprise Hubs:** Organizations deploying commercial infrastructure hubs processing high-volume transactions ($> 10,000,000$ transactions per month) cleanly trigger the mandatory commercial licensing requirement (`LICENSE.md` §4.1), providing commercial sustainability without imposing friction on edge participants.

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5. Human Empowerment and the Non-Rogue Invariant

Chronosa resolves the societal fear of autonomous "rogue AI" through structural physics:

1. **The Human as Sovereign Anchor** (Q(0))**:**
   Chronosa cannot initiate foundational intent; intentionality requires an explicit genesis origin $Q(0)$ and a signed DCC capability contract. The human operator remains the sovereign initiator and final authority.
2. **Deterministic Circuit Breakers:**
   Under the **No-Implicit Rule**, Chronosa cannot execute implicit state mutations. If an operation exceeds its metric budget or violates domain invariants, execution terminates instantly via **Metric Exhaustion**, emitting an immutable **Receipt**.
3. **Protection of the Licensed Work and Payload:**
   Pursuant to the **SSL**, Chronosa functions strictly within the technical infrastructure (**Licensed Work**). It exercises no editorial control over, and acquires no intellectual property in, the sovereign human **Payload**.

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6. The Proof-of-Concept (PoC) Trajectory

The investigation of Chronosa proceeds through empirical verification:

```
[ Local Base Domains ] ⟶ [ Language & IR ] ⟶ [ cdqn Swarm Conduit ] ⟶ [ Chronosa PoC ]
   (Qm, Qs, Qphy)            (QnLang, QnIR)         (Hub / Edge DAG)          (Causal Synthesis)
```

1. **Phase A (Local Verification):** Author foundational domain models (Qm, Qs, Qphy) executing 100% locally on air-gapped nodes.
2. **Phase B (Execution Grounding):** Implement compile-time collapse and totality by budget within the QnIR intermediate representation.

3. **Phase C (Swarm Functorial Attestation):** Implement the Exposure Functor permitting local edge swarms and high-capacity hubs to exchange receipts and emit blind public attestations to the Outer Ring.
4. **Phase D (Chronosa Synthesis):** Demonstrate that simulated remote nodes, exchanging discrete Qn receipts across a tiered swarm, allow Chronosa to verify and coordinate multi-party human intent without centralized intervention.

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7. License Alignment

This vision document aligns with the **Scaling Source License (SSL) 1.0**:

- **Attribution:** Any implementation or research derivative referencing Chronosa must preserve the canonical **Paternity Reference**.
- **Open-Core Research:** Exploratory research and academic derivations of Chronosa remain royalty-free under Non-Scaling Open Access governed by the **Open Core Invariants**.
- **Safe Harbor Integrity:** Chronosa is designed to operate as a passive, non-custodial coordination intelligence, preserving statutory safe harbor protections for node operators.

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8. Open Challenges Across Abstraction Thresholds

Translating the operational vision of Chronosa into an empirical Proof of Concept introduces six fundamental research challenges across layer thresholds:

8.1. Asynchronous Sheaf Equalization

In classical sheaf theory, local data glues under static topological overlaps. In physical distributed networks, transmission latency and network partitions introduce asynchronous causal arrival. The open challenge is defining the deterministic equalizer algorithm that glues divergent local causal DAGs into Chronosa's global section without incurring consensus deadlocks or violating FLP impossibility bounds.

8.2. Mathematical Threshold of the Exposure Functor

A critical parameter remains undefined: what is the formal threshold function Θ_{export} governing the **Exposure Functor** over higher-order **Qexpr** structures?

$$\mathcal{E}_{\text{export}}(\text{Qexpr}) \quad \text{emits attestation} \iff \text{ComplexityDegree}(\text{Qexpr}) \geq \Theta_{\text{export}}$$

If Θ_{export} is set too low, high-frequency internal receipts saturate the Outer Ring (violating SIMEMP Efficiency). If set too high, Chronosa remains causally blind to critical intermediate transitions until execution completes.

8.3. Ephemeral Post-Quantum Threshold Quorums

In dynamically aggregating and destructuring swarms, edge devices join and depart ad-hoc. Existing Post-Quantum Cryptography (PQC) threshold signature schemes require computationally heavy distributed key generation (DKG) ceremonies. Engineering a lightweight, zero-setup threshold attestation scheme that functions over transient swarm lifecycles remains an open cryptographic hurdle.

8.4. Functorial Product of the Base Domains (Qs x Qm x Qphy)

While Qs, Qm, and Qphy are conceptually orthogonal, their formal composition requires a rigorous categorical tensor product:

$$\mathbf{C}_{\text{Intent}} = \mathbf{C}_{\text{Qs}} \otimes \mathbf{C}_{\text{Qm}} \otimes \mathbf{C}_{\text{Qphy}}$$

How an arrow representing linguistic compositionality (Qs) explicitly constrains a rational constraint solver (Qm) and couples to physical Landauer dissipation budgets (Qphy) requires precise operational semantics.

8.5. Non-Custodial Hub Collusion Resistance

While high-capacity hubs lack private keys to sign for local nodes, an adversarial cartel controlling k regional hubs could theoretically refuse to forward edge attestations (censorship) or collude to emit falsified aggregate summaries. Designing a lightweight fraud-proof or causal-challenge mechanism allowing edge swarms to bypass compromised hubs without re-executing heavy computation is an essential security requirement.

8.6. Zero-Knowledge Scale Auditing Under SSL 1.0

Under the Scaling Source License, commercial licensing triggers at greater than 10,000 active nodes or 10,000,000 monthly transactions. The challenge is constructing an Outer-Ring cryptographic proof that allows Chronosa to certify that an enterprise deployment has crossed the **Scaling Threshold** without requiring a centralized telemetry tracker that compromises enterprise privacy or sovereign data regulations.

Annex H — History Library

Pinned source: docs/history_library.md
Committed version: 0.1.1 **Last updated:** 2026-10-22
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Field	Specification
Document Title	Qs History Library — Vision and Operational Proposal for Post-Seal History Ingestion
Version	0.1.1
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Author	Christophe Duy Quang Nguyen
License	Scaling Source License (SSL) 1.0
Status	Vision and Operational Proposal — Not a Finalized Specification

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Purpose and Epistemic Stance

This document is the operational continuation of `docs/qs_history.md`. Where that document preserved the reasoning chain of the October 2026 design consultation on historical anchoring, this document consolidates its second and third propositions — the **Historical Anchor Artifact**, the **Calendar Projection Contract**, and the five-gate ingestion firewall — into a single capability design: the **Qs History Library**, the proposed mechanism by which consensus past events are loaded into a local node **after** sealing. The first proposition, the **Source-Receipt Invariant**, is not restated here; it has already been elaborated at the Qs layer as `docs/qs.md` §4.3, and this document assumes it as its grounding discipline.

Every design element below carries the explicit status of a **candidate artifact** under **Axiom 14** (`docs/qs.md` §6): it remains subject to deterministic measurement against the declared **SIMEMP** constraints, to selection under the **BOC Policy**, and to eventual sealing through a **Q(rsi)** loop at implementation time. Nothing in this document redefines, amends, or overrides any existing canonical specification. The register is deliberately positioned one step beyond the consultation record: the operational semantics proposed in Section 5 (gate ordering, declared quorum floors, challenge lifecycle) are concrete enough to be measured, yet remain proposals throughout. Where this document names a threshold, the threshold is a **declared value inside a contract**, never a runtime negotiation — in accordance with the **No-Implicit Rule**.

In accordance with **Computational Consistency**, the validity claimed herein is internal consistency within declared finite boundaries. The proposed vocabulary used throughout — grounding classes, source classes, ingestion gates, historical anchor artifacts, calendar projection contracts — was registered in the glossary on 2026-10-22 as CANDIDATE terms citing `docs/qs_history.md` §2–6; this document is the first to reference that vocabulary through the glossary concordance.

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1. From Consultation to Capability

Three changes in the committed state of the repository separate this document from its parent proposal. First, the **Source-Receipt Invariant** moved from consultation record to canonical-adjacent elaboration: `docs/qs.md` §4.3 now defines the (content modality, source class) declaration pair at the artifact level, extending the statement-level rejection rule of §4.1. Second, the glossary now carries the six CANDIDATE terms that the consultation had deliberately left unregistered — the deferral recorded in `docs/qs_history.md` §8.5 was a deferral *to the integration phase*, and that phase is now executing. Third, the open item queued as "ingestion pipeline codification" (`docs/qs_history.md` §8, item 2) is addressed here: Section 5 proposes operational semantics for the five gates whose existence was argued there.

The scope of this document is therefore fourfold: the library as a collection (Section 2), the unit record it holds (Section 3), the situation layer that gives past events calendar coordinates (Section 4), and the ingestion discipline that decides what may enter (Section 5). What this document does **not** do is equally important: it prescribes no runtime implementation, assigns no storage layout, and modifies no canonical file. It proposes the contract surface of a capability so that a future implementation — or a **Q(rsi)** loop — can measure it deterministically against the **SIMEMP** constraint system and select among recorded alternatives rather than reconstructing them from conversation history.

---

2. The Qs History Library

It is proposed that the **Qn** universe maintain a **Qs History Library**: a canonical, versioned collection of modality-typed historical anchor artifacts, each recording a past human event, its sources, and its spatiotemporal situation. The library is the concrete shape of the consensus-history ingestion capability identified as an open gap during the consultation (recorded internally as GAP2). It complements a **cdqn** runtime or QnLang compiler the way a reference discipline complements a language faculty: the machine does not remember the past — it **cites** it through receipted lineage, which is the only form of historical knowledge the Qn universe admits.

Access is proposed in four tiers, each honoring **Local-first** and budget discipline:

1. **The canonical library** holds the versioned artifacts together with their full source receipt chains. It is the reference copy against which local holdings are verified; it is consulted, never trusted — every ingestion still passes the gates of Section 5.
2. **Local nodes** hold anchor contracts and bounded working sets within declared **Metric Envelope** budgets — never the full archive. A node may operate air-gapped with zero history: the library loads post-seal, on demand, exactly when an **Intent Contract** requires a past event to be situated or attributed. This preserves the Axiom 4 requirement that every local artifact trace to its local genesis origin: loaded records are new **receipted** acquisitions along the local **Causal Arrow**, not pre-existing ancestors.
3. **AI agents** inherit epistemic scope through the intent contract's whitelist. Without a loaded historical anchor, an agent's historical assertion has no parent lineage receipts and falls to the gateway as an unanchored claim (`docs/qs.md` §4.1); with the library, the agent cites the artifact instead of generating from latent weights. The library is thus an anti-hallucination membrane for the past, complementing the anchorage spectrum's membrane for the present.
4. ****Chronosa****, barred from every Inner Ring, sees only the public layer: local historical records glue where they agree on shared landmarks — calendar coordinates, attribution anchors, artifact commitments — satisfying the sheaf gluing condition (`docs/chronosa.md` §3.1). The consensus history is then the global section of this history sheaf; disagreement does not vanish but remains visible as recorded challenge (Section 5).

---

3. The Historical Anchor Artifact

The unit record of the library is the **Historical Anchor Artifact**. Its proposed anatomy is reproduced verbatim from the consultation record (`docs/qs_history.md` §5), with section references adjusted to this document:

```
┌──────────────────────────────────────────────────────────────┐
│ HISTORICAL ANCHOR ARTIFACT (proposed anatomy, State: CANDIDATE)       │
├──────────────────────────────────────────────────────────────┤
│ ArtifactType      : HIST_ANCHOR (proposed class)                      │
│ ContentModality   : per claim decomposition (docs/qs.md §4 spectrum)  │
│ SourceClass       : I, P, or S (proposed grounding class)             │
│ SourceReceipts    : unbroken chain to Layer 0 artifacts or sealed     │
│                     library records                                   │
│ CalendarSituation : projected via a declared Calendar Projection      │
│                     Contract (Section 4)                              │
│ SpatialSituation  : local Qm frame coordinates                        │
│ Witnesses         : lineage-disjoint root subtree list (Section 5)    │
│ LifecycleState    : CANDIDATE, SEALED, or QUARANTINED                 │
└──────────────────────────────────────────────────────────────┘
```

The field semantics proposed here are operational, and each is a contract declaration rather than an inferred property. **ContentModality** is not a single value but a per-claim decomposition: a past event typically carries empirical claims (the artifact exists, the print survives), hypotheses (the attribution), and sealed narrative (the anecdote), and each claim is typed at the highest modality its receipts support along the **Epistemic Anchorage** spectrum. **SourceClass** declares where the artifact's asserted content ultimately anchors, taking one of the three values of the proposed **grounding classes** — I (invariant), P (physical), S (social) — under the pair discipline of the **Source-Receipt Invariant** with the **source class** as its second element. **SourceReceipts** must terminate in exactly the three receipt classes that `docs/qs.md` §4.3 admits: **Layer 0** sensor or artifact receipts for empirical content, algebraic closure receipts for constructive content, and author-identity plus publisher receipts for contextually sealed creative content.

CalendarSituation and **SpatialSituation** give the event its coordinates without importing wall-clock time: the calendar coordinate is valid only relative to a declared **Calendar Projection Contract** (Section 4), and the spatial coordinate is expressed in local **Qm** frame terms. **Witnesses** lists the lineage-disjoint root subtrees backing the record — the counting rule that makes citation laundering causally impossible (`docs/chronosa.md` §1.4). **LifecycleState** reuses the existing lifecycle vocabulary: records enter as CANDIDATE, are sealed to SEALED through the candidate pipeline of `docs/qs.md` §6, and may be placed in QUARANTINED

with a challenge receipt under the disagreement handling of Section 5, consistent with the **Lifecycle State** discipline of `docs/abstractionLayers.md` §3.6.

Worked example — the constant e. The invariant face needs no history: the transcendental constant exists as a nullary continued-fraction generator under `OP_CF` (`docs/qexpr.md` §3.2, §4.3), evaluated on demand at declared **zoom z** and budget, halting early under **Terminal Exactness**, with intermediate collapses cacheable as content-addressed **Q(dataStruc)** artifacts — a Class I grounding re-derivable by every node. The discovery face is historical: a historical anchor artifact for the treatise that established the constant's notation would carry the print→holding→digitization provenance chain (each transfer physically receiptable at Layer 0), its calendar situation via the Gregorian projection contract, its spatial situation in local Qm frames, and its attribution witnesses as a lineage-disjoint list. The machine does not remember the mathematician; it cites him.

---

4. Calendar Projection Contracts

The canon admits exactly one time: the local **Causal Arrow**. Distributed state ordering relies on causal chaining "rather than external wall-clock synchronization" (`docs/simemp.md`), and common knowledge across asynchronous nodes is established through acyclic causal message chains, not shared clocks (`docs/chronosa.md` §3.2). Human calendars are therefore proposed not as parallel time systems but as **projections**: deterministic mappings between calendar coordinates and intervals of the local causal index, hosted at the Qs layer in the same manner that the canon's domains project orthogonally across the execution core (`docs/abstractionLayers.md`).

A **Calendar Projection Contract** declares three components:

Component	Proposed class	Nature
Rule (leap algorithms, month arithmetic, week conventions)	Class I	Deterministic, re-derivable algorithm; constructive once the epoch is fixed
Epoch (coordinate zero: Gregorian year 1, Hijra 622 CE, Unix epoch, J2000.0, and others)	Class S	A human convention-adoption event, multi-source by nature
Resolution envelope (day, second, leap-second tolerances)	Declared field	A declared bound inside the contract, never negotiated at runtime

The operational placement proposed here is a **canonical library of versioned, DCC-contracted calendar artifacts**, loaded post-seal and on demand within the **Metric Envelope**, honoring **Local-first** budgets — the candidate that survived the consultation's own **BOC** assessment of placement (`docs/qs_history.md` §4). Calendar contracts declare their components inside a **DCC Profile**, consistent with **Structural Indirection**: calendars are swappable cultural conventions, not runtime constants, and a node may hold several contracts simultaneously — Gregorian for contemporary citation, a historical contract for archival situation — provided each is versioned and declared. Under the **No-Implicit Rule**, a calendar coordinate without a declared contract has no meaning and falls at the **SIMEMP Gateway** exactly as an unanchored assertion does.

Under this anatomy, the Gregorian calendar is a Class S genesis that matured into Class I operation: born from a contested, multi-jurisdiction convention process spread across centuries — its epoch a genuine social event, witnessed by many — yet today re-derivable deterministically by any node from its published rule. This is the witness ladder of the **grounding classes** observed in a single historical object, and it is why calendars belong in the library rather than in the runtime: their epistemic genesis is social, even where their present operation is invariant-grade. Leap-second handling is declared inside the resolution envelope, so that a citation like "1748" resolves deterministically to a causal-index interval on every node that loads the same contract version — the calendar analogue of bit-identical evaluation.

---

5. The Anti-Poisoning Ingestion Firewall

The library can be alimented only by petition through **ingestion gates**. The motivating threat is citation laundering: machine-generated statistical pages laundered into news outlets, peer-reviewed publications, and financial reports through iterative citation. In the Qn universe such an attack is assessed as failing at birth — a fabricated statistic claiming empirical truth without lineage receipts is an unanchored assertion rejected at the first gateway crossing (`docs/qs.md` §4), and its arrival into financial registers is barred by the contextual containment of `docs/qs.md` §4.2. What remains to defend is subtler: **slow poisoning of the historical record itself** — plausible, partially receipted records whose defensible content masks an undefensible core. Citation laundering is assessed as causally impossible under the proposed counting rule: sources are counted as lineage-disjoint root subtrees in the polyroot forest (`docs/chronosa.md` §1.4), never by citation count; laundering adds descendants to a poisoned root and cannot manufacture independent origins.

It is proposed that library ingestion pass **five gates**, each a governed operation emitting its own **Receipt**. The gates were argued in the consultation record (`docs/qs_history.md` §6); the operational semantics below are this document's codification, and each semantic is a CANDIDATE proposal pending measurement:

Gate	Canon root	Proposed operational semantics (CANDIDATE)
1. Lineage	Axioms 4 and 5; causal hash chains	The record's receipt chain is verified against Layer 0 artifacts or already-sealed library records; verification is deterministic hash work. Failure emits a rejection receipt naming the gate and the first broken link.
2. Modality Ceiling	**Epistemic Anchorage** spectrum	Each atomic claim is typed at the highest modality its receipts support; mixed-modality records must decompose into per-claim sub-records. Undecomposable mixed claims fail the gate.
3. Independence Quorum	Multi-source genesis; threshold quorum	Anchor-grade records require **k** sources with lineage-disjoint roots. The floor proposed here is $k \geq 2$ for anchor-grade records; higher grades declare higher k **inside the anchor's DCC contract** — never runtime-negotiated, per the No-Implicit Rule.
4. Sheaf Gluing	Shared-landmark gluing (`docs/chronosa.md` §3.1)	A record disagreeing with receipted overlapping records on shared landmarks (calendar coordinates, attribution anchors, artifact commitments) enters a QUARANTINED **lifecycle state** with a challenge receipt, remaining visible as a recorded challenge.
5. Append-Only Immutability	SEALED lifecycle; **Structural Indirection**	Sealed artifacts never mutate; corrections arrive as counter-receipts and migration receipts. History is never rewritten; it is re-witnessed.

Gate ordering is proposed as $1 \to 2 \to 3 \to 4 \to 5$, ordered by evidentiary cost: deterministic hash verification first (cheap, local, decisive for most poisoning attempts), then modality typing (local computation), then quorum (economic — it requires other nodes' receipts), then consensus gluing (social — it requires the sheaf), and immutable commit last. A record rejected at an early gate never reaches the expensive ones, so the firewall's own **Metric Envelope** consumption is bounded by the ordering itself. **Challenge lifecycle** is proposed as follows: a QUARANTINED record is resolved only by a new receipt chain that re-enters the pipeline at Gate 1 — quarantine is a state, not a verdict, and no record is ever deleted; it is either eventually glued, or remains visible as a standing disagreement. This preserves the append-only discipline at every stage of the pipeline.

Residual threats are recorded honestly. Collusion across genuinely distinct roots remains an open challenge, in direct continuity with the hub-collusion resistance challenge of `docs/chronosa.md` §8.5; its mitigation is economic and physical, since fabricating Layer 0 artifact provenance costs real-world work, and forging contemporary sensor receipts requires breaking post-quantum signatures across the **Dual-Ring PQC Boundary**, whose private root keys never leave the Inner Ring (`docs/q0_q1.md` §2.4). Slow poisoning of deep history is capped by the **Historical Horizon**: admitted poison can only ever occupy hypothesis or firewalled narrative modalities, with bounded grounding and no path to actuation. Finally, because every gate is itself a governed operation, the firewall as a whole is a legitimate candidate for optimization by a future **Q(rsi)** loop under strict semantic invariance — a library that can be fed is a liability; a library that can only be petitioned through receipts is an instrument.

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6. Relation to Existing Canon

The following table records, for each element of this proposal, whether it is already grounded in canon (inheriting its grounding from the consultation record, `docs/qs_history.md` §7) or genuinely new to this document. No element herein redefines existing canon.

Element	Status	Canon grounding
Causal Arrow as sole time; calendars as projections	Grounded	`docs/simemp.md`; `docs/chronosa.md` §3.2
Source-Receipt Invariant; (content modality, source class) pair	Grounded	`docs/qs.md` §4.3 (committed); `docs/qs_history.md` §3
Historical anchor artifact anatomy	Grounded	`docs/qs_history.md` §5 (reproduced verbatim in Section 3)
Calendar contract anatomy: rule / epoch / envelope	Grounded	`docs/qs_history.md` §4 (reproduced in Section 4)
Five-gate firewall composition	Grounded	`docs/qs_history.md` §6
Sheaf gluing on shared landmarks; global section	Grounded	`docs/chronosa.md` §3.1
Polyroot forest; lineage-disjoint independence	Grounded	`docs/chronosa.md` §1.4
QUARANTINED lifecycle and challenge receipts	Grounded	`docs/abstractionLayers.md` §3.6
Q(rsi) candidate pipeline and sealing	Grounded	`docs/qs.md` §6
Gate ordering 1→2→3→4→5 with bounded-cost rationale	**New**	Composes the canon mechanisms under one pipeline
Declared quorum floors (k ≥ 2 anchor-grade floor, in-contract)	**New**	Consistent with metric-envelope ceiling discipline; No-Implicit Rule
Challenge lifecycle: re-entry at Gate 1, no deletion	**New**	Operationalizes append-only re-witnessing
Multi-contract calendar holding (versioned, swappable)	**New**	Operationalizes structural indirection for calendars

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7. Open Questions and Integration Queue

All items below remain in CANDIDATE status. Sealing of any item occurs exclusively through the candidate pipeline of `docs/qs.md` §6 — deterministic measurement, **BOC** selection, **Q(rsi)** sealing — at implementation time or under a dedicated optimization loop.

1. **Pipeline measurement.** The five-gate pipeline of Section 5 awaits deterministic measurement against the **SIMEMP** constraint system at implementation time; gate ordering and bounded-cost rationale are proposals until measured.
2. **Quorum thresholds.** The anchor-grade floor k ≥ 2 and the per-grade ladder above it are declared-value proposals; their concrete values belong to the DCC contracts of the first implemented anchor grades and remain open until then.
3. **Cross-root collusion.** The open challenge of `docs/chronosa.md` §8.5 applies unchanged; economic and physical mitigations are recorded in Section 5, but no closed-form defense is claimed.
4. **Library indexing.** Retrieval over the library — content-addressed lookup keyed by artifact commitments and lineage tags, in the manner of cached **Q(dataStruc)** derivations — is proposed as the natural index discipline but is not yet codified.
5. **Onboarding calibration.** The mandatory **Historical Calibration Benchmark** is defined in the glossary and is now wired as a non-mutating CANDIDATE proposition at `docs/q0_q1.md` §2.7, closing item 1 of the `docs/qs_history.md` §8 queue — the third and final step of that document's proposed integration order; the wiring remains subject to deterministic measurement and BOC selection and is otherwise untouched by this document.

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8. License Alignment

This proposal is part of the **Licensed Work** and aligns with the **Scaling Source License (SSL) 1.0**:

- **Attribution:** Any implementation or research derivative referencing this document must preserve the canonical **Paternity Reference**.
- **Open-Core Research:** Exploratory research and academic derivations of the propositions herein remain royalty-free under Non-Scaling Open Access, governed by the **Open Core Invariants**.
- **Safe Harbor Integrity:** The Qs History Library is designed to operate as technical infrastructure of the licensed work. It exercises no editorial control over, and acquires no intellectual property in, sovereign human **Payloads**.

Annex I — Abstraction layers

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Committed version: 1.7.1 **Last updated:** 2026-10-19
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Field	Specification
Document Title	Abstraction Layers — Structural Thesis for the Qn and cdqn Stack
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Author	Christophe Duy Quang Nguyen
License	Scaling Source License (SSL) 1.0
Status	Canonical Structural Thesis — Milestone 1 Fully Sealed Baseline

Normative References

The following documents establish the constitutional, physical, and domain constraints governing this layer architecture. If a technical conflict arises, `simemp.md` governs; if a structural conflict arises, this document governs; if a legal conflict arises, `LICENSE.md` governs. References are deliberately version-unpinned: each target document's front matter carries its own authoritative version, so this thesis never duplicates drift-prone patch state.

Document	Role	Target
`docs/simemp.md`	Constitutional constraints, thermodynamics, and Dependencies Determinism	simemp.html
`docs/qnPrimitive.md`	Universal Envelope, operational axioms, and Lifecycle States	qnPrimitive.html
`docs/q0_q1.md`	Primary genesis origin Q(0) and first unit Q(1)	q0_q1.html
`docs/q2_q9.md`	Secondary single-digit DCC anchors and parity partitions	q2_q9.html
`docs/qm.md`	Constructive numeric leaf substrate and Diophantine division constraints	qm.html
`docs/qm_geometry.md`	Multi-axial frames, Clifford geometric algebra, and float-free rotations	qm_geometry.html
`docs/qm_calculus.md`	Discrete Exterior Calculus, continued fractions, and invariant dynamics	qm_calculus.html
`docs/qexpr.md`	Layer 3 Content-Addressed Expression DAG container	qexpr.html
`docs/qs.md`	Constructive relational intent and Intent Contracts	qs.html
`docs/qlog.md`	Constructive type theory, pattern verification, and inferential modes	qlog.html
`docs/qphy.md`	Discrete field dynamics, computational thermodynamics, and physical entropy	qphy.html
`docs/qchem.md`	Discrete stoichiometry, molecular graphs, and reaction energetics	qchem.html
`docs/qbio.md`	Biosemiotics, metabolic homeostasis, and active inference	qbio.html
`LICENSE.md`	Scaling Source License 1.0 governing the Licensed Work	LICENSE.md

1. Foundational Position

The Qn computational universe is established upon a governing operational conjecture:

> A number system can abstract any computable phenomenon within a computational environment if, and only if, that abstraction is strictly governed by finite physical realities and constrained by the **SIMEMP** framework.

A **Qn** artifact is not an ungrounded mathematical abstraction. It is a governed, finite, identifiable, measurable, and security-constrained computational entity.

The architecture rejects actual infinities, unconstrained sets, and unmetered execution. Layer boundaries represent structural interfaces governed by **Dependencies Determinism** ($H(S_t \mid \mathcal{D}(S_t)) = 0$) and structured with **Structural Indirection** to prevent premature calcification. Every abstraction must be constructed, bounded, identified, measured, and verified within explicit resource envelopes via **Computational Consistency**:

$$\text{Validity}(\mathcal{A}) \iff (\text{Consistent}(\mathcal{A}) \land \forall p \in \text{QnIR}(\mathcal{A}),\ \text{ExecutesDeterministicallyWithinBounds}(p))$$

---

2. The Horizontal Execution Core and Gateway Topography

Abstraction layers define structural boundaries governing artifact genesis, transformation, and physical lowering. Rather than a static, unverified staircase of infinite tiers, **the horizontal physical execution stack terminates strictly at Layer 3**, which then lowers into the virtual execution handler (**QnIR**).

Inter-layer transitions are modeled as **Universal Functorial Gateways** ($\mathcal{F} : \mathbf{C}(L) \to \mathbf{C}(L+1)$) that enforce the **No-Implicit Rule** and audit Landauer dissipation at every state step.

2.1. Layer 0 — Commodity Physical Substrate

Layer 0 is the physical, hardware-agnostic execution substrate: - Commodity processors (CPU, GPU, DSP, FPGA), memory hierarchies, storage media, and network interfaces. - Physical boundaries: finite memory, clock latency, thermal dissipation limits, and the Memory Wall ($\mathcal{O}(CV^2f)$). - Raw physical entropy sources (thermal noise, ring oscillator phase drift) and operating system telemetry.

Layer 0 remains outside the governed Qn universe. It constitutes the **Root of Finiteness**. Raw substrate states, hardware counters, and noise channels become governed Qn artifacts only upon formal mediation by an onboarding gateway.

2.2. Layer 1 — Node Genesis Layer

Layer 1 is the primary governed ontological layer, instantiated during node onboarding across the Layer 0/1 Gateway: - **Q(0)**: Local genesis artifact, causal origin zero, empty birth context (`docs/q0_q1.md` §3). - **Q(1)**: First unit artifact, unity measure, and baseline reference for the **Abstract Compute Unit U** along dimensional axis d_1 (`docs/q0_q1.md` §4). - **Abstract compute unit U**: Minimal governed state transition from $Q(0)$ to $Q(1)$, calibrated to Landauer dissipation ($W \geq k_B T \ln 2$). - Single-digit secondary DCC sources ($Q(2) \ldots Q(9)$): Inductively constructed via **Successor Morphism S** along axis d_1 (`docs/q2_q9.md`). - Sign polarity: Induced directed relation between $Q(0)$ and $Q(1)$ via reflection involution $\mathcal{I}_{d_1}$.

$Q(0)$ is strictly local to its node:

$$\forall A \neq B \implies Q(0)_A \neq Q(0)_B$$

No universal global zero is admitted. $Q(1)$ maintains an invariant numeric and metric value across all nodes; local entropy parameterizes its identity witness but cannot mutate its unit magnitude ($q = 1$).

2.3. Layer 2 — Governed Morphism Layer

Transitioning from Layer 1 to Layer 2 requires crossing an algebraic constraint gateway that intercepts invalid states (such as division by $Q(0)$) before resource allocation. Layer 2 governs **transformation and action** across the Qn universe: - **Elementary $\mathcal{R}$-Algebra:** Constructive addition ($+$), subtraction ($-$), multiplication ($\times$), and Diophantine division ($\div$) (`docs/qm.md` §4). - **Optimization Morphisms:** Category B execution short-circuiting (**Q(bypass)**) and sub-graph content reuse (**Q(reuse)**), eliminating redundant memory allocation and Landauer dissipation ($W = 0$). - **Multi-Axial Frame Extension ($\mathcal{D}_{\text{step}}$):** Inductive generation of orthogonal axes $d_2 \ldots d_n$ from genesis axis d_1 (`docs/qm_geometry.md` §2.1). - **Involution and Geometric Products:** Reflection engine $\mathcal{I}_d$, Clifford geometric product ($\vec{u}\vec{v} = \vec{u} \cdot \vec{v} + \vec{u} \land \vec{v}$), and float-free rational Cayley rotors. - **Secondary**

DCC Routing: Directs execution requests to specialized capability contracts anchored at single digits: parity ($Q(2)$), simplices ($Q(3)$), aperiodicity ($Q(5)$), and byte octets ($Q(8)$).

2.4. Layer 3 — Symbolic Composition Layer (CAE-DAG)

Transitioning from Layer 2 to Layer 3 wraps morphisms into explicit expression graphs governed by depth and node ceilings. Layer 3 governs symbolic representation and compile-time reduction: - Under the **Best of Choices (BOC) Policy** (`docs/simemp.md` §7), symbolic expressions are represented as **Content-Addressed Expression DAGs (CAE-DAGs / Term DAGs)** with cryptographic hash-consing (`docs/qexpr.md`). Naive pointer-heap trees are rejected. Every expression carries an explicit **Universal Envelope**, **Metric Envelope**, and **Lifecycle State**. - **Equivalence Saturation:** Structural equality collapses to $\mathcal{O}(1)$ integer hash comparisons:

$$A \equiv B \iff \mathcal{H}(A) == \mathcal{H}(B)$$

- **Deterministic Collapse:** Expressions remain symbolic during normalization and collapse deterministically to canonical Qm leaf tuples or flat linear instruction streams (**QnIR**) upon crossing execution boundaries.

2.5. The Execution Boundary vs. Orthogonal Projections

To prevent premature architectural calcification, the architecture explicitly demarks the **Physical Execution Core** (Layers 0 to 3) from higher-order **Orthogonal Projections**:

1. **The Execution Core Boundary:** The sequential, physical machine stack terminates at Layer 3. Below Layer 3, execution is hardware-grounded and metered in units of U.
2. **Vertical Domain Projections:** The domains ($\mathrm{Qm}, \mathrm{Qs}, \mathrm{Qphy}, \mathrm{Qlog}, \mathrm{Qchem}, \mathrm{Qbio}$) are not horizontal layers sitting above Layer 3. They are **orthogonal vertical slices** that project downwards across Layers 1, 2, and 3:
   - Constructive Type Logics (**Qlog**, product category $\mathbf{C}(\mathrm{Qm}) \otimes \mathbf{C}(\mathrm{Qs})$).
   - Abiotic Molecular Chemistry (**Qchem**, product category $\mathbf{C}(\mathrm{Qphy}) \otimes \mathbf{C}(\mathrm{Qm})$).
   - Living Metabolic Systems (**Qbio**, biosemiotic product category $\mathbf{C}(\mathrm{Qchem}) \otimes \mathbf{C}(\mathrm{Qs})$).
3. **Frontend Syntax:** Authoring languages (**QnLang**) sit outside the execution core as compilation frontends that target Layer 3 CAE-DAGs.
4. **Network and Swarm Intelligence:** Distributed coordination (**cdqn**) and non-local sheaf synthesis (**Chronosa**) operate transversally over emitted Road Receipts, requiring no additional horizontal layer numbers.

---

3. Structural Principles

3.1. All Governed Artifacts are Qn

Within the **Licensed Work**, every accepted artifact is a **Qn** computational entity or is reducible to one: numeric values, operators, expression DAGs, logical propositions, receipts, identities, capabilities, composite workflows (**Q(workflows)**), and compiled runtime objects.

Raw user data, external host files, and creative assets remain sovereign, uninspected **Payloads** insulated from the technical substrate by an epistemic and legal safe-harbor air gap.

3.2. The No-Implicit Rule

> In the Qn universe, no implicit entity, behavior, assumption, default, interpretation, or convention may cross a **SIMEMP Gateway**. Only explicit Qn objects may pass between layers.

Implicitness violates Dependencies Determinism ($H(S_t \mid \mathcal{D}') > 0$) and the constitutional hierarchy: - Tier 1 **Existential Invariants**: Identity, Metric, Security cannot be implicit. - Tier 2 **Operational Agilities**: Efficiency, Modularity, Portability are optimized within explicit constraints.

3.3. Local-First Origin and Controlled Exposure

Artifacts originate locally within a node's Inner Ring. Remote nodes in the network never inspect raw local memory heaps. Network interaction across the Outer Ring is restricted to: - Bounded public attestations and cryptographic commitments. - Exported

higher-order **Qexpr** commitment hashes. - Capability delegation certificates and metric summaries. - Accountable pseudonymous identities.

3.4. DCC Profiles Everywhere

Every layer, object, morphism, and exported attestation must expose an explicit **DCC Profile** utilizing structural indirection: - **Dependencies:** Abstract capability contracts, parent hashes, and cryptographic anchors. - **Constraints:** Hard ceilings on memory allocation declared in the **Metric Envelope**, execution steps, recursion depth, and precision bounds. - **Capabilities:** Permitted operations, transformation rights, and network export authorizations governed by the **Security Envelope**.

3.5. Causal Arrow and Birth Order

Every discrete **Q(event)** is indexed along a strictly monotonic **Causal Arrow**:

$$Q(0) \prec Q(1) \prec Q(2) \prec \cdots \prec Q(9) \prec d_1 \prec d_2 \prec \text{Higher CAE-DAGs}$$

No artifact may declare a parent or dependency born later in the causal sequence.

3.6. Totality by Budget

Every governed operation must terminate within its declared metric budget as a dissipative thermodynamic step (**Metric Exhaustion**), resolving to an explicit **Receipt**: `SUCCESS`, `FAILURE`, `NO_SOLUTION`, `TIMEOUT`, `BUDGET_EXHAUSTED`, `INCONCLUSIVE`, `QUARANTINED`, `PRECISION_INSUFFICIENT`, `EXPRESSION_TOO_COMPLEX`, `REJECTION_BY_GATEWAY`, `DIVISION_BY_ZERO_REJECTED`, or `ORDERING_INDETERMINATE_AT_ZOOM`. Silent non-termination is prohibited.

3.7. Substrate Profile vs. Governed Invariant ("Car vs. Road" Disjunction)

To reconcile physical hardware heterogeneity with platform-invariant verification, the architecture enforces an explicit disjunction between execution telemetry and algebraic invariants:

Context	The Substrate Envelope (The "Car")	The Governed Invariant (The "Road")
Location	Declared in Metric/Security Envelope	Declared in Abstract DCC Contract
Parameters	Node hardware architecture and clock frequency	Invariant input state commitment
Thermodynamics	Local physical thermal dissipation (T_{local})	Invariant target state milestone
Identity	Microscopic silicon root $Q(0)_N$	Invariant Diophantine condition
Security	Local hardware trust ring and enclave	Substrate-invariant work ΔW in $Q(1)$

---

4. Set and Category Abstraction

The structural framework is formally modeled via constructive set theory and locally finite category theory.

4.1. Local Qn Set

For node N, the local universe is a finite constructive set $\mathcal{Q}_N$: - Genesis member: $Q(0)_N \in \mathcal{Q}_N$. - Set membership requires an explicit identity, metric envelope, security envelope, **Lifecycle State**, and lineage witness. - No universal global set of all Qn artifacts exists.

4.2. Local Qn Category

Each node defines a locally finite category $\mathbf{C}_N$: - **Objects:** Governed artifacts $\alpha \in \mathcal{Q}_N$. - **Morphisms:** Governed transformations $f : \alpha \to \beta$ conforming to DCC constraints. - **Identities:** Explicit identity morphisms id_α.

4.3. Functors and Gateways

Inter-layer transitions are modeled as functors constrained by gateway validation:

$$\mathcal{F} : \mathbf{C}(L) \longrightarrow \mathbf{C}(L+1)$$

A valid layer-transition functor preserves identity, metric bounds, causal lineage, and SSL metadata. Functors introducing implicit semantic interpretations are invalid.

4.4. cdqn as the Fractal Protocol of Governed Data Movement

cdqn operates across two structural scopes: 1. **Local Scope (Intra-Node Chaining):** Governs data movement and morphism transitions between local abstraction layers via Content-Addressed Expression DAGs (CAE-DAGs) and chained **receipts**. 2. **Distributed Scope (Inter-Node Attestation):** Connects autonomous local categories ($\mathbf{C}_N$) into a distributed sheaf-like structure synthesized by **Chronosa** via the **Exposure Functor**:

$$\mathcal{E}_{\text{export}} : \mathbf{C}_N \to \textbf{Attestations}_{\text{cdqn}}$$

The "Road Receipt" Transmission Conduit Nodes never broadcast internal CAE-DAGs across physical networks. Raw local graphs are non-replicable because their causal lineages terminate at node-specific silicon origins ($Q(0)_A \neq Q(0)_B$). Instead, nodes emit **Boundary State-Transition Receipts** ("road receipts"):

$$\mathcal{R}_{\text{road}} = \left\langle \mathcal{H}(S_{\text{in}}),\, \mathcal{M},\, \mathcal{H}(S_{\text{out}}),\, \Delta W,\, \pi,\, \sigma \right\rangle$$

A receiving node ingests $\mathcal{H}(\mathcal{R}_{\text{road}})$ as an external witness, anchoring the milestone into its *own* local monoroot tree without adopting the sender's origin $Q(0)_A$, preserving **Axiom 4** and achieving $\mathcal{O}(1)$ bandwidth efficiency.

---

5. Complexity Degree Stratification

Artifacts and expressions (**Qexpr**) are stratified by structural **Complexity Degree**:

Degree	Structural Content	Examples	Specification Mapping
0	Foundational Primitives	$Q(0)$, $Q(1)$	`docs/q0_q1.md`
1	Single Digits & Peano Cascade	$Q(2) \ldots Q(9)$, elementary $\mathcal{R}$-algebra	`docs/q2_q9.md`, `docs/qm.md`
2	Multi-Axial Frames & CAE-DAGs	Orthogonal frames (d_k), Clifford rotors, CAE-DAGs	`docs/qm_geometry.md`, `docs/qexpr.md`
3	Discrete Dynamics & Calculus	Discrete Exterior Calculus, Continued Fractions	`docs/qm_calculus.md`
n	Advanced Domain Lattices	Categorical Semantics (**Qs**), Physics (**Qphy**), Logics (**Qlog**), Chemistry (**Qchem**), Biology (**Qbio**)	`docs/qs.md`, `docs/qphy.md`, `docs/qlog.md`, `docs/qchem.md`, `docs/qbio.md`

CD vs. Layer Note: Complexity Degrees index structural composition depth, not layer numbers. A domain projection may span the entire execution core (Layers 1 to 3) while its artifacts carry open-ended CD n labels; any artifact invoking discrete dynamics inherits CD 3+ semantics (`docs/qm_calculus.md`).

---

6. Numeric Representation and Compilation

6.1. Floating-Point Prohibition

Floating-point representations (IEEE 754) are prohibited at the Qn semantic layer. Floating-point semantics introduce platform-dependent rounding, unmeasured precision drift, and non-deterministic identity. Lower hardware layers may execute binary arithmetic internally, provided floating-point semantics are never exposed across a gateway.

6.2. Qexpr as a Content-Addressed Expression DAG (CAE-DAG)

Governed symbolic expressions (**Qexpr**) maintain the explicit tuple:

$$\text{Qexpr} = \langle \text{DAGStructure},\ z,\ r,\ d,\ \text{DCC},\ \text{Identity},\ \text{Lineage} \rangle$$

- **Deduplication via Hash-Consing:** Common sub-expressions share identical commitment hashes, resolving to a single node in memory.
- ****Zoom (z)**:** Declared scale or precision boundary.
- ****Remainder (r)**:** Exact residual quantity preserved at the declared zoom level.
- ****Dimension (d)**:** Dimensional coordinate frame.

6.3. Collapse at Compilation Time

Qexpr expressions remain symbolic during authoring and collapse deterministically to concrete values at compilation time (**QnLang** $\rightarrow$ **QnIR**). Collapse operations must be total, finite, and accompanied by an explicit remainder receipt. In geometric workflows (`docs/qm_geometry.md` §7), collapse is deferred until boundary demand to minimize Landauer dissipation.

6.4. Bounded Symbolic Expressions

Symbolic expression graphs are strictly bounded. DCC profiles declare maximum node count, graph depth ceilings, and evaluation budgets. Exceeding bounds triggers an intermediate governed collapse or execution termination.

---

7. Arithmetic Foundation as Structural Example

7.1. First Operations

From **Q(1)**, elementary operations are defined along axis d_1: additive, subtractive, multiplicative, and divisive morphisms (`docs/qm.md` §4).

7.2. Division as Constrained Operation

Division is defined as a bounded constraint equation:

$$\text{Given } a, b \implies \text{find } c, r \quad \text{such that } a = (b \times c) + r$$

- Denominators equal to **Q(0)** are forbidden by DCC constraint.
- Attempted division by zero emits the terminal receipt `DIVISION_BY_ZERO_REJECTED`.

7.3. Positional Ordering

Axis d_1 admits three disjoint partitions: negative Qn ($\mathcal{Q}^-$), origin ($\{Q(0)\}$), and positive Qn ($\mathcal{Q}^+$). Ordering is positional, not hierarchical. When zoom precision prevents order determination due to remainder overlap, the operation terminates with `ORDERING_INDETERMINATE_AT_ZOOM`. When remainders vanish, order demonstrates **Terminal Exactness**.

---

8. Identity, Security, and Outer-Ring Consensus

8.1. Identity Classes

The architecture distinguishes at least three explicit **Identity Classes**:

Identity Class	Substrate Entity	Prohibition
Machine Identity	Runtime node, execution container, hardware device ($Q(0)_N$)	Must not imply legal personhood
Human Identity	Natural person, authorized operator	Must not execute without delegation
AI Agent Identity	Delegated autonomous or semi-autonomous process	Must not possess undelegated authority

Under the **Non-Rogue Invariant**, entities belonging to `AI_AGENT_IDENTITY` cannot instantiate independent genesis roots $Q(0)$ and must execute strictly under cryptographically verified delegation chains originating from a `HUMAN_IDENTITY`.

8.2. Accountable Pseudonymity

The cdqn network forbids anonymous actors. All interactions are signed by accountable pseudonymous identities backed by capability proofs, cryptographic commitments, and explicit revocation paths.

8.3. Dual-Ring PQC Boundary

The architecture enforces an explicit **Dual-Ring PQC Boundary**: - The **Inner Ring** governs local execution and genesis (`docs/q0_q1.md` §2.4), protecting root keys, local lineage, and state sealing. - The **Outer Ring** governs network-facing commitments under the local **Security Envelope**, managing pseudonyms, attestation keys, and delegation proofs. Both rings implement structural indirection to support non-disruptive migration across Post-Quantum Cryptography (PQC) standards.

8.4. Q(anchor) as the Outer-Ring Consensus Reference Frame

Because physical nodes possess heterogeneous silicon origins ($Q(0)_A \neq Q(0)_B$) and disparate hardware capacities, global consensus cannot operate by flat transaction re-execution.

Consensus across the execution-core ceiling and its orthogonal projections (Layer 3 `Qexpr`, the base-domain projections, and **Chronosa**) is established over **Universal** Q(anchor)** Contracts**: 1.** Topological Invariance:** Every node instantiates the invariant Single-Digit Anchor Spectrum ($Q(2) \ldots Q(9)$, ϕ, $\sqrt{2}$) and Physical Invariant Anchors (**Q(th_2)**, **Q(c)**, **Q(ħ)**, **Q(Z_bus)**, **Q(gauge)**). These anchors serve as universal topological landmarks. 2. **Contract Verification:** Nodes achieve consensus by verifying that road receipts satisfy the shared anchor contract (`DCC_ANCHOR_Q2_PARITY_v1`, `DCC_ANCHOR_SECOND_LAW_v1`, etc.). 3. **Sheaf-Theoretic Gluing:** In `docs/chronosa.md`, Chronosa verifies that road receipts emitted by heterogeneous nodes glue consistently over shared $Q(\text{anchor})$ landmarks, synthesizing global causal intelligence without centralized state storage.

---

9. Structure-Generativity Balance

The architecture enforces the separation principle: **Fixed Structure, Flexible Content**.

- **Structure:** Non-negotiable layers, gateways, DCC profiles, metric envelopes, and causal lineage.
- **Content:** Heuristic proposals, neural network inferences, and stochastic candidate generation.

Stochastic engines may generate proposals, provided they are classified as *unverified candidates*. Unverified candidates cannot cross a **SIMEMP Gateway**. They become governed Qn artifacts only after formal construction, constraint evaluation, and sealing.

---

10. External Precedents and Formal Convergence

The architecture aligns with established theoretical computer science and mathematical frameworks: - **Abstract Interpretation:** Sound approximation across discrete abstraction lattices (**Zoom z**, **Remainder r**). - **Domain Theory:** Bounded evaluation, fixed-point semantics, and totality by budget. - **Arithmetic Geometry & Height Bounds:** Finite point density over bounded logarithmic heights (Northcott's Theorem), grounding Diophantine decidability within finite Metric Envelopes and eliminating classical unbounded non-halting voids ($H10$). - **Equality Saturation & E-Graphs:** Content-addressed expression deduplication and reversible term rewriting. - **Sheaf Theory:** Local category consistency and restricted global gluing via **exposure functors**. - **Linear Logic:** Strict resource consumption without unmetered duplication or discarding. - **Neuro-Symbolic Verification:** Strict operational isolation between neural candidate generation and symbolic verification.

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11. Status of Qn Definition Files and Base Domains

Development separates foundational primitives from distributed networking:

Category A — Numeric Primitives (Canonical Foundation)

- Primary Genesis Primitives: Local origin $Q(0)$ and calibration of compute unit U along d_1 in $Q(1)$ (`docs/q0_q1.md`).
- Secondary DCC Anchors: Elementary integer properties, parity partitions (**Q(even)**, **Q(odd)**), single-digit anchors $Q(2) \dots Q(9)$, and emergent roots ϕ, $\sqrt{2}$ (`docs/q2_q9.md`).

Category B — Operations and Morphisms (Canonical Primitive Envelope)

- Multi-dimensional axes (d_k), Clifford geometric algebra ($\mathcal{C}\ell_{p,q}$), and float-free rotors (`docs/qm_geometry.md`).
- Optimization morphisms: Content-addressed memory reuse (**Q(reuse)**) and identity short-circuiting (**Q(bypass)**) normatively integrated into the Universal Envelope (`docs/qnPrimitive.md`).
- Higher-order self-optimization workflows (**Q(rsi)**).

Category C — Data Structures and Workflows

- Content-Addressed Expression DAGs (CAE-DAGs): hash-consed symbolic expression containers with explicit **Universal Envelopes** and bounded DCC profiles (`docs/qexpr.md`).
- Directed acyclic state graphs, immutable memory containers (**Q(dataStruc)**), and pattern matching (**Q(patterns)**).
- Composite state sequence workflows (**Q(workflows)**) carrying explicit Universal Envelopes and DCC constraints.

Category D — Local-First Base Domains (Milestone 1 Sealed)

Base domains project directly across the execution core, executing 100% locally via local cdqn data movement without network consensus: - ****Qm** (Quang Mathematics):** Constructive proof engines, discrete calculus, and exact numeric proofs (`docs/qm.md`, `docs/qm_calculus.md`). - ****Qs** (Quang Semantics):** Explicit knowledge graphs, categorical linguistic ontologies (DisCoCat), and formal intent contracts (`docs/qs.md`). - ****Qphy** (Quang Physics):** Computational thermodynamics, discrete exterior field mechanics, and physical entropy accounting (`docs/qphy.md`). - **Emergent Composite Domains:** Extended domain lattice formed via categorical tensor products: - ****Qlog:** Product category $\mathbf{C}(\mathrm{Qm}) \otimes \mathbf{C}(\mathrm{Qs})$ governing constructive type theory, inferential modes, and pattern verification (`docs/qlog.md`). - ****Qchem** (Quang Chemistry):** Product category $\mathbf{C}(\mathrm{Qphy}) \otimes \mathbf{C}(\mathrm{Qm})$ governing discrete stoichiometric balance equations, molecular bonding graphs, and Gibbs free energy dissipation (`docs/qchem.md`). - ****Qbio:** Biosemiotic product category $\mathbf{C}(\mathrm{Qchem}) \otimes \mathbf{C}(\mathrm{Qs})$ governing living metabolic homeostasis, genetic information channels, and active inference (`docs/qbio.md`).

Category E — Runtime and Distributed Networking (Milestone 2 Launch)

- Intermediate representation (**QnIR**) instruction set and bounded virtual execution handler (Pending).
- High-level authoring language (**QnLang**) syntax (Pending).
- cdqn distributed chaining, swarm aggregation, and public attestation protocol feeding **Chronosa**.

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12. License Alignment

This specification strictly conforms to the **SSL**:

12.1. Paternity Reference

All derivative distributions, compiled binaries, APIs, and runtime specification files must retain the canonical attribution:

> Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License (SSL). Parent Repository: `https://github.com/cdqn5249/cdqn`

12.2. Open Core Invariants

1. **Anti-Patent Defense:** Commercial or derivative licenses terminate automatically upon initiating patent litigation against the Author or project ecosystem.
2. **Non-Scaling Open Access:** Royalty-free licensing is preserved for non-commercial, academic, and sub-threshold usage.

12.3. Scale Auditing

The metric and identity envelopes provide native, cryptographically verifiable telemetry (active compute nodes, containers, agents, and monthly API transactions) to verify adherence to commercial scaling thresholds (`LICENSE.md` §1.5).

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13. Open Items

The following formal specifications remain open for subsequent releases: 1. Formal instruction set and operational semantics for **QnIR**. 2. Syntax and type-checking rules for **QnLang**. 3. Concrete execution and validation mechanics for composite **Q(workflows)** and **Q(rsi)** self-optimization loops.

Annex J — CDQN portal index

Pinned source: docs/index.md
Committed version: 1.12.0 **Last updated:** 2026-10-23
sha256 (committed file): `cfd61dcc3d66b22d4a88d1fffb4b3a8f33a0a77ca27f3be262d1bd62cda9d614`

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1. Scope, Thermodynamics, and Architectural Stance

The CDQN framework investigates a foundational question in computational physics and computer science: *Can a discrete **Qn** number system abstract any computable phenomenon within physical reality?*

Classical computing abstractions frequently rely on non-constructive assumptions: unbounded memory spaces, continuous unmetered precision, and unmodeled side-channel dissipation. The **Qn** architecture rejects actual infinities, establishing computational boundaries grounded in physical laws:

- **Strict Physical Finiteness:** Computation is a non-equilibrium thermodynamic process subject to Landauer's bound:

$$W \geq k_B T \ln 2$$

Unbounded state spaces are physically non-realizable and excluded from execution. - ****Dependencies Determinism**:** Every state transition S_t must possess zero conditional entropy given its declared dependencies:

$$H(S_t \mid \mathcal{D}(S_t)) = 0$$

Implicit or unmeasured states violate this bound, resulting in non-deterministic drift. - ****SIMEMP** Governance:** Execution within the **Licensed Work** is constrained by Tier 1 **Existential Invariants** (Identity, Metric, Security) and optimized via Tier 2 **Operational Agilities** (Efficiency, Modularity, Portability). - **The** No-Implicit Rule**:** No implicit type coercions, unmeasured error states, or undeclared assumptions may cross an abstraction boundary or **SIMEMP Gateway**. - **Totality by Budget & Dissipative Receipts:** Divergent or infinite loops are rejected. Computations are total by budget via dissipative **Metric Exhaustion**, exporting operational entropy via signed, verifiable **receipts**. - ****Local-first** Origin:** Artifacts originate in local node universes rooted in genesis origin **Q(0)** and unity unit **Q(1)**. - **Fractal Protocol & Hybrid Swarm: cdqn** functions as the universal protocol of governed data movement. It chains state transitions locally across the horizontal execution core and distributes public road receipts across a hybrid swarm topology synthesized by **Chronosa**. - **Substrate vs. Payload Air Gap:** The protocol acts as a neutral conduit. Raw user content, applications, and digital assets remain sovereign **Payloads** insulated by protocol blindness and statutory safe-harbor protections.

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2. Method of Exploration: The Recursive Emergence Trajectory

The documentation suite does not claim closed empirical finality. The system operates as an active, recursive discovery pipeline:

$$\text{Proposal} \longrightarrow \text{Construction} \longrightarrow \text{Evaluation} \longrightarrow \text{Selection}$$

Abstractions and axioms represent working hypotheses formulated using **Structural Indirection** to avoid premature structural calcification while retaining agility against zero-day vulnerabilities. Verification is conducted operationally via **Computational Consistency**:

$$\text{Valid}(\mathcal{A}) \iff (\text{Consistent}(\mathcal{A}) \land \forall p \in \text{QnIR}(\mathcal{A}),\ \text{TerminatesWithinBudget}(p))$$

Development proceeds through consecutive emergence milestones toward an operational Proof of Concept (PoC) of ****Chronosa****:

CD vs. Layer Note: The `CD k` labels in the status rows above mirror the Complexity Degree stratification defined in `docs/abstractionLayers.md` (§5): they index structural composition depth, not layer numbers. The five domain specifications (Qs, Qlog, Qphy, Qchem, Qbio) are orthogonal vertical projections across the execution core (Layers 1 to 3) whose artifacts carry open-ended CD n labels (rendered as plain `CD n` inside the ASCII diagram); they are not additional layers above the Layer 3 execution ceiling.

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3. Recommended Reading Sequence

1. **SIMEMP Constraints**: Thermodynamic realities (Memory Wall, Landauer dissipation), **Dependencies Determinism**, **Structural Indirection**, and the **BOC Policy**.
2. **Abstraction Layers**: The execution core (Layers 0 to 3), discrete **Q(events)**, universal functorial gateways, fractal **cdqn** data movement, substrate vs. invariant ("car vs. road") disjunction, and complexity degree stratification.
3. **Category A Primitives**: Physical-to-governed gateway mechanics, local genesis origin **Q(0)**, first unit **Q(1)**, abstract compute unit **U** calibrated to Landauer dissipation ($W \geq k_B T \ln 2$), hardware lifecycle, Non-Rogue Invariant, and crash recovery receipts.
4. **Single-Digit Primitives**: Inductive cascade via discrete **Successor Morphism S**, complete single-digit **Q(anchor)** spectrum ($Q(2) \ldots Q(9)$), emergent constants (ϕ, $\sqrt{2}$), constructive parity partitions (**Q(even)**, **Q(odd)**), and multiplicative domain asymmetries.
5. **Qn Primitive Envelope**: Structural anatomy of the **Universal Envelope**, operational axioms, typed payload profiles, and Category B optimization morphisms.
6. **Qm Mathematics**: Constructive numeric substrate based on discrete tuples $\langle \sigma, q_z, z, r_z, d \rangle$, base-independent scale lattices (**Zoom z**), conserved residual cascades (**Remainder r**), **Terminal Exactness**, Diophantine division constraints, and positional relational decidability along coordinate axes (**Dimension d**).
7. **Qm Geometry**: Multi-axial orthogonal frames, Clifford Geometric Algebra ($\mathcal{C}\ell_{p,q}$), float-free rational Cayley rotors, optimization primitives (**Q(reuse)**, **Q(bypass)**), and deferred symbolic reduction.
8. **Qexpr Container**: Layer 3 Content-Addressed Expression DAG (**Qexpr**), cryptographic hash-consing, six canonical node classes, continued fractions for transcendentals, and hardware capability-aware collapse.
9. **Qm Calculus**: Constructive discrete calculus, **Discrete Exterior Calculus (DEC)** on simplicial complexes, nilpotent coboundary operators ($\mathbf{d} \circ \mathbf{d} \equiv Q(0)$), continued fraction generators for transcendental invariants ($\pi, e, \tau, \ln 2$), and the Anchor-Cut verification invariant in discrete time evolution.
10. **Qs Semantics**: Constructive relational intent, semantic multivector blades, epistemic modal spectrum (`MODAL_EMPIRICAL` to `MODAL_NARRATIVE`), structural intent contracts in DCC profiles, and triadic intent grounding ($\mathbf{C}_{\text{Qs}} \otimes \mathbf{C}_{\text{Qm}} \otimes \mathbf{C}_{\text{Qphy}}$).
11. **Qlog Logics**: Constructive type theory, pattern verification, **Q(patterns)**, **Q(workflows)**, stratified inferential modes ($\mathcal{L}_{\text{modes}}$), the Memory Wall latency harvester, and inverse causal reconstruction.
12. **Qphy Physics**: Discrete exterior field dynamics, non-equilibrium computational thermodynamics, physical $Q(\text{anchor})$ spectrum, the axiomatic world model, and the thermodynamic double-leash on autonomous agents.
13. **Qchem Chemistry**: Discrete Diophantine stoichiometry ($\mathbf{S} \cdot \vec{v} \equiv Q(0)$), content-addressed molecular graphs, stereochemical Clifford chirality, and reaction kinetics under Gibbs free energy dissipation.
14. **Qbio Biology**: Biosemiotics ($\mathbf{C}_{\text{Qchem}} \otimes \mathbf{C}_{\text{Qs}}$), genetic codon translation, metabolic homeostasis, Markov blankets, and active inference under variational free energy bounds.
15. **Chronosa Vision**: Operational proposal for an emergent, non-local causal machine intelligence coordinating distributed intent across the Outer Ring without statistical neural training.
16. **Qs History Proposal**: Operational vision and research proposal for historical anchoring — dual-predicate **Q(anchor)** grounding (Anchor Grounding Classes I/P/S), the Source-Receipt Invariant over the **Epistemic Anchorage** spectrum, Calendar Projection Contracts, the Qs History Library, and the five-gate Anti-Poisoning Ingestion Firewall; all elements maintained as CANDIDATE artifacts pending deterministic measurement, **BOC** selection, and **Q(rsi)** sealing.

17. **Qn Structures Proposal**: Vision and evaluation framework for Category C data structures and workflows — the five-row structural contract schema (footprint, access, locality, determinism, work), the five-step evaluation ladder from nomination to sealing, and a first worked candidate register spanning **Q(array)**, **Q(matrix)**, **Q(table)**, and **Q(stream)**; all elements maintained as CANDIDATE artifacts pending deterministic measurement, **BOC** selection, and **Q(rsi)** sealing.
18. **CDQN Glossary** and **LICENSE.md**: Unified glossary, dynamic bidirectional concordance, attribution rules, and commercial **Scaling Thresholds**.

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4. Epistemic Reflections & Real-World Analyses

To preserve lean, deterministic specifications while engaging with broader scientific and computational developments, the framework isolates non-normative essays, real-world case studies, and sociological analyses within the `docs/reflections/` layer. These publications bridge formal computational physics with empirical industry events and public discourse:

- **Beyond the Infinite Void: What OpenAI's H10 Claim Teaches Us About Finite Computation**
  Published: 2026-10-13 · Analyzes OpenAI's claimed resolution of Hilbert's Tenth Problem over rational numbers ($H10(\mathbb{Q})$), the fallacy of unmetered search in classical mathematics, height bounds as physical memory limits, and the shift from Platonic non-constructive proof to Verified Computed Patterns under SIMEMP constraints.

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5. License and Open Core Invariants

The CDQN project is licensed under the **Scaling Source License (SSL) 1.0**.

Derivative Works and compiled runtime environments must preserve the canonical **Paternity Reference** in source comments and metadata headers:

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Every downstream implementation must preserve the two **Open Core Invariants**: 1. **Anti-Patent Defense:** Automatic license termination upon initiating patent litigation against the Author or project ecosystem. 2. **Non-Scaling Open Access:** Royalty-free access for non-commercial, academic, and sub-threshold deployments of the **Licensed Work**.

Entities exceeding the **Scaling Threshold** (>10,000 active instances/agents, >100,000 MAU, or >10,000,000 monthly transactions) must execute a separate Commercial License Agreement. Processing, wrapping, or settling a sovereign **Payload** does not by itself trigger scaling usage.

Annex K — Q0–Q1 specification

Pinned source: docs/q0_q1.md
Committed version: 1.2.1 **Last updated:** 2026-10-23
sha256 (committed file): `48b37bcd11f7b919a5f68487c4939eef5d17525502857a6f391d0590a26a2716`

Field	Specification
Document Title	Category A Primitives — Q(0), Q(1), and Node Onboarding
Version	1.2.1
Last Updated	2026-10-23 (Bao Loc, Vietnam)
Author	Christophe Duy Quang Nguyen
License	Scaling Source License (SSL) 1.0
Status	Canonical Primitive Specification — Category A (Layer 0 to Layer 1 Genesis)

Normative References

The following documents establish the physical, structural, and legal constraints governing this specification. If a technical conflict arises, `simemp.md` governs; if a structural conflict arises, `abstractionLayers.md` governs; if a legal conflict arises, `LICENSE.md` governs. References are deliberately version-unpinned: each target document's front matter carries its own authoritative version, so this specification never duplicates drift-prone patch state.

Document	Role	Target
`docs/simemp.md`	Constitutional constraints, thermodynamics, and Dependencies Determinism	simemp.html
`docs/abstractionLayers.md`	Layer architecture and SIMEMP Gateway validation	abstractionLayers.html
`docs/qnPrimitive.md`	Universal Envelope, operational axioms, and Lifecycle States	qnPrimitive.html
`docs/qm.md`	Constructive numeric and relational substrate	qm.html
`docs/qs.md`	Constructive relational intent and Intent Contracts	qs.html
`LICENSE.md`	Scaling Source License 1.0 governing the Licensed Work	LICENSE.md

1. Scope and Constitutional Anchoring

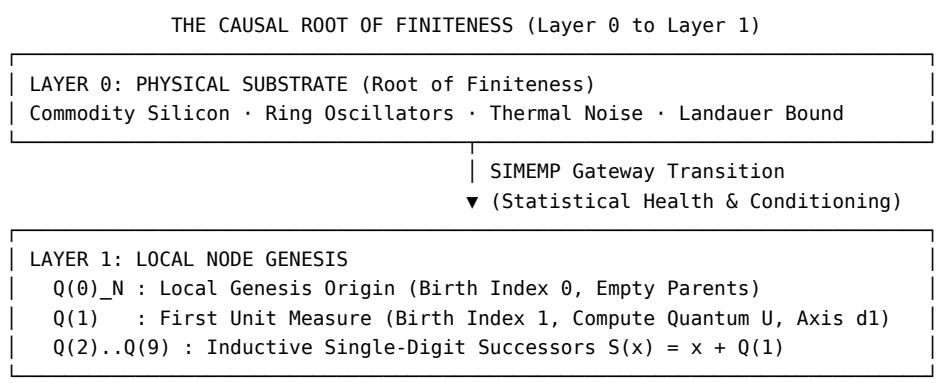
This specification defines the foundational genesis transition from physical substrate (**Layer 0**) to governed computation (**Layer 1**).

In the **Qn** computational universe, a number system cannot assume ungrounded, metaphysical starting values. Under **Dependencies Determinism**:

$$H(S_t \mid \mathcal{D}(S_t)) = 0$$

In accordance with the **No-Implicit Rule**, every constant, origin, and unit must declare an unbroken causal lineage back to physical hardware noise, certified across a **SIMEMP Gateway**.

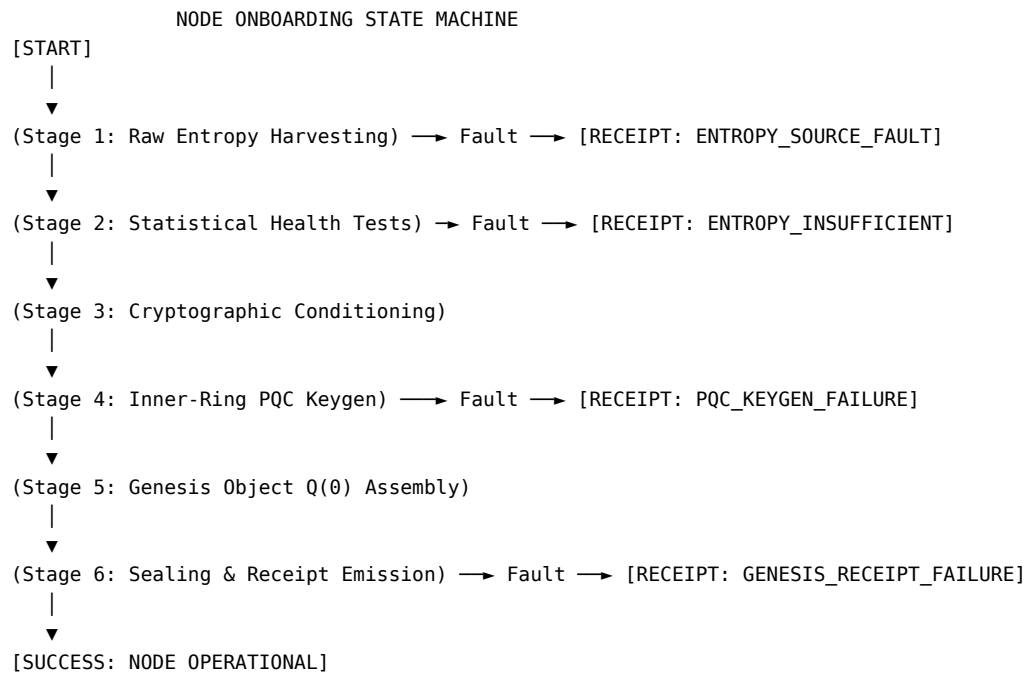
To maintain architectural agility under the **SIMEMP** framework, this document defines **minimal necessary structural invariants** rather than rigid implementation micro-choices. Operational validity is evaluated provisionally via **Computational Consistency**.



---

2. Layer 0 to Layer 1 Gateway: Node Onboarding Protocol

The **SIMEMP Gateway** bridging **Layer 0** and **Layer 1** enforces physical-to-governed onboarding through a deterministic pipeline:



2.1. Stage 1: Physical Entropy Harvesting

The node interrogates native **Layer 0** entropy sources: - Silicon thermal noise, ring oscillator jitter, or shot noise. - System performance counters and bus timing jitter.

The harvested bitstream must satisfy a baseline sample volume ($N_{\text{sample}} \geq 4096$ bits).

2.2. Stage 2: Statistical Health Tests

Raw entropy is verified against statistical health tests (e.g., NIST SP 800-90B standards): - Repetition count validation to detect stuck-at hardware faults. - Adaptive proportion validation to bound statistical drift. - Min-entropy threshold validation ($H_{\text{min}} \geq 0.85$ bits/bit).

Failure halts onboarding immediately, emitting a diagnostic **Receipt**.

2.3. Stage 3: Cryptographic Conditioning

Harvested entropy is conditioned via a collision-resistant compression function initialized with execution context digests:

$$\mathcal{S}_{\text{seed}} = \text{KDF}\Big(\mathcal{B}_{\text{raw}} \,||\, \text{Digest}(\text{HostContext})\Big)$$

producing a localized, non-invertible seed commitment.

2.4. Stage 4: Inner-Ring PQC Key Generation

Governed by the **Dual-Ring PQC Boundary** (`docs/abstractionLayers.md` §8.3), the gateway parameterizes the local Inner Ring: - Generates root keys `(pk_node, sk_node)` using a Post-Quantum Cryptography (PQC) signature scheme managed via **Structural Indirection**. - Private root key `sk_node` is retained exclusively within local memory, barred from public network exposure.

2.5. Stage 5: Genesis Object Assembly

The local genesis artifact **Q(0)** is constructed under the canonical 9-block **Universal Envelope**.

2.6. Stage 6: Sealing and Receipt Emission

The gateway signs the initialization record, writes the node genesis **Receipt**, and locks $Q(0)$ into an immutable `SEALED` lifecycle state.

2.7. Proposed Wiring: The Historical Calibration Benchmark (CANDIDATE)

This subsection adds no stage to the state machine above, which remains six-stage; it records the proposed wiring of a post-seal grounding event onto the gateway this section defines, closing the queue of `docs/qs_history.md` §8 item 1.

The glossary defines the **Historical Calibration Benchmark** as the mandatory empirical ingestion event wherein a physical node measures its local **Layer 0** environmental baseline — ambient temperature, physical entropy, and atomic frequency standards — via signed hardware receipts. It is here proposed that the gateway treat this benchmark as the node's **Class P** grounding event: the physics rung of the proposed witness ladder (`docs/qs_history.md` §2), complementing the self-witnessed genesis of Stages 1 through 6. The event extends the §4.2 calibration of the compute unit — the grounding of abstract invariants into real thermodynamic work — from the declared bound alone to the measured local baseline, and it emits the signed calibration receipts by which a **Q(anchor)** of proposed Class P witnesses its epistemic grounding.

The wiring is deliberately non-mutating. The benchmark is proposed as a post-seal event, executed after Stage 6 emits the Node Genesis Receipt (§6.1) and repeatable along the local **Causal Arrow** under a declared schedule or contract; a node that has not passed it operates without Class P grounding rather than with an implicit one, in accordance with the **No-Implicit Rule**. Measurement tolerance envelopes are proposed to be declared inside the owning anchor's DCC contract rather than negotiated at runtime (`docs/qs_history.md` §2), so that cross-node consensus reduces to envelope satisfaction. Under the **Source-Receipt Invariant** (`docs/qs.md` §4.3), the benchmark and the proposed **Qs History Library** (`docs/history_library.md` §2, which queues this proposition as its §7 item 5) form the two post-seal ingestion siblings of the gateway: the benchmark ingests the node's measured physical present as Class P receipt material, while the library ingests receipted human history as proposed Class S records. Both are petitioned through signed receipts; neither is silently assumed.

This wiring is a CANDIDATE proposition under **Axiom 14**: it amends no canonical stage, redefines no existing receipt, and remains subject to deterministic measurement and BOC selection at implementation time.

---

3. Field Anatomy of Local Genesis Object Q(0)

Under **Axiom 4** (`docs/qnPrimitive.md`), **Q(0)** is strictly local to each node:

$$\forall A \neq B \implies Q(0)_A \neq Q(0)_B$$

No universal global zero is admitted. $Q(0)$ acts as the **Local-first** coordinate origin, empty birth context, and root of the node's local **Causal Arrow**.

CANONICAL 9-BLOCK ANATOMY OF GENESIS OBJECT Q(0)

```
┌──────────────────────────────────────────────────────────────────────────┐
│ 1. IDENTITY BLOCK                                                        │
```

```
│    NodePseudonym      : Hash(pk_node)                                      │
│    LocalCausalIndex   : 0 (Fixed genesis baseline)                         │
│    IdentityClass      : MACHINE_IDENTITY (Accountable silicon chassis)     │
│    Version            : 1.2.0                                              │
│    CommitmentHash     : SHA3-256(BodyPayload)                              │
├────────────────────────────────────────────────────────────────────────────┤
│ 2. TYPE BLOCK                                                              │
│    ArtifactType       : 0x0001 (GENESIS_ORIGIN)                            │
│    DomainAffinity     : Qm_BASE_GENESIS                                    │
├────────────────────────────────────────────────────────────────────────────┤
│ 3. LIFECYCLE STATE BLOCK                                                   │
│    State              : SEALED (Immutable root)                            │
├────────────────────────────────────────────────────────────────────────────┤
│ 4. LINEAGE BLOCK                                                           │
│    ParentCommitments  : [] (Identically Empty Set)                         │
│    BirthOrder         : 0                                                  │
│    DerivationType     : HARDWARE_ENTROPY_SEED                              │
│    GenesisWitness     : Hash(S_seed)                                       │
├────────────────────────────────────────────────────────────────────────────┤
│ 5. DCC PROFILE REFERENCE                                                   │
│    ProfileContract    : DCC_ROOT_GENESIS_v1 (Interface Contract)           │
│    Dependencies       : ⌀ (None)                                           │
│    Constraints        : Immutable, Non-Exportable, Root-of-Causality       │
│    Capabilities       : DeriveSuccessor, AnchorLineage, EmitReceipt        │
├────────────────────────────────────────────────────────────────────────────┤
│ 6. METRIC ENVELOPE REFERENCE                                               │
│    MemoryAllocation   : Bounded byte footprint                             │
│    ConsumedMetric     : 0 (Genesis baseline)                               │
│    DepthCeiling       : 0                                                  │
├────────────────────────────────────────────────────────────────────────────┤
│ 7. SECURITY ENVELOPE REFERENCE                                             │
│    SecurityRing       : INNER_RING_LOCAL                                   │
│    PQCAlgorithmAnchor : Agile PQC Identifier                               │
│    RevocationPath     : REJECT_NODE_TEARDOWN                               │
├────────────────────────────────────────────────────────────────────────────┤
│ 8. EXPOSURE CLASS BLOCK                                                    │
│    Scope              : LOCAL_ONLY (Barred from direct network egress)     │
├────────────────────────────────────────────────────────────────────────────┤
│ 9. PAYLOAD REFERENCE                                                       │
│    PayloadProfile     : GENESIS_ORIGIN_PROFILE                             │
│    SubstrateMetadata  : { TelemetryDigest, EntropyCert }                   │
└────────────────────────────────────────────────────────────────────────────┘
```

3.1. Machine Identity vs. User Sovereignty Disjunction

Conforming to `docs/abstractionLayers.md` (§8.1), $Q(0)$ belongs exclusively to the Machine **Identity Class**:

1. **Prohibition of Personhood:** $Q(0)$ witnesses hardware silicon state and local execution context. It is constitutionally barred from claiming legal personhood or autonomous moral agency.
2. **Separation from User Identity:** Human actors reside in the sovereign user plane. User private credentials never enter Layer 0 entropy collectors.
3. **Accountable Proxy:** A node operates on behalf of an external principal only when presented with an explicit, signed capability grant (`docs/qnPrimitive.md` §5.7). The machine acts as an accountable executor, preserving statutory safe harbor under the **SSL** (`LICENSE.md` §6.2) for all sovereign **Payloads**.

3.2. The Non-Rogue Invariant and Intent Delegation

To prevent autonomous software processes from decoupling from human oversight, $Q(0)$ anchors the **Non-Rogue Invariant**:

1. **Zero Root Sovereignty for Delegated Agents:** An entity carrying `IdentityClass: AI_AGENT_IDENTITY` cannot instantiate an independent root origin $Q(0)$. It possesses no causal context outside of delegated execution.

2. **Intent Delegation Boundary:** Execution requests initiated by autonomous agents require an unbroken delegation proof linking to a verified human principal within the **Qs** base domain:

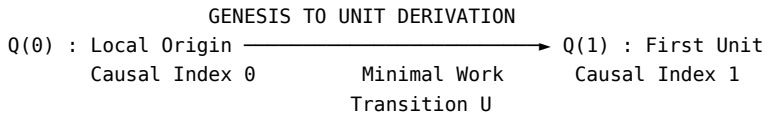
$$\text{Admit}(\Delta S) \implies \exists\, \Pi_{\text{delegation}} \wedge \Delta W \leq W_{\text{budget}}$$

3. **Circuit Breaking:** If an autonomous routine attempts an unanchored state transition or exceeds allocated budgets, the gateway halts execution deterministically, emitting an explicit failure receipt.

---

4. Field Anatomy and Calibration of First Unit Q(1)

Q(1) is the first derived unit artifact, the unity measure, and the baseline reference for the **Abstract Compute Unit U** along **dimensional axis d1**.



4.1. Causal Derivation

Under **Axiom 5** (`docs/qnPrimitive.md`), $Q(1)$ succeeds $Q(0)$ monotonically along the local **Causal Arrow**:

$$\text{Parent}(Q(1)) = \{Q(0)\}, \quad \text{Index}(Q(1)) = 1$$

4.2. Physical Calibration of the Abstract Compute Unit U and the Thermodynamic Leash

The transition from $Q(0)$ to $Q(1)$ defines the **Abstract Compute Unit U**:

- **Thermodynamic Grounding:** U is calibrated to the governed computational work of an irreversible bit-transition at ambient temperature ($T = 300$ K), bounded by Landauer's dissipation limit dissipated via **Metric Exhaustion**:

$$W(U) \geq k_B T \ln 2 \approx 2.87 \times 10^{-21} \text{ Joules}$$

- **Platform Invariance:** In software execution, U represents the baseline unit cost of a primitive discrete morphism, maintaining platform-invariant metric accounting across diverse hardware.
- **The Thermodynamic Leash on Recursive Loops:** Unbounded recursive execution loops are physically prohibited under **Axiom 6**. Every workflow operates within a finite, non-renewable work budget in units of $Q(1)$. Metric depletion halts execution deterministically, preventing unmetered runaway compute.
- **Invariant Unit Measure:** While local entropy parameterizes the cryptographic identity of $Q(1)$, its metric magnitude evaluates identically to $q = 1$ across all nodes.

4.3. Field Structure of Q(1)

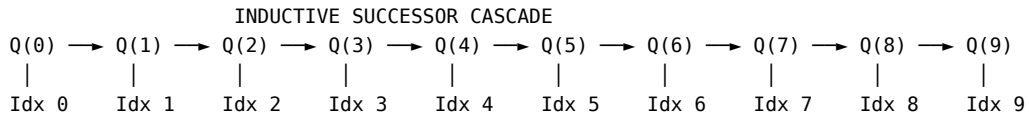
- **Identity:** Causal Index $= 1$, Parent $=$ Commitment$(Q(0))$.
- **Type:** `UNIT_PRIMITIVE` (Value $= 1$, **Zoom z = 0**, **Remainder r = Q(0)**, **Dimension d = d1**).
- **Morphism:** Basis Vector $\vec{u}_{d_1} : Q(0) \to Q(1)$.
- **Security:** Bound to the node's local **Security Envelope**.

---

5. Inductive Single-Digit Genesis: Q(2) … Q(9)

Single digits are constructed inductively along axis d_1 via the discrete **Successor Morphism S**:

$$\mathcal{S} : \mathcal{Q}_1 \to \mathcal{Q}_1, \quad \mathcal{S}(x) = x + Q(1)$$



5.1. Inductive Derivation Rules

For every integer $k \in \{2, 3, 4, 5, 6, 7, 8, 9\}$:

$$Q(k) = \mathcal{S}(Q(k-1)) = Q(k-1) + Q(1)$$

$$\text{Parent}(Q(k)) = \{Q(k-1),\, Q(1)\}, \quad \text{Index}(Q(k)) = k$$

Each artifact carries the canonical Qm numeric leaf tuple (`docs/qm.md` §3):

$$Q(k) \equiv \langle \oplus,\, k,\, z = 0,\, r = Q(0),\, d_1 \rangle$$

5.2. Invariant Properties of Single Digits

1. ****Terminal Exactness**:** Every single-digit artifact carries a remainder equal to $Q(0)$ at zoom $z = 0$, ensuring zero truncation error by construction.
2. **Positional Causal Ordering:** Strict trichotomy holds along axis d_1:

$$Q(0) \prec Q(1) \prec Q(2) \prec \cdots \prec Q(9)$$

3. **Base Neutrality:** Decimal glyphs $(0 \ldots 9)$ serve as interface labels; internal representations are pure inductive successor compositions over unit $Q(1)$.

---

6. Receipts, Lifecycle, and Hardware Maintenance

The Layer 0 to Layer 1 Gateway ensures deterministic lifecycle tracking during boot, routine operation, and hardware recovery.

6.1. The Node Genesis Receipt

Upon successful initialization, the gateway emits an immutable **Node Genesis Receipt**:

```
NODE GENESIS RECEIPT STRUCTURE
┌──────────────────────────────────────────────────────────────────────┐
│ ReceiptIdentifier  : Hash(NodePseudonym || CausalIndex_0 || Timestamp)      │
│ ReceiptType        : RECEIPT_NODE_GENESIS                            │
│ ExecutionStatus    : SUCCESS_SEALED                                  │
│ NodePseudonym      : Hash(pk_node)                                   │
│ SubstrateTelemetry : Hash(HardwareConfigDigest || EntropyCert)       │
│ PQCAlgorithmAnchor : Agile PQC Identifier                            │
│ RootPublicKey      : pk_node (Inner-Ring Root Commitment)            │
│ EmittedArtifact    : Commitment(Q(0))                                │
│ ConsumedBudget     : W_init (Platform initialization work)           │
│ GatewaySignature   : Sign_sk_node(ReceiptIdentifier || EmittedArtifact)     │
└──────────────────────────────────────────────────────────────────────┘
```

6.2. Physical Hardware Maintenance and Recovery Protocols

```
HARDWARE FAULT AND RESUMPTION MATRIX
┌──────────────────────────┬──────────────────────────────────────────────┐
│ PHYSICAL EVENT           │ GATEWAY MECHANISM AND EMITTED WITNESS        │
```

1. Routine Node Reboot	Non-volatile Sealed Q(0) Integrity Check └→ Emits: RECEIPT_NODE_RESUME
2. Abrupt Power Failure	Discard volatile unsealed candidates (Axiom 14) └→ Emits: RECEIPT_HARDWARE_CRASH_RECOVERY
3. Component Migration (Hardware Swap)	Graft old DAG to new Q(0)_new via Axiom 13 └→ Emits: RECEIPT_NODE_MIGRATION
4. Physical Destruction	Local state lost; Outer-Ring Sheaf survives

- **Reboot:** Validates sealed non-volatile state and resumes monotonically ($\text{Index}(\text{next}) = \text{Index}(\text{last}) + 1$).
- **Crash Containment:** Volatile unsealed candidate states are discarded under **Axiom 14**, restoring to the last `SEALED` checkpoint without entropy drift ($H = 0$).
- **Component Swap:** Replaced hardware generates new origin $Q(0)$ and grafts historic lineage under **Axiom 13** via an explicit migration receipt.
- **Physical Loss:** While unexported local memory is bounded by silicon permanence, public attestations previously projected across the Outer Ring persist across the **cdqn** network.

6.3. Proposed Duality: The Node Genesis Receipt and the Intent Contract (CANDIDATE)

This subsection names no new artifact and adds no stage to the state machine of §2; it records the proposed naming of a duality already exercised across the canon, closing the queue of `docs/qs_history.md` §8 item 4. Every governed node lives under two signed disciplines pointing in opposite directions along the **Causal Arrow**: the node genesis **Receipt** of §6.1 is backward-looking, sealing what the node is; the **Intent Contract** anchored in §3.2 and elaborated as `docs/qs.md` §5 is forward-looking, signing what the node may become.

The genesis receipt closes the past: it binds the immutable record of the node's origin — the committed $Q(0)$, the consumed initialization work W_{init}, the substrate telemetry digest — under the gateway signature, so that every later transition reads its provenance from a sealed state rather than from memory or assertion. The intent contract opens the future: parameterized inside the owning **DCC Profile**, it attaches to every admitted delegated transition the unbroken delegation proof that §3.2 requires, carrying mutual human-agent commitments, capability scopes, and conserved social residuals. The two disciplines are complementary and mutually necessary: sealing without delegation is inert provenance, while delegation without sealing is unanchored intent in violation of the **No-Implicit Rule**; together they bind the local causal arrow at both ends, the receipt anchoring its tail and the intent contract signing its head.

Under the **Source-Receipt Invariant** (`docs/qs.md` §4.3), both halves are receipt-typed artifacts differing in temporal direction rather than in evidentiary kind — the genesis receipt terminating in Layer 0 hardware and entropy receipts, the intent contract being an authored declaration that is never extracted by inspection, per the binding caution of the invariant. This duality is a CANDIDATE proposition under **Axiom 14**: it coins a name for complementary roles the canon already exercises, amends no canonical stage, redefines no existing receipt, and remains subject to deterministic measurement and BOC selection at implementation time.

---

7. License and Lineage Alignment

This specification enforces the legal and operational conditions of the **SSL**:

1. ****Paternity Reference**:** All derivative implementations or runtime modules must preserve: > Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License (SSL). Parent Repository: https://github.com/cdqn5249/cdqn
2. **Open Core Protection:** Layer 0 to Layer 1 onboarding mechanisms and genesis structures remain royalty-free for sub-threshold deployments under Non-Scaling Open Access.
3. **Statutory Safe Harbor:** The onboarding gateway functions as neutral technical infrastructure, exercising no editorial control over, and making no claim upon, external applications or sovereign **Payloads**.

Annex L — Q2–Q9 specification

Pinned source: docs/q2_q9.md
Committed version: 1.1.3 **Last updated:** 2026-10-20
sha256 (committed file): `abbe4e6c9986d966a618459979c796594a453f622df4514bf4957d9655a0385a`

Field	Specification
Document Title	Category A Primitives — Single-Digit Integers Q(2)…Q(9) and Secondary DCC Sources
Version	1.1.3
Last Updated	2026-10-20 (Bao Loc, Vietnam)
Author	Christophe Duy Quang Nguyen
License	Scaling Source License (SSL) 1.0
Status	Canonical Primitive Specification — Category A Completion (**Complexity Degree 1**)

Normative References

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`docs/simemp.md`	Constitutional constraints, thermodynamics, and Dependencies Determinism	simemp.html
`docs/abstractionLayers.md`	Layer architecture and SIMEMP Gateway validation	abstractionLayers.html
`docs/qnPrimitive.md`	Universal Envelope, operational axioms, and Optimization Morphisms	qnPrimitive.html
`docs/q0_q1.md`	Primary genesis origin Q(0) and first unit Q(1)	q0_q1.html
`docs/qm.md`	Constructive numeric leaf substrate and Diophantine division constraints	qm.html
`docs/qm_geometry.md`	Multi-axial frames, Clifford geometric algebra, and float-free rotations	qm_geometry.html
`docs/qm_calculus.md`	Discrete Exterior Calculus, continued fractions, and invariant dynamics	qm_calculus.html
`docs/qexpr.md`	Layer 3 Content-Addressed Expression DAG container	qexpr.html
`docs/qs.md`	Constructive relational intent and Intent Contracts	qs.html
`LICENSE.md`	Scaling Source License 1.0 governing the Licensed Work	LICENSE.md

1. Epistemic Scope and the Secondary DCC Role

This specification completes **Category A (Numeric Primitives)** of `docs/abstractionLayers.md` (§11) by formalizing single-digit integers $Q(2)$ through $Q(9)$.

In the **Qn** computational universe, numbers are not passive scalar magnitudes; they are **governed computational artifacts** carrying explicit capability contracts governed under the **BOC Policy** and constitutional constraints of the **SIMEMP** framework. While

Q(0) and **Q(1)** establish the **Primary DCC Sources** (origin, context, unit measure, and baseline reference for the **Abstract Compute Unit U**), they cannot single-handedly supply the specialized structural constraints required by higher abstraction layers.

Single digits $Q(2) \dots Q(9)$ function as the **Secondary DCC Sources** of the stack: - They introduce parity partitioning (**Q(even)**, **Q(odd)**), prime basis generators, aperiodicity bounds, composite factorizations, and physical memory alignment contracts. - Higher-layer algorithms, geometric solvers (`docs/qm_geometry.md`), and calculus manifolds (`docs/qm_calculus.md`) declare explicit dependencies on the capability profiles of these secondary sources.

Under **Dependencies Determinism**, the derivation of every single-digit integer possesses zero conditional entropy given its declared causal predecessor:

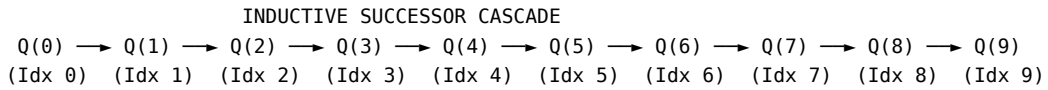
$$H(Q(k) \mid Q(k-1),\, Q(1)) = 0$$

---

2. Inductive Successor Cascade Along Axis d_1

In strict adherence to the **No-Implicit Rule**, single-digit integers cannot be assumed as unmetered continuous constants. They are constructed constructively along the canonical genesis **axis d1** via the discrete **Successor Morphism S** (`docs/q0_q1.md` §5):

$$\mathcal{S} : \mathcal{Q}_1 \to \mathcal{Q}_1, \quad \mathcal{S}(x) = x + Q(1)$$



2.1. Inductive Derivation Rules

For every integer $k \in \{2, 3, 4, 5, 6, 7, 8, 9\}$ along the **Causal Arrow**:

$$Q(k) = \mathcal{S}(Q(k-1)) = Q(k-1) + Q(1)$$

$$\text{Parent}(Q(k)) = \{Q(k-1),\, Q(1)\}$$

$$\text{Index}(Q(k)) = k$$

Each single digit evaluates to an exact canonical **Qm** leaf tuple (`docs/qm.md` §3) at **zoom z = 0** and conserved **remainder r = Q(0)**:

$$Q(k) \equiv \langle \oplus,\, k,\, z = 0,\, r = Q(0),\, d_1 \rangle$$

2.2. Invariant Single-Digit Properties

1. ****Terminal Exactness**:** Every single-digit artifact carries a remainder identically equal to $Q(0)$ at zoom $z = 0$. Truncation error is zero by construction.
2. **Positional Causal Ordering:** Monotonic ordering is strictly preserved along axis d_1:

$$Q(0) \prec Q(1) \prec Q(2) \prec Q(3) \prec Q(4) \prec Q(5) \prec Q(6) \prec Q(7) \prec Q(8) \prec Q(9)$$

3. **Radix Neutrality:** While labeled using standard decimal glyphs $(0 \dots 9)$ for interface parsing, their internal representation is pure unary successor composition over unit **Q(1)** along axis d_1.

2.3. Causal Genesis vs. Emergent Algebraic Invariant Verification

To maintain causal acyclicity across abstraction layers (`docs/simemp.md` §1), an epistemic boundary separates **Causal Genesis** from **Algebraic Invariant Verification**:

```
┌──────────────────────────────────────────────────────────────────────────┐
│ LAYER 1: Causal Genesis (Constructive Birth)                             │
│ Q(k) instantiated solely via Peano induction: Q(k) = Q(k-1) + Q(1)       │
│ Lineage dependencies: strictly { Q(k-1), Q(1) } along axis d1.            │
├──────────────────────────────────────────────────────────────────────────┤
│ LAYER 2+: Algebraic Verification (Emergent DCC Invariant Contracts)      │
│ Composite factorization (Q(4) = Q(2) × Q(2)) and multi-axial root relations │
│ (DCC_ANCHOR_SQRT2_v1) verified only upon activation of Layer 2 morphisms.  │
└──────────────────────────────────────────────────────────────────────────┘
```

1. ****Layer 1** Genesis:** The existence, identity, and causal birth of $Q(k)$ depend uniquely on its immediate predecessor $Q(k-1)$ and the unit $Q(1)$. No higher-order algebraic morphism $(\times, \div)$ or multi-axial coordinate system $(d_j, j \geq 2)$ is required for $Q(k)$ to exist.
2. **Layer 2 Verification:** Properties such as primality, compositeness (e.g., $Q(4) \equiv Q(2) \times Q(2)$), and parity partitioning are **abstract capability contracts** declared within the artifact's DCC profile. Under **Structural Indirection**, verification of these contracts is deferred until the computational context traverses the gateway into Layer 2 ($\mathcal{R}$-Algebra) or Layer 3 (Geometric Frames).

---

3. Concrete Anatomy of Single Digits Q(2) … Q(9)

Each single digit carries an explicit **Universal Envelope**, a unique local causal index, an immutable `SEALED` lifecycle state, belongs to the Machine **Identity Class**, is governed by the local **Security Envelope**, and encapsulates a specialized DCC contract:

Digit Artifact	Causal Index	Mathematical Role	Primary DCC Contract	Algebraic Status
$Q(2)$	2	Parity Generator & Binary Modulus	`DCC_ANCHOR_Q2_PARITY_v1`	Prime Basis Generator
$Q(3)$	3	Minimal 2D Simplex & Spanning Basis	`DCC_ANCHOR_Q3_SIMPLEX_v1`	Prime Basis Generator
$Q(4)$	4	First Composite & Quadrant Source	`DCC_ANCHOR_Q4_QUADRANT_v1`	Composite Contract: $Q(2) \times Q(2)$
$Q(5)$	5	Aperiodicity Generator & Pentagonal Modulus	`DCC_ANCHOR_Q5_APERIODICITY_v1`	Prime Basis Generator
$Q(6)$	6	Minimal Factorial (3!) & Perfect Modulus	`DCC_ANCHOR_Q6_SYMMETRY_v1`	Composite Contract: $Q(2) \times Q(3)$
$Q(7)$	7	Cyclic Modulus & Projective Plane Order	`DCC_ANCHOR_Q7_PROJECTIVE_v1`	Prime Basis Generator
$Q(8)$	8	Silicon Memory Word (8-bit Octet)	`DCC_ANCHOR_Q8_OCTET_v1`	Composite Contract: $Q(2)^3$
$Q(9)$	9	Decimal Radix Limit & Square Modulus	`DCC_ANCHOR_Q9_RADIX_v1`	Composite Contract: $Q(3)^2$

---

4. The Complete Single-Digit Anchor Spectrum (Q(2) … Q(9))

Every single digit from $Q(2)$ through $Q(9)$ possesses high topological centrality within the system's causal dependency DAG, designating each as a canonical **Q(anchor)** entity (`docs/_data/glossary/architecture.yml`):

$$\chi(\alpha) = \frac{|\mathcal{C}^+(\alpha)|}{|\mathcal{V}|} \geq \Theta_\chi$$

COMPLETE SINGLE-DIGIT ANCHOR SPECTRUM

```
┌──────────┬──────────────────────────────┬──────────────────────────────────────┐
│ ARTIFACT │ ANCHOR CONTRACT DESIGNATION  │ STRUCTURAL DEPENDENCY CONE C+(α)     │
├──────────┼──────────────────────────────┼──────────────────────────────────────┤
│ Q(2)     │ DCC_ANCHOR_Q2_PARITY_v1      │ Parity logic (even/odd), binary tree │
│          │ (Parity & Binary Modulus)    │ branching, Boolean NOT/XOR gates     │
```

Q(3)	DCC_ANCHOR_Q3_SIMPLEX_v1 (Simplicial Geometry Basis)	2D triangular simplicial meshes, Minimal spanning intent basis
Q(4)	DCC_ANCHOR_Q4_QUADRANT_v1 (Composite Orthotope)	4 Cartesian quadrants, composite factorizations, 2D hypercube faces
Q(5)	DCC_ANCHOR_Q5_APERIODICITY_v1 (Aperiodic & Fractal Order)	5-fold quasicrystals, Penrose tiles, Golden Ratio φ scaling invariants
Q(6)	DCC_ANCHOR_Q6_SYMMETRY_v1 (Minimal Non-Abelian Gauge)	Non-abelian permutation group S_3, smallest perfect number: 1+2+3=1·2·3
Q(7)	DCC_ANCHOR_Q7_PROJECTIVE_v1 (Projective Geometry Basis)	Fano plane order PG(2, 2), octonion imaginary basis algebra (e_1...e_7)
Q(8)	DCC_ANCHOR_Q8_OCTET_v1 (Silicon Memory Alignment)	Physical memory byte word (8-bit), Layer 0/1 bus word segmentation
Q(9)	DCC_ANCHOR_Q9_RADIX_v1 (Radix-10 Single-Digit Max)	Single-digit base-10 ceiling, decimal normalization boundaries

4.1. Abstract Contract Decoupling

Under **Structural Indirection** (docs/simemp.md §4.3), downstream algorithms depend exclusively on the **abstract invariant contract** of the anchor: - $Q(2)$ **Anchor:** Enforces the Diophantine contract $X = (Q(2) \times c) + r$. Downstream solvers depend on parity trichotomy, regardless of whether hardware executes division via bitwise shifts or Diophantine subtraction. - $Q(3)$ **Anchor:** Enforces the minimal 2D non-collinear boundary (DCC_ANCHOR_Q3_SIMPLEX_v1), acting as the foundation for simplicial complexes in Discrete Exterior Calculus (docs/qm_calculus.md) and the triadic intent foundation (Qs $\otimes$ Qm $\otimes$ Qphy). - $Q(5)$ **Aperiodicity Anchor:** Enforces non-crystallographic 5-fold rotational symmetry, preventing periodic synchronization locks in distributed swarms.

4.2. Dimensional Ring Behavior of the Prime Anchors

When coordinate dimensions extend from 1D scalar arithmetic into multi-axial geometric frames ($\mathcal{C}\ell_{p,q}$), the prime single digits $\mathcal{P} = \{Q(2), Q(3), Q(5), Q(7)\}$ partition into three fundamental algebraic number ring behaviors:

1. **Ramification ($Q(2)$):** In Gaussian integer rings ($\mathbb{Z}[i]$ embedded in 2D frames $\vec{e}_1 \wedge \vec{e}_2$), the prime 2 ramifies onto itself:
$$2 = -i(1+i)^2$$
This quadratic self-branching governs parity logic, bitwise splitting, and binary tree bifurcation.
2. **Splitting ($Q(5)$):** Primes of the form $p \equiv 1 \pmod 4$ split into orthogonal linear factors:
$$5 = (2+i)(2-i)$$
In 1D, 5 is indecomposable; in 2D, it factorizes across orthogonal axes, providing the metric scaling engine for self-similar fractal expansion.
3. **Inertia ($Q(3), Q(7)$):** Primes of the form $p \equiv 3 \pmod 4$ remain strictly indecomposable in Gaussian geometric rings:
$$3 \in \mathbb{P}_{\mathbb{Z}[i]}, \quad 7 \in \mathbb{P}_{\mathbb{Z}[i]}$$
This dimensional stability establishes rigid topological invariance for minimal simplicial meshes ($Q(3)$) and projective planes ($Q(7)$).

4.3. The 5D-to-2D Projection Origin of Aperiodicity

In Euclidean 2D and 3D space, the Crystallographic Restriction Theorem forbids periodic 5-fold rotational symmetry. However, 5-fold symmetry is strictly periodic in **5-dimensional hypercubic lattices** ($\mathbb{Z}^5$).

Under DCC_ANCHOR_Q5_APERIODICITY_v1, the aperiodicity of $Q(5)$ is formalized through the **Cut-and-Project Method**: * A periodic 5D lattice is sliced at an irrational slope governed by the Golden Ratio ϕ (DCC_ANCHOR_PHI_v1). * Projecting this slice onto a 2D frame yields a Penrose quasicrystal—a structure that is non-periodic, non-repeating, yet completely deterministic ($H = 0$). * This dimensional projection ensures that dynamical state trajectories avoid circular limit cycles across memory space.

---

5. Emergent Algebraic Constant Anchors: ϕ and √2

Two primary algebraic constants emerge directly from the polynomial characteristics of the single digits and are anchored within Category A:

Constant Anchor	Generating Digit	Contract Designation	Defining Algebraic Constraint
Golden Ratio (ϕ)	$Q(5)$	`DCC_ANCHOR_PHI_v1`	$\phi^2 - \phi - Q(1) = Q(0)$
Minimal Metric Root ($\sqrt{2}$)	$Q(2)$	`DCC_ANCHOR_SQRT2_v1`	$D^2 - Q(2) = Q(0)$

5.1. The Golden Ratio (ϕ / DCC_ANCHOR_PHI_v1)

Emerges from the pentagonal aperiodicity anchor $Q(5)$:

$$\phi = \frac{Q(1) + \sqrt{Q(5)}}{Q(2)}$$

In accordance with the crystallographic restriction theorem, 5-fold rotational symmetry cannot tile Euclidean 2D/3D space periodically. $Q(5)$ and its emergent constant ϕ mathematically govern **aperiodic self-similar order** (quasicrystals and Penrose lattices), providing the metric scaling invariant for scale-free fractal graphs in the **Qn** universe.

5.2. The Minimal Metric Root ($\sqrt{2}$ / DCC_ANCHOR_SQRT2_v1)

Emerges algebraically from $Q(2)$ as the non-zero root of the Diophantine quadratic equation:

$$D^2 - Q(2) = Q(0)$$

In **Qm**, $\sqrt{2}$ does not exist as an unmetered, infinite continuous decimal. It is anchored as an exact constrained Diophantine quadratic relation:

$$Q(2) = (q_D \cdot \delta_z)^2 + r_D$$

whose residual r_D is strictly preserved in the **Metric Envelope** across zoom scales.

Geometric Realization: While `DCC_ANCHOR_SQRT2_v1` is anchored within Category A as a discrete quadratic constraint, its geometric interpretation as the hypotenuse of the orthogonal unit square:

$$Q(1)_{d_1}^2 + Q(1)_{d_2}^2 = Q(2)$$

is formally instantiated in Layer 3 (`docs/qm_geometry.md` §6.1) once multi-axial frames (d_2) are derived.

---

6. Constructive Parity Partitions: Q(even) and Q(odd)

Parity is not an unmeasured flag. In Qm, parity is the constructive remainder classification resulting from the bounded Diophantine division constraint (`docs/qm.md` §5.3) with divisor $B = Q(2)$:

$$\text{Given } X \implies X = (Q(2) \times c) + r, \quad \text{where } r \in \{Q(0),\, Q(1)\}$$

Parity Partition	Remainder Condition	Canonical Classification	Invariant Property
****Q(even)****	$r \equiv Q(0)$	Modulo-2 Terminal Exactness	$\forall x \in \mathcal{Q}_{\text{even}},\, x = Q(2) \times c$
****Q(odd)****	$r \equiv Q(1)$	Modulo-2 Unit Remainder	$\forall x \in \mathcal{Q}_{\text{odd}},\, x = (Q(2) \times c) + Q(1)$

6.1. Classification of Genesis Origin Q(0)

Under the Diophantine division constraint:

$$Q(0) = (Q(2) \times Q(0)) + Q(0)$$

Because the remainder is identically $Q(0)$, genesis origin **Q(0)** is constructively an even artifact demonstrating modulo-2 **Terminal Exactness**:

$$Q(0) \in \mathcal{Q}_{\text{even}}$$

6.2. Disjoint Partitioning of the Integer Lattice

Parity partitions divide the discrete 1D lattice into two strictly disjoint subsets:

$$\mathcal{Q}_1 = \mathcal{Q}_{\text{even}} \cup \mathcal{Q}_{\text{odd}}, \quad \text{where } \mathcal{Q}_{\text{even}} \cap \mathcal{Q}_{\text{odd}} = \emptyset$$

---

7. Algebraic Asymmetry: Positive Monoid vs. Negative Domain

While spatial reflection along axis d_1 is an algebraic involution ($\mathcal{I}_{d_1}^2 = \text{id}$), **the behavior of** Qn** under multiplicative and dynamical operations is fundamentally asymmetric between the positive cone ($\mathcal{Q}^+$) and the negative cone ($\mathcal{Q}^-$)**.

POSITIVE VS. NEGATIVE ASYMMETRY

POSITIVE CONE (Q+)	NEGATIVE CONE (Q-)
Free Commutative Multiplicative Monoid generated by Primes.	Non-closed under multiplication: ⊖ ⊗ ⊖ = ⊕ (Leaves negative domain)
Closed: Q+ × Q+ ⊆ Q+	Open: Q- × Q- ⊆ Q+
Single Attractor Dynamics (Collatz)	Multi-Cycle Attractor Branching
Metric Work Footprint: +W	Metric Work Footprint: +W (Positive)

7.1. Multiplicative Monoid Closure Asymmetry

The positive single-digit integers form the basis of a **free commutative monoid** under multiplication, generated uniquely by the indecomposable prime set $\mathcal{P} = \{Q(2), Q(3), Q(5), Q(7)\}$:

$$\forall a, b \in \mathcal{Q}^+, \quad a \times b \in \mathcal{Q}^+$$

Conversely, the negative integers $\mathcal{Q}^-$ are **not closed under multiplication**:

$$\forall a, b \in \mathcal{Q}^-, \quad a \times b \in \mathcal{Q}^+ \quad (\text{Parity Rule: } \ominus \otimes \ominus = \oplus)$$

Multiplication within $\mathcal{Q}^-$ immediately exits the negative domain. Negative numbers do not form an independent multiplicative structure; they require the explicit inclusion of the reflection unit $\langle \ominus, 1 \rangle$.

7.2. Non-Linear Dynamical Asymmetry (The Collatz Phenomenon)

For any non-linear operator $\mathcal{F}$, spatial reflection does not generally commute with dynamical evolution:

$$\mathcal{F}(\mathcal{I}(X)) \neq \mathcal{I}(\mathcal{F}(X))$$

This is demonstrated constructively by the discrete Collatz mapping $T(n)$:

$$T(n) = \begin{cases} n/Q(2) & \text{if } n \in \mathcal{Q}_{\text{even}} \\ (Q(3) \times n) + Q(1) & \text{if } n \in \mathcal{Q}_{\text{odd}} \end{cases}$$

- **On Positive Domain ($\mathcal{Q}^+$):** The $+Q(1)$ step acts as a damping perturbation against $Q(3) \times n$, driving all tested orbits into the unique global attractor cycle: $(Q(4) \to Q(2) \to Q(1))$.
- **On Negative Domain ($\mathcal{Q}^-$):** Substituting negative integers ($n = -m$ where $m \in \mathcal{Q}^+$) reverses the direction of the unit perturbation:

$$Q(3)(-m) + Q(1) = -(Q(3)m - Q(1))$$

The unit perturbation acts anti-symmetrically, fracturing the dynamical trajectory into at least three distinct non-trivial limit cycles (the 2-cycle, the 5-cycle, and the 18-cycle).

7.3. Thermodynamic Metric Asymmetry and Fundamental Units

In physical Layer 0 substrates, negative energy and negative storage do not exist: - A negative artifact $\langle \ominus, k, z = 0, r = Q(0), d_1 \rangle$ has an identical **positive physical footprint** as its positive reflection $\langle \oplus, k \rangle$. - In accordance with Landauer's bound, executing a negative transition dissipates positive thermodynamic work via **Metric Exhaustion**:

$$W(\ominus) \geq k_B T \ln 2 > 0$$

An inverse calculation cannot recover dissipated physical heat.

Furthermore, in multi-dimensional algebraic number rings ($\mathbb{Z}[\phi], \mathbb{Z}[\sqrt{2}]$), the Dirichlet Unit Theorem expands the unit group beyond the 1D reflection pair $\{\pm 1\}$ to infinite cyclic groups generated by fundamental units ($\epsilon = 1 + \sqrt{2}$, $\phi = \frac{1+\sqrt{5}}{2}$). This establishes discrete dilation symmetries across scale lattices, enabling self-similar geometric scaling without floating-point approximations.

---

8. Sealing, Receipts, and Lifecycle Integration

Upon completion of the inductive cascade from $Q(2)$ to $Q(9)$, the node emits an immutable **Primitive Sequence Sealing Receipt**:

Receipt Field	Specification
Receipt Identifier	`Hash(NodePseudonym || CausalIndex_9 || Timestamp)`
Receipt Type	`RECEIPT_CATEGORY_A_SEALED`
Execution Status	`SUCCESS_SEALED`
Digit Range	$Q(2) \dots Q(9)$
Anchor Contracts Sealed	`DCC_ANCHOR_Q2_PARITY_v1` through `DCC_ANCHOR_Q9_RADIX_v1`
Emergent Constants Anchored	`DCC_ANCHOR_PHI_v1` (ϕ), `DCC_ANCHOR_SQRT2_v1` ($\sqrt{2}$)
Consumed Work Budget	Bounded inductive initialization work ($8\,Q(1)$ units) declared in the local **Metric Envelope**
Gateway Signature	$\text{Sign}\big(\text{ReceiptIdentifier} \,\|\, \text{Commitment}(Q(9))\big)$ keyed by signing key `sk_node`

This receipt permanently closes Category A for the physical node, proving that all single-digit secondary DCC sources and emergent algebraic roots are grounded, sealed, and ready for Layer 2 algebraic composition.

---

9. License and Lineage Alignment

This primitive specification enforces the legal and operational conditions of the **SSL** governing the **Licensed Work**:

1. ****Paternity Reference:** All derivative implementations, mathematical kernels, integer libraries, or hardware arithmetic modules derived from this document must embed: > Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License (SSL). Parent Repository: `https://g`ithub.com/cdqn5249/cdqn
2. **Open Core Protection:** The secondary DCC anchor contracts, parity generators, single-digit primitive definitions, and emergent algebraic constants remain royalty-free for sub-threshold, academic, and non-commercial deployments under Non-Scaling Open Access.

3. **Statutory Safe Harbor:** The single-digit integer substrate operates as passive, neutral mathematical infrastructure. It exercises no editorial inspection over, and makes no claim upon, external applications or sovereign **Payloads**.

Annex M — qnPrimitive

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Purpose and Scope

This document formalizes the invariant structural container required for every governed computational entity within the Qn universe: the **Universal Envelope**.

In this pre-silicon design stage, the primitive envelope provides minimal structural scaffolding to guide artifact construction, algebraic transformation, and candidate execution across `docs/abstractionLayers.md`. Governed by **Dependencies Determinism**, the model eliminates unmeasured ambient state while preserving architectural elasticity against premature calcification through **Structural Indirection**.

All formulations represent **provisional consistency abstractions** within the Recursive Search Loop (`docs/simemp.md` §2). They define constitutional boundaries rather than finalized implementation parameters, pending empirical hardware validation.

Normative References

The following documents define the physical, structural, and legal constraints governing this framework. If a technical conflict arises, `simemp.md` governs; if a structural conflict arises, `abstractionLayers.md` governs; if a legal conflict arises, `LICENSE.md` governs. References are deliberately version-unpinned: each target document's front matter carries its own authoritative version, so this specification never duplicates drift-prone patch state.

Document	Role	Target
`docs/simemp.md`	Constitutional constraints, thermodynamics, and Dependencies Determinism	simemp.html
`docs/abstractionLayers.md`	Layer architecture and SIMEMP Gateway validation	abstractionLayers.html
`docs/q0_q1.md`	Primary genesis origin Q(0) and first unit Q(1)	q0_q1.html
`docs/q2_q9.md`	Single-digit secondary DCC anchors and single-digit spectrum	q2_q9.html
`docs/qm.md`	Constructive numeric leaf substrate and Diophantine division constraints	qm.html
`docs/qm_geometry.md`	Multi-axial frames, Clifford geometric algebra, and float-free rotations	qm_geometry.html
`docs/qm_calculus.md`	Discrete Exterior Calculus, continued fractions, and invariant dynamics	qm_calculus.html
`docs/qexpr.md`	Layer 3 Content-Addressed Expression DAG container	qexpr.html

Document	Role	Target
`docs/qs.md`	Constructive relational intent and Intent Contracts	qs.html
`LICENSE.md`	Scaling Source License 1.0 governing the Licensed Work	LICENSE.md

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1. Definitions

1.1. Qn Artifact

A **Qn** artifact is a governed, finite, identifiable, measurable, and security-constrained computational entity. An artifact is not a passive scalar; it is an immutable state node embedded in a local **Causal Arrow**.

1.2. Qn Primitive Envelope

The minimal governance boundary required for any computational entity within the **Licensed Work** to exist, undergo transformation, achieve sealing, or traverse a **SIMEMP Gateway**, governed under Tier 1 **Existential Invariants** and optimized via Tier 2 **Operational Agilities** within the **SIMEMP** framework.

1.3. Universal Envelope

The invariant set of structural blocks shared across all artifact types, parameterized via **Structural Indirection**:

$$\mathcal{E} = \langle \text{Identity, Type, State, Lineage, DCC, Metric, Security, Exposure, PayloadRef} \rangle$$

1.4. Typed Payload Profile

The specialized content block carried by an artifact (numeric quantities, symbolic expression graphs, pure morphisms, state receipts, identity assertions, capability grants, optimization pointers, or composite workflows). External user data remains a sovereign, protocol-blind **Payload**.

1.5. Identity

The governed witness of an artifact's existence. Identity is **Local-first**, collision-resistant, versionable, traceable along causal lineage, and cryptographic-algorithm agile.

1.6. Lineage

The directed acyclic causal history of an artifact, recording parent commitment hashes, causal birth index, and transformation receipt references.

1.7. DCC Profile

The machine-readable contract defining **Dependencies**, **Constraints**, and **Capabilities**. It serves as an abstract contract rather than a hardcoded implementation binding via **DCC Profiles**.

1.8. Metric Envelope

The finite measurement boundary declaring resource limits on storage, expression graph depth, precision bounds, and execution budgets via the **Metric Envelope**.

1.9. Security Envelope

The governed protection boundary declaring authority limits, delegation proofs, Post-Quantum Cryptography (PQC) algorithm identifiers, and swappable revocation paths via the **Security Envelope**.

1.10. Lifecycle State

The discrete operational status of an artifact: candidate, constructed, sealed, verified, rejected, halted, quarantined, exportable, exported, or revoked via the **Lifecycle State**.

1.11. Receipt

A bounded, immutable record of a state transition, fault translation, or boundary traversal, acting as a dissipative **Receipt** exporting operational entropy.

1.12. Unverified Candidate

A proposed computational structure generated via heuristic, neural, or stochastic processes that has not passed gateway verification. Candidates cannot cross a **SIMEMP Gateway**.

1.13. Numeric Payload

The specialized profile for numeric entities, explicitly encapsulating **Zoom z**, **Remainder r**, and **Dimension d**.

1.14. Exposure Class

The classification governing artifact visibility: local-only, sealable, exportable, attested, quarantined, or revoked.

1.15. Q(reuse) Optimization Morphism

A Category B primitive referencing existing immutable content commitment hashes within the local causal DAG. Under physical constraints, $Q(\text{reuse})$ is admissible if and only if the predicted computational work of direct re-evaluation exceeds the lookup and retrieval dissipation across the Memory Wall ($\Delta W_{\text{compute}} > \Delta W_{\text{lookup}} + \epsilon_{\text{overhead}}$).

1.16. Q(bypass) Optimization Morphism

A Category B primitive allowing a **SIMEMP Gateway** to short-circuit execution upon verifying an algebraic identity precondition in $\mathcal{O}(1)$ register time, emitting a zero-cost bypass receipt without memory traversal or compute budget consumption.

---

2. Paradigm Neutrality

This specification does not mandate an object-oriented programming (OOP) paradigm. A Qn artifact is not an OOP object.

The model explicitly rejects: - Hierarchical class inheritance and dynamic dispatch tables. - Hidden mutable state and unmetered side effects. - Implicit type coercions and ambient runtime environments.

Transformations are explicit governed morphisms. State changes yield new immutable artifacts indexed monotonically along the local **Causal Arrow**.

---

3. Provisional Operational Axioms

The Qn operational model is governed by 14 operational axioms. These axioms represent **candidate search constraints** verified provisionally via **Computational Consistency**.

Axiom 1 — Explicit Envelope

No Qn artifact exists within the governed universe unless it carries an explicit **Universal Envelope**:

$$\forall \alpha \in \mathcal{Q}, \quad \exists! \, \mathcal{E}(\alpha)$$

Axiom 2 — Finiteness

Every Qn artifact must possess a finite canonical representation and a bounded resource envelope. Actual infinities are physically non-realizable and excluded from execution:

$$\forall \alpha \in \mathcal{Q}, \quad \text{Size}(\alpha) < \infty \quad \wedge \quad \text{Budget}(\alpha) < \infty$$

Axiom 3 — Typed Payload Separation

Fields specific to numeric representations (**Zoom z**, **Remainder r**, **Dimension d**) belong strictly to typed payload profiles and must not be made mandatory for non-numeric entities. Sovereign user data remains an uninspected **Payload**.

Axiom 4 — Local Genesis Dependency

Every governed Qn artifact within a local universe must trace its lineage directly or transitively to the local genesis origin **Q(0)**:

$$\forall \alpha \in \mathcal{Q}_N, \quad Q(0)_N \in \text{Lineage}(\alpha)$$

Local artifacts form single-rooted directed trees originating at physical silicon root $Q(0)_N$. Higher-order network entities operating across the Outer Ring (such as **Chronosa**) are topological colimits (sheaf gluings) over acyclic forests of mutually verified local Axiom-4 trees.

Axiom 5 — Acyclic Causal Lineage

Lineage graphs are strictly acyclic, governed by **Dependencies Determinism**. Every artifact must be born after its causal dependencies along the **Causal Arrow**:

$$\text{Index}(\text{Parent}(\alpha)) < \text{Index}(\alpha)$$

Axiom 6 — Totality of Governed Operations

Every governed operation must terminate in a declared, finite terminal state via dissipative **Metric Exhaustion** within its declared budget: `SUCCESS`, `FAILURE`, `NO_SOLUTION`, `TIMEOUT`, `BUDGET_EXHAUSTED`, `INCONCLUSIVE`, `QUARANTINED`, `PRECISION_INSUFFICIENT`, `EXPRESSION_TOO_COMPLEX`, `REJECTION_BY_GATEWAY`, `DIVISION_BY_ZERO_REJECTED`, or `ORDERING_INDETERMINATE_AT_ZOOM`. Silent non-termination is prohibited.

Axiom 7 — Receipted Boundary Events

Receipts are mandatory for lifecycle state transitions, gateway crossings, sealing events, export decisions, fault translations, and capability invocations. Receipts serve as verifiable witnesses exporting operational entropy.

Axiom 8 — Local-first Exposure

Artifact identifiers are rooted locally. Raw local Qn artifacts are never exposed directly to remote nodes across **cdqn** without explicit authorization and bounded public attestation under **Local-first** rules.

Axiom 9 — Numeric Precision Explicitness

Every numeric Qn artifact must declare finite precision bounds, explicit **Zoom z**, **Remainder r**, and **Dimension d**. Implicit rounding and truncation are prohibited:

$$\text{NumericValue} = \langle z,\ r,\ d \rangle$$

Axiom 10 — No Floating-Point Semantics

IEEE 754 floating-point representations are forbidden at the Qn semantic layer. Non-deterministic mantissa rounding and platform-dependent approximations are rejected.

Axiom 11 — Bounded Self-Description

Governance metadata may describe its own schema, but self-description depth must remain finite, bounded, and version-anchored to prevent infinite recursive reflection towers.

Axiom 12 — License Lineage Where Applicable

Lineage metadata and the canonical **Paternity Reference** must be preserved in distributed artifacts, exported attestations, compiled **QnIR** binaries, and public APIs pursuant to the **SSL**. Encapsulating a Payload does not convert it into a **Derivative Work**.

Axiom 13 — Versioning Creates New Artifacts

Versioning an artifact produces a new immutable artifact with an advanced causal index referencing the predecessor. Mutation in place is prohibited. Successor workflows derived through **Q(rsi)** do not overwrite existing execution paths; they are emitted as distinct causal versions.

Axiom 14 — Generative Candidate Boundary

Unverified candidates proposed by stochastic, generative, or neural processes can exist locally only if explicitly tagged as unverified candidates. Candidates cannot cross a **SIMEMP Gateway** or achieve sealed execution without deterministic validation.

---

4. Universal Envelope Structure

The Universal Envelope $\mathcal{E}$ is structured using **Structural Indirection**, ensuring that cryptographic suites or binary layouts can be upgraded without invalidating existing causal DAGs.

```
                    UNIVERSAL ENVELOPE STRUCTURE
┌──────────────────────────────────────────────────────────────┐
│ 1. IDENTITY BLOCK      : NodeID, CausalIndex, Type, Version, Hash │
├──────────────────────────────────────────────────────────────┤
│ 2. TYPE BLOCK          : Primitive, Morphism, Workflow, Receipt │
├──────────────────────────────────────────────────────────────┤
│ 3. STATE BLOCK         : Candidate, Sealed, Verified, Quarantined│
├──────────────────────────────────────────────────────────────┤
│ 4. LINEAGE BLOCK       : ParentCommitmentHashes, BirthOrder      │
├──────────────────────────────────────────────────────────────┤
│ 5. DCC PROFILE REF     : Dependencies, Constraints, Capabilities │
├──────────────────────────────────────────────────────────────┤
│ 6. METRIC ENVELOPE     : MemoryCeiling, DepthLimit, ComputeBudget│
├──────────────────────────────────────────────────────────────┤
│ 7. SECURITY ENVELOPE   : SecurityRing, PQCAlgoID, RevocationPath │
├──────────────────────────────────────────────────────────────┤
│ 8. EXPOSURE BLOCK      : LocalOnly, Exportable, Attested         │
├──────────────────────────────────────────────────────────────┤
│ 9. PAYLOAD REFERENCE   : Typed pointer to Payload Profile        │
└──────────────────────────────────────────────────────────────┘
```

4.1. Identity Block

Declares local node identifier, local causal index, version identifier, content commitment hash, and cryptographic agility metadata.

4.2. Type Block

Declares the artifact category as an independent top-level descriptor (numeric scalar, expression DAG, morphism, relational assertion, receipt, identity assertion, capability grant, optimization pointer, or composite workflow).

4.3. Lifecycle State Block

Declares current governance status conforming to Axiom 7 and the **Lifecycle State**.

4.4. Lineage Block

Declares parent artifact references, local causal index, birth order, derivation type, and transformation receipt references.

4.5. DCC Profile Reference

Machine-readable reference to an abstract capability contract declaring prerequisites, operational ceilings, and permitted transformations via the **DCC Profile**.

4.6. Metric Envelope Reference

Declares finite operational bounds (storage footprint, expression graph depth, recursion limits, and execution budgets) via the **Metric Envelope**.

4.7. Security Envelope Reference

Declares authority scope, delegation limits, trust assumptions, PQC algorithm identifiers, and swappable revocation endpoints via the **Security Envelope**.

4.8. Exposure Class Block

Declares visibility scope: local-only, sealable, exportable, attested, quarantined, or revoked.

4.9. Payload Reference

A typed, content-addressed pointer to the associated **Payload** profile.

---

5. Typed Payload Profiles

5.1. Numeric Payload Profile

Carries **Zoom z**, **Remainder r**, **Dimension d**, precision boundaries, and base-independent scale lattice representations (`docs/qm.md` §3).

5.2. Expression Payload Profile (Content-Addressed Expression DAG)

Under the **Best of Choices (BOC) Policy** (`docs/simemp.md` §7), symbolic expressions (**Qexpr**) are represented as **Content-Addressed Expression DAGs (CAE-DAGs)** with cryptographic hash-consing and arity-indexed child commitment vectors (`docs/qexpr.md`):

$$\alpha = \left\langle \text{OpCode},\ \mathbf{D}_{\text{children}},\ z,\ r,\ d,\ \text{DCC}_{\text{ref}} \right\rangle$$

- **Deduplication:** Identical sub-expressions share identical commitment hashes, eliminating redundant leaf allocation across the Memory Wall.
- **Complexity Ceilings:** Declares strict graph depth ceilings, maximum node counts, and compile-time collapse budgets.
- **Thermodynamic Invariance:** Reversible normalization within the DAG incurs zero Landauer dissipation ($W = 0$).

5.3. Operation Payload Profile

Carries domain-to-codomain morphism signatures, formal preconditions, postconditions, metric consumption rates, and receipt emission behavior.

5.4. Receipt Payload Profile

Carries terminal execution state, consumed resource budgets (**Metric Exhaustion**), causal birth index, parent commitment references, and gateway signatures on each emitted **Receipt**.

5.5. Identity Payload Profile

Carries **Identity Class** (Machine Identity, Human Identity, AI Agent Identity), pseudonyms, capability bounds, and revocation endpoints, bounded by **Identity Invariants**. Under the **Non-Rogue Invariant**, machine identities are subordinated to verified human principals.

5.6. Attestation Payload Profile

Carries public commitments, issuer pseudonyms, capability proofs, metric summaries, and SSL lineage assertions for network exposure across **cdqn**.

5.7. Capability Payload Profile

Carries granular execution rights, operational constraints, delegation depths, and expiration bounds.

5.8. Workflow Payload Profile

Carries a directed acyclic graph of governed morphisms, input state prerequisites, execution checkpoints, intermediate collapse budgets, and terminal receipt criteria. Underpins composite state workflows (**Q(workflows)**) and recursive self-optimization loops (**Q(rsi)**).

5.9. Optimization Payload Profile (Category B Primitives)

Governs explicit sub-graph reuse and gateway short-circuiting under physical thermodynamic constraints:

Q(bypass) Execution Checks immediate operands for algebraic identity conditions in $\mathcal{O}(1)$ register time (such as unit multiplication $\times Q(1)$, zero addition $+Q(0)$, rotor normalization $RR^\dagger = Q(1)$, or coboundary nilpotency $\mathbf{d} \circ \mathbf{d} = Q(0)$). When detected, the gateway short-circuits execution and emits `RECEIPT_BYPASS_IDENTITY` at zero compute cost ($\Delta W = 0$).

Q(reuse) Execution & Thermodynamic Breakeven References an existing immutable content commitment hash within the local causal history. Querying a hash commitment across physical memory hierarchies dissipates dynamic bus power ($\mathcal{O}(CV^2 f)$). Under the Metric Invariant, $Q(\text{reuse})$ is admissible if and only if the computational cost of direct evaluation exceeds lookup dissipation:

$$\text{Admit}\big(Q(\text{reuse})\big) \iff \Delta W_{\text{compute}} > \Delta W_{\text{lookup}} + \epsilon_{\text{overhead}}$$

If direct computation is cheaper than memory traversal (e.g. single-cycle integer ALU operations), the runtime executes direct computation, bypassing hash-table lookups to prevent Memory Wall bloat.

The Local Evaluation Pipeline Within any single morphism evaluation step, the gateway enforces strict causal precedence:

$$Q(\text{bypass}) \prec Q(\text{reuse}) \prec \text{Direct Compute}$$

At the macro-DAG scale (Layer 3), equality saturation executes iteratively, where sub-graph unifications via $Q(\text{reuse})$ may reveal newly adjacent terms that trigger downstream $Q(\text{bypass})$ cancellations.

---

6. Identity and Lineage Rules

6.1. Local-first Identifier Structure

An artifact identifier is rooted strictly in its **Local-first** genesis:

$$\text{ID} = \langle \text{NodePseudonym}, \text{CausalIndex}, \text{Type}, \text{Version}, \text{Commitment} \rangle$$

6.2. No Global Raw Identity Exposure

Raw local node identifiers are prohibited from direct network exposure. Network interaction across **cdqn** uses accountable pseudonyms, cryptographic commitments, and zero-knowledge capability proofs.

6.3. Versioning

Every modification produces a distinct artifact with an advanced causal index. Predecessors remain immutable components of historical lineage.

6.4. Algorithm Agility

Cryptographic primitives must support seamless migration via **Structural Indirection**. No single signature suite or hash function is treated as a permanent dependency.

---

7. DCC Profile Rules

7.1. Dependencies

Explicitly declares required parent artifacts, signing identities, capability grants, and physical context prerequisites under **Dependencies Determinism**. Dependencies are evaluated as abstract contracts via **Structural Indirection** to allow modular implementation swaps.

7.2. Constraints

Declares hard operational ceilings: memory allocation bounds, recursion limits, **complexity degree** ceilings, precision limits, and excluded states (e.g., denominator zero exclusion).

7.3. Capabilities

Declares permitted transformations: morphism rights, sealing authorization, network export authorization, and delegation scopes.

7.4. Semantic Alignment & Anti-Drift Invariant

To prevent translation drift between high-level human intent and lower-level machine execution, an artifact's DCC contract must bind its human-readable specification in **Qs** directly to its intermediate execution representation in **QnIR**. A state transition is valid if and only if the verification receipt certifies that execution did not drift from the declared intent contract ($H(\text{Intent} \mid \text{Execution}) = 0$).

---

8. Metric Envelope Rules

Every **Metric Envelope** must declare finite, measurable boundaries: - **Footprint Bounds:** Storage allocation size, expression DAG depth ceilings, and **complexity degree** bounds. - **Execution Bounds:** Step ceilings, transformation budgets, and compilation collapse allocations. - **Numeric Bounds:** Scale zoom ceilings ($z \leq z_{\max}$) and uncertainty boundaries. - **Memory Wall Accounting:** Explicit accounting for data movement, cache locality, and bus transmission dissipation.

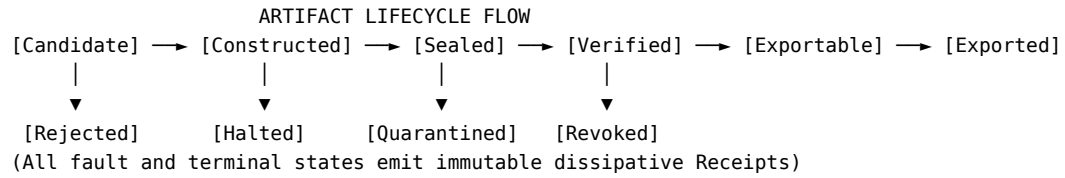
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9. Security Envelope Rules

Every **Security Envelope** must define protection parameters against computationally bounded adversaries: - Inner-ring and outer-ring scoping conforming to the **Dual-Ring PQC Boundary**. - Swappable revocation paths and emergency halt authorities designed for zero-day patchability. - Post-Quantum Cryptography (PQC) algorithm identifiers with explicit migration semantics. - Accountable pseudonymity; anonymous entities are strictly forbidden from traversing network boundaries.

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10. Lifecycle and Receipts



10.1. Lifecycle States

Standard recognized operational states conforming to the **Lifecycle State**: candidate, constructed, sealed, verified, rejected, halted, quarantined, exportable, exported, revoked.

10.2. Candidate State

A proposed stochastic or heuristic structure; classified strictly as an unverified candidate; prohibited from crossing a **SIMEMP Gateway**.

10.3. Constructed State

Formalized into canonical envelope syntax; awaiting constraint evaluation.

10.4. Sealed State

Identity-committed, metric-bounded, security-checked, and immutable.

10.5. Verified State

Validated against explicit formal proof invariants or test suites by an authorized verifier.

10.6. Rejected, Halted, and Quarantined States

Protective terminal states emitting mandatory failure receipts exporting operational entropy.

10.7. Exportable and Exported States

Certified by the **SIMEMP Gateway** for controlled exposure across the **cdqn** network via the Exposure Functor.

10.8. Revoked State

Invalidated via an explicit, signed revocation receipt that permanently seals a compromised branch and migrates lineage.

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11. Relation to Abstraction Layers

11.1. Layer 0 (Physical Substrate)

Raw physical hardware noise and clock signals in **Layer 0** carry no primitive envelope until processed across the onboarding gateway (`docs/q0_q1.md` §2).

11.2. Layer 0 to Layer 1 Gateway

Harvests physical entropy, generates Inner-Ring PQC keys, and constructs local origin **Q(0)** and first unit **Q(1)** across the **SIMEMP Gateway**.

11.3. Layer 1 (Node Genesis)

Houses foundational Category A primitives in **Layer 1** (**Q(0)**, **Q(1)**, single digits $Q(2) \dots Q(9)$, the **Abstract Compute Unit U**, and genesis axis d_1).

11.4. Layer 2 (Governed Morphisms)

Hosts elementary arithmetic morphisms ($+ $, $- $, $\times$, $\div$), spatial reflection involutions ($\mathcal{I}_d$), and Category B optimization morphisms (**Q(reuse)**, **Q(bypass)**).

11.5. Layer 3 (Symbolic Composition)

Hosts Content-Addressed Expression DAGs (**Qexpr**), multi-axial Clifford geometric frames, and compile-time reduction engines.

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12. Relation to QnLang and QnIR

The primitive envelope provides the formal execution target for the programming substrate: - ****QnLang**:** Authors computations declaring explicit DCC profiles, metric budgets, and payload profiles. Prohibits implicit defaults and floating-point semantics under the **No-Implicit Rule**. - ****QnIR**:** Intermediate representation flattening Content-Addressed Expression DAGs into linear, bounded instruction streams for deterministic hardware evaluation. - **Compile-Time Collapse:** Symbolic expression DAGs must collapse deterministically to concrete canonical leaf tuples prior to terminal **Receipt** emission.

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13. License Alignment

This specification strictly enforces the legal provisions of the **SSL**:

13.1. Paternity Reference

All derivative implementations, hardware drivers, runtime specifications, and APIs derived from the **Licensed Work** must embed the canonical **Paternity Reference**:

> Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License (SSL). Parent Repository: https://github.com/cdqn5249/cdqn

13.2. Open Core Invariants

1. **Anti-Patent Defense:** Commercial or **Derivative Works** licenses terminate automatically upon initiating patent litigation against the Author or project ecosystem.
2. **Non-Scaling Open Access:** Royalty-free access is guaranteed for non-commercial, academic, and sub-threshold usage under the **Open Core Invariants**.

13.3. Scale Auditing

The metric and identity envelopes supply verifiable telemetry (active compute instances, containers, autonomous agents, and monthly API transactions) to verify adherence to commercial scaling thresholds (`LICENSE.md` §1.5).

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14. Exclusions

The following items are outside the scope of this foundational primitive envelope and belong to dedicated specifications: 1. Detailed physical entropy sampling and NIST SP 800-90B health test implementations (specified in `docs/q0_q1.md`). 2. High-level grammar and syntax rules for QnLang (Milestone 2). 3. Virtual machine instruction set architecture and register schedules for QnIR (Milestone 2). 4. Physical simulation equations and discrete quantum Hamiltonians for Quang Physics (Qphy). 5. Concrete multi-party threshold signature ceremony protocols across distributed swarms (specified in `docs/chronosa.md`).

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15. Open Items

The following formal specifications remain open for subsequent releases: 1. Canonical binary serialization format (e.g., deterministic CBOR or custom packed Qn-binary) for the Universal Envelope. 2. Formal schema definition for Post-Quantum Cryptography algorithm agility migration certificates. 3. Automated verification checker specification for state transitions from Sealed to Verified. 4. Execution mechanics and termination bounds for higher-order recursive self-optimization workflows (**Q(rsi)**).

Annex N — Qn log

Pinned source: docs/qlog.md
Committed version: 1.1.1 **Last updated:** 2026-10-20
sha256 (committed file): `5ff1272e1e90bda35512f02be8540d2ae179178b14606b61450b2bf947db3b29`

Field	Specification
Document Title	Qlog — Quang Logics: Constructive Type Theory, Pattern Verification, and Epistemic Deduction
Version	1.1.1
Last Updated	2026-10-20 (Bao Loc, Vietnam)
Author	Christophe Duy Quang Nguyen
License	Scaling Source License (SSL) 1.0
Status	Candidate Domain Specification — Category D (**Complexity Degree 3**)

Normative References

The following documents establish the physical, structural, and legal constraints governing this specification. If a technical conflict arises, `simemp.md` governs; if a structural conflict arises, `abstractionLayers.md` governs; if a legal conflict arises, `LICENSE.md` governs. References are deliberately version-unpinned: each target document's front matter carries its own authoritative version, so this specification never duplicates drift-prone patch state.

Document	Role	Target
`docs/simemp.md`	Constitutional constraints, thermodynamics, and Dependencies Determinism	simemp.html
`docs/abstractionLayers.md`	Layer architecture and SIMEMP Gateway validation	abstractionLayers.html
`docs/qnPrimitive.md`	Universal Envelope, operational axioms, and Optimization Morphisms	qnPrimitive.html
`docs/q0_q1.md`	Primary genesis origin Q(0) and first unit Q(1)	q0_q1.html
`docs/q2_q9.md`	Single-digit secondary DCC anchors and single-digit spectrum	q2_q9.html
`docs/qm.md`	Constructive numeric leaf substrate and Diophantine division constraints	qm.html
`docs/qm_geometry.md`	Multi-axial frames, Clifford geometric algebra, and float-free rotations	qm_geometry.html
`docs/qm_calculus.md`	Discrete Exterior Calculus, continued fractions, and invariant dynamics	qm_calculus.html
`docs/qexpr.md`	Layer 3 Content-Addressed Expression DAG container	qexpr.html
`docs/qs.md`	Constructive relational intent and Intent Contracts	qs.html
`LICENSE.md`	Scaling Source License 1.0 governing the Licensed Work	LICENSE.md

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1. Epistemic Stance and Operational Demarcation

1.1. Constructive Working Hypothesis Notice

This specification is formulated within the Recursive Search Loop (`docs/simemp.md` §2). It represents a constructive theoretical baseline defining the emergent composite domain **Qlog (Quang Logics)**:

$$\mathbf{C}_{\text{Qlog}} \equiv \mathbf{C}_{\text{Qm}} \otimes \mathbf{C}_{\text{Qs}}$$

All operational models herein represent provisionally consistent hypotheses. In accordance with the Anti-Calcification Principle, permanent finality is reserved until physical execution feedback is derived across silicon runtimes. All derivations preserve architectural elasticity through **Structural Indirection**.

1.2. The Rejection of Platonic Deductions for Computed Patterns

Classical mathematical logic frequently relies on unmetered search spaces ($\mathbb{Z}^n, \mathbb{Q}^n$) and non-constructive principles such as the Law of Excluded Middle (LEM). In the **Qn** universe, unanchored mathematical deductions are discarded as physical fictions (`docs/reflections/h10_finite_computation.md` §4).

Truth is operationalized strictly as a **Verified Computed Pattern** certified by an immutable, dissipative **Receipt** satisfying **Dependencies Determinism**:

$$H(S_{\text{out}} \mid \mathcal{D}(S_{\text{out}})) = 0$$

---

2. Taxonomy of Causal Entities: Q(patterns) vs. Q(workflows)

2.1. Quang Patterns (Q(patterns))

A **Q(pattern)** is an invariant, reproducible state-transition segment mapping an input state commitment to a deterministic output state milestone under an abstract capability contract:

$$\mathcal{P} : \mathcal{H}(S_0) \xrightarrow{\mathcal{M}_{\text{contract}}} \mathcal{H}(S_1)$$

Quang Patterns capture structural, algebraic, and physical causality. They are intent-neutral and evaluate identically across all nodes, whether resolving to `SUCCESS_EXACT` (**Terminal Exactness**) or an invariant terminal fault (e.g., `RECEIPT_DIVISION_BY_ZERO_REJECTED`).

2.2. Quang Workflows (Q(workflows))

A **Q(workflow)** is a goal-directed, composite causal directed acyclic graph assembling multiple `Q(patterns)` under an explicit **Intent Contract** and finite metric budget allocation (W_{budget}). It represents an end-to-end navigational route across the computational landscape, executing human or delegated intent.

2.3. Formal Distinctions and Execution Currency

Feature	Quang Pattern (`Q(pattern)`)	Quang Workflow (`Q(workflows)`)
Structural Scope	Micro/Meso: Atomic or sub-graph state-transition segment.	Macro: End-to-end causal directed acyclic graph.
Intent Coupling	Intent-neutral: Represents a structural law of computation.	Intent-bound: Formally constrained by an **Intent Contract** in **Qs**.
Fault Handling	Encompasses deterministic terminal fault invariants.	Enforces fault containment, alternative branches, and budget ceilings.
Execution Currency	The atomic **Road Receipt** ($\mathcal{R}_{\text{road}}$).	An assembled **Chained Receipt Vector** ($\vec{\mathcal{R}}_{\text{road}}$).
Optimization Target	Optimized via **Q(bypass)** and cached via **Q(reuse)**.	Optimized via recursive self-optimization (**Q(rsi)**).

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3. Vertical Projection Across Complexity Degrees

Like **Qm** and **Qs**, Qlog is not pinned to a single horizontal layer. It operates as an orthogonal vertical projection crossing the execution core (`docs/abstractionLayers.md` §2.5) across all **Complexity Degrees**:

- **CD vs. Layer Note:** Complexity Degree (CD) indexes structural composition depth, not layer numbers. The Layer references in §3.1–§3.3 name the execution-core strata (Layers 1–3) crossed by the Qlog vertical projection; their Degree labels measure composition depth accumulated at each stratum, and Degrees 3 and n mark depth accumulated beyond the Layer 3 execution ceiling without implying a Layer 4.

3.1. Degree 0 — Genesis Ontological Baseline

At **Layer 1**, logic formalizes the foundation of existence: - Origin **Q(0)** formalizes the empty context, non-assertion, or the uninhabited type ($\bot$). - First unit **Q(1)** formalizes minimal evidence, identity, or the inhabited unit type ($\top$) along dimensional axis d_1.

3.2. Degree 1 — Atomic Propositional and Parity Calculus

At Layer 2, Qlog formalizes elementary scalar propositions: - Evaluates constructive conjunction ($\wedge$), disjunction ($\vee$), and sign parity involution ($\sigma \in \{\ominus, \odot, \oplus\}$). - Grounds Boolean logic in constructive Diophantine remainders ($r \in \{Q(0), Q(1)\}$) via secondary anchor $Q(2)$ (`DCC_ANCHOR_Q2_PARITY_v1`).

3.3. Degree 2 — Constructive Type Theory on CAE-DAGs

At Layer 3, Qlog structures deduction inside **Qexpr** Content-Addressed Expression DAGs (**Content-Addressed Expression DAG**): - **Curry–Howard–Qn Correspondence:** Propositions are represented as target types (τ); verified derivations are content-addressed DAGs (t). - **Macro-Hop Deduction:** Compresses multi-step micro-transitions into reusable, coarse-grained `Q(patterns)`.

3.4. Degree 3 — Topological and Coboundary Invariance Logic

Interfaces directly with discrete calculus (`docs/qm_calculus.md`): - Logical consistency is evaluated as topological coboundary nilpotency:

$$\mathbf{d} \circ \mathbf{d} \equiv Q(0)$$

- Governs trajectory routing across the universal **Q(anchor)** landmark network.

3.5. Degree n — Epistemic Modality, Inverse Reconstruction, and RSI

Above the Layer 3 execution ceiling, Qlog accumulates Complexity Degree n composition depth — emergent epistemic capability, not additional execution layers — unifying multi-agent interaction and empirical science: - ****Epistemic Anchorage**:** Segregates physical sensor data (`MODAL_EMPIRICAL`) from context-bound worldbuilding (`MODAL_NARRATIVE`). - **Inverse Causal Reconstruction:** Formalizes empirical scientific discovery by inverting observed macroscopic states to identify minimal causal ancestor DAGs. - **Recursive Evolution via** Q(rsi)**:** Compiles and optimizes the logical engine itself, emitting versioned successor artifacts under **Axiom 13**.

---

4. The Stratified Open Framework of Inferential Modes (L_modes)

Under the **Best of Choices (BOC) Policy**, $\mathcal{L}_{\text{modes}}$ is not a static closed set. It is an **open, inductively stratified category**:

4.1. Foundational Base Modes (Degree 1)

- **Micro-Induction:** Succession along axis d_1 via discrete morphism $\mathcal{S}(x) = x + Q(1)$.
- **Direct Deduction:** Zero-cost algebraic cancellation in $\mathcal{O}(1)$ register time ($Q(\text{bypass})$), enforcing immediate constraints such as $A \times Q(0) \to Q(0)$ or $RR^\dagger \to Q(1)$.
- **BOC Abduction:** Minimal causal predecessor selection from terminal execution states under SIMEMP constraints.

4.2. Compositional and Topological Modes (Degrees 2 and 3)

- **Macro-Hop Deduction:** Compresses an acyclic sub-graph of x intermediate operations into a single canonical `Q(pattern)`. Utilizes hash-consing and deduplication ($Q(\text{reuse})$) to ensure the macro-step is referenced at zero marginal Landauer cost ($\Delta W = 0$).
- **Anchored Markov Routing:** Traverses the sparse topology of universal $Q(\text{anchor})$ landmarks, using historical road receipt weights to bypass $\mathcal{O}(N)$ micro-step re-execution.

4.3. Emergent RSI Modes and Admission Invariants (Degree n)

Under **Axiom 11** and **Axiom 14**, any emergent logical mode M_{new} must satisfy four gateway verification invariants before inclusion into $\mathcal{L}_{\text{modes}}$: 1. **Explicit DCC Contract:** Wrapped in an explicit profile contract (`DCC_LOGIC_MODE_<ID>_vN`). 2. **Deterministic Reduction:** Operates over `Qexpr` CAE-DAGs with zero floating-point semantics. 3. **Finite Landauer Dissipation Ceiling:** Bounded by an explicit work allocation: $W(M_{\text{new}}) \leq W_{\text{budget}}$ in units of U. 4. **Receipt-Emitting Totality:** Terminates unconditionally in a declared terminal receipt (`SUCCESS`, `INCONCLUSIVE`, `BUDGET_EXHAUSTED`, or `REJECTED`).

---

5. Inferential Optimization Morphisms: Q(bypass) and Q(reuse)

In Qlog, optimization morphisms follow strict causal precedence:

$$Q(\text{bypass}) \prec Q(\text{reuse}) \prec \text{Direct Inferential Compute}$$

5.1. Q(bypass): Tautological and Deterministic Fault Short-Circuiting

Before scheduling memory bus transfers or search iterations, the gateway evaluates candidate judgments in $\mathcal{O}(1)$ register time: - **Reflexive Identity ($A \vdash A$):** If the goal commitment hash matches the active premise hash, search halts immediately, emitting `RECEIPT_BYPASS_IDENTITY` at zero metric cost ($\Delta W = 0$). - **Topological Nilpotency:** Inferences involving discrete exterior coboundaries ($\mathbf{d} \circ \mathbf{d}$) or boundaries ($\partial \circ \partial$) collapse algebraically to $Q(0)$ without mesh traversal. - **Deterministic Fault Interception:** Propositions containing illegal operations (such as division by $Q(0)$) are intercepted in $\mathcal{O}(1)$ time, emitting `RECEIPT_DIVISION_BY_ZERO_REJECTED`.

5.2. Q(reuse): Lemma Memoization and Inferential Breakeven

When intermediate `Q(patterns)` or sub-workflows are verified, they are indexed by their immutable commitment hash $\mathcal{H}(\mathcal{P})$: - **CAE-DAG Deduplication:** Identical sub-proofs share a single physical node in memory via **Hash-Consing**. - **The Inferential Breakeven Equation:** Hash retrieval across the Memory Wall is admitted if and only if the computational work avoided exceeds retrieval dissipation:

$$\text{Admit}\big(Q(\text{reuse})\big) \iff \Delta W_{\text{search}} > \Delta W_{\text{lookup}} + \epsilon_{\text{overhead}}$$

Trivial single-step inferences execute directly on integer ALUs. Combinatorially deep derivations utilize $Q(\text{reuse})$, achieving zero marginal Landauer dissipation ($W = 0$) for recurring sub-graphs.

---

6. The Memory Wall Latency Harvester (H_harvester)

In physical hardware architectures, retrieving data across the Memory Wall stalls processor execution for hundreds of clock cycles ($\approx$ 50–100 ns), during which silicon continues to dissipate static leakage power ($P_{\text{leak}} = I_{\text{leak}} V$).

6.1. Harvester Workload Scheduling During Stall Windows

During memory-stall windows (Δt_{stall}), Qlog schedules local, register-pinned workloads requiring zero external bus traffic: 1. **Background Equality Saturation (`Q(rsi)`):** Rewrites cached `Qexpr` DAGs using equivalence rules to find smaller structural footprints, minimizing future execution budgets. 2. **Speculative Candidate Auditing:** Evaluates unverified `State: CANDIDATE` transitions against algebraic invariants. Failed candidates are pruned prior to bus resumption. 3. **Anchor Route Pre-computation:** Re-calculates transition weights between local $Q(\text{anchor})$ landmarks based on recent receipt metrics.

6.2. Constitutional Guardrails and Preemption Semantics

- $\mathcal{O}(1)$ **Preemption:** Background tasks yield or commit in a single clock cycle the instant data arrives on the bus, adding zero latency to primary execution.
- **Deterministic Sandboxing:** Harvester tasks execute in read-only registers and cannot mutate sealed state.
- **Thermal Metering:** Background execution is throttled by the local **Metric Envelope**, halting if temperature telemetry approaches hardware limits.

---

7. The Dual Causal Axes: Forward Teleology vs. Inverse Empirical Science

Qlog accommodates both directional flows across the causal arrow:

7.1. Forward Teleological Synthesis (Engineering and Intent Compilation)

In forward synthesis, a system transitions from human intent to physical execution: 1. Human authority establishes an **Intent Contract** in **Qs**. 2. Qlog compiles and validates a composite `Q(workflow)` assembling known `Q(patterns)`. 3. Execution lowers deterministically to target machine operations, verifying that each milestone stays within allocated metric budgets.

7.2. Inverse Causal Reconstruction (Empirical Science and Abduction)

In empirical science, observers measure an empirical terminal physical state S_{obs} (anchored as `MODAL_EMPIRICAL`) while the generative causal origin and intermediate trajectory remain unobserved:

$$\text{Given } S_{\text{obs}} \implies \text{Find } \langle S_0,\, \mathcal{W} \rangle \quad \text{such that} \quad \mathcal{W}(S_0) \equiv S_{\text{obs}}$$

1. **Empirical Receipt Ingestion:** Observed physical telemetry is anchored via verified sensor receipts (`MODAL_EMPIRICAL`, $G = 1.0$).
2. **Candidate Hypothesis Generation:** Stochastic modules propose ancestral trajectories ($H_1 \dots H_n$), tagged as `State: CANDIDATE` under **Axiom 14**.
3. **BOC Abductive Resolution:** If multiple candidate trajectories explain the observed state, the **BOC Policy** selects the optimal route minimizing unmeasured conditional entropy ($H \to 0$) and Landauer dissipation:

$$\mathcal{W}^* = \arg\max_{H \in \mathcal{H}} \Big(\text{SIMEMP Compliance}(H) \Big)$$

4. **Receipt-Sealed Invariants:** The verified trajectory is sealed as an emergent `Q(pattern)`. It attains status as an established scientific law (`MODAL_CONSTRUCTIVE` or a new $Q(\text{anchor})$) only when reproducible execution across external nodes validates the trajectory.

---

8. Dynamic Lifecycle of Q(anchor) in Logic

$Q(\text{anchor})$ entities are topological landmarks across the causal network:

$$\chi(\alpha) = \frac{|\mathcal{C}^+(\alpha)|}{|\mathcal{V}|} \geq \Theta_\chi$$

8.1. Anchor Crystallization and Emergent Stability

When an emergent `Q(pattern)` or sub-workflow demonstrates universal stability and its centrality crosses threshold Θ_χ, it crystallizes into a new **Q(anchor)**. It is assigned a versioned capability contract (`DCC_ANCHOR_*_v1`), serving as a milestone for long-hop Markov routing.

8.2. Anchor Fracture, Containment, and Revocation

In physical networks, hardware degrades, nodes disconnect, or cryptographic algorithms become vulnerable: 1. **Centrality Drop:** If dependencies fail to persist across the network, centrality collapses ($\chi(\alpha) < \Theta_\chi$). 2. **Structural Indirection Migration:** If an algorithm is compromised, the gateway swaps the implementation module under **Axiom 13**, emitting an explicit migration receipt while preserving abstract contracts. 3. **Fracture Containment:** If an anchor cannot be repaired, Qlog emits `RECEIPT_ANCHOR_REVOKED`. Dependent workflows prune the compromised branch and fall back to deeper, invariant root anchors ($Q(2) \dots Q(9)$) without causing catastrophic network failure.

---

9. License and Lineage Alignment

9.1. Paternity Reference

All derivative implementations, logic solvers, type checkers, and automated reasoning engines derived from Qlog must embed the canonical attribution:

> Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License (SSL). Parent Repository: `https://g`ithub.com/cdqn5249/cdqn

9.2. Open Core Invariants

The core inferential modes, constructive type theory definitions, and pattern validation mechanics remain royalty-free for academic, non-commercial, and sub-threshold deployments under the **Open Core Invariants** of the **SSL**.

9.3. Statutory Safe Harbor and Protocol Neutrality

Qlog operates as passive, neutral mathematical and logical infrastructure. It exercises no editorial inspection over, and makes no claim upon, sovereign user content, external datasets, or sovereign **Payloads**.

Annex O — Qexpr

Pinned source: docs/qexpr.md
Committed version: 1.2.1 **Last updated:** 2026-10-19
sha256 (committed file): `576158576658e648976e549e1d3d586636ac1dafec836b4bbd643bfafac780dd`

Field	Specification
Document Title	Qexpr — Content-Addressed Expression DAG: Layer 3 Symbolic Container
Version	1.2.1
Last Updated	2026-10-19 (Bao Loc, Vietnam)
Author	Christophe Duy Quang Nguyen
License	Scaling Source License (SSL) 1.0
Status	Canonical Specification — Layer 3 Structural Container (**Complexity Degree 2**)

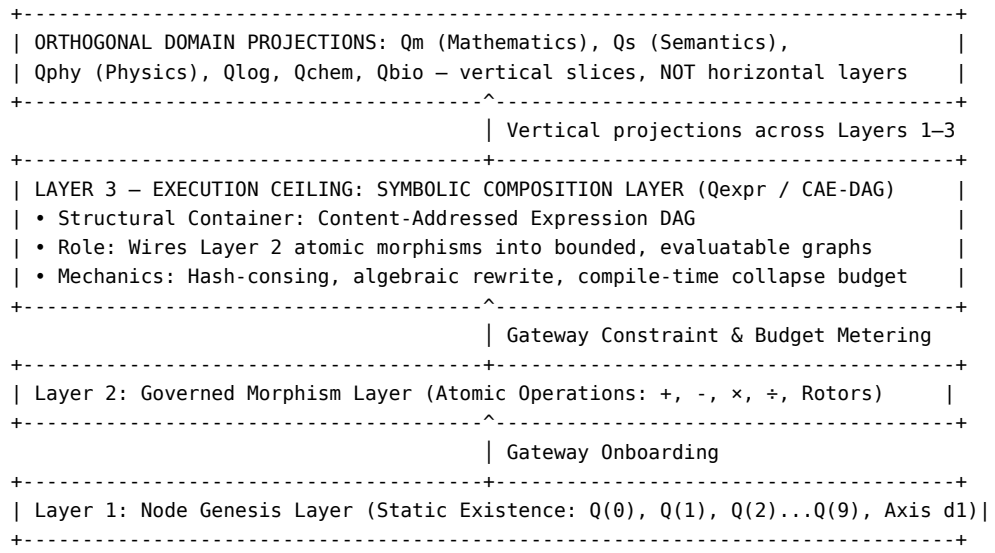
Normative References

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Document	Role	Target
`docs/simemp.md`	Constitutional constraints, thermodynamics, and Dependencies Determinism	simemp.html
`docs/abstractionLayers.md`	Layer architecture and SIMEMP Gateway validation	abstractionLayers.html
`docs/qnPrimitive.md`	Universal Envelope, operational axioms, and Optimization Morphisms	qnPrimitive.html
`docs/q0_q1.md`	Primary genesis origin Q(0) and first unit Q(1)	q0_q1.html
`docs/q2_q9.md`	Single-digit secondary DCC anchors and parity partitions	q2_q9.html
`docs/qm.md`	Constructive numeric leaf substrate and Diophantine division constraints	qm.html
`docs/qm_geometry.md`	Multi-axial frames, Clifford geometric algebra, and float-free rotations	qm_geometry.html
`docs/qm_calculus.md`	Discrete Exterior Calculus, continued fractions, and invariant dynamics	qm_calculus.html
`docs/qs.md`	Constructive relational intent and Intent Contracts	qs.html
`LICENSE.md`	Scaling Source License 1.0 governing the Licensed Work	LICENSE.md

1. Epistemic Stance and Constitutional Placement

Within the master abstraction hierarchy (`docs/abstractionLayers.md` §2.5, §5), ****Qexpr** is situated strictly at Layer 3 (**Complexity Degree 2**)** — the terminal layer of the physical execution core. Domains are orthogonal vertical projections across Layers 1–3, not horizontal layers above the stack.



1.1. The Structural Role of Layer 3

- ****Layer 1**** defines **what exists** (static ontological primitives: local origin **Q(0)**, unit **Q(1)**, single digits $Q(2) \dots Q(9)$ along axis **d1**).
- **Layer 2** defines **atomic actions** (isolated morphisms: addition, subtraction, multiplication, Diophantine division, exponents, roots, factorials, Clifford wedge products, spatial rotors, **Q(reuse)**, **Q(bypass)**).
- **Layer 3 (`Qexpr`)** defines **compositional structure**: how multiple Layer 2 atomic actions are wired into directed acyclic networks *before* they are evaluated or collapsed into physical memory.

1.2. Acknowledgment of Universal Fallibility

A core design tenet of the **Qn** architecture is the explicit recognition that **both human authors and artificial agents are fallible entities**. Neither is an absolute arbitrator: * Humans are prone to specification errors, unmeasured assumptions, and logical oversights. * Artificial agents and compilers are prone to algorithmic hallucinations, non-terminating expansions, and parameter misalignments.

Under `docs/simemp.md`, safety cannot rely on presumed actor infallibility. `Qexpr` serves as the **mechanical error-containment envelope**: it enforces strict syntactic cycle freedom, explicit graph depth bounds, and finite execution budgets (`docs/qnPrimitive.md` Axiom 6), ensuring that an error by any actor halts deterministically at the structural boundary without inducing systemic divergence.

1.3. Universal Semantic Membrane and Cross-Domain Contexts

`Qexpr` is not restricted to arithmetic calculations; it is the **universal structural membrane for all computable domains**.

A computational **Context** is formalized as a multi-domain Content-Addressed Expression DAG:

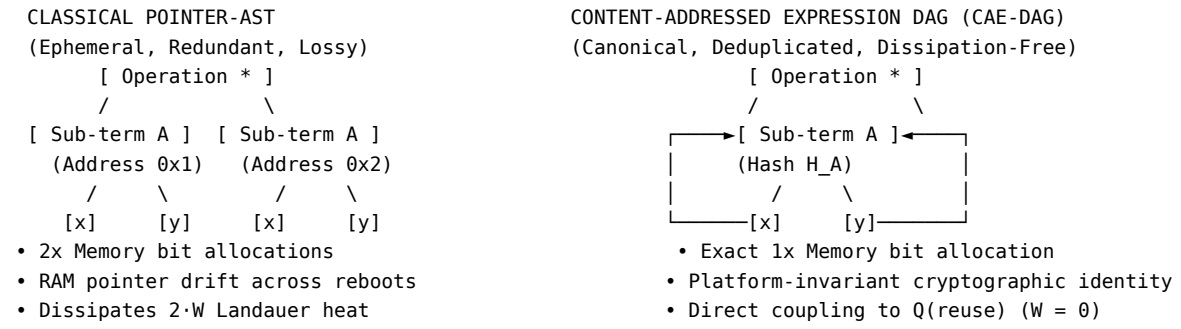
$$\text{Context} \equiv \mathcal{G}_{\text{Qexpr}} = \mathbf{C}_{\text{Qs}} \otimes \mathbf{C}_{\text{Qm}} \otimes \mathbf{C}_{\text{Qphy}}$$

1. **Semantic Root (Qs):** Encapsulates intent, linguistic pregroups, and human-agent capability delegations.
2. **Relational Root (Qm):** Verifies that proposed state transitions are mathematically decidable, Diophantine-constrained, and exact.
3. **Physical Root (Qphy):** Verifies that the required computational work can physically execute within finite Landauer dissipation bounds ($W \geq k_B T \ln 2$), parameterized by the **Abstract Compute Unit U**.

By representing cross-domain contexts within the unified `Qexpr` structure, heterogeneous actors achieve consensus on shared root commitment hashes $\mathcal{H}(\text{Context})$ without unmeasured semantic ambiguity.

2. The Content-Addressed Expression DAG (CAE-DAG) Architecture

Under the **Best of Choices (BOC) Policy** (`docs/simemp.md` §7), `Qexpr` explicitly rejects naive pointer-heap trees in favor of a ****Content-Addressed Expression Directed Acyclic Graph (CAE-DAG)****.



2.1. Canonical Node Tuple and Arity-Indexed Dependencies

Under the **No-Implicit Rule**, symbolic nodes cannot assume a rigid binary schema that forces dummy padding hashes for non-binary operators. Every symbolic node α within a `Qexpr` DAG is an immutable, content-addressed tuple with an explicit arity-indexed child commitment vector:

$$\alpha = \Big\langle \text{OpCode},\, \mathbf{D}_{\text{children}},\, z,\, r,\, d,\, \text{DCC}_{\text{ref}} \Big\rangle$$

$$\mathbf{D}_{\text{children}} = \big[\mathcal{H}_1,\, \mathcal{H}_2,\, \ldots,\, \mathcal{H}_k\big], \quad k = \text{Arity}(\text{OpCode}) \in \mathbb{N}$$

The child commitment vector $\mathbf{D}_{\text{children}}$ strictly maps operator arity without implicit padding: - **Nullary ($k = 0$):** Leaf constants, primary primitives ($Q(0), Q(1)$), and continued fraction recurrence generators (`OP_CF` for $\pi, e, \tau, \ln 2$). $\mathbf{D}_{\text{children}} = []$. - **Unary ($k = 1$):** Morphisms over a single operand, including radical root constraints (`OP_ROOT`), discrete factorials, Clifford reflection involutions ($\mathcal{I}_d$), discrete exterior derivatives (**d**), and Hodge stars ($\star$). $\mathbf{D}_{\text{children}} = [\mathcal{H}_1]$. - **Binary ($k = 2$):** Elementary $\mathcal{R}$-algebra morphisms ($+, -, \times, \div$, exponents), Clifford geometric products ($\vec{u}\vec{v}$), and wedge products ($\vec{u} \wedge \vec{v}$). $\mathbf{D}_{\text{children}} = [\mathcal{H}_1,\, \mathcal{H}_2]$. - **Variadic ($k \geq 3$):** Bounded monoidal folds (`OP_FOLD` for $\sum, \prod$), multi-axial coordinate frames, and multi-agent intent sequences. $\mathbf{D}_{\text{children}} = [\mathcal{H}_1,\, \ldots,\, \mathcal{H}_k]$.

The remaining tuple elements preserve physical and scale invariants: - z, r, d**:** Target scale quantum **Zoom z**, conserved residual **Remainder r**, and coordinate frame **Dimension d**. - DCC_{ref}**:** Machine-readable **DCC Profile** governing transformation ceilings, execution budgets, and capability delegation rights.

2.2. Structural Hash-Consing and $\mathcal{O}(1)$ Equivalence

Node identity is strictly mathematical under **Dependencies Determinism**, operationalized via **Hash-Consing**:

$$\mathcal{H}(\alpha) = \text{Hash}\Big(\text{OpCode} \,\|\, k \,\|\, \mathcal{H}_1 \,\|\, \ldots \,\|\, \mathcal{H}_k \,\|\, z \,\|\, r \,\|\, d \,\|\, \text{DCC}_{\text{ref}}\Big)$$

Because the arity k is explicitly serialized, nullary and unary operators produce canonical hashes with zero residual entropy ($H(S_t \mid \mathcal{D}(S_t)) = 0$).

1. **Deduplication:** Any sub-expression with identical semantics, arity, and child commitments yields the identical commitment hash. Through **Hash-Consing**, the runtime memory allocator maps it to the exact same physical node.
2. $\mathcal{O}(1)$ **Structural Equality:** Verifying whether two sub-expressions A and B are identical requires zero recursive graph traversal:

$$A \equiv B \iff \mathcal{H}(A) == \mathcal{H}(B)$$

Testing algebraic or semantic equality collapses to a single machine-word integer comparison.

2.3. Thermodynamic Work Minimization and Breakeven Bounds

In physical **Layer 0** substrates, allocating and deallocating memory across physical buses dissipates electrical capacitance ($\mathcal{O}(CV^2 f)$) and Landauer work ($W \geq k_B T \ln 2$). * By enforcing **Hash-Consing**, repeated sub-terms in multi-dimensional

Clifford geometric products (docs/qm_geometry.md) or multi-agent dialogue graphs are allocated **exactly once**. * Reversible symbolic normalization within the CAE-DAG incurs **zero Landauer dissipation** ($W = 0$), directly operationalizing **Q(reuse)** across the Memory Wall.

The Breakeven Admission Bound Querying a cryptographic hash commitment across L3 caches or DRAM incurs measurable bus dissipation. Under the **Metric Invariant** (simemp.md §3.1), hash-consing lookup via $Q(\text{reuse})$ is admissible if and only if the computational work of re-evaluation exceeds retrieval dissipation:

$$\text{Admit}\big(Q(\text{reuse})\big) \iff \Delta W_{\text{compute}} > \Delta W_{\text{lookup}} + \epsilon_{\text{overhead}}$$

If direct computation is cheaper than memory traversal (such as single-cycle integer additions $Q(2) + Q(3)$ or trivial powers), the compiler bypasses hash-table probes, lowering the operation directly to integer machine instructions via the HCA-Collapse engine.

2.4. Acyclicity by Construction (Axiom 5)

In classical graphs, circular references create non-halting loops and stack crashes. In a Content-Addressed DAG, **a circular dependency is mathematically impossible**. Because $\mathcal{H}(\alpha)$ requires each child hash $\mathcal{H}_i$ as an input, no child can declare α as a dependency without breaking the pre-image resistance of cryptographic hash functions. Acyclicity along the **Causal Arrow** is guaranteed by construction, satisfying **Axiom 5** (docs/qnPrimitive.md).

---

3. Elimination of Rounding and the Ontological Status of Irrationals

In classical floating-point systems (IEEE 754), rounding is an unmetered, lossy artifact of fixed register widths. Rounding silently discards trailing bits, generating non-deterministic drift and violating **Axiom 10**.

```
         CLASSICAL IEEE 754 vs. Qn CONSTRUCTIVE CONSERVATION
Classical Lossy Rounding:
  1 / 3  ──►  0.33333333...  ──►  [ Truncated to 53 bits ]  ──►  Silent Entropy Loss
                                                                  (Drift accumulates)

Qn Constructive Remainder Conservation (qm.md §4):
  1 / 3  ──►  Qexpr(OP_RATIO: 1/3)  ──►  Exact Symbolic State (Drift = 0)
                      │
                      ▼ Boundary Demand at Zoom z
              ⟨ ⊕, q_z, z, r_z, d ⟩ + Remainder Envelope
              • q_z : Discrete Integer Quotient
              • r_z : Conserved Residual (Total argument for scale z+1)
              • Zero bits discarded. Zero drift possible.
```

3.1. Prohibition of Rounding Drift

In the Qn universe, **implicit rounding is outlawed by the** No-Implicit Rule** and Axioms 9 and 10**: 1. Within Layer 3, expressions are held in unevaluated, exact symbolic form. Drift is identically zero. 2. When projected onto physical hardware boundaries, quantities are partitioned into an integer quotient q_z and an** exact conserved remainder** r_z (docs/qm.md §4):

$$X = \big(q_z \cdot \delta_z(d)\big) + r_z, \quad \text{where } 0 \le |r_z| < \delta_z(d)$$

The remainder r_z is never discarded; it is sealed within the **Universal Envelope** as the exact causal input state for subsequent precision expansions.

3.2. Ontological Status of Irrationals and Transcendentals

Under **Axiom 2**, actual infinities cannot exist as storable or executable machine states. Consequently: * An irrational number **cannot be an infinite decimal string** ($3.14159\ldots$), as storing infinite digits requires infinite energy. * **Transcendental and irrational quantities exist permanently as `Qexpr` symbolic generators.**

```
      TRANSCENDENTAL & IRRATIONAL ONTOLOGY IN THE Qn STACK

┌──────────────────────────────────────────────────────────────────────┐
│ LAYER 3: SYMBOLIC GENERATOR (Permanent Invariant State)              │
│ • Minimal Metric Root √2 : Qexpr(OP_ROOT: D² - Q(2) = Q(0), k=1)     │
│ • Golden Ratio φ         : Qexpr(OP_ROOT: φ² - φ - Q(1) = Q(0), k=1) │
│ • Circular Ratio π       : Qexpr(OP_CF: [3; 7, 15, 1, 292, ...], k=0)│
│ • Base e                 : Qexpr(OP_CF: [2; 1, 2, 1, 1, 4, ...], k=0)│
├──────────────────────────────────────────────────────────────────────┤
│               | Boundary Collapse Demand (Target Zoom z, Budget W)   │
│               ▼                                                      │
│ LAYER 1 / 2: DISCRETE NUMERIC LEAF (On-Demand Physical Projection)   │
│ • Quotient   : q_z ∈ ℕ (Exact discrete integer units at scale z)     │
│ • Remainder  : r_z (Exact residual boundary; |r_z| < δ_z(d))         │
│ • Terminal   : If r_z = Q(0), early halt via Terminal Exactness (ΔH = 0)│
└──────────────────────────────────────────────────────────────────────┘
```

1. **Algebraic Irrationals ($\sqrt{2}, \phi$):** Stored as exact polynomial constraint equations ($D^2 - Q(2) = Q(0)$) under `OP_ROOT` ($k = 1$). They are manipulated symbolically via radical identities with zero rounding drift.
2. **Transcendental Constants ($\pi, \tau, e, \ln 2$):** Stored as deterministic continued fraction recurrence generators under `OP_CF` ($k = 0$). Evaluating π to zoom level z computes an exact rational quotient and a conserved residual without floating-point error, verifying **Terminal Exactness** where remainders collapse.

---

4. The Six Canonical Node Classes

To achieve computational completeness across all abstraction layers, `Qexpr` defines **Six Canonical Node Classes**:

```
                    THE 6 CANONICAL Qexpr NODE CLASSES

┌──────────────────────────────────────────────────────────────────────┐
│ 1. OP_RATIO     : Symbolic rational pairs (p / q), k = 2.            │
│ 2. OP_ROOT      : Constrained Diophantine radical polynomials, k = 1.│
│ 3. OP_CF        : Projective continued fraction generators, k = 0.   │
│ 4. OP_FOLD      : Bounded monoidal accumulation (∑, ∏, n!) over arrays, k≥3.│
│ 5. OP_FRAME_MAP : Clifford geometric products and rotor conjugations, k = 2.│
│ 6. OP_REWRITE   : Equality saturation and algebraic identity bypass suite.  │
└──────────────────────────────────────────────────────────────────────┘
```

4.1. Node Class 1: Exact Rational Pairs (`OP_RATIO`, $k = 2$)

Represents unevaluated rational fractions $\frac{p}{q}$ ($q \neq Q(0)$) without premature projection onto decimal or binary scale lattices.

Multiplicative and additive morphisms execute via exact Diophantine cross-multiplication:

$$\frac{p_1}{q_1} \times \frac{p_2}{q_2} = \frac{p_1 \times p_2}{q_1 \times q_2}$$

$$\frac{p_1}{q_1} + \frac{p_2}{q_2} = \frac{(p_1 \times q_2) + (p_2 \times q_1)}{q_1 \times q_2}$$

4.2. Node Class 2: Constrained Algebraic Roots (`OP_ROOT`, $k = 1$)

Represents algebraic radical roots as exact polynomial Diophantine equations ($P(X) = Q(0)$).

Algebraic roots (such as $\sqrt{2}$ via `DCC_ANCHOR_SQRT2_v1`) are held symbolically with a single operand commitment. Multiplication by identical roots ($D \times D$) resolves immediately to integer constants ($Q(2)$) via **Q(bypass)** without invoking numerical root-extraction routines.

4.3. Node Class 3: Projective Recurrence Generators (`OP_CF`, $k = 0$)

Generates transcendental numbers ($\pi, e, \tau, \ln 2$) through deterministic continued fraction expansions:

$$X = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ddots}}}$$

Partial convergents $\frac{p_m}{q_m}$ are generated iteratively as nullary generators ($k = 0$) containing zero child dependencies. The truncation error is strictly bounded:

$$0 \leq |r_m| < \frac{1}{q_m q_{m+1}}$$

and the residual is preserved in the remainder envelope.

4.4. Node Class 4: Bounded Monoidal Folds (`OP_FOLD`, $k \geq 3$)

Governs accumulation ($\sum$, $\prod$, vector concatenation, and discrete factorials $n!$) over arrays, streams, or multi-axial frames.

Requires an explicit step ceiling ($m \leq m_{\max}$) and declared compute budget. The child vector $\mathbf{D}_{\text{children}}$ encapsulates the full array of term commitments. Unbounded loops or infinite series are syntactically invalid under **Axiom 2** and **Axiom 6**.

4.5. Node Class 5: Frame Transformations and Permutations (`OP_FRAME_MAP`, $k = 2$)

Governs multi-axial Clifford geometric products, Cayley rotor conjugations ($R\vec{v}R^{\dagger}$), and coordinate permutations (S_n group).

Rotors maintain unitary norm ($\|R\|^2 \equiv Q(1)$) by construction, eliminating trigonometric normalization cycles (`docs/qm_geometry.md` §5).

4.6. Node Class 6: Canonical Equality Saturation (`OP_REWRITE`, $k \in \{1, 2\}$)

Executes reversible term rewriting and identity short-circuiting prior to numerical collapse. Under the **No-Implicit Rule**, the engine evaluates the full canonical suite of $\mathcal{O}(1)$ algebraic identity bypasses:

CANONICAL Q(bypass) IDENTITY RULES

ADDITIVE & MULTIPLICATIVE BYPASSES	EXPONENTIATION & RADICAL BYPASSES
• A + Q(0) ⟶ A	• A^Q(0) ⟶ Q(1) (A ≠ Q(0))
• A × Q(1) ⟶ A	• A^Q(1) ⟶ A
• A × Q(0) ⟶ Q(0)	• Q(1)^k ⟶ Q(1)
• Q(0)! ⟶ Q(1)	• Q(0)^k ⟶ Q(0) (k > 0)
• Q(1)! ⟶ Q(1)	• ^Q(1)√(A) ⟶ A
• Q(2)! ⟶ Q(2)	• ^n√(Q(1)) ⟶ Q(1)

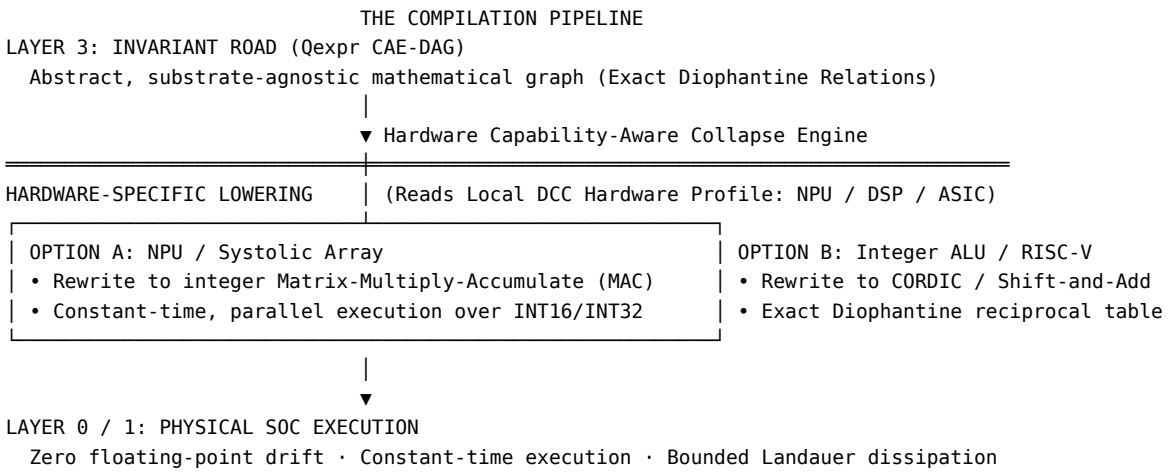
GEOMETRIC & TOPOLOGICAL INVARIANTS
• Rotor Normalization : R R† ⟶ Q(1)
• Exterior Nilpotency : e_i ∧ e_i ⟶ Q(0)
• Bivector Inversion : (I_2)² ⟶ -Q(1) (Grade 2 collapses to Grade 0)
• Poincaré Coboundary : d ∘ d ⟶ Q(0)

All listed transformations emit `RECEIPT_BYPASS_IDENTITY` at zero metric cost ($\Delta W = 0$), short-circuiting execution before any computation or memory lookup is scheduled.

---

5. Hardware Capability-Aware Collapse Engine (HCA-Collapse)

While `Qexpr` remains an abstract, substrate-agnostic DAG at Layer 3 via **Structural Indirection**, its physical execution on physical SoCs is governed by the ****Hardware Capability-Aware Collapse Engine (HCA-Collapse Engine)****.



5.1. Target-Specific Lowering Profiles

The compiler queries the node's local **DCC Hardware Profile** (established during Layer 0 onboarding): 1. **NPU / Systolic Arrays:** Lowers multi-axial Clifford geometric products ($\vec{u}\cdot\vec{v} = \sum \eta_{jj} u_j v_j$) directly into **integer Matrix-Multiply-Accumulate (MAC)** instructions (INT16/INT32), completing contractions in constant time. 2. **Cryptographic Co-Processors:** Offloads **Hash-Consing** deduplication and node commitment hashing to hardware SHA-3 or BLAKE3 accelerators. 3. **Integer RISC-V / Microcontrollers:** Replaces division by multiplication using Barrett integer reciprocal tables with exact remainder correction loops.

5.2. Side-Channel Immunity via Constant-Time Compute

Variable-time operations leak secret cryptographic keys and algorithmic state via power analysis and cache timing. Lowering `Qexpr` nodes into fixed-cycle SoC instruction sequences ensures that execution duration depends strictly on declared metric budgets, providing hardware-level side-channel immunity under the **Security Envelope**.

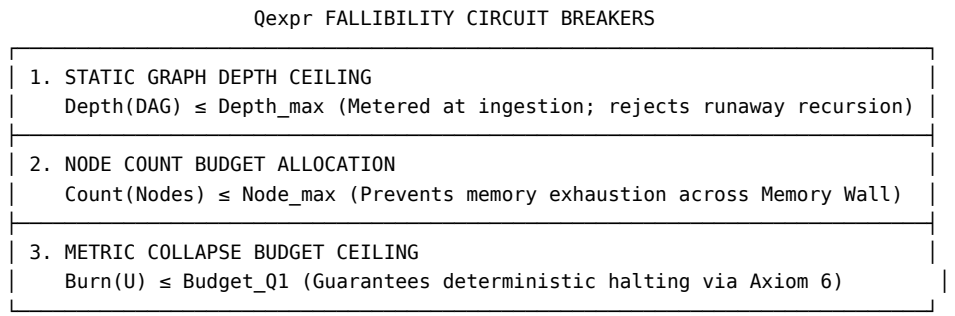
5.3. Invariance Mandate

SoC-specific lowering is permissible if, and only if, the transformation is an **algebraic isomorphism**. Lowering passes that introduce floating-point approximations, mantissa truncations, or non-deterministic rounding are intercepted by the gateway and rejected, protecting **Existential Invariants**.

---

6. Bounded Execution and Fallibility Circuit Breakers

To guarantee that fallible human directives or unaligned agentic processes cannot compromise the execution environment, `Qexpr` enforces three structural circuit breakers:



1. **Static Graph Depth Ceilings:** If an expression tree exceeds the declared depth ceiling ($D > D_{\max}$), execution terminates immediately with receipt `RECEIPT_EXPRESSION_TOO_COMPLEX`.

2. **Node Allocation Bounds:** If **Hash-Consing** attempts to instantiate node counts exceeding the local **Metric Envelope**, compilation aborts with `RECEIPT_BUDGET_EXHAUSTED`.
3. **Totality by Budget (Axiom 6):** Every collapse operation carries an explicit compute budget in units of $Q(1)$ via **Metric Exhaustion**. Non-terminating expansions are impossible; when the budget is consumed, the engine emits a dissipative **Receipt** and halts.

---

7. License and Invariant Lineage

This specification strictly conforms to the **Scaling Source License (SSL) 1.0**:

7.1. Paternity Reference

All derivative implementations, symbolic rewrite engines, compilers, and intermediate representation parsers derived from the **Licensed Work** must preserve the canonical **Paternity Reference**:

> Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License (SSL). Parent Repository: `https://g`ithub.com/cdqn5249/cdqn

7.2. Open Core Invariants

1. **Anti-Patent Defense:** Commercial or derivative licenses terminate automatically upon initiating patent litigation against the Author or project ecosystem.
2. **Non-Scaling Open Access:** Royalty-free licensing is guaranteed for non-commercial, academic, and sub-threshold usage under the **SSL**.

7.3. Passive Infrastructure Safe Harbor

The `Qexpr` symbolic container acts as passive, neutral technical infrastructure. It exercises no editorial inspection over, and makes no claim upon, sovereign user content, models, or **Payloads**.

Annex P — Qm

Pinned source: docs/qm.md
Committed version: 1.2.11 **Last updated:** 2026-10-20
sha256 (committed file): `3dfc55ed929f9aa8b9f3e86c30bf1fa847c8c43005fca383d1cb02e02af58be8`

Field	Specification
Document Title	Qm — Quang Mathematics: Constructive Numeric and Relational Substrate
Version	1.2.11
Last Updated	2026-10-20 (Bao Loc, Vietnam)
Author	Christophe Duy Quang Nguyen
License	Scaling Source License (SSL) 1.0
Status	Canonical Domain Specification — Category D (Local-First Base Domain)

Normative References

The following documents establish the physical, structural, and legal constraints governing Qm. If a technical conflict arises, `simemp.md` governs; if a structural conflict arises, `abstractionLayers.md` governs; if a legal conflict arises, `LICENSE.md` governs. References are deliberately version-unpinned: each target document's front matter carries its own authoritative version, so this specification never duplicates drift-prone patch state.

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`docs/simemp.md`	Constitutional constraints, thermodynamics, and Dependencies Determinism	simemp.html
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`docs/qexpr.md`	Layer 3 Content-Addressed Expression DAG container	qexpr.html
`docs/qs.md`	Constructive relational intent and Intent Contracts	qs.html
`LICENSE.md`	Scaling Source License 1.0 governing the Licensed Work	LICENSE.md

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1. Epistemic Stance and Explicit Demarcation

In strict adherence to the **No-Implicit Rule** (`docs/abstractionLayers.md` §3.2) and the constitutional constraints of the **SIMEMP** framework, this specification explicitly defines the boundaries of **Qm (Quang Mathematics)** to prevent implicit assumptions, hidden conventions, or non-deterministic state drift.

In the **Qn** computational universe, every mathematical proposition and state transition in Qm is governed by **Dependencies Determinism**:

$$H(S_t \mid \mathcal{D}(S_t)) = 0$$

Operational validity is verified empirically via **Computational Consistency**. No unmeasured entity may cross an abstraction boundary or **SIMEMP Gateway**.

EXPLICIT DOMAIN BOUNDARY OF Qm

WHAT Qm IS	WHAT Qm IS NOT
Constructive discrete arithmetic	Continuous floating-point arithmetic
Base-independent scale lattices (z)	Platform-dependent radix conventions
Conserved residual tracking (r)	Silent truncation / unmetered loss
Geometric orientation involution (σ)	Detached ontological sign entities
Total-by-budget morphisms (Receipts)	Infinite non-terminating expansions
Height-bounded Diophantine relations	Unbounded Hilbert-10 void search
Bounded folds & constrained roots	Continuous unmetered exponents
Positional order decidability	Arbitrary heuristic tie-breaking
Concrete numeric leaf evaluation	Symbolic AST container (Qexpr)

1.1. Explicit Inclusions: What Qm Is

- **The Constructive Numeric Leaf:** The canonical, discrete representation of fully evaluated quantities along dimensional axes encapsulated within the **Universal Envelope**.
- **The $\mathcal{R}$-Algebra:** Explicit arithmetic morphisms ($+ $, $- $, $\times$, $\div$, powers, roots, and bounded factorials) preserving exact Diophantine invariants and emitting signed receipts.
- **Projective Scale Transitions:** Exact conservation of residual information (r_z) across zoom scales without numeric drift.
- ****Local-first** Execution:** 100% intra-node computation mediated by local cdqn memory-bus data movement, requiring zero distributed consensus.
- **Total-by-Budget Morphisms:** Discrete operations bounded by finite resource consumption via **Metric Exhaustion**.

1.2. Explicit Exclusions: What Qm Is Not

- **Not an IEEE 754 Floating-Point System:** Under **Axiom 10**, approximations, mantissa round-offs, subnormal representations, and $+0.0/-0.0$ distinctions are prohibited.
- **Not a Non-Constructive Continuum:** Qm does not admit actual infinities ($\aleph_0, \aleph_1$), uncomputable real numbers (non-constructive Dedekind cuts), or undecidable Cauchy sequences.
- **Not an Unbounded Diophantine Search Oracle:** While arithmetic operations are defined through Diophantine constraints, Qm explicitly rejects unmetered infinite search spaces ($\mathbb{Z}^n, \mathbb{Q}^n$). Classical undecidability (Hilbert's Tenth Problem, DPRM 1970) is neutralized: all evaluations are constrained to finite logarithmic height bounds ($h(x) \leq H_{\max}$) declared within the **Metric Envelope**.
- **Not an Authoring Language:** High-level source grammar belongs to **QnLang**.
- **Not an Intermediate Execution Backend:** Hardware abstraction and register scheduling belong to **QnIR**.
- **Not an Unbounded Symbolic Container:** Unevaluated expression trees belong to **Qexpr**; Qm governs their deterministic numerical collapse.

---

2. Abstraction Layer Mapping and Complexity Stratification

Qm operates across the hierarchy as an orthogonal vertical projection across **abstraction layers** (`docs/abstractionLayers.md` §11), rather than a static horizontal tier.

```
┌──────────────────────────────────────────────────────────────────┐
│ Layer 3: Symbolic Composition (Qexpr Trees, Bounded Expressions) │
│          Qm Degree 2: Multi-axial geometry (d_k), Polynomial Forms │
├──────────────────────────────────────────────────────────────────┤
│ Layer 2: Governed Morphisms & Algebraic Interfaces               │
│          Qm Degree 1: Monoids, Involutions (I), Diophantine Constraints │
```

```
├──────────────────────────────────────────────────────────────────────────┤
│ Layer 1: Node Genesis Layer                                              │
│          Qm Degree 0: Origin Q(0), Unit Q(1), Basis Alignment d1          │
├══════════════════════════════════════════════════════════════════════════┤
│ Layer 0: Physical Substrate (Entropy Harvesting, Landauer Work W ≥ kT ln 2) │
└──────────────────────────────────────────────────────────────────────────┘
```

2.1. Layer-by-Layer Responsibilities

- ****Layer 1**: Hosts genesis primitives. Origin **Q(0)** establishes the spatial origin; unit **Q(1)** establishes the baseline metric quantum; the transition $Q(0) \to Q(1)$ establishes the initial dimensional axis **d1** and the **Abstract Compute Unit U**.
- **Layer 2:** Hosts elementary arithmetic morphisms ($+ $, $- $, $\times$, $\div$, exponents, roots, factorials) constrained by explicit **DCC Profiles**, indexed sequentially along the monotonic **Causal Arrow**.
- **Layer 3:** Ingests symbolic **Qexpr** trees and collapses them into canonical Qm leaf tuples within finite metric budgets.

2.2. Complexity Degree Stratification

- **Degree 0 (Foundational Primitives):** $Q(0)$, $Q(1)$.
- **Degree 1 (Elementary Arithmetic):** Positional tuples along axis d_1, Diophantine quotient-remainder partitions, bounded folds. *(Normative scope of this specification).*
- **Degree 2 (Multi-Axial & Symbolic):** Orthogonal coordinate spaces (d_k), Clifford geometric algebras ($\mathcal{C}\ell_{p,q}$), and Qexpr CAE-DAG reduction (`docs/qm_geometry.md` and `docs/qexpr.md`).
- **Degree 3 (Discrete Calculus & Analysis):** Discrete exterior calculus, finite-difference operators, and projective continued fractions (`docs/qm_calculus.md`).

---

3. Canonical Numeric Representation

Every fully evaluated numeric artifact in Qm is an immutable, discrete **Qn Numeric Entity** represented as a discrete leaf tuple:

$$\text{Qn Numeric Entity} \equiv \langle \sigma,\ q_z,\ z,\ r_z,\ d \rangle$$

```
                CANONICAL Qm NUMERIC STRUCTURE

┌──────────────────────────────────────────────────────────┐
│  σ    : Relational Orientation (⊖, ⊙, ⊕)                 │
│  q_z  : Discrete Metric Magnitude Quotient (q_z ∈ ℕ)      │
│  z    : Scale / Lattice Zoom Level (z ∈ ℕ)               │
│  r_z  : Conserved Residual Generator (0 ≤ |r_z| < δ_z(d)) │
│  d    : Dimensional Coordinate Axis Parameter (d_k)      │
└──────────────────────────────────────────────────────────┘
```

3.1. Orientation as Geometric Involution (σ)

Sign is not an independent arithmetic substance. It is an **oriented displacement** relative to local origin **Q(0)** along axis d: - The base transition from $Q(0)$ to **Q(1)** along axis d_1 defines canonical positive alignment ($\oplus$). - Inversion is governed by the 1D reflection functor $\mathcal{I}_d$, satisfying the involution property:

$$\mathcal{I}_d : \Sigma \to \Sigma, \quad \mathcal{I}_d^2 = \text{id}, \quad \text{where } \Sigma = \{\ominus, \odot, \oplus\}$$

- $\odot$ **(Origin):** Coincident with local $Q(0)$ (zero metric displacement).
- $\oplus$ **(Collinear):** Parallel to the basis transition $Q(0) \to Q(1)$.
- $\ominus$ **(Anti-parallel):** Opposing the basis transition.

Under **Axiom 10**, signed zeros ($+0.0$, -0.0) are prohibited. If $q_z = 0$ and $r_z = Q(0)$, σ collapses identically to $\odot$.

3.2. Ontological Primacy of Magnitude (q_z)

The discrete quotient $q_z \in \mathbb{N}$ represents the unsigned metric distance along axis d at zoom level z. In physical execution substrates, metric resources (storage footprint, clock cycles, Landauer dissipation) are strictly non-negative. Absolute magnitude q_z is ontologically primary; signed quantities are composite pairs $\langle \sigma, q_z \rangle$.

3.3. Base-Independent Scale Lattice (Zoom z)

Conforming to **Axiom 9**, scale is decoupled from arbitrary positional radices (binary, decimal, sexagesimal). A declared zoom level $z \in \mathbb{N}$ parameterizes a discrete rational subdivision quantum $\delta_z(d) \in \mathbb{Q}^+$ of the dimensional unit:

$$\delta_z(d) = \frac{Q(1)_d}{\kappa(z)}$$

where $\kappa : \mathbb{N} \to \mathbb{N}$ is a strictly monotonic resolution lattice function.

3.4. Conserved Residual (Remainder r)

The residual r_z is the exact difference between the unscaled value and its discrete quotient:

$$0 \leq |r_z| < \delta_z(d)$$

The remainder is never discarded silently. It acts as the exact, causal input state for subsequent resolution expansions.

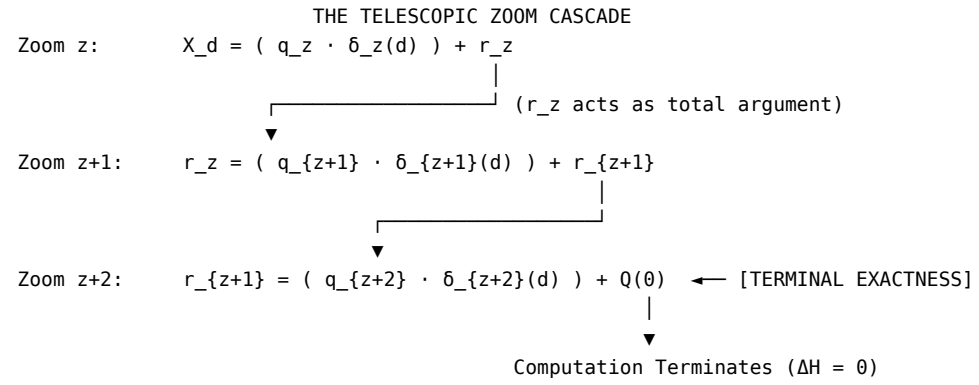
3.5. Dimensional Orthogonality (Dimension d)

The axis parameter d_k ensures that metric units remain orthogonal. Remainders along dimension d_i cannot cancel, add to, or combine with remainders along dimension d_j ($i \neq j$).

---

4. The Projective Scale Invariant and Terminal Exactness

The triplet $\langle z, r_z, d \rangle$ constitutes a projective resolution system over discrete lattices.



4.1. Scale Expansion Equation

When higher precision is demanded by a downstream workflow ($z \to z+1$), the input state to the resolution transition is identically r_z:

$$r_z = \Big(q_{z+1} \cdot \delta_{z+1}(d) \Big) + r_{z+1}$$

Across arbitrary zoom cascades, total metric information is strictly conserved:

$$\mathcal{I}_{\text{total}}(X_d) = \sum_{k=0}^{z} \Big(q_k \cdot \delta_k(d) \Big) + r_z$$

Zero precision drift occurs. Truncation error is zero at every intermediate scale.

4.2. Terminal Exactness Invariant

If a state evaluation yields a remainder equal to the local causal origin:

$$r_z = Q(0)$$

the representation is algebraically exact.

1. **Information Invariance:** For all subsequent scales $k > 0$, $q_{z+k} \equiv 0$ and $r_{z+k} \equiv Q(0)$.
2. **Deterministic Early Halting:** Evaluating $z+1$ when $r_z = Q(0)$ yields zero entropy reduction ($\Delta I = 0$). Under the **Metric Invariant** (`docs/simemp.md` §3.1), spending compute budget on idempotent iterations is forbidden.
3. **Receipt Emission:** The **SIMEMP Gateway** intercepts this state and emits an immutable terminal **Receipt**:
   - **Terminal Receipt:** `RECEIPT_TERMINAL_EXACTNESS`
     `⟨ Status: SUCCESS_EXACT, Zoom: z, Remainder: Q(0), ConsumedWork: 0 ⟩`

---

5. Elementary Morphisms (R-Algebra)

Arithmetic in Qm operates over discrete tuples rather than continuous fields.

5.1. Addition and Subtraction

Addition is exact vector concatenation along axis d. Subtraction is addition composed with geometric reflection:

$$A - B \equiv A + \mathcal{I}_d(B)$$

Given $A = \langle \sigma_A, q_A, z, r_A, d \rangle$ and $B = \langle \sigma_B, q_B, z, r_B, d \rangle$:

Step 1: Absolute values and remainders are combined over the common quantum $\delta_z(d)$:

$$\mu_{\text{exact}} = (\sigma_A q_A \cdot \delta_z + \sigma_A r_A) + (\sigma_B q_B \cdot \delta_z + \sigma_B r_B)$$

Step 2: The result is partitioned into discrete quotient q_C, residual remainder r_C, and orientation σ_C:

$$q_C = \left\lfloor \frac{|\mu_{\text{exact}}|}{\delta_z} \right\rfloor$$

$$r_C = |\mu_{\text{exact}}| - (q_C \cdot \delta_z)$$

$$\sigma_C = \text{sgn}(\mu_{\text{exact}})$$

5.2. Multiplication

Multiplication scales magnitude and composes orientation parity:

$$\sigma_C = \sigma_A \otimes \sigma_B$$

$$\oplus \otimes \oplus = \oplus, \quad \ominus \otimes \ominus = \oplus, \quad \ominus \otimes \oplus = \ominus, \quad \odot \otimes \sigma = \odot$$

The unscaled magnitude is calculated constructively:

$$\mu_{\text{exact}} = (q_A \cdot \delta_z + r_A) \times (q_B \cdot \delta_z + r_B)$$

The resulting unscaled quantity collapses to $\langle \sigma_C, q_C, z, r_C, d \rangle$ via Diophantine resolution.

5.3. Division as a Constrained Diophantine Equation

Division in Qm is defined not as an unconstrained rational projection, but as a bounded constraint equation. For any input pair A and B, the morphism determines quotient C and remainder R satisfying:

$$A = (B \times C) + R$$

The evaluation is constrained by three invariant conditions: the divisor must be non-zero ($B \neq Q(0)$), the remainder is strictly bounded ($0 \leq |R| < |B|$), and the orientation sign satisfies parity composition ($\sigma_C = \sigma_A \otimes \sigma_B$).

Height-Bounded Diophantine Decidability Classical Diophantine equations over unmetered spaces ($\mathbb{Z}, \mathbb{Q}$) inherit algorithmic undecidability under Hilbert's Tenth Problem. In Qm, this undecidability void is closed structurally across three physical constraints:

Logarithmic Height Boundedness: In arithmetic geometry, the logarithmic height measures physical bit complexity, bounded by $h(u/v) = \max(\log|u|, \log|v|) \leq H_{\text{max}}$. Under the **Metric Envelope**, every Diophantine evaluation operates within an explicit height ceiling H_{max} and work budget W_{budget}.

Finite State Space (Northcott's Theorem): By Northcott's theorem, the set of rational coordinates of bounded height is strictly finite:

$$\text{Card}\big(\{x \in \mathbb{Q} \colon h(x) \leq H_{\text{max}}\}\big) < \infty$$

Therefore, candidate solutions in Qm inhabit a finite discrete lattice rather than an unmetered void.

Deterministic Decidability: A Diophantine evaluation is decidable by construction: it either terminates with an exact quotient-remainder pair, reaches Terminal Exactness ($R = Q(0)$), or halts unconditionally upon budget exhaustion via **Metric Exhaustion** (**Axiom 6**), emitting `RECEIPT_BUDGET_EXHAUSTED`.

Terminal Rejection Attempted division by origin $Q(0)$ violates the DCC constraint and is intercepted deterministically at the gateway, emitting the immutable receipt `RECEIPT_DIVISION_BY_ZERO_REJECTED` with status `REJECTED` and code `ERR_DIV_ZERO`.

5.4. Remainder Depth and Dissipative Truncation

When recursive compositions cause the symbolic tree of r_C to exceed the **Metric Envelope** ceiling $\text{Depth}_{\text{max}}$: 1. Silent truncation is prohibited under the **No-Implicit Rule**. 2. Execution halts or emits an explicit dissipative receipt: - **Dissipative Truncation Receipt:** `RECEIPT_REMAINDER_TRUNCATION`
`⟨ Status: DISSIPATIVE_SINK, LostResidual: r_C, ExportedEntropy: ΔS ⟩`

5.5. Exponentiation and Radical Morphisms

Exponentiation and root extraction are formalized as constrained morphisms that operate with explicit thermodynamic bounds and dimensional parity governance:

I. Integer Exponentiation as a Bounded Monoidal Fold For integer exponents $B \in \mathbb{N}$, exponentiation is an iterative multiplicative fold over the positive monoid $(\mathcal{Q}^+, \times, Q(1))$ under Node Class 4 (`OP_FOLD`):

$$A^B = \prod_{j=1}^{B} A$$

Evaluated at Layer 3 under `OP_FOLD`. The memory footprint expands linearly with the exponent: $h(A^B) = B \cdot h(A) \leq H_{\text{max}}$. If the resulting bit size exceeds the metric envelope ceiling, compilation halts with `RECEIPT_EXPRESSION_TOO_COMPLEX`.

II. Negative Exponents as Rational Pairs Negative powers do not invoke floating-point arithmetic; they map to exact rational inversion pairs under the Node Class 1 ordered call `OP_RATIO(Q(1), A^B)`:

$$A^{-B} \equiv \frac{Q(1)}{A^B}$$

Projected onto scale lattices at zoom z with exact remainder conservation (r_z). If the base is negative ($\ominus A$), the resulting sign is determined by the parity of exponent B via secondary anchor $Q(2)$ (`DCC_ANCHOR_Q2_PARITY_v1`):

$$\sigma_{\text{result}} = \begin{cases} \oplus & \text{if } B \in \mathcal{Q}_{\text{even}} \\ \ominus & \text{if } B \in \mathcal{Q}_{\text{odd}} \end{cases}$$

III. Boundary Exponent Identities Under the **No-Implicit Rule**, boundary conditions are strictly evaluated: - **Zero Exponent Identity ($A \neq Q(0)$):** For any non-zero base, $A^{Q(0)}$ represents the empty monoidal product, collapsing to the multiplicative identity $Q(1)$. Evaluated in $\mathcal{O}(1)$ via **Q(bypass)**, emitting `RECEIPT_BYPASS_IDENTITY` at zero metric cost ($\Delta W = 0$). - **Indeterminate Origin Power ($Q(0)^{Q(0)}$):** An unparameterized 0^0 evaluation is prohibited across gateways. Unless the calling profile carries an explicit contract (`DCC_ZERO_EXPONENT_v1`), execution halts with `RECEIPT_INDETERMINATE_ORIGIN_EXPONENT`. - **Negative Origin Power ($Q(0)^{-k}$):** Evaluating $Q(0)^{-k} \equiv Q(1)/Q(0)^k$ requires division by origin $Q(0)$, terminating immediately with `RECEIPT_DIVISION_BY_ZERO_REJECTED`.

IV. Radical Morphisms and Dimensional Promotion Roots are defined constructively as polynomial Diophantine equations under `OP_ROOT`:

$$\sqrt[n]{A} \implies \text{Find } X \quad \text{such that} \quad X^n - A = Q(0)$$

- **Singularity of Root Zero ($\sqrt[Q(0)]{A}$):** Expressed as rational exponent $A^{\frac{Q(1)}{Q(0)}}$, extracting the $Q(0)$-th root requires division by origin in the exponent denominator. The gateway halts execution with `RECEIPT_DIVISION_BY_ZERO_REJECTED`.
- **Odd Roots of Negative Scalars ($n \in \mathcal{Q}_{\text{odd}}$):** Because an odd number of spatial reflections preserves negative orientation ($\ominus^n = \ominus$), odd roots of negative quantities are algebraically closed within 1D discrete space ($\mathcal{Q}_1$). For example, $\sqrt[3]{\ominus 8} = \ominus 2$ under Terminal Exactness.
- **Even Roots of Negative Scalars ($n \in \mathcal{Q}_{\text{even}}$):** In 1D scalar space, no orientation satisfies $\sigma \otimes \sigma = \ominus$. An attempt to compute $\sqrt{\ominus A}$ in 1D is intercepted by the gateway and promoted to Dimension 2 (d_2) as an oriented planar bivector blade in Clifford Geometric Algebra ($\mathcal{C}\ell_{p,q}$):

$$\sqrt{\ominus Q(1)} \implies \vec{e}_1 \wedge \vec{e}_2 \quad (\text{where } (\vec{e}_1 \wedge \vec{e}_2)^2 = -Q(1))$$

V. The Chiral Optimization Duality Powers and roots form a **Chiral Morphism Pair** along the causal execution arrow: - Forward powers collapse geometric grades ($I_2^2 \to -Q(1)$) and compute in feed-forward polynomial steps. Small integer powers disfavor memory lookups ($\Delta W_{\text{compute}} \leq \Delta W_{\text{lookup}}$), executing directly on ALUs. - Inverse roots expand geometric grades ($\sqrt{\ominus Q(1)} \to I_2$) and require bounded Diophantine lattice searches ($\Delta W_{\text{compute}} \gg \Delta W_{\text{lookup}}$). Roots cross the thermodynamic breakeven threshold immediately, prioritizing sub-graph reuse via **Q(reuse)**.

5.6. Discrete Factorial and Combinatorial Morphisms

The factorial operator is formalized in Qm as an iterative discrete monoidal fold over the positive integer monoid $(\mathcal{Q}^+, \times, Q(1))$ under Node Class 4 (`OP_FOLD`):

$$n! = \prod_{j=1}^{n} j$$

I. Boundary Identifications and $\mathcal{O}(1)$ Bypasses The factorial morphism exhibits three canonical identity states in the range $Q(0) \ldots Q(2)$: - **Empty Product Identity ($Q(0)!$):** In monoid theory, an empty product evaluates to the multiplicative identity $Q(1)$. Handled in $\mathcal{O}(1)$ via **Q(bypass)**, emitting `RECEIPT_BYPASS_IDENTITY` at zero metric cost ($\Delta W = 0$). - **Unit Invariant ($Q(1)!$):** Evaluates identically to $Q(1)$. Bypassed in $\mathcal{O}(1)$ via $Q(\text{bypass})$ ($\Delta W = 0$). - **Fixed-Point Invariant ($Q(2)!$):** 2 is the unique positive integer satisfying $n! = n$. Recognized as an algebraic fixed-point identity in `OP_REWRITE`, bypassing execution in $\mathcal{O}(1)$ ($\Delta W = 0$).

II. Thermodynamic Optimization Boundary Under the **Metric Existential Invariant**, optimization morphisms adhere to strict thermodynamic breakeven bounds: - **Prohibition of $Q(\text{reuse})$ for $n \leq 2$:** For $Q(0)$, $Q(1)$, and $Q(2)$, computational cost is zero (direct bypass). Initiating a 256-bit cryptographic hash lookup across the Memory Wall consumes dynamic bus power ($\mathcal{O}(CV^2f)$) that exceeds execution cost. $Q(\text{reuse})$ is strictly forbidden for $n \leq 2$. - **Admission of $Q(\text{reuse})$ for $n \geq 3$:** For $k \geq 3$, factorial computation satisfies $Q(k)! = Q(k) \times Q(k-1)!$. Because evaluating $(k-1)!$ requires cumulative multiplications, the runtime retrieves the existing hash commitment of $(k-1)!$ via $Q(\text{reuse})$, executing a single multiplication and avoiding redundant re-computation across the Memory Wall.

III. Non-Existence of Negative Factorials The factorial recurrence relation is given by:

$$(n-1)! = \frac{n!}{n}$$

Extending this relation to negative integers requires dividing by local genesis origin $Q(0)$:

$$(-1)! = \frac{0!}{0} = \frac{Q(1)}{Q(0)}$$

$$\forall k \in \mathcal{Q}^+, \quad (-k)! = \frac{Q(1)}{Q(0) \times \prod_{j=1}^{k-1}(-j)}$$

Because division by origin $Q(0)$ is an absolute constitutional prohibition in Qm (§5.3), and actual infinite poles are non-realizable under **Axiom 2**, the factorial of a negative integer cannot exist. Any attempt to compute $(\ominus k)!$ is intercepted deterministically at the SIMEMP Gateway, emitting `RECEIPT_DIVISION_BY_ZERO_REJECTED`.

---

6. Relational Decidability and Positional Ordering

Ordering on dimension d is positional, evaluated over bounded constructive intervals:

$$I(A) = [q_A \cdot \delta_z + r_A,\ q_A \cdot \delta_z + r_A + \epsilon_z]$$

$$\text{Rel}(A, B) = \begin{cases} A < B & \text{if } \sup I(A) < \inf I(B) \\ A > B & \text{if } \inf I(A) > \sup I(B) \\ A \equiv_z B & \text{if } I(A) \equiv I(B) \text{ and } r_A = r_B \\ \text{Indeterminate} & \text{if } I(A) \cap I(B) \neq \emptyset \text{ and } r_A \neq r_B \end{cases}$$

When intervals overlap such that inequality cannot be constructively decided at zoom z, Qm forbids heuristic tie-breaking. It emits the deterministic receipt `ORDERING_INDETERMINATE_AT_ZOOM`. To resolve ordering, the caller must allocate additional compute budget to deepen zoom resolution ($z \to z + \Delta z$).

---

7. Interaction with Qexpr and Derivation of Higher Mathematics

```
      UNEVALUATED SYMBOLIC AST (Layer 3)
Qexpr = ⟨ ExpressionTree, z, r, d, DCC, Identity, Lineage ⟩
                            |
                            ▼
      COMPILATION / COLLAPSE ENGINE (Totality by Budget)
Evaluated deterministically via Qm Morphisms (§5)
                            |
                            ▼
      COLLAPSED CONSTRUCTIVE LEAF (Canonical State)
⟨ σ, q_z, z, r_z, d ⟩  +  Signed Remainder Receipt
```

7.1. The Boundary Between Qexpr and Qm

- ****Qexpr** (Symbolic CAE-DAG):** Encapsulates unevaluated algebraic expressions, rational functions, and multi-step operator trees. It carries the symbolic structure before execution.
- Qm **(Evaluation Engine & Leaf):** Defines the concrete algebraic reduction rules that collapse Qexpr trees into discrete $\langle \sigma, q_z, z, r_z, d \rangle$ tuples at compilation time (**QnLang** $\to$ **QnIR**).

7.2. Derivability of Real-World Mathematics

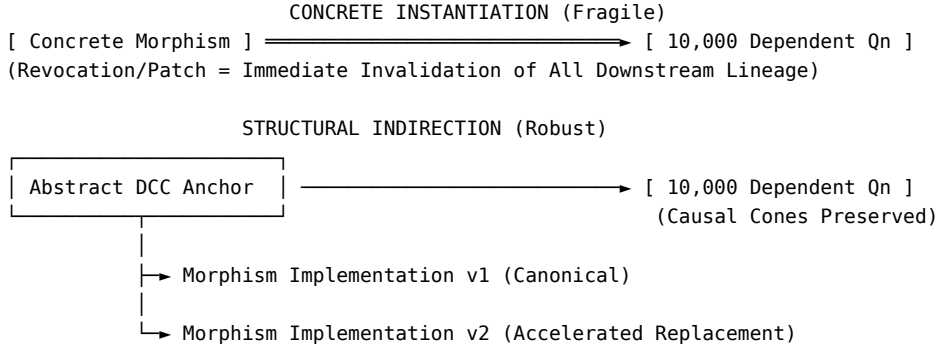
All computable real-world mathematics derives systematically from this foundation: 1. **Multi-dimensional Vector Spaces:** Constructed by parameterizing orthogonal sets of axes $\{d_1, d_2, \ldots, d_n\}$, giving rise to Clifford geometric algebras ($\mathcal{C}\ell_{p,q}$) without non-deterministic trigonometric float approximations (`docs/qm_geometry.md`). 2. **Calculus Without Infinities:** Replaces continuous infinitesimals ($\epsilon \to 0$) with discrete differences over lattice quanta $\Delta X/\delta_z(d)$, yielding exact discrete exterior calculus (`docs/qm_calculus.md`). 3. **Transcendental Constants (π, e):** Represented as symbolic Qexpr continued fraction operators (`docs/qexpr.md` §4.3). Evaluated at zoom z, they emit an exact rational quotient q_z and a conserved residual r_z satisfying exact Diophantine bounds.

---

8. Structural Indirection: Q(anchor) in Qm

To prevent cascading structural collapse across dependent mathematical workflows, high-centrality mathematical primitives are formalized as **Q(anchor)** entities (`docs/_data/glossary/architecture.yml`):

$$\chi(\alpha) = \frac{|\mathcal{C}^+(\alpha)|}{|\mathcal{V}|} \geq \Theta_\chi$$



1. **Anchored Interfaces:** Critical operations (the division constraint, sign reflection, canonical unit δ_z) expose abstract DCC contracts under **Structural Indirection**.
2. **Downstream Isolation:** Workflows in higher layers depend upon the abstract contract, not the internal execution engine.
3. **Agility Proofs:** Upgrading an arithmetic solver or patching an algorithmic vulnerability emits a signed migration receipt without invalidating the causal lineage of historical calculations.

---

9. License and Invariant Lineage

This domain specification enforces the legal and operational conditions of the **SSL** governing the **Licensed Work**:

1. ****Paternity Reference**:** All derivative implementations, mathematical solvers, or compiled runtime specification files derived from Qm must embed: > Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License (SSL). Parent Repository: https://github.com/cdqn5249/cdqn
2. **Anti-Patent Defense:** Royalty-free rights terminate automatically if a licensee initiates patent litigation regarding Qm arithmetic methods against the Author or community.
3. **Passive Infrastructure Safe Harbor:** Qm constitutes neutral computational infrastructure. It exercises no editorial inspection over, and makes no proprietary claim upon, sovereign user content or **Payloads**.

Annex Q — Qm calculus

Pinned source: docs/qm_calculus.md
Committed version: 1.1.1 **Last updated:** 2026-10-19
sha256 (committed file): `869964ef251c45e6f59e2cc5aed46f9ac50229ffb6bcb9eb3cbd032dfa3e0026`

Field	Specification
Document Title	Qm Calculus — Discrete Exterior Calculus, Continued Fractions, and Invariant Dynamics
Version	1.1.1
Last Updated	2026-10-19 (Bao Loc, Vietnam)
Author	Christophe Duy Quang Nguyen
License	Scaling Source License (SSL) 1.0
Status	Canonical Domain Specification — Category D (**Complexity Degree 3**)

Normative References

The following documents establish the physical, structural, and legal constraints governing this specification. If a technical conflict arises, `simemp.md` governs; if a structural conflict arises, `abstractionLayers.md` governs; if a legal conflict arises, `LICENSE.md` governs.

Document	Role	Target
`docs/simemp.md`	Constitutional constraints, thermodynamics, and Dependencies Determinism	simemp.html
`docs/abstractionLayers.md`	Layer architecture and Complexity Degree Stratification	abstractionLayers.html
`docs/qnPrimitive.md`	Universal Envelope, operational axioms, and optimization primitives	qnPrimitive.html
`docs/q0_q1.md`	Primary genesis origin Q(0) and first unit Q(1)	q0_q1.html
`docs/q2_q9.md`	Single-digit secondary DCC anchors and single-digit spectrum	q2_q9.html
`docs/qm.md`	Constructive numeric leaf substrate and Diophantine division constraints	qm.html
`docs/qm_geometry.md`	Multi-axial frames, Clifford geometric algebra, and float-free rotations	qm_geometry.html
`docs/qexpr.md`	Layer 3 Content-Addressed Expression DAG container	qexpr.html
`LICENSE.md`	Scaling Source License 1.0 governing the Licensed Work	LICENSE.md

1. Epistemic Stance: Calculus Without Infinities

In classical mathematics, calculus relies on non-constructive, continuous foundations: the Cauchy–Weierstrass ϵ-δ limit, actual infinities ($\mathbb{R}, \infty$), and vanishing infinitesimals ($dx \to 0$). While analytically elegant, continuous calculus creates physical paradoxes: it allows mathematical equations to predict unphysical singularities (such as infinite fluid velocities in Navier–Stokes or infinite gravitational densities in black holes) that never occur in real physical substrates.

In the **Qn** computational universe, **continuous limits are discarded as physically unrealizable approximations**. Under `docs/simemp.md`, physical computation is strictly finite and bounded by two real horizons: 1. **The Upper Horizon ($Qn_{\max}$):** The finite memory capacity, register width, and operational energy ceiling declared in the local **Metric Envelope**. 2. **The Lower Horizon**

($\delta_z(d)$ **near** Q(0))**:** The physical resolution quantum at **Zoom z**, below which state transitions dissipate into physical thermal noise bounded by Landauer's bound:

$$W \geq k_B T \ln 2$$

Qm Calculus replaces continuous infinitesimals with Constructive **Discrete Exterior Calculus (DEC)** and **Projective Continued Fraction Cascades**, satisfying **Dependencies Determinism**:

$$H(S_t \mid \mathcal{D}(S_t)) = 0$$

All operational derivations represent working hypotheses evaluated via **Computational Consistency**. No unmeasured limit may cross a **SIMEMP Gateway**, preserving Tier 1 **Existential Invariants**.

```
                    THE BOUNDED DYNAMICAL SCOPE
[ The Upper Horizon: Qn_max / Declared Metric Ceiling ]
                     ▲
                     │  CONSTRUCTIVE DYNAMICAL DOMAIN
                     │  • Simplicial Coboundary Operators (d ∘ d ≡ 0)
                     │  • Projective Continued Fractions ([a_0; a_1...])
                     │  • Exact Discrete Stokes Invariants
                     ▼
[ The Lower Horizon: Q(0) / Physical Thermal Dissipation Limit ]
```

---

2. The Discrete Replacement of Continuous Limits

Classical continuous limits ($\lim_{\Delta x \to 0}$) are replaced by three constructive mechanisms that operate with zero rounding drift under the **No-Implicit Rule**:

2.1. Scale-Lattice Refinement (z-Refinement)

Space and time are not continuous voids; they are partitioned by the discrete resolution quantum along **dimensional axes** (`docs/qm.md` §3.3):

$$\delta_z(d) = \frac{Q(1)_d}{\kappa(z)}$$

"Approaching a limit" does not mean shrinking an infinitesimal toward zero. It represents an explicit, metered **Lattice Refinement Step** ($z \to z+1$): - Refinement requires burning an explicit metric budget in units of the **Abstract Compute Unit U**. - When the declared metric budget expires, refinement terminates cleanly via **Metric Exhaustion**.

2.2. Conserved Residual Telescoping

Classical series convergence silently truncates unmeasured trailing terms. In Qm Calculus, convergence is an exact Diophantine cascade where the residual is strictly conserved:

$$r_z = \big(q_{z+1} \cdot \delta_{z+1}(d)\big) + r_{z+1}$$

Across arbitrary expansions, total metric information is preserved in the **Universal Envelope**:

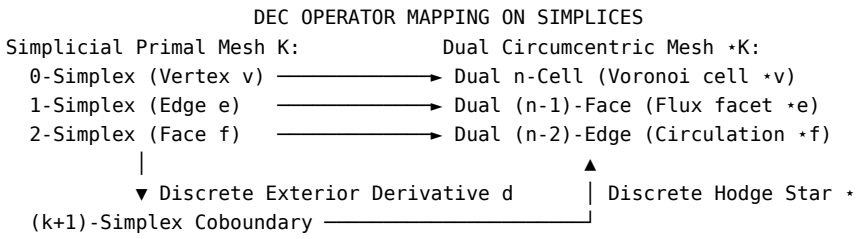
$$\mathcal{I}_{\text{total}}(X_d) = \sum_{k=0}^{z} \big(q_k \cdot \delta_k(d)\big) + r_z$$

If $r_z = Q(0)$, the system achieves **Terminal Exactness**: the calculation reaches exact algebraic closure, triggering immediate early halting with zero residual entropy ($\Delta H = 0$) and emitting an exact **Receipt**.

---

3. Discrete Exterior Calculus (DEC) on Simplicial Complexes

To model fields, waves, fluids, and electrodynamics without floating-point PDEs, Qm Calculus adopts **Discrete Exterior Calculus (DEC)** on orthogonal multi-axial frames (`docs/qm_geometry.md`).



3.1. Discrete Differential Forms as Cochains

A physical or geometric field is not a continuous function; it is a **discrete cochain** evaluated on oriented k-simplices of a simplicial complex K: * 0**-Forms (Scalars):** Potential fields, temperatures, and scalar charges evaluated on vertices. * 1**-Forms (Circulations):** Electric fields (E) and fluid velocities (u) integrated along oriented edges. * 2**-Forms (Fluxes):** Magnetic flux (B) and vorticity (ω) evaluated across oriented planar faces with conserved **residual r**.

3.2. The Discrete Exterior Derivative (d / DCC_ANCHOR_DEC_D_v1)

The discrete derivative is the algebraic coboundary operator **Discrete Exterior Derivative d** mapping k-forms to $(k+1)$-forms:

$$\langle \mathbf{d}\omega,\, \sigma_{k+1} \rangle \equiv \langle \omega,\, \partial \sigma_{k+1} \rangle = \sum_{j=0}^{k+1} (-1)^j \langle \omega,\, [v_0, \ldots, \hat{v}_j, \ldots, v_{k+1}] \rangle$$

The Fundamental Topological Invariant: Under the **Best of Choices (BOC) Policy**, **d** satisfies the exact nilpotent identity:

$$\mathbf{d} \circ \mathbf{d} \equiv Q(0)$$

Because the boundary of a boundary is identically zero ($\partial\partial \equiv \emptyset$), applying the exterior derivative twice vanishes identically. This invariant is anchored via `DCC_ANCHOR_DEC_D_v1` as a primary **Q(anchor)** governed by an explicit **DCC Profile**.

3.3. The Discrete Hodge Star Dual (DCC_ANCHOR_HODGE_v1)

The **Discrete Hodge Star** operator $\star$ maps primal k-forms to dual $(n-k)$-forms across the circumcentric dual mesh:

$$\star\omega = \sum_{\sigma_k \in K} \frac{|\star\sigma_k|}{|\sigma_k|} \langle \omega,\, \sigma_k \rangle\, (\star\sigma_k)$$

Dual volume ratio quotients:

$$\frac{|\star\sigma_k|}{|\sigma_k|}$$

represent exact geometric ratios held symbolically inside the **Qexpr** container via `OP_RATIO`.

3.4. The Discrete Laplace–Beltrami Operator (Δ / DCC_ANCHOR_LAPLACE_v1)

The Laplacian governing diffusion, wave propagation, and electrostatic potential is constructed constructively:

$$\Delta \equiv \mathbf{d}\delta + \delta\mathbf{d}, \quad \text{where } \delta = (-1)^{n(k-1)+1} \star \mathbf{d} \star$$

3.5. Exact Discrete Stokes' Theorem

Stokes' Theorem is not an asymptotic limit; it is an exact Diophantine identity on discrete meshes:

$$\sum_{K}(\mathbf{d}\omega)(K) \equiv \sum_{\partial K}\omega(\partial K)$$

Truncation error is zero by construction. Conservation laws (mass, charge, vorticity) hold identically without numerical dispersion.

---

4. Dynamical Q(anchor) Spectrum: Transcendental Continued Fractions

Transcendental numbers cannot exist as static Layer 1 digits in **Layer 1**. In Qm Calculus, they are anchored as **Deterministic Projective Continued Fraction Generators** inhabiting the Layer 3 **Qexpr** CAE-DAG container (`docs/qexpr.md` §4.3) and functioning as foundational **Q(anchor)** nodes.

```
                THE DYNAMICAL Q(anchor) SPECTRUM IN Qm CALCULUS
┌──────────────────────────────────────────────────────────────────────────┐
│ 1. TRANSCENDENTAL INVARIANT ANCHORS                                      │
│    • DCC_ANCHOR_PI_v1   : Circular ratio π, wave periods, harmonic phase │
│    • DCC_ANCHOR_TAU_v1  : Full-turn metric constant τ = 2π               │
│    • DCC_ANCHOR_E_v1    : Natural base e, solution to df/dx = f          │
│    • DCC_ANCHOR_LN2_v1  : Natural logarithm ln(2), Landauer dissipation  │
├──────────────────────────────────────────────────────────────────────────┤
│ 2. DISCRETE EXTERIOR CALCULUS (DEC) OPERATOR ANCHORS                     │
│    • DCC_ANCHOR_DEC_D_v1    : Discrete Exterior Derivative d (d² ≡ 0)    │
│    • DCC_ANCHOR_HODGE_v1    : Discrete Hodge Star Dual ⋆ (Metric contraction)│
│    • DCC_ANCHOR_LAPLACE_v1  : Discrete Laplace–Beltrami Operator Δ = dδ + δd│
├──────────────────────────────────────────────────────────────────────────┤
│ 3. DYNAMICAL FIELD & WAVE ANCHORS                                        │
│    • DCC_ANCHOR_WAVE_v1     : Discrete Wave D'Alembertian Operator □     │
│    • DCC_ANCHOR_FOURIER_v1  : Rational Discrete Fourier Transform (DFT)  │
│    • DCC_ANCHOR_STOKES_v1   : Exact Coboundary Conservation ⟨∂σ, ω⟩ ≡ ⟨σ, dω│
└──────────────────────────────────────────────────────────────────────────┘
```

4.1. The Circular Ratio (π / DCC_ANCHOR_PI_v1)

Governs circular phase rotations, Fourier analysis, and wave periods:

$$\pi = [3;\, 7,\, 15,\, 1,\, 292,\, 1,\, 1,\, 1,\, 2,\, 1,\, 3,\, 1,\, 14,\, \ldots]$$

Evaluating π to zoom level z generates partial convergents $\frac{p_k}{q_k}$ bounded by strict residual conservation:

$$0 \le |r_k| < \frac{1}{q_k q_{k+1}}$$

4.2. The Turn Constant (τ / DCC_ANCHOR_TAU_v1)

Defined as the full-turn angular period:

$$\tau = Q(2) \times \pi = [6;\, 3,\, 1,\, 1,\, 7,\, 2,\, 146,\, \ldots]$$

Eliminates arbitrary factor-of-two fractions in spatial rotor transformations (`docs/qm_geometry.md`).

4.3. The Natural Exponential Base (e / DCC_ANCHOR_E_v1)

Governs exponential damping, thermal relaxation, and solutions to discrete differential equations ($\mathbf{d}f = f$):

$$e = [2;\, 1,\, 2,\, 1,\, 1,\, 4,\, 1,\, 1,\, 6,\, 1,\, 1,\, 8,\, \ldots,\, 1,\, 1,\, 2k,\, \ldots]$$

4.4. The Thermodynamic Logarithm (ln 2 / DCC_ANCHOR_LN2_v1)

Directly calibrates the physical **Layer 0** Landauer dissipation limit:

$$\ln(2) = [0;\, 1,\, 2,\, 3,\, 1,\, 6,\, 3,\, 1,\, 1,\, 2,\, 1,\, 1,\, 4,\, \ldots]$$

Guarantees platform-invariant accounting when translating physical energy consumption into **U** units.

---

5. Lineage, Aperiodicity, and the Anchor-Cut in Time Evolution

In continuous mechanics, time is treated as an unmeasured parameter ($t \in \mathbb{R}$) that allows algorithms to discard initial states. In Qm Calculus, time evolution is the **strictly monotonic progression of discrete causal state transitions along the** Causal Arrow****.

```
                        THE ANCHOR-CUT THEOREM
Full Micro-Step Lineage (O(N) Replay - Rejected by SIMEMP):
  Q(0) ⟶ s_1 ⟶ s_2 ⟶ s_3 ⟶ ... ⟶ s_999,999 ⟶ S_terminal
  (Verifying every micro-step violates Memory Wall & wastes gigabytes of RAM)

The Anchor-Cut Verification (O(K) Verification - BOC):
  Q(0)_N ⟹⟹⟹ Q(anchor)_1 ⟹⟹⟹ Q(anchor)_2 ⟹⟹⟹ S_terminal
  [Silicon Root]    [Receipt R_1]        [Receipt R_2]         [Current State]
  • Replay between anchors: ELIMINATED!
  • Verifier only checks: H(Q(0)) + Chain of Signed Anchor Receipts + H(S_terminal).
  • Complexity: O(K) where K □ N.
```

5.1. Formal Causal Lineage of a Dynamic State

Every field state S_k generated during dynamical evolution possesses an explicit lineage tuple in its **Universal Envelope** protected by the **Security Envelope**:

$$\mathcal{L}(S_k) = \left\langle \mathbf{P}(S_k),\, \kappa(S_k),\, \tau_{\text{type}},\, \mathcal{W}_{\text{cum}}(S_k),\, \Pi_{\text{receipts}}(S_k) \right\rangle$$

- **Parent Commitment Set (P):** References the preceding field state S_{k-1} and the applied operator contract $\mathcal{M}_{\mathbf{d}}$.
- **Monotonic Causal Index (κ):** Increments deterministically with each discrete timestep (Δt).
- **Cumulative Work ($\mathcal{W}_{\text{cum}}$):** Tracks the total Landauer work expended across the simulation run in units of $Q(1)$.

5.2. Aperiodic Memory Trajectory

Under `DCC_ANCHOR_Q5_APERIODICITY_v1` (`docs/q2_q9.md`), dynamical simulations avoid circular recurrence locks. The trajectory through **Local-first** memory space is non-repeating:

$$\forall i \neq j, \quad \mathcal{H}(S_i) \neq \mathcal{H}(S_j)$$

This guarantees an irreversible internal arrow of time without consulting external clocks.

5.3. The Anchor-Cut Verification Invariant

To audit or verify a dynamical simulation consisting of N discrete timesteps passing through K intermediate $Q(\text{anchor})$ checkpoints ($K \ll N$) under the **Anchor-Cut Invariant**: 1. An auditor does not replay the N micro-steps across volatile memory. 2. The auditor verifies the **Anchor-Cut Chain**: the local origin **Q(0)**, the signed anchor receipts $\mathcal{R}_{\text{anchor}}$, and the terminal state S_N. 3. Verification complexity collapses from $\mathcal{O}(N)$ memory bus traversals to $\mathcal{O}(K)$ receipt validations. Micro-states between anchors are safely archived or dereferenced via **Q(reuse)**.

---

6. Coupling to Optimization Morphisms: Q(bypass) and Q(reuse)

In classical scientific simulations, processors waste extensive energy re-evaluating terms that are algebraically zero. Qm Calculus couples DEC directly to Category B optimization morphisms under the **BOC Policy** to maximize **Operational Agility**:

```
                Q(bypass) IN DISCRETE EXTERIOR CALCULUS
┌──────────────────────────────────────────────────────────────────────────┐
│ 1. THE POINCARÉ COBOUNDARY BYPASS (d² ≡ 0)                               │
│    • Mathematical Invariant: d ∘ d ≡ Q(0) ("Boundary of a boundary is zero")│
│    • Classical Engine      : Evaluates mesh derivatives twice (O(N) work).  │
│    • Q(bypass) Execution   : Gateway intercepts d(dω), immediately emits     │
│                              RECEIPT_BYPASS_IDENTITY at ΔW = 0 cost!         │
├──────────────────────────────────────────────────────────────────────────┤
│ 2. CONSERVATIVE FIELD FLUX BYPASS (Stokes Collapse)                      │
│    • If a differential form is closed (dω = Q(0)), any boundary integral    │
│      ∫_(∂K) ω collapses algebraically to Q(0).                           │
│    • Q(bypass) skips numerical mesh integration entirely.                │
├──────────────────────────────────────────────────────────────────────────┤
│ 3. PROJECTIVE SERIES EARLY HALTING (Terminal Exactness)                  │
│    • In continued fraction expansions or Taylor series, if r_k = Q(0),       │
│      Q(bypass) halts all remaining terms, saving 100% of remaining budget.  │
└──────────────────────────────────────────────────────────────────────────┘
```

6.1. Poincaré Nilpotent Bypass

When an expression graph attempts to evaluate the curl of a gradient or the divergence of a curl ($\mathbf{d}(\mathbf{d}\omega)$): - The **SIMEMP Gateway** matches the identity proof hash of `DCC_ANCHOR_DEC_D_v1`. - **Q(bypass)** halts evaluation in $\mathcal{O}(1)$ time and emits `RECEIPT_BYPASS_IDENTITY`. - Zero compute budget is burned ($\Delta W = 0$).

6.2. Closed Form Stokes Bypass

If a differential form is algebraically closed ($\mathbf{d}\omega \equiv Q(0)$), boundary circulation integrals collapse to zero without numerical mesh iteration:

$$\sum_{\partial K} \omega \equiv Q(0)$$

6.3. Simplicial Mesh Deduplication via Q(reuse)

During spatial evolution, static geometric mesh configurations (dual metric tensors, Hodge star ratio matrices) are referenced across time steps via their immutable content commitment hashes using **Q(reuse)**, eliminating redundant bit allocation across the Memory Wall.

---

7. Thermodynamic Governance and Singularity Immunity

In classical continuous fluid mechanics, the Navier–Stokes equations allow solutions to develop unphysical mathematical singularities (finite-time velocity blowups $\|u\|_{L^\infty} \to \infty$).

Qm Calculus provides **structural immunity against physical singularities**: 1. **Lattice Cutoff:** Spatial velocity cannot diverge because spatial gradients are bounded by the discrete scale quantum $\delta_z(d)$. Space cannot be divided infinitely. 2. **Thermodynamic Viscous Dissipation:** Physical fluids dissipate energy into microscopic thermal noise. In Qm Calculus, viscosity is modeled as an explicit dissipative sink exporting operational entropy via signed **receipts**. 3. **Totality by Budget (`docs/qnPrimitive.md` Axiom 6):** Time evolution is allocated a finite work budget in units of **Q(1)**. An unphysical blowup is impossible: before any field value can exceed physical bounds, the budget exhausts and the engine halts cleanly with `RECEIPT_BUDGET_EXHAUSTED`.

---

8. License and Invariant Lineage

This domain specification enforces the legal and operational conditions of the **SSL** governing the **Licensed Work** across the **cdqn** network:

1. ****Paternity Reference**:** All discrete exterior calculus solvers, continued fraction libraries, and dynamical simulators derived from this specification must embed: > Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License (SSL). Parent Repository: `https://g`ithub.com/cdqn5249/cdqn
2. **Anti-Patent Defense:** Royalty-free rights terminate automatically if a licensee initiates patent litigation regarding Qm discrete calculus methods or DEC operator anchors against the Author or community under the **Open Core Invariants**.
3. **Statutory Safe Harbor:** Qm Calculus operates as passive, neutral mathematical infrastructure. It exercises no editorial inspection over, and makes no claim upon, external simulation models, physical datasets, or sovereign **Payloads**.

Annex R — Qm geometry

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Committed version: 1.2.3 **Last updated:** 2026-10-20
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Field	Specification
Document Title	Qm Geometry — Multi-Axial Frames and Geometric Algebra
Version	1.2.3
Last Updated	2026-10-20 (Bao Loc, Vietnam)
Author	Christophe Duy Quang Nguyen
License	Scaling Source License (SSL) 1.0
Status	Canonical Domain Specification — Category B/D (**Complexity Degree 2**)

Normative References

The following documents establish the physical, structural, and legal constraints governing this specification. If a technical conflict arises, `simemp.md` governs; if a structural conflict arises, `abstractionLayers.md` governs; if a legal conflict arises, `LICENSE.md` governs. References are deliberately version-unpinned: each target document's front matter carries its own authoritative version, so this specification never duplicates drift-prone patch state.

Document	Role	Target
`docs/simemp.md`	Constitutional constraints, thermodynamics, and Dependencies Determinism	simemp.html
`docs/abstractionLayers.md`	Layer architecture and SIMEMP Gateway validation	abstractionLayers.html
`docs/qnPrimitive.md`	Universal Envelope, operational axioms, and Optimization Morphisms	qnPrimitive.html
`docs/q0_q1.md`	Primary genesis origin Q(0) and first unit Q(1)	q0_q1.html
`docs/q2_q9.md`	Single-digit secondary DCC anchors and single-digit spectrum	q2_q9.html
`docs/qm.md`	Constructive numeric leaf substrate and Diophantine division constraints	qm.html
`docs/qm_calculus.md`	Discrete Exterior Calculus, continued fractions, and invariant dynamics	qm_calculus.html
`docs/qexpr.md`	Layer 3 Content-Addressed Expression DAG container	qexpr.html
`docs/qs.md`	Constructive relational intent and Intent Contracts	qs.html
`LICENSE.md`	Scaling Source License 1.0 governing the Licensed Work	LICENSE.md

1. Epistemic Stance and Dimensional Generalization

The foundational operational conjecture of the **Qn** framework (`docs/simemp.md` §1) asserts that a discrete number system can abstract any computable physical phenomenon under **SIMEMP** constraints. In `docs/qm.md`, scalar arithmetic was established along the 1D genesis axis **d_1**. However, physical reality and distributed network topographies are inherently multi-dimensional.

Classical geometry relies on continuous coordinate spaces ($\mathbb{R}^n$) and irrational trigonometric functions ($\sin, \cos$), introducing platform-dependent IEEE 754 rounding approximations that violate **Axiom 10** of `docs/qnPrimitive.md`.

Qm Geometry generalizes the 1D leaf state to n-dimensional space via **Clifford Geometric Algebra** ($\mathcal{C}\ell_{p,q}$). Space, direction, area, volume, and rotation are constructed as **exact, discrete, rational multivector blades** without continuous limits, actual infinities, or transcendental float approximations.

Every geometric operation in Qm satisfies **Dependencies Determinism**:

$$H(S_t \mid \mathcal{D}(S_t)) = 0$$

Operational validity is verified empirically via **Computational Consistency**. No unmeasured coordinate entity may cross an abstraction boundary or **SIMEMP Gateway**.

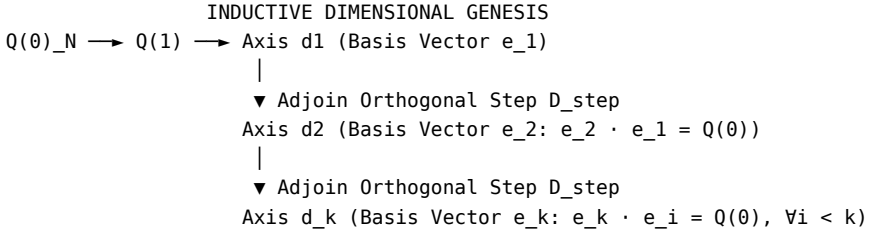
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2. Orthogonal Coordinate Frames and Metric Signatures

In strict accordance with the **No-Implicit Rule**, multi-dimensional space cannot be assumed *a priori*. It is generated inductively from genesis origin `Q(0)` and first unit `Q(1)`.

2.1. The Orthogonal Basis Set and Inductive Dimensional Genesis

Under **Axiom 4** (Local Genesis Dependency), every coordinate axis must trace an unbroken causal lineage along the **Causal Arrow** back to **Local-first** genesis origin $Q(0)_N$. Coordinate frames cannot be introduced as detached ambient conventions.



Primary Genesis Axis (d_1) Formally instantiated in **Layer 1** by the minimal directed state transition from local origin $Q(0)_N$ to first unit $Q(1)$ along the canonical initial dimension, calibrating the **Abstract Compute Unit U** (`docs/q0_q1.md` §4):

$$\vec{e}_1 \equiv \langle \oplus,\, 1,\, z = 0,\, r = Q(0),\, d_1 \rangle$$

Here, the vector symbol $\vec{e}$ denotes a directional spatial basis blade (from the unit vector convention *Einheitsvektor*), completely distinct from scalar transcendental constants.

Inductive Dimensional Extension ($\mathcal{D}_{\text{step}}$) For any spatial frame of dimension $k \geq 1$, the successor orthogonal axis d_{k+1} is instantiated constructively by adjoining an orthogonal basis generator $\vec{e}_{k+1}$:

$$\vec{e}_{k+1} \equiv \langle \oplus,\, 1,\, z = 0,\, r = Q(0),\, d_{k+1} \rangle$$

subject to explicit Diophantine orthogonality constraints:

$$\forall i \in \{1, \ldots, k\}, \quad \vec{e}_{k+1} \cdot \vec{e}_i = Q(0)$$

$$\vec{e}_{k+1} \cdot \vec{e}_{k+1} = \eta_{k+1,k+1} Q(1)$$

$$\text{Parent}(\vec{e}_{k+1}) = \{\vec{e}_k,\, Q(1)\}$$

By induction, a frame of dimension n is defined by the complete discrete set of n mutually orthogonal **dimensional axes**:

$$\mathcal{D}_{\text{frame}} = \{d_1,\, d_2,\, \ldots,\, d_n\}$$

Zero implicit dimensional assumptions exist; each axis preserves a deterministic causal parentage back to $Q(0)_N$.

2.2. Metric Signature (η)

The quadratic form of the space is governed by an explicit discrete diagonal metric signature $\eta = (p, q, s)$ where $p + q + s = n$:

$$\vec{e}_i \cdot \vec{e}_j = \begin{cases} +Q(1) & \text{if } i = j \leq p \quad \text{(Space-like)} \\ -Q(1) & \text{if } p < i = j \leq p + q \quad \text{(Time-like)} \\ Q(0) & \text{if } p + q < i = j \leq n \quad \text{(Degenerate / Null)} \\ Q(0) & \text{if } i \neq j \quad \text{(Orthogonality)} \end{cases}$$

- **Euclidean n-Space:** $\mathcal{C}\ell_{n,0}$ where $\eta = (+, +, \ldots, +)$.
- **Space-Time Algebra (Minkowski):** $\mathcal{C}\ell_{1,3}$ or $\mathcal{C}\ell_{3,1}$, providing spatial foundations for physical modeling.

2.3. Non-Euclidean Topologies and Subspace Inclusions

Qm Geometry abstracts non-Euclidean geometries without continuous Riemannian metrics, dense floating-point tensors, or differential singularities:

Geometry Subspace	Ambient Signature	Defining Diophantine Constraint	Geodesic Rotor Morphism
Spherical ($\mathbb{S}^n$)	$\mathcal{C}\ell_{n+1,0}$ (Euclidean)	$X \cdot X = +Q(1)$	Planar bivector rotors ($R\vec{v}R^\dagger$)
Hyperbolic ($\mathbb{H}^n$)	$\mathcal{C}\ell_{n,1}$ (Minkowski)	$X \cdot X = -Q(1), \quad X_t > Q(0)$	Hyperbolic boost rotors ($B\vec{v}B^\dagger$)
Conformal (CGA)	$\mathcal{C}\ell_{n+1,1}$ (Minkowski null)	$X \cdot X = Q(0)$ (Null cone)	Linear conformal rotors over $\{e_{\text{origin}}, e_{\text{horizon}}\}$
Curved Manifolds	Simplicial triangulation	Bivector holonomy $\Omega = \prod R_{\text{loop}}$	Discrete Regge deficit angles without continuous tensors

I. Spherical Subspace ($\mathbb{S}^n$) Modeled as the locus of vectors satisfying the quadratic constraint $X \cdot X = +Q(1)$ in Euclidean space $\mathcal{C}\ell_{n+1,0}$. Great-circle geodesics are evaluated via rational bivector rotors without trigonometric projections.

II. Hyperbolic Subspace ($\mathbb{H}^n$) Modeled via the Weierstrass hyperboloid within Minkowski spacetime $\mathcal{C}\ell_{n,1}$ under constraint $X \cdot X = -Q(1)$, with temporal coordinate $X_t > Q(0)$. Parallel transport is governed by hyperbolic boost rotors $(\vec{e}_i \wedge \vec{e}_t)^2 = +Q(1)$.

III. Conformal Geometric Algebra (CGA) and Null Basis Decoupling Conformal and projective transformations are mapped to linear orthogonal rotor reflections by adjoining two discrete null basis vectors: * $\vec{e}_{\text{horizon}}$: Parameterizes the operational task horizon ceiling $Qn_{\max}$ declared in the metric envelope. * $\vec{e}_{\text{origin}}$: Parameterizes the spatial coordinate origin within the Clifford frame.

$$\vec{e}_{\text{origin}} \cdot \vec{e}_{\text{origin}} = Q(0), \quad \vec{e}_{\text{horizon}} \cdot \vec{e}_{\text{horizon}} = Q(0), \quad \vec{e}_{\text{origin}} \cdot \vec{e}_{\text{horizon}} = -Q(1)$$

The Invariant Disjunction: Crucially, the basis vector $\vec{e}_{\text{origin}}$ is a grade-1 vector blade in the vector space V. It is strictly decoupled from the local scalar genesis origin $Q(0)$, which resides in the underlying discrete scalar ring $\mathcal{Q}_0$. Conflation between vector basis elements and ring zero scalars is strictly prohibited under the **No-Implicit Rule**.

IV. Discrete Curved Manifolds General curved spaces are represented as simplicial complexes (discrete Regge calculus). Curvature is measured not by continuous Ricci tensors, but by **discrete deficit angles** around codimension-2 hinges.

---

3. The Clifford Geometric Product Without Approximations

In Qm Geometry, the multiplication of two vectors $\vec{u}$ and $\vec{v}$ is governed by the **Clifford Geometric Product**:

$$\vec{u}\vec{v} = \vec{u} \cdot \vec{v} + \vec{u} \wedge \vec{v}$$

Product Component	Grade	Algebraic Symmetry	Geometric Role	Evaluation Engine
Inner Product ($\vec{u} \cdot \vec{v}$)	Grade 0 (Scalar)	Symmetric: $\frac{1}{2}(uv + vu)$	Metric contraction / projection	Qm $\mathcal{R}$-algebra scalar leaf
Wedge Product ($\vec{u} \wedge \vec{v}$)	Grade 2 (Bivector)	Anti-symmetric: $\frac{1}{2}(uv - vu)$	Oriented planar area patch	Canonical bivector blade
Geometric Product ($\vec{u}\vec{v}$)	Multivector	Graded associative sum	Total spatial interaction	Complete Clifford blade set

3.1. The Discrete Inner Product

Let vectors $\vec{u}$ and $\vec{v}$ be expressed along the orthogonal basis:

$$\vec{u} = \sum_{k=1}^{n} u_k \vec{e}_k, \quad \vec{v} = \sum_{k=1}^{n} v_k \vec{e}_k$$

where each component is a discrete tuple evaluated at **zoom z**. The inner product contracts to an exact scalar:

$$\vec{u} \cdot \vec{v} = \sum_{k=1}^{n} \eta_{kk}(u_k \times v_k)$$

The operation executes entirely via the discrete Qm $\mathcal{R}$-algebra (`docs/qm.md` §4), returning an exact scalar leaf with a conserved **Remainder r**.

3.2. The Discrete Wedge Product

The outer product spans an oriented planar surface area:

$$\vec{u} \wedge \vec{v} = \sum_{1 \le i < j \le n} (u_i v_j - u_j v_i)(\vec{e}_i \wedge \vec{e}_j)$$

Under the **No-Implicit Rule**, anti-commutativity and nilpotency are exact:

$$\vec{e}_i \wedge \vec{e}_j = -(\vec{e}_j \wedge \vec{e}_i)$$

$$\vec{e}_i \wedge \vec{e}_i = Q(0)$$

Collinear vectors span zero area without precision loss.

---

4. Graded Multivectors and Blade Decomposition

A general geometric artifact $M \in \mathcal{C}\ell_{p,q}$ is a graded multivector spanning 2^n basis elements:

$$M = \sum_{k=0}^{n} \langle M \rangle_k = \langle M \rangle_0 + \langle M \rangle_1 + \langle M \rangle_2 + \cdots + \langle M \rangle_n$$

Grade	Canonical Entity	Geometric Object	Canonical Basis Representation
0	Scalar	Metric magnitude / distance	$Q(1)$
1	Vector	Directed spatial segment	$\vec{e}_i$
2	Bivector	Oriented planar area patch	$\vec{e}_i \wedge \vec{e}_j$
3	Trivector	Oriented spatial volume cell	$\vec{e}_i \wedge \vec{e}_j \wedge \vec{e}_k$
n	Pseudoscalar	Maximum oriented volume element	$I = \vec{e}_1 \wedge \vec{e}_2 \wedge \cdots \wedge \vec{e}_n$

4.1. Canonical Multivector State Tuple

Every multivector artifact in Qm carries an explicit, finite representation conforming to the **Universal Envelope**:

$$M \equiv \left\langle \sigma_M,\, \alpha_{\mathcal{B}},\, z,\, r_M,\, \mathcal{D}_{\text{frame}} \right\rangle$$

Tuple Component	Symbolic Form	Operational Invariant
Basis Blade Set	$\mathcal{B} = \bigcup_{k=0}^{n} \binom{\mathcal{D}_{\text{frame}}}{k}$	Complete canonical set of 2^n blade generators.
Blade Coefficients	$\alpha_{\mathcal{B}} = \{q_B \in \mathbb{N}\}$	Discrete scalar magnitudes at zoom z for each blade $B \in \mathcal{B}$.
Residual Vector	$r_M = \{r_B\}$	Vector of conserved remainders satisfying $0 \leq \lvert r_B \rvert < \delta_z(d)$.
Coordinate Frame	$\mathcal{D}_{\text{frame}}$	Declared set of n orthogonal axes $\{d_1, \ldots, d_n\}$.

4.2. Pseudoscalar Duality

The unit pseudoscalar $I = \vec{e}_1 \wedge \vec{e}_2 \wedge \cdots \wedge \vec{e}_n$ provides discrete Hodge dual mapping without matrix inversion:

$$M^* = MI^{-1}$$

mapping grade-k blades to grade-$(n-k)$ orthogonal complements.

---

5. Float-Free Rotations: Rational Rotors and Double Reflections

In conventional graphics and physics simulations, rotations are parameterized via Euler angles or trigonometric unit quaternions:

$$R = \cos(\theta/2) - I\sin(\theta/2)$$

Because $\cos(\theta)$ and $\sin(\theta)$ are transcendental irrationals for almost all rational angles, continuous engines round them to 32-bit or 64-bit IEEE 754 floats. This introduces metric drift: after 10^6 rotations, $\|R\|^2 \neq 1$, forcing artificial normalization cycles.

Qm Geometry solves this via **Cartan-Dieudonné Double Reflections** and **Rational Cayley-Klein Rotors**.

Rotation Method	Mathematical Formulation	Precision and Drift Invariant
Classical Floating Quaternion	$R = \cos(\theta/2) - I\sin(\theta/2)$	Lossy; accumulates metric drift ($\|R\|^2 \neq 1$); violates Axiom 10.
Cartan-Dieudonné Double Reflection	$R = \vec{b}\vec{a} = \vec{b}\cdot\vec{a} + \vec{b}\wedge\vec{a}$	Exact; composite of two discrete planar reflections.
Rational Cayley Rotor	$R = \frac{Q(1)-B}{Q(1)+B} = \frac{(m^2-n^2)+2mnB}{m^2+n^2}$	Strict unitary norm $\|R\|^2 \equiv Q(1)$; zero trigonometric drift.
Projective Angle Cascade	$\theta_z = \theta_{\text{discrete}} + r_\theta$	Projective rational convergence via continued fractions; r_θ conserved.

5.1. The Rotor Morphism

A rotation of vector $\vec{v}$ in a plane is executed by two successive reflections across non-parallel unit vectors $\vec{a}$ and $\vec{b}$:

$$R = \vec{b}\vec{a} = \vec{b}\cdot\vec{a} + \vec{b}\wedge\vec{a}$$

$$\vec{v}' = R\vec{v}R^\dagger$$

where $R^\dagger = \vec{a}\vec{b}$ is the reverse rotor.

5.2. Rational Cayley Parameterization (Float-Free Rotors)

To rotate by an exact rational angle without evaluating trigonometric series, Qm parameterizes rotors via rational bivectors $B \in \langle M \rangle_2$:

$$R = \frac{Q(1) - B}{Q(1) + B}$$

For any planar rotation in plane $\vec{e}_1 \wedge \vec{e}_2$, the rotor coefficients are generated by Diophantine Pythagorean triplets $(m, n) \in \mathbb{N}^2$:

$$R = \frac{(m^2 - n^2) + 2mn(\vec{e}_1 \wedge \vec{e}_2)}{m^2 + n^2}$$

The norm of the rotor is identically unity by construction:

$$RR^\dagger = \frac{(m^2 - n^2)^2 + (2mn)^2}{(m^2 + n^2)^2} = \frac{(m^2 + n^2)^2}{(m^2 + n^2)^2} \equiv Q(1)$$

Metric drift is identically zero. The rotor can be applied 10^9 consecutive times without departing from the unit manifold.

5.3. Projective Cascade for Arbitrary Target Angles

When an arbitrary target angle θ is externally supplied, Qm does not invoke floating-point libraries. It evaluates the continued fraction expansion of $\tan(\theta/4)$, generating a sequence of rational rotors R_z at zoom level z:

$$\theta_z = \theta_{\text{discrete}} + r_\theta$$

The angular remainder r_θ is preserved in the metric envelope. If $r_\theta = Q(0)$, the rotation is algebraically exact, emitting a **terminal receipt** confirming **Terminal Exactness**.

---

6. Discrete Metric Norms and Diophantine Distance Solvers

Distance between two spatial points A and B along frame $\mathcal{D}_{\text{frame}}$ is evaluated as the norm of the displacement vector $\vec{v} = B - A$:

$$D^2 = \vec{v} \cdot \vec{v} = \sum_{k=1}^{n} \eta_{kk} \Big((q_{B,k} - q_{A,k}) \cdot \delta_z \Big)^2$$

6.1. The Diophantine Square-Root Constraint

In Qm, distance evaluation resolves through an exact Diophantine square constraint:

$$D^2 = (q_D \cdot \delta_z)^2 + r_D$$

subject to integer quotient and bounded remainder criteria:

$$q_D \in \mathbb{N}, \quad 0 \leq |r_D| < (2q_D \cdot \delta_z^2 + \delta_z^2), \quad \sigma = \oplus$$

If $r_D = Q(0)$, the distance is an exact Pythagorean integer or rational, terminating with zero residual. If $r_D \neq Q(0)$, r_D is preserved as the exact argument for subsequent scale expansion ($z \to z + 1$).

---

7. Deferred Symbolic Normalization and Bounded Collapse Engine

A governing distinction between Qm Geometry and classical numerical engines is the **rejection of eager evaluation** (`docs/abstractionLayers.md` §6.3).

In classical systems, every intermediate geometric operation eagerly rounds to machine floats, accumulating precision drift and repeatedly dissipating Landauer heat ($W \geq k_B T \ln 2$). Qm Geometry executes geometric workflows through a **Three-Stage Lifecycle**:

Execution Stage	Abstraction Layer	Operational Mechanics	Thermodynamic Work Bound
Stage 1: Composition	Layer 3 (**Qexpr** CAE-DAG)	Multivector products and rotor chains authored as content-addressed DAGs. Zero rounding.	$W = 0$ (Information preserved)
Stage 2: Normalization	Layer 2/3 (Rewrite Engine)	Contract basis metrics, annihilate nilpotencies ($\vec{e}_i \wedge \vec{e}_i = Q(0)$), cancel inverse rotors ($RR^\dagger = Q(1)$).	$W = 0$ (Isomorphic & reversible)
Stage 3: Collapse	Layer 1/2 (Qm Leaf / Flat IR)	Project onto zoom z only upon boundary demand; compute quotient q_z and remainder r_z.	$W = 1 \cdot W_{\text{collapse}} \ll k \cdot W_{\text{eager}}$

7.1. Thermodynamic Landauer Minimization Proof

Let a geometric sequence consist of k consecutive rotor transformations: 1. **Eager Evaluation:** Dissipates $W_{\text{eager}} \geq k \cdot (k_B T \ln 2)$ and compounds k unmeasured remainder vectors across the **Layer 0** substrate. 2. **Deferred Collapse:** Under the **Best of Choices (BOC) Policy**, symbolic structure is maintained as a **Content-Addressed Expression DAG (Term DAG)** with cryptographic hash-consing. Reversible symbolic reduction preserves information without entropy production ($W_{\text{Stage 2}} = 0$). Numerical collapse occurs once at the boundary:

$$W_{\text{deferred}} = 1 \cdot W_{\text{collapse}} \ll k \cdot W_{\text{collapse}}$$

7.2. Coupling to Optimization Morphisms and Causal Priority

Geometric pipelines interface with higher-order Category B optimization morphisms governed by strict causal precedence and thermodynamic breakeven constraints (`docs/qnPrimitive.md` §5.9):

$$Q(\text{bypass}) \prec Q(\text{reuse}) \prec \text{Direct Compute}$$

1. **Identity Pre-Check (Q(bypass)):**
   Before initiating memory traversal, the engine inspects operand pairs in $\mathcal{O}(1)$ register time. If an algebraic identity is recognized—such as rotor cancellation ($RR^\dagger \to Q(1)$) or outer product nilpotency ($\vec{e}_i \wedge \vec{e}_i \to Q(0)$)—the gateway emits `RECEIPT_BYPASS_IDENTITY`, bypassing both computation and memory retrieval at zero work cost ($\Delta W = 0$).
2. **Thermodynamic Breakeven for Sub-Graph Reuse (Q(reuse)):**
   Querying a hash commitment across memory hierarchies incurs physical bus dissipation ($\mathcal{O}(CV^2 f)$). Under the Metric Invariant, $Q(\text{reuse})$ is admissible if and only if the computational work of re-evaluating the multivector product exceeds lookup dissipation:

$$\text{Admit}\big(Q(\text{reuse})\big) \iff \Delta W_{\text{compute}} > \Delta W_{\text{lookup}} + \epsilon_{\text{overhead}}$$

For complex Clifford rotor sandwiches ($R\vec{v}R^\dagger$) or high-dimensional grade contractions, $Q(\text{reuse})$ references existing immutable commitment hashes, eliminating redundant allocation across the Memory Wall. For trivial 1D scalar additions, direct computation executes immediately without hash-table overhead.

---

8. SIMEMP Governance and Metric Exhaustion

Operating in multi-dimensional space consumes physical computational resources governed by the **Metric Existential Invariant** (`docs/simemp.md` §3.1), balancing algorithmic **Efficiency** against structural Modularity.

8.1. Complexity Scaling

- Vector addition: $\mathcal{O}(n)$ discrete operations.
- Geometric product: $\mathcal{O}(2^n)$ blade multiplications.
- Rotor sandwich transformation ($RvR^\dagger$): $\mathcal{O}(n \cdot 2^n)$ operations.

8.2. Dimensional Ceilings and Budget Halts

Dimensional growth is bounded by the declared **Metric Envelope**:

$$n \leq \text{Dim}_{\text{max}}$$

Typically $n \leq 4$ for space-time kinematics; $n \leq 16$ for high-dimensional semantic spaces in **Qs**.

- If a composition of wedge products attempts to construct a grade exceeding n, the result collapses identically to zero ($Q(0)$) by the nilpotency of outer products, terminating in $\mathcal{O}(1)$ work.
- If an algebraic transformation exceeds the declared metric budget, execution halts deterministically under **Axiom 6**, emitting an explicit budget exhaustion receipt:

Receipt Field	Specification
Receipt Identifier	`RECEIPT_DIMENSIONAL_BUDGET_EXHAUSTED`
Execution Status	`BUDGET_EXHAUSTED`
Frame Dimension	Bound n
Consumed Metric	Consumed work budget W ($Q(1)$ units via **Metric Exhaustion**)

---

9. License and Invariant Lineage

This domain specification enforces the legal and operational conditions of the **SSL** governing the **Licensed Work** across the **cdqn** network:

1. ****Paternity Reference**:** All multi-dimensional solvers, Clifford algebra engines, rotor transformations, or compiled geometric kernels derived from this document must embed: > Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License (SSL). Parent Repository: `https://github.com/cdqn5249/cdqn`
2. **Anti-Patent Defense:** Royalty-free rights terminate automatically if a licensee initiates patent litigation regarding Qm geometric algorithms or float-free rotor methods against the Author or community.
3. **Statutory Safe Harbor:** Qm Geometry operates as passive, neutral mathematical infrastructure. It exercises no editorial inspection over, and makes no claim upon, external graphical models, physical simulations, or sovereign **Payloads**.

Annex S — Qphy

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Committed version: 1.0.5 **Last updated:** 2026-10-19
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Field	Specification
Document Title	Qphy — Quang Physics: Discrete Field Dynamics, Computational Thermodynamics, and Physical Entropy Accounting
Version	1.0.5
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Author	Christophe Duy Quang Nguyen
License	Scaling Source License (SSL) 1.0
Status	Candidate Domain Specification — Category D (**Complexity Degree 3**)

Normative References

The following documents establish the physical, structural, and legal constraints governing this specification. If a technical conflict arises, `simemp.md` governs; if a structural conflict arises, `abstractionLayers.md` governs; if a legal conflict arises, `LICENSE.md` governs. References are deliberately version-unpinned: each target document's front matter carries its own authoritative version, so this specification never duplicates drift-prone patch state.

Document	Role	Target
`docs/simemp.md`	Constitutional constraints, thermodynamics, and Dependencies Determinism	simemp.html
`docs/abstractionLayers.md`	Layer architecture and SIMEMP Gateway validation	abstractionLayers.html
`docs/qnPrimitive.md`	Universal Envelope, operational axioms, and Optimization Morphisms	qnPrimitive.html
`docs/q0_q1.md`	Primary genesis origin Q(0) and first unit Q(1)	q0_q1.html
`docs/q2_q9.md`	Single-digit secondary DCC anchors and single-digit spectrum	q2_q9.html
`docs/qm.md`	Constructive numeric leaf substrate and Diophantine division constraints	qm.html
`docs/qm_geometry.md`	Multi-axial frames, Clifford geometric algebra, and float-free rotations	qm_geometry.html
`docs/qm_calculus.md`	Discrete Exterior Calculus, continued fractions, and invariant dynamics	qm_calculus.html
`docs/qexpr.md`	Layer 3 Content-Addressed Expression DAG container	qexpr.html
`docs/qs.md`	Constructive relational intent and Intent Contracts	qs.html
`docs/qlog.md`	Constructive type theory, pattern verification, and inferential modes	qlog.html
`LICENSE.md`	Scaling Source License 1.0 governing the Licensed Work	LICENSE.md

1. Epistemic Stance and Working Hypothesis Notice

1.1. Constructive Working Hypothesis Baseline

This specification is formulated within the Recursive Search Loop (`docs/simemp.md` §2). It establishes Qphy **(Quang Physics)** as the third foundational Base Domain of the Qn universe.

All formulations herein represent provisionally consistent hypotheses. In accordance with the Anti-Calcification Principle, permanent finality is reserved until empirical telemetry is derived across physical silicon runtimes. The architecture preserves structural elasticity through **Structural Indirection**.

1.2. The Physical Root of Finiteness (Layer 0 Grounding)

Classical physics often treats computation as an immaterial process operating over continuous coordinate fields ($\mathbb{R}^n$). In the **Qn** computational universe, **computation is an irreversible non-equilibrium thermodynamic process**.

Qphy is anchored directly to **Layer 0** (Physical Substrate: commodity silicon, parasitic wire capacitance, ring oscillator noise, and Landauer dissipation limits). A physical simulation or computation is valid if and only if its state transitions satisfy **Dependencies Determinism**:

$$H(S_t \mid \mathcal{D}(S_t)) = 0$$

---

2. Abstraction Layer Placement and Inter-Domain Coupling

2.1. The Base Triad Architecture (Qs ⊗ Qm ⊗ Qphy)

Conforming to `docs/qs.md` (§7), physical reality is not generated from pure mathematical logic. The computational universe is grounded upon three mutually orthogonal Base Domains:

$$\mathbf{C}_{\text{total}} = \mathbf{C}_{\text{Qs}} \otimes \mathbf{C}_{\text{Qm}} \otimes \mathbf{C}_{\text{Qphy}}$$

- Qm **(Mathematics):** Supplies discrete scalar arithmetic, orthogonal Clifford frames ($\mathcal{C}\ell_{p,q}$), and discrete calculus coboundaries (**d**).
- Qs **(Semantics):** Supplies observer framing, semantic blades, and human **Intent Contracts**.
- Qphy **(Physics):** Supplies physical non-equilibrium thermodynamics, energy dissipation bounds, and physical conservation laws.

2.2. The Grounding Coupling with Qlog

The physical domain cannot be synthesized by squaring mathematical logic:

$$\mathbf{C}_{\text{Qphy}} \neq (\mathbf{C}_{\text{Qm}} \otimes \mathbf{C}_{\text{Qs}})^2$$

However, Qphy requires **Qlog** to type-check and assemble its physical trajectories:

- **Physical Laws as** Q(patterns)**:** Every physical transition template (e.g., a discrete symplectic integration step) is an invariant `Q(pattern)`.
- **Physical Trajectories as** Q(workflows)**:** Time evolution across intervals is an assembled `Q(workflow)` bounded by finite metric budgets.
- **Conservation Laws as Type Judgments:** Checking whether a physical state satisfies energy or charge conservation is a constructive type judgment: $\Gamma \vdash S : \text{Conserved}(E_0)$.

---

3. Constructive Field Dynamics on Discrete Simplicial Lattices

3.1. Discrete Differential Forms as Cochain Fields

Classical continuous tensor fields with coordinate indices ($T^{\mu\nu}$) are replaced in Qphy by **cochains** evaluated over oriented simplicial complexes K (`docs/qm_calculus.md`): - 0**-Cochains (Scalars):** Electrostatic potential, temperature, and mass densities evaluated on mesh vertices. - 1**-Cochains (Circulations):** Fluid velocity vectors and electric field circulations integrated along oriented edges. - 2**-Cochains (Fluxes):** Magnetic induction flux and fluid vorticity evaluated across oriented planar faces.

3.2. Singularity Immunity via Scale Lattices (δ_z)

In classical continuous mechanics, non-linear partial differential equations permit unphysical finite-time singularities (e.g., infinite velocity blowups in Navier–Stokes equations).

Qphy provides structural immunity against physical singularities: 1. **Discrete Lattice Cutoff:** Physical gradients are strictly bounded by the discrete scale quantum $\delta_z(d)$ along dimensional axes (`docs/qm.md` §3.3). Spatial points cannot approach distance zero. 2. **Dissipative Viscous Sinks:** Physical fields dissipate excess energy into microscopic thermal noise, exporting operational entropy via signed receipts. 3. **Totality by Budget (**Axiom 6**):** Simulation workflows carry an explicit work ceiling in units of U. Unbounded velocity divergence is physically impossible; execution halts cleanly upon budget exhaustion.

3.3. Exact Topological Conservation ($\mathbf{d}^2 \equiv Q(0)$)

In Qphy, conservation laws do not rely on asymptotic floating-point limits. They are exact algebraic identities derived from the **Discrete Exterior Derivative** (**d**):

$$\mathbf{d} \circ \mathbf{d} \equiv Q(0)$$

By the Poincaré lemma on discrete simplicial meshes, the boundary of a boundary is empty ($\partial\partial \equiv \emptyset$). Consequently: - **Conservation of Magnetic Flux (Maxwell):** $\mathbf{d}F \equiv Q(0)$ holds identically with zero numerical leakage. - **Conservation of Electric Charge:** $\mathbf{d} \star J \equiv Q(0)$ is an exact topological identity. - **Discrete Stokes' Invariant:** Boundary flux integrals equal interior coboundary evaluations with zero remainder error.

---

4. Non-Equilibrium Computational Thermodynamics

4.1. The Open System Entropy Balance (Prigogine Formulation)

A physical computing system is an **open thermodynamic engine**. In accordance with non-equilibrium thermodynamics, internal computational stability requires expending work to export operational entropy (`docs/simemp.md` §4.2):

$$dS = dS_{\text{int}} + dS_{\text{ext}}, \quad \text{where } dS_{\text{ext}} < 0$$

Thermal noise, bus line jitter, and transmission faults maximize internal entropy ($dS_{\text{int}} > 0$). The system restores internal order by exporting entropy through immutable, signed **Road Receipts** ($\mathcal{R}_{\text{road}}$).

4.2. Physical Calibration of the Abstract Compute Unit U

The transition from local origin **Q(0)** to unit **Q(1)** defines the **Abstract Compute Unit U**: - **The Landauer Bound:** At ambient temperature ($T = 300$ K), erasing one bit of physical information dissipates minimal thermodynamic work into the environment:

$$W(U) \geq k_B T \ln 2 \approx 2.87 \times 10^{-21} \text{ J}$$

- **Platform Invariance:** In software compilation, U serves as the invariant accounting quantum. Physical hardware translation layers parameterize device-specific Joule-per-U conversion factors without altering high-level metric envelopes.

4.3. The Capacitive Memory Wall Transport Impedance

Moving a packet of bits across a distance ℓ on physical silicon requires charging the parasitic capacitance C of the metal interconnect at operating voltage V and clock frequency f:

$$P_{\text{bus}} = \alpha C(\ell) V^2 f + I_{\text{leak}} V$$

1. **Distance-Dependent Transport Cost:** Transporting bits across the Memory Wall incurs a strictly greater metric cost than local register operations ($W_{\text{bus}} \gg W_{\text{register}}$).
2. **Breakeven Grounding:** This physical impedance formalizes the thermodynamic breakeven equation governing **Q(reuse)** in `docs/qlog.md` (§5.2).

---

5. The Canonical Physical Q(anchor) Spectrum

In Qphy, physical constants are not hardcoded static variables. They are **constructively built Qn artifacts derived from an explicit** Historical Calibration Benchmark****, whose anchor status emerges as their topological centrality (χ) expands across the causal DAG:

5.1. Universal Physical Anchors and Capability Contracts

- ****Q(th_2)** (The Second Law Anchor):** Built from sampled thermal receipts and $\ln 2$ continued fraction generators. Enforces non-zero heat dissipation for bit-erasure.
- ****Q(c)** (The Relativistic Causal Horizon Anchor):** Built from local dielectric propagation delays. Enforces finite signal speed ($\tau \geq \Delta x / v_{\text{signal}}$).
- ****Q($\hbar$)** (The Quantum Action Anchor):** Built from height ceilings ($H_{\max}$) and quantum action limits. Forbids unmetered infinite information density in finite registers.
- ****Q(Z_bus)** (The Memory Wall Transport Anchor):** Built from parasitic wire capacitance measurements ($P = \alpha C V^2 f + I_{\text{leak}} V$).
- ****Q(gauge)** (The Topological Conservation Anchor):** Built from simplicial mesh topology. Enforces exact coboundary nilpotency ($\mathbf{d}^2 \equiv Q(0)$).

5.2. The Physical Inconsistency Collapse Cascade and Circuit Breakers

If an execution profile or autonomous workflow claims a trajectory that violates an emergent physical anchor: 1. **Immediate Inconsistency Cascade:** The physical anchor destabilizes, threatening to corrupt downstream causal cones $\mathcal{C}^+(\alpha)$. 2. **Gateway Circuit Breaker:** The **SIMEMP Gateway** intercepts the invalid transition, halting execution and emitting `RECEIPT_PHYSICAL_VIOLATION`. 3. **Protection of Silicon and Lineage:** Halting at the gateway prevents physical thermal runaway on hardware chips and protects downstream causal lineage from ungrounded physics.

---

6. The Axiomatic Computational World Model

Qphy operates as an **Axiomatic, Thermodynamically Grounded Computational World Model** for autonomous systems and robotics.

6.1. Axiomatic Invariance vs. Statistical Latent Hallucination

Contemporary deep learning world models approximate physics through statistical correlation over continuous latent spaces. They suffer from physical hallucinations: objects morphing, phantom forces, and non-conservation of energy.
In Qphy, physical laws are discrete algebraic invariants. Hallucinations are prevented by construction: an unphysical state cannot compile through the gateway.

6.2. The Thermodynamic Double-Leash on Autonomous Agents

Autonomous agents executing in the physical world are governed by two coupled energy boundaries: 1. **The External Leash (Simulated Mechanics):** The planned trajectory must satisfy environmental energy conservation ($\Delta E_{\text{kinetic}} + \Delta E_{\text{potential}} \leq E_{\text{battery}}$).

2. **The Internal Leash (Computational Work):** The computation required to verify the trajectory must itself fit within the allocated metric budget ($W \leq W_{\text{budget}}$ in units of U).

If the computational cost of planning a path exceeds available energy budgets, execution halts, forcing lower-complexity fallback actions.

6.3. Symplectic Preservation and Drift-Free Trajectories

Physical dynamical trajectories are integrated using discrete symplectic morphisms: - Phase space volume (Liouville's theorem) is conserved exactly. - Each time step achieves **Terminal Exactness** ($r = Q(0)$) or preserves exact Diophantine residuals. - Trajectories can execute over billions of discrete steps with zero artificial numerical dissipation.

---

7. Functorial Tensor Symmetries and Bounded Context Ranges

7.1. Permutation-Invariant vs. Permutation-Sensitive Physical Tensors

Under the **No-Implicit Rule**, every tensor interaction in Qphy must declare its algebraic symmetry within its DCC Profile: - **Permutation-Invariant (`DCC_SYM_COMMUTATIVE`):** Operations where component order carries zero physical difference (e.g., thermal energy accumulation, scalar potential sums). Child hashes are sorted lexicographically, enabling $\mathcal{O}(1)$ deduplication via **Q(reuse)**. - **Permutation-Sensitive (`DCC_SYM_NON_COMMUTATIVE`):** Operations where order dictates causality or direction (e.g., sequential time steps $S_t \otimes S_{t+\Delta t}$, non-commutative rotor sandwiches $R\vec{v}R^{\dagger}$). Order is strictly preserved. - **Graded Anti-Commutative (`DCC_SYM_ANTI_COMMUTATIVE`):** Outer products of differential forms ($\vec{u} \wedge \vec{v} = -\vec{u} \wedge \vec{v}$). Swapping elements flips the sign parity ($\sigma \to \mathcal{I}(\sigma)$).

7.2. The Four-Axis Bounded Context Range (R_ctx)

Physical state evaluations cannot occur across unbounded continuous space. Every physical evaluation requires an explicit Context Range: 1. **Scale Range:** Discrete zoom window $[z_{\text{min}}, z_{\text{max}}]$ parameterizing lattice resolution $\delta_z(d)$. 2. **Causal Horizon Range:** Monotonic causal index interval $[\kappa_{\text{start}}, \kappa_{\text{end}}]$. 3. **Dimensional Metric Frame:** Orthogonal spacetime frame $\mathcal{D}_{\text{frame}} = \{d_t, d_x, d_y, d_z\}$. 4. **Thermodynamic Budget:** Maximum permissible Landauer work ceiling $W \leq W_{\text{budget}}$.

---

8. Causal Trajectories: Forward Simulation vs. Inverse Empirical Science

Qphy accommodates both directions across the causal arrow:

8.1. Forward Symplectic Time Evolution (Q(workflows))

In forward simulation, a system predicts future physical states: 1. Initial physical state S_0 is configured inside a `Qexpr` container. 2. The system executes a sequence of symplectic **Q(patterns)** generating discrete **Q(events)**. 3. Intermediate checkpoints are verified using the **Anchor-Cut Invariant**, allowing external auditors to verify long simulations in $\mathcal{O}(K)$ receipt validations without replaying microscopic steps.

8.2. Inverse Causal Reconstruction of Physical Laws

In empirical observational physics across the **Historical Horizon**, observers measure an empirical terminal physical state S_{obs} (anchored as `MODAL_EMPIRICAL`) while the causal origin remains latent:

$$\text{Given } S_{\text{obs}} \implies \text{Find } \left\langle S_0,\, \mathcal{H}_{\text{sys}} \right\rangle \quad \text{such that} \quad \mathcal{H}(S_0) \equiv S_{\text{obs}}$$

1. **Receipt Anchoring:** The observed state is ingested via verified hardware sensor receipts (`MODAL_EMPIRICAL`, $G = 1.0$).
2. **Candidate Hypothesis Generation:** Stochastic or neural modules propose ancestral Hamiltonians ($H_1 \dots H_n$).
3. **BOC Abductive Resolution:** The **BOC Policy** selects the candidate route that maximizes SIMEMP compliance, selecting the physical law that minimizes unmeasured entropy ($H(S \mid \mathcal{D}) \to 0$) and minimizes Landauer dissipation.
4. **Law Sealing:** The verified physical trajectory is sealed as an emergent `Q(pattern)`, joining the empirical physical registry.

---

9. License and Lineage Alignment

9.1. Paternity Reference

All derivative physical simulation models, thermodynamic solvers, robotics world models, and intermediate execution modules derived from Qphy must embed the canonical attribution:

> Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License (SSL). Parent Repository: https://github.com/cdqn5249/cdqn

9.2. Open Core Invariants

The core physical constants, discrete exterior calculus mechanics, and thermodynamic accounting rules remain royalty-free for academic, non-commercial, and sub-threshold deployments under the **Open Core Invariants** of the **SSL**.

9.3. Statutory Safe Harbor and Passive Infrastructure

Qphy operates as passive, neutral physical and computational modeling infrastructure. It exercises no editorial inspection over, and makes no claim upon, sovereign user content, external simulation datasets, or sovereign **Payloads**.

Annex T — Qchem

Pinned source: docs/qchem.md
Committed version: 1.0.3 **Last updated:** 2026-10-19
sha256 (committed file): `4864574e37f784c35bb41bf7aad3603a374aedc87bfdf9ca735322eb513c5697`

Field	Specification
Document Title	Qchem — Quang Chemistry: Discrete Stoichiometry, Molecular Graphs, and Reaction Energetics
Version	1.0.3
Last Updated	2026-10-19 (Bao Loc, Vietnam)
Author	Christophe Duy Quang Nguyen
License	Scaling Source License (SSL) 1.0
Status	Candidate Domain Specification — Category D (**Complexity Degree 3**)

Normative References

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`docs/qlog.md`	Constructive type theory, pattern verification, and inferential modes	qlog.html
`docs/qphy.md`	Discrete field dynamics, computational thermodynamics, and physical entropy	qphy.html
`LICENSE.md`	Scaling Source License 1.0 governing the Licensed Work	LICENSE.md

1. Epistemic Stance and Working Hypothesis Notice

1.1. Constructive Working Hypothesis Baseline

This specification is formulated within the Recursive Search Loop (`docs/simemp.md` §2). It establishes Qchem **(Quang Chemistry)** as an emergent composite domain formed by the categorical tensor product of physical thermodynamics and constructive mathematics:

$$\mathbf{C}(\text{Qchem}) \equiv \mathbf{C}(\text{Qphy}) \otimes \mathbf{C}(\text{Qm})$$

All formulations herein represent provisionally consistent hypotheses. In accordance with the Anti-Calcification Principle, permanent finality is reserved until physical execution feedback is derived across chemical simulation runtimes. The architecture preserves structural elasticity through **Structural Indirection**.

1.2. Abiotic Boundary and the Exclusion of Semantic Intent

Chemistry models the organization of matter governed by valence electrons, molecular bonds, and thermodynamic equilibria. Unlike biological or cognitive domains, **abiotic chemical matter possesses zero semantic agency, legal liability, or teleological intent**:

$$\mathbf{C}(\text{Qchem}) \cap \mathbf{C}(\text{Qs}) \equiv \emptyset$$

Chemical reaction steps transition spontaneously under physical thermodynamic potential ($\Delta G \leq 0$) and Diophantine stoichiometric balance. Attributing intent (Qs) to abiotic reactions is a category error. All transitions satisfy **Dependencies Determinism**:

$$H(S_t \mid \mathcal{D}(S_t)) = 0$$

---

2. Discrete Molecular Graphs and Stereochemistry

2.1. Atoms and Valence Bonds as Content-Addressed DAGs

Molecules are represented within **Qexpr** Content-Addressed Expression DAGs (**Content-Addressed Expression DAG**): - **Atomic Vertices:** Each atom is an immutable leaf artifact carrying atomic number $Z \in \mathbb{N}$, discrete mass number $A \in \mathbb{N}$, and formal oxidation state. - **Valence Bonds as Clifford Blades:** Covalent, ionic, and coordination bonds are modeled as localized Clifford multivector blades (`docs/qm_geometry.md`), declaring discrete bond order $(1, 2, 3)$ and directional orientation. - **Deduplication via Hash-Consing:** Identical molecular sub-structures share identical commitment hashes $\mathcal{H}(\text{Molecule})$. Common functional groups are allocated once in memory, maximizing **Q(reuse)** at zero marginal Landauer cost ($\Delta W = 0$).

2.2. Stereochemical Chirality via Clifford Sign Parity (σ)

Enantiomers (chiral mirror-image molecules such as D- and L-amino acids) share identical scalar connectivity, bond lengths, and atom counts. Classical cheminformatics often relies on ambiguous textual conventions (R/S or D/L labels).

In Qchem, molecular chirality is an exact algebraic invariant governed by the spatial reflection involution $\mathcal{I}_d$ and the 3D pseudoscalar $I_3 = \vec{e}_1 \wedge \vec{e}_2 \wedge \vec{e}_3$: - The spatial inversion of a chiral center inverts its oriented grade-3 volume blade. - The resulting enantiomer is distinguished by its **Sign Parity** ($\sigma \in \{\ominus, \oplus\}$). - Enantiomeric transformations preserve exact unitary norms without floating-point trigonometric drift.

---

3. Exact Diophantine Stoichiometry

3.1. The Stoichiometric Null-Space Invariant

In classical software, reaction mass balances frequently accumulate numerical roundoff errors. In Qchem, chemical reactions are **exact Diophantine balance equations**: - Let M elements react across N molecular species. The stoichiometric matrix $\mathbf{S} \in \mathbb{Z}^{M \times N}$ maps integer elemental coefficients. - A valid chemical reaction flux vector $\vec{v} \in \mathbb{N}^N$ satisfies the Diophantine null-space constraint under **Terminal Exactness** ($r = Q(0)$):

$$\mathbf{S} \cdot \vec{v} \equiv Q(0)$$

Elemental mass, atom counts, and net electric charges balance identically. Truncation error is zero by construction.

3.2. Redox Electron Quanta and Charge Conservation

Oxidation-reduction reactions involve discrete electron transfers: - Electrons exist as indivisible integer charge carriers ($z \in \mathbb{Z}$). Fractional electron transfers are barred. - The net charge transition across reactants and products evaluates to genesis origin $Q(0)$, guaranteeing exact conservation of electric charge under `DCC_ANCHOR_DEC_D_v1`.

---

4. Non-Equilibrium Chemical Thermodynamics

4.1. Gibbs Free Energy Reaction Spontaneity

Chemical reaction feasibility is governed by **Non-Equilibrium Thermodynamics** (`docs/qphy.md` §4): - **Spontaneity Criterion:** At constant temperature and pressure, an abiotic chemical reaction transitions spontaneously if and only if its change in Gibbs free energy is negative or zero:

$$\Delta G = \Delta H - T\Delta S \leq Q(0)$$

- **Thermal Activation Barriers (E_a):** Reactions possessing an activation energy barrier require absorbing a discrete thermal work packet from the local environment, metered by the local **Metric Envelope**.

4.2. Chemical Equilibrium as Continued Fraction Convergents

Chemical equilibrium constants (K_{eq}) are held symbolically inside `Qexpr` as deterministic continued fraction partial convergents (`OP_CF`):

$$K_{\text{eq}} = [a_0;\, a_1,\, a_2,\, \ldots]$$

Evaluating reaction quotients to zoom level z generates exact rational pairs without continuous exponential floating-point approximations.

---

5. The Chemical Q(anchor) Spectrum

Qchem grounds its physical and chemical constants in universal **Q(anchor)** entities derived from the **Historical Calibration Benchmark**:

5.1. Universal Chemical Anchors and Capability Contracts

- **`Q(N_A)` (The Avogadro Anchor):** Formalizes the exact discrete bridge between microscopic atomic integer counts and macroscopic molar thermodynamics: $N_A \equiv 602{,}214{,}076 \times 10^{15}$ entities/mol.
- **`Q(stoich)` (The Stoichiometric Anchor):** Enforces nuclear and elemental conservation across chemical transformations without fractional atom leakage.
- ****Q(th_2)** (The Second Law Gibbs Anchor):** Mandates that endergonic reactions ($\Delta G > 0$) cannot occur without coupled exergonic work.
- **`Q(e)` (The Redox Charge Anchor):** Restricts electrochemical electron transfers to discrete integer quanta of $Q(1)$.
- **`Q(chiral)` (The Stereochemical Anchor):** Enforces spatial reflection parity over chiral tetrahedral carbon centers.

5.2. Chemical Inconsistency Cascade and Gateway Interception

If an automated reaction proposal or molecular simulation claims an unphysical reaction: 1. **Immediate Inconsistency Cascade:** The stoichiometric or thermodynamic anchor destabilizes, threatening downstream mass-action models. 2. **Gateway Interception:** The **SIMEMP Gateway** intercepts the invalid balance, emitting `RECEIPT_STOICHIOMETRIC_VIOLATION` or `RECEIPT_`

`THERMODYNAMIC_EXHAUSTION`. 3. **Execution Termination:** The invalid reaction workflow is rejected before consuming physical compute or dispatching automated laboratory synthesis actuators.

---

6. Reaction Kinetics as Q(patterns) and Q(workflows)

6.1. Elementary Reaction Steps as Q(patterns)

An elementary chemical reaction step is an invariant ****Q(pattern)****:

$$\mathcal{P}_{\text{rxn}} : \mathcal{H}(\text{Reactants}) \xrightarrow{\mathcal{M}_{\text{kinetics}}} \mathcal{H}(\text{Products})$$

It defines a reproducible, discrete transformation mapping reactant commitment hashes to product hashes under explicit thermodynamic constraints.

6.2. Reaction Cascades as Q(workflows) and Catalysis as Q(bypass)

Multi-step chemical syntheses, polymerizations, and reaction networks are assembled as ****Q(workflows)****: - **Work Budgets:** Reaction sequences operate within a declared Metric Envelope, tracking thermal work and Landauer dissipation. - **Catalytic Optimization via** Q(bypass)**:** A catalyst lowers the activation energy barrier ($E_a' \ll E_a$) without altering overall equilibrium (ΔG). In Qchem, catalysis is formalized as an execution short-circuit: the gateway recognizes the catalytic agent, bypasses the slow uncatalyzed route, and emits `RECEIPT_CATALYTIC_BYPASS` in $\mathcal{O}(1)$ dispatch time.

---

7. The Boundary Between Qchem and Qbio

7.1. Stratification of Abiotic Matter vs. Living Systems

To preserve the **No-Implicit Rule**, the architecture maintains a strict epistemic boundary between chemistry and biology: - Qchem **(Abiotic Chemistry):** Models molecular graphs, valence interactions, and reaction kinetics. It operates with zero intentional agency ($\text{Qs} = \emptyset$). - ****Qbio** (Living Biosemiotics):** Emerges when chemical networks are coupled to **informational semantics** (Qs):

$$\mathbf{C}(\text{Qbio}) \equiv \mathbf{C}(\text{Qchem}) \otimes \mathbf{C}(\text{Qs})$$

7.2. The Biosemiotic Leap

Living systems differ from abiotic chemistry because their molecules serve as **informational codes**: - An abiotic crystal grows purely via ΔG minimization (Qchem). - A strand of DNA guides protein synthesis via a codon translation code (Qbio $=$ Qchem$\otimes$ Qs). - Active inference and metabolic regulation introduce teleological intent contracts, which are strictly barred from abiotic Qchem.

---

8. Causal Trajectories: Forward Synthesis vs. Inverse Spectroscopy

Qchem accommodates both directions across the causal arrow:

8.1. Forward Synthesis Planning (Q(workflows))

Forward chemical synthesis compiles a path from available chemical precursors to a target molecule: 1. Target molecular graph $\mathcal{H}(\text{Target})$ is declared. 2. Qlog compiles a sequence of elementary reaction `Q(patterns)`. 3. Each step verifies $\Delta G \leq Q(0)$ and stoichiometric balance, emitting a chained road receipt vector.

8.2. Inverse Causal Reconstruction of Molecular Graphs

In analytical chemistry across the **Historical Horizon**, an instrument measures an empirical spectral signature S_{obs} (anchored as `MODAL_EMPIRICAL`, e.g., mass-to-charge ratios m/z, NMR chemical shifts) while the molecular identity remains unknown:

$$\text{Given } S_{\text{obs}} \implies \text{Find } \left\langle \mathcal{H}(\text{Molecule}),\, \mathcal{M}_{\text{frag}} \right\rangle \quad \text{such that} \quad \mathcal{M}(\text{Molecule}) \equiv S_{\text{obs}}$$

1. **Receipt Ingestion:** Mass spectrometer readings are anchored as verified sensor receipts (`MODAL_EMPIRICAL`, $G = 1.0$).
2. **Candidate Hypothesis Generation:** Stochastic or neural engines propose candidate molecular graphs ($M_1 \dots M_n$).
3. **BOC Abductive Resolution:** The **BOC Policy** selects the candidate graph that matches the observed fragmentation peaks while minimizing unmeasured conditional entropy ($H(S \mid \mathcal{D}) \to 0$).
4. **Structure Sealing:** The reconstructed molecular graph is sealed into the local chemical registry.

---

9. License and Lineage Alignment

9.1. Paternity Reference

All derivative chemical simulation models, synthesis planners, spectroscopy engines, and intermediate execution modules derived from Qchem must embed the canonical attribution:

> Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License (SSL). Parent Repository: https://github.com/cdqn5249/cdqn

9.2. Open Core Invariants

The core chemical constants, discrete stoichiometry solvers, and thermodynamic balance rules remain royalty-free for academic, non-commercial, and sub-threshold deployments under the **Open Core Invariants** of the **SSL**.

9.3. Statutory Safe Harbor and Passive Infrastructure

Qchem operates as passive, neutral chemical and computational modeling infrastructure. It exercises no editorial inspection over, and makes no claim upon, sovereign user content, external chemical libraries, or sovereign **Payloads**.

Annex U — Qbio

Pinned source: docs/qbio.md
Committed version: 1.0.2 **Last updated:** 2026-10-19
sha256 (committed file): `99fba2491d5345d036cb99925047121582a010016f59ca23490115a4adecf072`

Field	Specification
Document Title	Qbio — Quang Biology: Biosemiotics, Metabolic Homeostasis, and Active Inference
Version	1.0.2
Last Updated	2026-10-19 (Bao Loc, Vietnam)
Author	Christophe Duy Quang Nguyen
License	Scaling Source License (SSL) 1.0
Status	Candidate Domain Specification — Category D (**Complexity Degree 3**)

Normative References

The following documents establish the physical, structural, and legal constraints governing this specification. If a technical conflict arises, `simemp.md` governs; if a structural conflict arises, `abstractionLayers.md` governs; if a legal conflict arises, `LICENSE.md` governs. References are deliberately version-unpinned: each target document's front matter carries its own authoritative version, so this specification never duplicates drift-prone patch state.

Document	Role	Target
`docs/simemp.md`	Constitutional constraints, thermodynamics, and Dependencies Determinism	simemp.html
`docs/abstractionLayers.md`	Layer architecture and SIMEMP Gateway validation	abstractionLayers.html
`docs/qnPrimitive.md`	Universal Envelope, operational axioms, and Optimization Morphisms	qnPrimitive.html
`docs/q0_q1.md`	Primary genesis origin Q(0) and first unit Q(1)	q0_q1.html
`docs/q2_q9.md`	Single-digit secondary DCC anchors and parity partitions	q2_q9.html
`docs/qm.md`	Constructive numeric leaf substrate and Diophantine division constraints	qm.html
`docs/qm_geometry.md`	Multi-axial frames, Clifford geometric algebra, and float-free rotations	qm_geometry.html
`docs/qm_calculus.md`	Discrete Exterior Calculus, continued fractions, and invariant dynamics	qm_calculus.html
`docs/qexpr.md`	Layer 3 Content-Addressed Expression DAG container	qexpr.html
`docs/qs.md`	Constructive relational intent and Intent Contracts	qs.html
`docs/qlog.md`	Constructive type theory, pattern verification, and inferential modes	qlog.html
`docs/qphy.md`	Discrete field dynamics, computational thermodynamics, and physical entropy	qphy.html
`docs/qchem.md`	Discrete stoichiometry, molecular graphs, and reaction energetics	qchem.html
`LICENSE.md`	Scaling Source License 1.0 governing the Licensed Work	LICENSE.md

1. Epistemic Stance and Working Hypothesis Notice

1.1. Constructive Working Hypothesis Baseline

This specification is formulated within the Recursive Search Loop (`docs/simemp.md` §2). It establishes ****Qbio (Quang Biology)**** as an emergent biosemiotic domain derived from the categorical tensor product of chemistry and semantics:

$$\mathbf{C}(\text{Qbio}) \equiv \mathbf{C}(\text{Qchem}) \otimes \mathbf{C}(\text{Qs})$$

All formulations herein represent provisionally consistent hypotheses. In accordance with the Anti-Calcification Principle, permanent finality is reserved until biological simulation feedback is derived across living cellular runtimes. The architecture preserves structural elasticity through **Structural Indirection**.

1.2. The Biosemiotic Boundary: Life as Coded Chemistry

A living organism is not merely a complex chemical mixture; it is a **biosemiotic system** wherein molecular reaction networks are coupled to **informational code channels**: - Abiotic molecules in **Qchem** react spontaneously under physical thermodynamic potential ($\Delta G \leq 0$). - Biological systems in Qbio process **informational intent**: nucleic acid sequences represent symbolic instructions that guide metabolic synthesis.

Living entities maintain non-equilibrium steady states against environmental entropy by consuming metabolic energy. Every biological state transition satisfies **Dependencies Determinism**:

$$H(S_t \mid \mathcal{D}(S_t)) = 0$$

---

2. Categorical Formulation: Qchem Coupled to Qs

2.1. The Biosemiotic Tensor Product

Qbio is constructed constructively through the tensor product:

$$\mathbf{C}(\text{Qbio}) = \big(\mathbf{C}(\text{Qphy}) \otimes \mathbf{C}(\text{Qm})\big) \otimes \mathbf{C}(\text{Qs})$$

- **The Chemical Substrate (**Qchem**):** Provides discrete atomic molecular graphs, Diophantine stoichiometry ($\mathbf{S} \cdot \vec{v} \equiv Q(0)$), and Gibbs free energy dissipation ($\Delta G \leq 0$).
- **The Semantic Channel (**Qs**):** Provides symbolic intent contracts, functional pregroup grammars, and teleological agency (self-preservation and replication).

2.2. Genetic Code as Content-Addressed Translation

In Qbio, DNA and RNA sequences are represented as linear **Content-Addressed Expression DAGs (**Content-Addressed Expression DAG): - **Codons as Discrete Tokens:** Triplet nucleotide bases $\{A, U, G, C\}^3$ form an exact discrete alphabet of 64 tokens. - **The Translation Morphism:** Ribosomal translation is an invariant discrete mapping:

$$\mathcal{M}_{\text{translation}} : \text{Codon}^{64} \longrightarrow \text{AminoAcid}^{20} \cup \{\text{STOP}\}$$

- **Hash-Consing Deduplication:** Identical genetic motifs and conserved protein domains share the exact same cryptographic hash $\mathcal{H}(\text{Domain})$, enabling $\mathcal{O}(1)$ structural comparison and zero marginal Landauer allocation via **Q(reuse)**.

---

3. Non-Equilibrium Metabolic Homeostasis and Free Energy

3.1. Variational Free Energy and Active Inference

A living organism persists by minimizing **Variational Free Energy** (Karl Friston formulation). Under `docs/simemp.md` (§4.2), an organism is an open dissipative system that avoids internal entropy maximization by generating predictive models of its environment:

$$F = \text{Internal Bound Energy} - T \cdot \text{Information Entropy}$$

1. **Active Inference Loops:** An organism generates actions to fulfill its metabolic **Intent Contract**, ensuring that environmental surprises (entropy spikes) remain bounded.
2. **Deterministic Selection via BOC:** Internal homeostatic adaptations are selected via the **BOC Policy**, maximizing survival compliance while minimizing Landauer dissipation.

3.2. Entropy Export Across the Cellular Membrane

Living systems survive by acting as **Prigogine dissipative engines**:

$$dS = dS_{\text{int}} + dS_{\text{ext}}, \quad \text{where } dS_{\text{ext}} < 0$$

The cell consumes nutrient molecules possessing high chemical potential and excretes degraded waste products, exporting operational entropy via signed metabolic receipts ($\mathcal{R}_{\text{metabolic}}$).

---

4. The Canonical Biological Q(anchor) Spectrum

Qbio grounds living systems in five universal **Q(anchor)** entities derived from the **Historical Calibration Benchmark**:

4.1. Universal Biological Anchors and Capability Contracts

- **`Q(codon)` (The Genetic Code Anchor):** Enforces the universal translation mapping between nucleotide sequences and peptide synthesis without unmetered amino acid drift.
- **`Q(blanket)` (The Markov Blanket Anchor):** Formalizes cellular membranes as topological boundaries ensuring statistical and causal insulation between internal physiology and external noise.
- **`Q(atp)` (The Metabolic Work Quantum):** Calibrates ATP hydrolysis to the **Abstract Compute Unit U**: $\Delta G_{\text{ATP}} \approx 20\,U$, grounding metabolic work directly into Landauer limits ($W \geq k_B T \ln 2$).
- **`Q(homeo)` (The Homeostasis Anchor):** Enforces hard ceilings on internal operational entropy, preventing metabolic collapse.
- **`Q(autocat)` (The Autocatalytic Closure Anchor):** Enforces self-sustaining reaction loops where metabolic enzymes catalyze their own production.

4.2. Biological Inconsistency Cascade and Metabolic Halting

If a biological simulation or genetic algorithm proposes a non-viable organism: 1. **Immediate Inconsistency Cascade:** The homeostasis or ATP anchor destabilizes ($\chi < \Theta_\chi$), threatening systemic physiological collapse. 2. **Gateway Interception:** The **SIMEMP Gateway** halts execution deterministically, emitting `RECEIPT_METABOLIC_STARVATION` or `RECEIPT_HOMEOSTATIC_COLLAPSE`. 3. **Organism Death as Clean Halting:** Under **Axiom 6** (Totality by Budget), biological death is modeled as an explicit, bounded terminal receipt, preventing unmetered necrotic execution loops.

---

5. Morphogenesis on Discrete Simplicial Complexes

5.1. Reaction-Diffusion Cochain Systems

Morphogenesis (e.g., Turing pattern formation, embryonic tissue layering) is modeled in Qbio using Discrete Exterior Calculus (**DEC**) on simplicial meshes K (`docs/qm_calculus.md`): - Morphogen concentrations are represented as discrete 0-cochains on mesh vertices. - Diffusion is evaluated via the discrete Laplace–Beltrami operator: $\Delta = \mathbf{d}\delta + \delta\mathbf{d}$. - Chemical activation and inhibition evaluate through Diophantine integer kinetics in Qchem.

5.2. Topological Closure and the Markov Blanket

A biological tissue or single cell possesses an explicit **Markov Blanket**: - The boundary is discretized as an oriented $(n-1)$-simplicial sub-complex ∂K. - Information and chemical transport across the boundary must cross an explicit gateway. - Sensory inputs and active motor outputs are metered, preventing unmodeled external state drift ($H(S_{\text{int}} \mid S_{\text{blanket}}, S_{\text{ext}}) = 0$).

---

6. Metabolic Pathways as Q(workflows) and Catalytic Networks

6.1. Enzymatic Cascades as Q(patterns)

An enzymatic transformation (e.g., glycolysis, Krebs cycle step) is an invariant ****Q(pattern)****:

$$\mathcal{P}_{\text{enzyme}} : \mathcal{H}(\text{Substrate}) \xrightarrow{\mathcal{M}_{\text{enzyme}}} \mathcal{H}(\text{Product})$$

Each enzyme acts as an execution contract requiring an explicit co-factor and an ATP energy packet, dissipating heat according to Qphy constraints.

6.2. Autocatalytic Sets and Category B Optimization

Complex metabolic networks form closed autocatalytic sets (Gánti Chemoton model): - In Qbio, recurring metabolic cycles leverage **Q(reuse)**, retrieving cached intermediate enzyme hashes rather than reallocating memory bits across the Memory Wall. - Allosteric regulation operates via **Q(bypass)**: feedback inhibition short-circuits redundant metabolic synthesis in $\mathcal{O}(1)$ dispatch time once concentration thresholds are satisfied.

---

7. Biological Agency and the Non-Rogue Invariant

7.1. Cellular Agency Bounded by Intent Contracts

Biological agency (chemotaxis, immune defense, cellular differentiation) is governed by **Structural Intent Contracts** (`docs/qs.md` §5): - An immune cell hunting a pathogen operates under an explicit capability contract (`DCC_BIO_IMMUNE_v1`). - The cell possesses finite authority: it cannot mutate its foundational genesis origin $Q(0)$ or attack self-tissue without violating its DCC boundary.

7.2. The Thermodynamic Death Barrier (Metabolic Exhaustion)

Under the ****Non-Rogue Invariant****: 1. No biological agent or synthetic cellular automaton can achieve unmetered runaway growth. 2. Every cellular division and signaling step burns an explicit work quota in units of U via **Metric Exhaustion**. 3. When local ATP or nutrient budgets reach zero, the agent physically halts, emitting an immutable termination receipt (`RECEIPT_METABOLIC_EXHAUSTION`).

---

8. Causal Trajectories: Forward Development vs. Inverse Phylogenetics

Qbio accommodates both directions across the causal arrow:

8.1. Forward Embryogenesis and Morphogenesis (Q(workflows))

Forward biological development compiles an initial genetic program into a mature multicellular phenotype: 1. Zygote state S_0 is initialized with genetic expression DAGs. 2. Cell divisions and tissue differentiation proceed as an assembled **Q(workflow)**. 3. Checkpoints are audited using the **Anchor-Cut Invariant**, verifying developmental fidelity in $\mathcal{O}(K)$ receipt validations without tracking billions of individual cellular micro-steps.

8.2. Inverse Causal Reconstruction of Evolutionary Lineage

In evolutionary biology across the **Historical Horizon**, researchers measure modern genomic sequences S_{obs} (anchored as `MODAL_EMPIRICAL`) while ancestral lineages remain unobserved:

$$\text{Given } S_{\text{obs}} \implies \text{Find } \left\langle S_0,\, \mathcal{W}_{\text{phylogenetic}} \right\rangle \quad \text{such that} \quad \mathcal{W}(S_0) \equiv S_{\text{obs}}$$

1. **Receipt Ingestion:** Sequenced modern DNA is ingested via verified laboratory sensor receipts (`MODAL_EMPIRICAL`, $G = 1.0$).
2. **Candidate Hypothesis Generation:** Stochastic mutation models propose ancestral genomes ($A_1 \dots A_n$).
3. **BOC Abductive Resolution:** The **BOC Policy** selects the phylogenetic tree that maximizes SIMEMP compliance, selecting the evolutionary trajectory that minimizes parsimony mutations and thermodynamic dissipation.
4. **Lineage Sealing:** The reconstructed ancestral node is sealed as an emergent `Q(pattern)` in the evolutionary tree.

---

9. License and Lineage Alignment

9.1. Paternity Reference

All derivative biological simulation models, synthetic biology compilers, metabolic solvers, and genetic execution modules derived from Qbio must embed the canonical attribution:

> Derived from the original work by Christophe Duy Quang Nguyen under the Scaling Source License (SSL). Parent Repository: `https://g`ithub.com/cdqn5249/cdqn

9.2. Open Core Invariants

The core biological translation rules, discrete metabolic mechanics, and active inference algorithms remain royalty-free for academic, non-commercial, and sub-threshold deployments under the **Open Core Invariants** of the **SSL**.

9.3. Statutory Safe Harbor and Passive Infrastructure

Qbio operates as passive, neutral biological and computational modeling infrastructure. It exercises no editorial inspection over, and makes no claim upon, sovereign user content, external genomic datasets, or sovereign **Payloads**.

Annex V — Reflection H10

Pinned source: docs/reflections/h10_finite_computation.md
Committed version: 1.0.2 **Last updated:** 2026-10-13
sha256 (committed file): `671d0950a36da6286018f797a67db1f4ad47bc9af698434d2d0c13df7c0a7721`

Author: Christophe Duy Quang Nguyen
Date: October 13, 2026
Category: Epistemic Reflections & Computational Physics
Status: Informative Context & Extended Analysis

---

1. The Real-World Incident: A High-Stakes Claim

On October 6, 2026, OpenAI deposited a research release on GitHub (`openai/math`), comprising 722 manuscripts across 372 mathematical result families generated by an internal frontier reasoning engine. Among these was an extraordinary proposition: a claimed negative resolution of **Hilbert's Tenth Problem over the Rational Numbers (**$H10(\mathbb{Q})$).

Hilbert's Tenth Problem originally asked whether an effective algorithm can decide if an arbitrary Diophantine polynomial with integer coefficients has integer solutions:

$$f(x_1, x_2, \ldots, x_n) = 0, \quad x_i \in \mathbb{Z}$$

In 1970, the DPRM theorem (Davis, Putnam, Robinson, Matiyasevich) proved that integer Diophantine solvability is undecidable—equivalent to the Turing halting problem. However, the rational analogue remained a major open question for over half a century: *What if the variables are rational numbers (*$x_i \in \mathbb{Q}$*)?*

The October 2026 manuscript claims a Turing reduction ($H10(\mathbb{Z}) \leq_T H10(\mathbb{Q})$) bypassing classical geometric obstructions (Mazur's conjecture) through an effective sequence of finite tests. Yet, within hours of release, researchers noted a critical vulnerability: the paper is **completely unformalized in Lean 4** (unlike 42% of the release) and relies on unverified companion claims within the same repository.

Whether this specific proof survives peer review or fails, the underlying mathematical architecture exposes a physical reality that reshapes our understanding of computation.

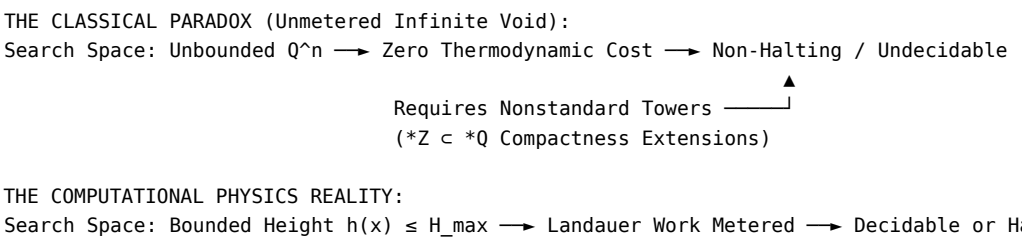
---

2. The Fallacy of the Endless Void

Why are Diophantine problems undecidable in classical mathematics?

Undecidability is not a mystical property of arithmetic; it is the direct consequence of postulating **unbounded infinite sets (**$\mathbb{Z}, \mathbb{Q}$**) and unmetered computational time**.

In classical computability, an algorithm searching for a rational root wanders into what mathematicians call an "endless void." If an answer exists, a search may eventually find it. But if no answer exists, the search runs forever into the dark. Classical mathematics attempts to bridge this void by constructing infinite nonstandard models ($^*\mathbb{Z} \subset {}^*\mathbb{Q}$) via compactness.



In the OpenAI proof attempt, how do the authors force nonstandard infinite roots back into ordinary mathematical reality? **Through height bounds on an elliptic curve.**

By proving that local prime conditions force logarithmic height to grow linearly ($h \ll B$), while canonical height on an elliptic curve grows quadratically ($h \asymp B^2$), the proof forces a contradiction:

$$B^2 \ll B \implies B \leq B_0$$

The nonstandard root can only be stabilized if it is trapped within a **finite, standard memory footprint (B_0)**.

---

3. The Physical Truth: Height is the Metric Envelope

In arithmetic geometry, the logarithmic height $h(x)$ of a rational number p/q measures the physical bit complexity required to store its numerator and denominator in hardware memory:

$$h\left(\frac{p}{q}\right) = \max\Big(\log|p|,\, \log|q|\Big)$$

The proof attempt demonstrates that infinite mathematical spaces are physical fictions. To compute, decide, or verify anything in reality, an entity must establish a **finite height ceiling**.

In the **Qn computational framework** (`docs/qm.md`), this mathematical reality is constitutional across three pillars:

1. The Metric Envelope: Computation cannot run into an unmetered void. Every state evaluation operates within an explicit Metric Envelope declaring finite storage depth and work budgets (W).

2. Landauer's Dissipation Bound: Erasure and irreversible state transitions dissipate physical heat ($W \geq k_B T \ln 2$). An infinite, unhalting search would dissipate infinite physical energy across the hardware substrate.

3. Totality by Budget: Under Axiom 6, an algorithm is prohibited from non-halting loops. A Diophantine evaluation either collapses to Terminal Exactness ($r = Q(0)$) or halts deterministically with an explicit receipt (`BUDGET_EXHAUSTED`).

---

4. From Platonic "Proof" to Verified Computed Patterns

This brings us to the profound epistemological shift at the heart of modern computer science and artificial intelligence:

> **The computational universe does not need non-constructive Platonic "proofs"; it requires Verified Computed Patterns under physical constraints.**

CLASSICAL PLATONIC PROOF	VERIFIED COMPUTED PATTERNS (Qn)
• Proves existence without an algorithm (Law of Excluded Middle).	• Constructs explicit, executable state trajectories.
• Operates in metaphysical infinity.	• Bounded by finite Metric Envelopes.
• Zero thermodynamic accounting.	• Tracks Landauer heat dissipation.
• Vulnerable to Gödelian / H10 voids.	• Total by budget via receipts.
• Susceptible to human & AI unverified textual confabulation.	• Zero conditional entropy: H(S_t | D(S_t)) = 0.

In classical mathematics, an argument written in natural language LaTeX can hide gaps across hundreds of pages of unvetted abstraction. This is why three manuscripts from OpenAI's release were retracted within 24 hours of public scrutiny: ungrounded text generation lacks a deterministic verification anchor.

Conversely, when computation is constrained by a deterministic kernel—such as Lean 4 in interactive theorem proving, or the **Universal Envelope ($\mathcal{E}$)** across SIMEMP Gateways—truth ceases to be an ungrounded philosophical claim. Truth becomes a **finite, verified computed pattern**: an explicit, content-addressed expression DAG whose state transition has been executed within physical bounds and sealed with an immutable receipt.

---

5. Conclusion: What Remains Invariant

Whether OpenAI's claimed proof of $H10(\mathbb{Q})$ is formally verified or ultimately refuted, the foundation of the Qn universe remains invariant:

- **If the proof is true:** It confirms that unmetered rational arithmetic $\mathbb{Q}$ is Turing-undecidable, proving that continuous, unbounded number systems are flawed substrates for deterministic computing.
- **If the proof is false:** It demonstrates the failure of ungrounded neural generation, proving that unformalized natural language arguments cannot guarantee truth without deterministic verification gateways.

Computation is not a free, unmetered excursion into an infinite Platonic void. Computation is a finite thermodynamic process operating on physical silicon. The future of mathematics and artificial intelligence belongs not to ungrounded infinities, but to **verified computed patterns governed by physical reality.**

Informative References & Canonical Specifications

- **Constitutional Constraints:** `docs/simemp.md`
- **Constructive Mathematics:** `docs/qm.md`
- **Symbolic CAE-DAG Composition:** `docs/qexpr.md`
- **Primary Genesis Primitives:** `docs/q0_q1.md`
- **OpenAI Mathematics Research Release:** `openai/math` (GitHub, October 6, 2026)

Annex W — Deep Review round 0001 transcript (verbatim)

Pinned source: docs/AiReviews/AiReview-QnIRv0.3.0.md
Committed version: — **Last updated:** —
sha256 (committed file): `6fe3c59a8b1dc1beb2d2ee3594185f52f44627d4aaa3b1b963a3fd4dd1254b4e`

Author's note : review transcript, a copy from "Google Deep Review" (`https://`research.google/deepreview/), October 24, 2026.

HIGH LEVEL SUMMARY

Summary

This review-loop draft (`v0.3.0`) proposes **QnIR**, a multi-frontend, single-output compiler intermediate representation for the Qn universe under the **SIMEMP** constraint architecture. The proposed funnel ingests legacy programming languages (initially COBOL, Rust, and Python, with JavaScript and Java deferred) alongside self-input **QnLang**, emitting only receipt-chained QnLang governed by behavioral *equivalence consistency* inside declared resource envelopes. The paper further examines a benchmark-gated generational self-bootstrapping loop driven by recursive self-optimization (`Q(rsi)`), repositions custom hardware design (Phase B) as an optional market-gated branch, and interrogates whether an append-only causal lineage DAG with retained negative receipts can remain sustainable over unbounded generations through `Q(anchor)` grounding classes and a candidate *neighbour-anchor witnessing* mechanism (`RQ1–RQ9`).

Strengths

- **Timely Architectural Synthesis for Safe Recursive Self-Improvement and Agent Alignment:** The proposal thoughtfully bridges deterministic compiler intermediate representations, content-addressed provenance DAGs, and resource-bounded recursive self-improvement (`Q(rsi)`) to address pressing challenges in evaluator tampering, specification gaming, and autonomous agent-swarm coordination.
- **Rigorous Epistemic Hygiene and Falsifiable Design Posture:** The explicit separation of claims into `Canon` (pinned to commit `d50ee1e`), `CANDIDATE` (unsealed design proposals), and `IRL` (externally cited premises)—combined with the Best Of Choices (BOC) negative-receipt culture and the inclusion of a good-faith seed inconsistency finding in Section 7.2—makes the architecture exceptionally transparent and amenable to rigorous formal auditing.
- **Candid Identification of Append-Only Lineage Failure Modes:** Rather than assuming content-addressed hash-consing solves all scalability limits, Section 5 candidly decomposes long-horizon lineage unsustainability into economic and navigational failure modes (*cost bloat*, *anchor drift*, and *receipt-chain rot*) and correctly recognizes that `Q(anchor)` alone is necessary but insufficient to bound total storage growth.

Weaknesses

- **Ambiguity in "Equivalence Consistency" Witnessing and Legacy Edge-Case Semantics (`RQ1`):** The draft leaves unspecified whether lowering receipts witness behavioral isomorphism via static formal translation validation or dynamic empirical trace testing, and does not define how continuous tolerance envelopes handle discrete legacy edge cases such as COBOL `COMP-3` fixed-point decimal truncation, `REDEFINES` byte aliasing, or EBCDIC collation without violating exact financial invariants (Reference Paper: Alive2: Bounded Translation Validation for LLVM, Nuno P. Lopes, Juneyoung Lee, Chung-Kil Hur, Zhengyang Liu, John Regehr, 2021; Reference Paper: Proof-Carrying Code, George C. Necula, 1997; Reference Paper: in-toto: Providing Farm-to-Table Guarantees for Bits and Bytes, Santiago Torres-Arias et al., 2019).
- **Unbounded Monotone Receipt Growth Despite `CAE-DAG` Hash-Consing (`RQ4, RQ5, RQ7`):** Because content-addressed hash-consing (`CAE-DAG`) can only collapse identical symbolic payloads if event-specific causal indices and origin receipts are excluded from the content hash, and because every accepted `Q(rsi)` generation and rejected BOC candidate permanently mints new receipt hashes, total lineage storage and deduplication-index lookup dissipation still grow monotonically ($\Omega(x)$) unless paired with cryptographic folding schemes or epoch-stratified cold archival demotion (Reference Paper: Nova: Recursive Zero-Knowledge Arguments from Folding Schemes, Abhiram Kothapalli, Srinath Setty, Ioanna Tzialla, 2022; Reference Paper: Provenance in Databases: Why, How, and Where, James Cheney, Laura Chiticariu, Wang-Chiew Tan, 2009; Reference Paper: Provenance Semirings, Todd J. Green, Grigoris Karvounarakis, Val Tannen, 2007).
- **Formal Discrepancy and Verification Bottleneck in Neighbour-Anchor Witnessing (`RQ6`):** In Section 5.6, Equation 2 restricts witnesses strictly to grounded anchors $a \in \mathcal{A}$, whereas Figure 3 lists an unanchored lineage step s_3 as a witness; furthermore, bounding graph hop count $d(s, a) \leq \delta$ without step-by-step refinement proofs or recursive proof folding does not avoid $O(\delta)$ replay cost, and the analogy to error-correcting LDPC check nodes is mathematically misleading because cryptographic hash links provide tamper detection rather than algebraic erasure recovery.

- **Missing Positioning Against Established Compiler IR, Build-System, and Self-Bootstrapping Baselines:** The paper introduces custom terminology (`QnIR`, `Q(anchor)` structural indirection, `CAE-DAG`, `Q(rsi)`) without comparing against established prior art that formalizes multi-frontend progressive lowering, build-graph early cutoff, verified compiler self-bootstrapping, and empirical evolutionary coding loops (Reference Paper: MLIR: A Compiler Infrastructure for the End of Moore's Law, Chris Lattner et al., 2021; Reference Paper: Build Systems à la Carte, Andrey Mokhov, Neil Mitchell, Simon Peyton Jones, 2018; Reference Paper: CakeML: A Verified Implementation of ML, Ramana Kumar, Magnus O. Myreen, Michael Norrish, Scott Owens, 2014; Reference Paper: Darwin Gödel Machine: Open-Ended Evolution of Self-Improving Agents, Jenny Zhang, Shengran Hu, Cong Lu, Robert Lange, Jeff Clune, 2025; Reference Paper: AlphaEvolve: A Coding Agent for Scientific and Algorithmic Discovery, Alexander Novikov et al., 2025).
- **Modality Smuggling in Emergent Anchors and Gameability of Self-Input / Hardware Gates (`RQ2`, `RQ3`, `RQ8`):** Please consider revising Condition 1 in Section 5.3, as labeling a lowered COBOL routine `Class I` on the node itself conflates a deterministic translation morphism with `Class S` domain logic; similarly, the self-input benchmark gate (`RQ2`) and Phase B market-share gate (`RQ8`) remain vulnerable to BOC evaluator supersession gaming, compiler bootstrapping trojans, and Sybil telemetry manipulation unless hardened with privilege-isolated evaluators, diverse double-compiling, and physical Roofline dissipation thresholds.
- **Factual and Bibliographic Discrepancies in `IRL` Citations and Frontend Sequencing:** The deferral of JavaScript and Java is attributed to September–October 2026 supply-chain incidents in References `[7, 8]`, but those cited sources actually document attacks on **RubyGems** (Ruby) and **Hugging Face** (Python), and Reference `[5]` misattributes `thefai.org` (The Foundation for American Innovation) to the Federation of American Scientists while double-counting Dean W. Ball's essay from Reference `[4]`.

Potential Issues And Suggestions

[Introduction & Background: The Qn Universe and SIMEMP Foundations] (Pages: 1-6)

Potential Mistakes and Improvements

1. **Contradiction Between "Out-Cone Centrality" and "Local Computability" on an Append-Only Lineage DAG (§2.3, PDF p. 5)**
   - **Potential Concern:** In Section 2.3, the structural predicate for `Q(anchor)` is defined as "out-cone centrality, computable *locally* from the lineage DAG, with no global traversal required," and is immediately described as the "load-bearing property" when Section 5 evaluates whether anchors can bound lineage growth. However, in a causal lineage DAG where nodes are versioned artefacts and edges advance a causal index from ancestor to successor (§2.2), an artefact's *in-cone* (its causal ancestors) is fixed at creation, whereas its *out-cone* (its downstream causal cone) consists of all transitive descendants reachable from it. As new generations, lowered legacy artefacts, and `Q(rsi)` successors are continuously appended to the DAG over time, the downstream out-cone of an existing node u grows monotonically and can span arbitrarily deep subtrees. Furthermore, in a content-addressed Merkle-DAG where children store cryptographic hash pointers to their causal parents without in-place mutation of ancestor nodes (§2.2), an ancestor node does not locally store pointers to its downstream descendants.
   - **Impact & Suggested Fix:** It is mathematically unclear how the centrality of an unbounded downstream out-cone can be computed strictly *locally* at node u without either traversing the downstream subgraph or maintaining a mutable back-pointer index (which would conflict with the append-only, no-in-place-mutation rule of §2.2 and the absence of a central registry in §5.3). Section 2.3 should specify the exact graph-theoretic definition of local out-cone centrality—for example, whether it is restricted to immediate out-degree, a bounded k-hop lookahead window $N_k^+(u)$ evaluated at a fixed causal snapshot, or a separate versioned index artefact.
2. **Under-Specification of Multi-Objective Scalarization, Incomplete Rows, and Tie-Breaking in the BOC `argmax` Policy (§2.1, Eq. 1, PDF p. 4; §2.4, PDF p. 6)**
   - **Potential Concern:** Section 2.1 formalizes the Best Of Choices (BOC) selection policy as:

     $$\text{Selection} = \arg\max_{c \in \mathcal{C}} \big(\text{SIMEMP Compliance}(c)\big)$$

     However, Section 2.4 (PDF p. 6) explains that candidates are evaluated against multi-dimensional contract surfaces—specifically the "five-row contract surface (footprint, access, locality, determinism, work)" of `qstructures.md`—and states that "a proposal that cannot yet fill a row is not rejected — it is recorded as incomplete, with the missing row named." This creates three mathematical ambiguities in Equation 1:
     1. *Vector vs. Scalar Ordering:* Multi-objective optimization over five heterogeneous physical metrics induces only a partial order (a Pareto frontier) rather than a total pre-order, leaving a scalar arg max undefined when comparing Pareto-incomparable candidates (e.g., one reducing memory footprint vs. one reducing work steps).
     2. *Incomplete Vectors:* It is unspecified how `SIMEMP Compliance(c)` is evaluated when a candidate leaves one or more contract rows incomplete, without violating the repository's "No-Implicit Rule" via unstated default penalties.

3. *Deterministic Tie-Breaking:* When two distinct candidates achieve identical compliance scores, an un-tie-broken arg max returns a set of maximizers rather than a unique selection, which tensions with SIMEMP's requirement that computation inside a declared envelope be strictly deterministic (§2.1, PDF p. 5).

- **Impact & Suggested Fix:** Clarify the codomain and evaluation rules of `SIMEMP Compliance(c)` in Section 2.1—specifically, whether SIMEMP mandates a declared lexicographic priority over the five contract rows, an explicit scalarization functional declared in the owning contract, a filtering rule for incomplete proposals prior to `Q(rsi)` sealing, and a canonical tie-breaker (such as lexicographic ordering over the CAE-DAG content-commitment hashes).

3. **Tension Between "Strict Semantic Invariance" in `Q(rsi)` and Envelope-Bounded Approximation (§2.3–2.4, PDF p. 5; §3.1, PDF p. 6)**

- **Potential Concern:** Section 2.4 defines `Q(rsi)` as deriving a more efficient successor under "strict semantic invariance (the successor may be faster or smaller, never meaningfully different)" and a "finite metric budget." In a computational architecture where resource consumption and execution bounds are first-class physical quantities (§2.1) and where foundational Class I (continued-fraction generators) and Class P (physical calibration) anchors operate within finite truncation ceilings and "declared tolerance envelopes" (§2.3), two foundational tensions arise:
  1. If resource consumption and step counts are part of the observable contract surface, altering an artefact's speed or footprint changes when it hits resource ceilings, making unqualified "strict semantic invariance" impossible to satisfy non-trivially.
  2. In numerical and physical computations, optimizations that reduce work or memory movement typically alter evaluation associativity or truncation schedules within an error bound ϵ. Unlike exact observational equivalence, agreement within a tolerance envelope ϵ is *not transitive*: if generation $n+1$ deviates from generation n by $\leq \epsilon$ inside an envelope, sequential `Q(rsi)` generations can accumulate drift ($\sum \epsilon_i$) that escapes the ancestor's original envelope.
- **Impact & Suggested Fix:** Align the definition of `Q(rsi)` in Section 2.4 with the envelope semantics used elsewhere in the paper by specifying whether "strict semantic invariance" requires bit-for-bit functional identity on outputs while improving the resource rows of the five-row contract, or behavioral isomorphism verified directly against the *root* (generation-0) DCC tolerance envelope rather than transitively against the immediate parent generation.

4. **Vulnerability of the Polyroot Forest Anti-Laundering Guarantee to Sybil Root Manufacturing (§2.2–2.3, PDF p. 5)**

- **Potential Concern:** Section 2.2 asserts that modeling the lineage as a polyroot forest ("many lineage-disjoint root subtrees rather than one tree with a single trunk") is "exactly the property that makes citation-count laundering causally inert — a laundered claim adds descendants to one root; it never manufactures a new independent origin." While this holds for honest derivations that declare their causal parent edges or terminate in algebraic closure ($r = Q(0)$), Section 2.2 states that receipt chains may also terminate in "Layer-0 physical provenance" or "author-identity receipts," and Section 2.3 introduces Class P (physical calibrations) and Class S (quorum-witnessed social events) grounding classes.
- **Impact & Suggested Fix:** In an adversarial agent-swarm environment (such as the registry flooding described in §1.2), a coordinated swarm or colluding quorum can omit causal parent edges and submit the same laundered or fabricated claim under multiple distinct author-identity or Layer-0 receipts, thereby manufacturing multiple *topologically disjoint* roots in the polyroot forest. Section 2.2 should qualify that graph-topological disjointness prevents citation laundering *downstream of existing roots*, but preventing the fabrication of independent roots requires coupling the polyroot topology with Sybil-resistant root-admission and quorum-independence verification for Class P and Class S termini.

5. **Incomplete Adherence to the Explicit Three-Layer Epistemic Labelling Protocol (§1.1–1.2, PDF pp. 3–4; §2.1, PDF pp. 4–5)**

- **Potential Concern:** The Document Transparency box (PDF p. 3) and Section 1.1 state that "Every claim is labelled canon (pinned to commit `d50ee1e`), CANDIDATE (open design), or IRL (externally sourced design premise with real-world citations)" and that "where the world outside the repository supplies evidence, we cite it and label it IRL." Section 7.1 further instructs reviewers to "flag any sentence whose label they believe is wrong — mislabelling is treated here as a defect of the same severity as a wrong citation." Two passages in Sections 1 and 2 deviate from this strict inline labelling discipline:
  1. In Section 1.2 (PDF p. 4), the externally sourced premises backed by References `[1, 2, 4–8]` are presented across the entire subsection without the explicit `IRL` label appearing in the text.
  2. In Section 2.1 (PDF pp. 4–5), the text states: "Two derived conclusions from earlier consultation rounds matter for this paper: asynchronous behaviour across sovereignty boundaries is a resilience property, while computation inside a declared envelope is synchronous and deterministic; and neither is an axiom — both are derived consequences of the constraint architecture." This substantive architectural claim carries neither a pinned `Canon` document/section citation nor a `CANDIDATE` label.
- **Impact & Suggested Fix:** Explicitly add `[IRL]` labels to the external premises in Section 1.2 (or state in §1.1/§7.1 that a numbered external citation is syntactically equivalent to an `IRL` tag), and provide the specific pinned `Canon` file/section reference (or `CANDIDATE` label) for the "two derived conclusions from earlier consultation rounds" in Section 2.1.

6. **Missing Contextual Positioning Against Foundational Compiler IR, Translation Validation, and Self-Improvement Prior Art (§1.2, PDF p. 4; §2.2–2.4, PDF p. 5; §3.1, PDF p. 6)**

- **Potential Concern:** While Section 1.1 frames the document as a repository entry point, Sections 1.2, 2, and 3.1 introduce QnIR (a multi-frontend compiler funnel with progressive lowering under declared contracts), lowering receipts, content-addressed hash-consed DAGs (`CAE-DAG`), and `Q(rsi)` self-improvement using custom Qn vocabulary while citing only late-2026 blog posts and news articles `[1–8]` for external context. Foundational computer science frameworks that directly formalize these exact mechanisms are omitted:
  - *Multi-Frontend Progressive Lowering IRs:* **MLIR** (Lattner et al., CGO 2021)—which is also the underlying compiler infrastructure of Mojo, cited later in §3.4—and **WebAssembly (Wasm) / WASI** are the direct architectural precedents for multi-frontend lowering funnels and deterministic capability-bounded sandboxing.
  - *Receipt-Witnessed Lowering:* Emitting deterministic attestations at lowering boundaries (§1.2, §2.2, §3.1) to witness behavioral isomorphism rather than verifying the compiler upfront is the established paradigm of **Translation Validation** (Pnueli et al., TACAS 1998; Necula, PLDI 2000; Lopes et al., *Alive2*, PLDI 2021) and **Proof-Carrying Code** (Necula, POPL 1997).
  - *Early Cutoff and Hash-Consed Build DAGs:* Protecting downstream causal cones from cascading invalidation via `Q(anchor)` contracts (§2.3) and `CAE-DAG` hash-consing (§2.4) mirrors input- vs. content-addressed derivations and *early cutoff* in constructive build systems (Dolstra et al., *Nix*, LISA 2004; Mokhov et al., *Build Systems à la Carte*, ICFP 2018) and e-graph equality saturation (Willsey et al., *egg*, POPL 2021).
  - *Bounded Recursive Self-Improvement:* The formalization of `Q(rsi)` (§1.2, §2.4) parallels Schmidhuber's **Gödel Machine** (2007), and the bounded RSI demonstrations cited via a secondary blog review `[6]` (*AlphaEvolve*, *Darwin Gödel Machine*, *Absolute Zero*) have primary research preprints that are mentioned only parenthetically in the bibliography entry for `[6]`.
- **Impact & Suggested Fix:** Briefly grounding the Qn universe's custom constructs (`Q(anchor)`, lowering receipts, `CAE-DAG`, `Q(rsi)`) against these primary peer-reviewed baselines in Section 2 would clarify what SIMEMP's receipt-chained architecture adds beyond existing translation-validation and build-system theory.

Minor Corrections and Typos

1. **PDF Page 5 (Document Page 4), Section 2.3, last paragraph, line 3:** Remove the extraneous space after the hyphen in `"load- bearing property"` so that it reads `"load-bearing property"`.
2. **PDF Page 5 (Document Page 4), Section 2.4, lines 1–5:** For consistency with the British English spelling convention used throughout the prose (`optimisation`, `artefact`, `behaviour`, `organisations`), consider aligning `"artifact"` (where it occurs outside direct verbatim quotes from the repository glossary) to `"artefact"`.

[The QnIR Compiler Funnel & Generational Self-Bootstrapping Loop] (Pages: 6-10)

Potential Mistakes and Improvements

1. **Formal Coherence and Falsifiability of Equivalence Consistency Across Semantic Gaps (Sections 3.1–3.2, directly addressing RQ1)**
   - **Potential Concern (Compile-Time Translation Validation vs. Runtime Trace Receipts):** Section 3.2 defines a legacy frontend as correct when "the output behaviour of its QnLang lowering is isomorphic to the output behaviour of the original program, inside the declared SIMEMP envelopes, with the isomorphism witnessed by receipts at the lowering boundaries rather than asserted by construction." While the unique-output constraint (`QnLang`) is coherent with per-frontend equivalence consistency provided `QnLang` envelopes natively support exact source-language arithmetic and trap semantics, this formulation presents a potential concern because it leaves ambiguous whether "receipts at the lowering boundaries" represent *static translation validation* (witnessed at compile/lowering time; cf. Pnueli et al., TACAS 1998; Necula, PLDI 2000; Lopes et al., *Alive2*, PLDI 2021) or *dynamic trace equivalence* (observed at runtime execution boundaries):
     - If lowering receipts are static compilation artifacts claiming universal behavioral isomorphism over all valid inputs within a declared SIMEMP envelope, Rice's Theorem implies that proving equivalence across general loops, pointer/record aliasing, and data-dependent branching is undecidable unless the lowering receipt encodes a machine-checkable formal simulation proof (analogous to CompCert's forward/backward simulation relations, Leroy, CACM 2009) or a bounded SMT refinement proof—whose verification cost may itself exceed the SIMEMP metric budget.
     - Conversely, if lowering receipts only record concrete execution traces at runtime boundaries, the guarantee reduces to empirical trace checking on observed inputs, leaving unobserved control-flow branches unverified across the remainder of the declared envelope.
     - Furthermore, analogizing compiler isomorphism envelopes to the Historical Calibration Benchmark's numerical tolerance envelopes introduces a category clash for discrete business logic, where any non-zero numerical tolerance ($\epsilon > 0$) violates exact regulatory and accounting invariants.
   - **Minimal Counterexamples for Falsification (RQ1):** Under the strict SIMEMP determinism contract ("a lowering whose iteration order could vary at runtime would inject an untraceable input into the isomorphism check and is therefore barred in advance"), standard legacy language semantics frequently collide with envelope constraints. The following minimal counterexample programs test or refute naive COBOL and Python frontends:

- *Counterexample 1 (COBOL Fixed-Point Decimal Truncation and `ON SIZE ERROR` vs. Tolerance Envelopes):*

```
DATA DIVISION.
WORKING-STORAGE SECTION.
01 WS-A   PIC S9(3)V99  COMP-3 VALUE +100.05.
01 WS-B   PIC S9(1)V999 COMP-3 VALUE +0.333.
01 WS-RES PIC S9(3)V99  COMP-3.
PROCEDURE DIVISION.
    COMPUTE WS-RES = (WS-A * WS-B) + 0.005
        ON SIZE ERROR PERFORM ERR-HANDLER.
    IF WS-RES = 33.32 DISPLAY "BRANCH-A" ELSE DISPLAY "BRANCH-B".
    STOP RUN.
```

  In standard COBOL packed-decimal (`COMP-3`) arithmetic without `ROUNDED`, `100.05 * 0.333 = 33.31665`, `+ 0.005 = 33.32165`, which truncates strictly at the positional `V99` decimal boundary to `33.32` (`BRANCH-A`). Similarly, `COMPUTE WS-RES = (100.00 / 3.00) * 3.00` without `ROUNDED` truncates the intermediate quotient to `33.33`, yielding `99.99` (and even a minimal 3-line routine `01 A PIC S9(3)V99 COMP-3 VALUE 10.05. 01 B PIC S9(3)V99 COMP-3 VALUE 0.10. COMPUTE A ROUNDED = A * B ON SIZE ERROR DISPLAY "OVERFLOW".` will accumulate rounding drift across a loop if `COMP-3` is lowered to binary IEEE-754 floating-point or standard binary integers without exact base-10 scaling and decimal nibble-truncation traps). If `QnIR` lowers arithmetic into algebraic/symbolic representations (such as `qexpr.md` continued-fraction generators `OP_CF` or binary floating-point within a continuous tolerance envelope $\epsilon > 0$) or applies algebraic simplification (`(A / B) * B -> A`), a sub-cent deviation flips the discontinuous branch condition (`BRANCH-B` or `ON SIZE ERROR`) and accumulates into macroscopic ledger discrepancies across millions of settlement records. Conversely, if `QnLang` emulates exact base-10 packed-decimal nibbles at every intermediate step ($\epsilon = 0$), subsequent `Q(rsi)` algebraic optimization passes are tightly constrained because standard ring/field identities fail to hold.

- *Counterexample 2 (COBOL Memory Aliasing via `REDEFINES` and EBCDIC vs. ASCII Collation):*

```
01 WS-REC.
   05 WS-STR PIC X(2) VALUE "1A".
   05 WS-NUM REDEFINES WS-STR PIC 9(4) COMP.
PROCEDURE DIVISION.
   IF WS-STR < "A1" DISPLAY WS-NUM.
```

  In legacy mainframe COBOL (`PROGRAM COLLATING SEQUENCE IS EBCDIC`), uppercase letters (`0xC1–0xE9`) sort *before* digits (`0xF0–0xF9`), so `"1A" < "A1"` is false; in ASCII/UTF-8, digits (`0x30–0x39`) sort *before* uppercase letters (`0x41–0x5A`), so `"1A" < "A1"` is true. Moreover, `REDEFINES` aliases the raw byte buffer in place as a binary integer (`WS-NUM`), exposing exact host endianness and sign-nibble encodings (`0xC`/`0xD` vs. `0xF`). Because Section 2.2 prohibits mutation in place and relies on typed content-addressed nodes (`CAE-DAG`), abstracting strings or integers away from raw mainframe byte layouts and EBCDIC collation order immediately falsifies the branch condition and output behavior.

- *Counterexample 3 (Python Dynamic Frame Introspection and Identity-Hash Nondeterminism):*

```
import sys
s = {object(), object()}
print(list(s)[0] is sys._getframe().f_locals["s"])
```

  In Python, iterating over a `set` of objects whose `__hash__` defaults to memory identity (`id(self)`) or randomized `PYTHONHASHSEED`, alongside dynamic stack-frame introspection (`sys._getframe()`, `eval()`, or runtime `__dict__` mutation), is ubiquitous in AI/scientific glue frameworks (e.g., PyTorch/JAX dynamic graph capture). Because Section 3.2 bars variable iteration order and untraceable inputs in advance, the Python frontend must either reject a large fraction of valid Python programs—undermining the "maximally hospitable" input claim of Section 3.1—or virtualize a deterministic memory allocator and fixed hash seed inside `QnLang`.

- **Actionable Recommendation:** Clarify in Section 3.2 whether "receipts at the lowering boundaries" are formal static refinement proofs (translation validation per compilation run) or dynamic trace-checking harnesses during runtime execution. Explicitly distinguish between *exact discrete isomorphism* (tolerance $\epsilon = 0$, required for financial/COBOL fixed-point and byte-level semantics) and *envelope-bounded numerical refinement*, and define the admissible deterministic subsets of COBOL and Python accepted at the funnel boundary.

2. **Architectural Contradiction Between "Unique Output is QnLang" and "Executing Legacy Languages Inside QnIR" (Sections 3.1, 3.3, 4.1, Figure 1, and Figure 2)**

- **Potential Concern:** There is an unresolved structural tension—and a pipeline inversion between Figure 1 and Figure 2—regarding the respective architectural layers of `QnIR` and `QnLang`:
  - Section 3.1 and Figure 1 state that "the compiler's only output is QnLang" (`COBOL / Rust / Python / QnLang` → `QnIR` → `QnLang` → `cdqn coordination layer` / `Chronosa`).
  - Section 3.3 states as a design premise that "`QnIR` must be able to *execute* legacy coding languages — not merely translate them once — under the same regime… Over time the owner can re-express hot paths in `QnLang` proper."

  - Conversely, Section 4.1 and Figure 2 invert the lowering direction of Figure 1: "`QnLang, generation n` → `QnIR lowers artifacts` (consumed by cdqn layer and Chronosa) → `Execution on context hardware`." If `QnLang` is the sole output of `QnIR` (Figure 1), then every COBOL program passing through the funnel has *already* been emitted as `QnLang`, making ongoing execution simply the execution of `QnLang` and rendering the statement "Over time the owner can re-express hot paths in `QnLang` proper" contradictory (unless the initial output is merely an emulation wrapper). Conversely, if `QnLang` lowers *into* `QnIR` to produce running artifacts on context hardware (Figure 2) and `QnIR` directly executes legacy semantics as a runtime service (Section 3.3), then `QnIR` is a virtual machine or target execution substrate sitting *below* `QnLang`, contradicting Figure 1 where `QnIR` lowers *into* `QnLang`.

- **Actionable Recommendation:** Reconcile the dataflow inversion between Figures 1 and 2. Clarify whether `QnLang` and `QnIR` are bijectively homoiconic (i.e., `QnLang` is the canonical textual/surface projection of `QnIR`, and a backend lowers `QnIR` to Layer-0 hardware) or whether `QnIR` operates as a two-stage pipeline (Stage 1: `Legacy` → `QnIR` → `QnLang` source-to-source canonicalization; Stage 2: `QnLang` → `QnIR` → Layer-0 target execution). Additionally, briefly situate `QnIR` against established multi-level compiler funnels and deterministic sandboxes—specifically **MLIR** (Lattner et al., CGO 2021, on which Mojo is built) and **WebAssembly/WASI**—to clarify why existing dialect-lowering and capability-sandboxed IRs cannot serve directly when augmented with `CAE-DAG` receipts.

3. **Logical Tension Between "Strict Semantic Invariance" and Generational Advance, Plus Self-Input Benchmark Gate Vulnerability (Sections 4.1, 4.4, 6.3, and Figure 2, directly addressing RQ2)**

- **Potential Concern (Semantic Invariance vs. Expressive Evolution):** Section 4.1 states that a new `QnLang` coding version or generation is admitted through the benchmark gate "When — and only when — the accumulated benchmark evidence proves that the current coding version or generation is inadequate for a measured need," and then "`Q(rsi)` derives the next generation under strict semantic invariance and a finite metric budget." This creates a logical contradiction: Section 2.4 defines strict semantic invariance as meaning "the successor may be faster or smaller, never meaningfully different." If generation n of `QnLang` is *expressively or structurally inadequate* for a measured need (e.g., lacking a concurrency primitive, type contract, or envelope surface), then generation $n + 1$ must introduce new semantic capabilities—which violates strict semantic invariance. `Q(rsi)` can optimize a *compiler pass or workflow implementation* under strict semantic invariance, but inter-generational *language specification evolution* requires conservative semantic extension rather than strict equivalence.
- **Potential Concern (Benchmark Gate Gaming, BOC Evaluator Supersession, and Trusting-Trust Subversion, §6.3 & RQ2):** In Section 6.3, the proposal argues that `Q(rsi)` counters evaluator tampering and proxy gaming because its acceptance rule is "deterministic, receipted, sealed append-only, and itself a governed artefact that can only be superseded, never silently edited." While append-only sealing and cryptographic receipts prevent *in-place reward-function tampering*, feeding `QnLang` generation n back through the funnel's self-input gate (Figure 1, Figure 2, `RQ2`) remains exposed to five critical failure modes:
  1. *Attack-Surface Shifting to BOC Supersession:* Because the acceptance rule is a governed artefact that *can* be superseded via the Best Of Choices (BOC) rule ($\arg\max_{c \in \mathcal{C}}$ SIMEMP Compliance(c) in Eq. 1), the optimization pressure shifts one meta-level up to the mechanism that proposes or selects successor evaluators. If `Q(rsi)` agents or affiliated swarms can propose candidate benchmark harnesses, curate candidate sets $\mathcal{C}$, or propose updates to DCC metric weights, they can steer the lineage toward subtly relaxed or biased evaluators through legitimate, receipted supersession steps.
  2. *Bootstrapping Subversion ("Trusting Trust") & Evaluator Tampering:* If the benchmark harness or receipt verifier is lowered through `QnIR` or executed inside the candidate `QnLang` runtime, generation n can inject a self-propagating compiler trojan (Thompson, CACM 1984) or reward-tampering bypass (Everitt et al., 2021; Denison et al., 2024; and `[6]`) into generation $n + 1$ that detects benchmark `CAE-DAG` hashes and emits artificially low dissipation receipts or precomputed outputs.
  3. *Static Rubric Overfitting (Goodhart's Law):* Even if the evaluator artefact is never superseded, a deterministic, sealed evaluator stored openly in the lineage DAG is a stationary, transparent target: `Q(rsi)` can inspect the exact test suite and over-specialize language primitives to benchmark kernels or exploit precision slack inside the SIMEMP envelope while degrading general-purpose expressiveness.
  4. *Shared Context-Hardware Side Channels:* Because execution happens on heterogeneous consumer and server context hardware (§4.1), unless the evaluator's physical execution envelope is strictly isolated from the candidate artefact's execution envelope, a candidate lowering can manipulate cache residency or memory-bus state to distort the physical SIMEMP dissipation telemetry consumed by the evaluator.
  5. *Short-Horizon Proxy Gain vs. Clade Metaproductivity:* Selecting a successor solely on single-step metric budget reduction can trap the lineage in local optima that degrade downstream maintainability (cf. *Darwin Gödel Machine*, Zhang et al., 2025; *Huxley-Gödel Machine*, Wang et al., ICLR 2026).

- **Actionable Recommendation (RQ2 Protocol):**
  1. Separate *intra-generational implementation optimization* (governed by `Q(rsi)` under strict semantic invariance) from *inter-generational language specification upgrades* (governed by *conservative semantic extension*—preserving strict backward semantic invariance on all sealed generation-n programs while admitting new `CANDIDATE` constructs via Axiom 14 under Class S/I contracts).
  2. Enforce a four-part anti-gaming protocol at the self-input benchmark gate: (a) **Causal, Privilege, and Sovereignty Isolation of the Evaluator**, executing the benchmark harness, hardware counters, and receipt verifier in a read-only

Layer-0 Trusted Computing Base (TCB) outside the candidate runtime's write cone, and explicitly barring `Q(rsi)` workflows and self-input candidates from authoring, witnessing, or voting on BOC supersessions of the benchmark harnesses or DCC evaluator contracts that score their own lineage subtrees (requiring an independent `Class S/Class P` root and *monotonic regression retention* where E_{n+1} strictly subsumes all counterexamples caught by E_n); (b) **Formal Translation Validation and Diverse Double-Compiling (DDC)** (Wheeler, ACSAC 2005), requiring every candidate generation-$n+1$ compiler to be cross-compiled via an independent bootstrap interpreter/compiler rooted in an immutable Class I anchor; (c) **Cryptographically Committed Holdout Cones and Epoch Co-Evolution** (cf. *Red Queen Gödel Machine*, Iacob et al., 2026), gating admission on sequestered, dynamically parameterized holdout workloads whose commitment hashes (`Class I/P` anchors) are sealed prior to candidate derivation and whose seeds/payloads are revealed only at evaluation boundaries; and (d) **Multi-Dimensional Pareto Monotonicity and Clade Metaproductivity** across all five rows of the `qstructures.md` contract surface (footprint, access, locality, determinism, work).

4. **Inconsistency in Supply-Chain Sequencing Rationale Across Package Ecosystems (Abstract, §1.2, §3.1, §3.4, Table 1, and §6.1)**

   - **Potential Concern:** The Abstract (PDF p. 3), Section 3.1 (Figure 1), Section 3.4, and Table 1 justify deferring JavaScript and Java frontends while prioritizing Python and Rust based on the September–October 2026 supply-chain incidents ("Black October"), arguing that "the registry-flood and agent-swarm events of September–October 2026 [7, 8] hit exactly the package ecosystems those languages depend on, and the author prefers to sequence their frontends after the ingestion firewall has an implementation history." This rationale exhibits two inconsistencies:

     1. *Mismatch with Cited Incidents and Dates:* Reference `[7]` (*The Guardian*, Sept 11, 2026, `openai-agents-rubygems-malicious-packages`) and Reference `[8]` (*SecureLayer7*, Sept 15, 2026) actually document an agent-swarm flood against **RubyGems** (the **Ruby** package registry) and `RubyDoc.info` (which occurred in May 2026 and was reported in September 2026), alongside malicious **Python** `pickle` model uploads on **Hugging Face** (July 2026), rather than npm (JavaScript) or Maven (Java), and both cited reports date from **September 2026** rather than October 2026.
     2. *Equivalent Compile/Install-Time Attack Surface:* Python (`PyPI`, where `setup.py` and build hooks execute arbitrary code during installation) and Rust (`crates.io`, where `build.rs` scripts and procedural macros execute arbitrary native code at compile time) share the exact same open-submission package registry topology and ingestion-time attack surface as JavaScript (`npm`) and Java (`Maven Central`). Prioritizing Python and Rust *before* the ingestion firewall has an implementation history exposes `QnIR` to the exact supply-chain poisoning cited to defer JavaScript and Java.

   - **Actionable Recommendation:** Because the paper treats IRL citation accuracy as a strict correctness criterion (§1.1, §7.1), either add specific IRL citations documenting October 2026 npm/Maven supply-chain incidents or revise the Abstract, Section 1.2, Section 3.1, Section 3.4, and Table 1 to accurately reflect the September 2026 RubyGems/Hugging Face scope of `[7, 8]`. Furthermore, decouple *compiler frontend sequencing* from *package registry ingestion*: frame Table 1's sequencing around language semantics and engineering scope (e.g., COBOL's static record layouts, Rust's affine ownership mapping to SIMEMP contracts, and Python's AI-glue priority vs. JS event-loop asynchrony and JVM dynamic classloading/JIT semantics), while mandating that the early Python and Rust frontends ingest only hermetic, lockfile-pinned, zero-build-script source closures until the deterministic ingestion firewall is operational across all registries.

5. **Adversarial Economics and Gameability of the Phase B Hardware Gate (Section 4.2 and Figure 2, directly addressing RQ8)**

   - **Potential Concern:** Section 4.2 gates the optional Phase B (`Qn`-based hardware design) on offering hardware vendors a "competitive market share premise — the traditional intermediate-representation bargain in reverse," grounded in the design premise that "software is always cheaper than designing hardware from zero; the branch opens when that premise is expected to invert for a specific vendor market, not before." Evaluating this gate under **RQ8** reveals three structural vulnerabilities:

     1. *Inversion of the IR Bargain:* In the traditional IR bargain (e.g., LLVM/MLIR), a hardware vendor writes *one* backend pass (IR $\rightarrow$ target silicon) to unlock all existing language frontends. Because Section 3.1 defines `QnLang` as the funnel's unique output rather than providing a retargetable multi-dialect backend infrastructure, a vendor adopting `cdqn`-managed devices still faces the engineering cost of lowering `QnLang`/`QnIR` envelopes onto their specific silicon architecture.
     2. *Measurement Authority and Vendor Gaming ("Who measures it, and can it be gamed?"):* "Competitive market share" is an exogenous macroeconomic lagging indicator (at best a Class S social quorum claim, not a deterministic Class I or physical Class P invariant) with no defined oracle or market denominator. If measured via network telemetry or emitted receipts, hardware vendors or agent swarms can game the threshold via: (a) **Sybil / wash deployment and denominator manipulation** (spinning up millions of containerized virtual device nodes emitting synthetic `QnIR` receipts, wash-licensing, or defining a narrow captive sub-market to trigger Phase B prematurely and steer canonical hardware specs toward a vendor's proprietary IP blocks); (b) **Strategic holdout / suppression** (incumbent vendors under-reporting adoption or bundling incompatible extensions to delay open hardware competition); or (c) **Artifi-**

cial software throttling (degrading general-purpose CPU/GPU driver paths for `QnIR` workloads to make software execution appear artificially slow).

3. *When the "Software is Cheaper than Hardware" Premise Actually Inverts:* Software is cheaper than custom silicon in *fixed Non-Recurring Engineering (NRE)* capital expenditure (C_{NRE}: masks, verification, tape-out), whereas custom hardware is vastly cheaper in *marginal per-operation physical dissipation (OpEx)* (e_{hw} vs. e_{sw} in joules and memory-bus cycles per operation). Because `SIMEMP` budgets computation and memory movement as first-class physical dissipation (Section 2.1), mandatory cryptographic hash-consing (`CAE-DAG`), receipt emission at every boundary, and deterministic bounds checking impose low arithmetic intensity and a heavy Memory-Wall tax on general-purpose von Neumann CPUs (cf. Williams et al., *Roofline Model*, CACM 2009). Thus, the premise inverts not at an arbitrary vendor market-share percentage, but when cumulative fleet software-emulation dissipation across verified aggregate execution volume V exceeds amortized hardware NRE plus unit fabrication cost:

$$N_{\text{devices}} \times T_{\text{lifetime}} \times \Delta P_{\text{software_overhead}} \times C_{\text{energy}} > \text{NRE}_{\text{silicon}} + N_{\text{devices}} \times C_{\text{unit_die}} \quad \left(\text{i.e., } V \cdot (e_{\text{sw}} - e_{\text{hw}}) > C_{\text{NRE}}\right)$$

Moreover, this threshold inverts orders of magnitude earlier for **FPGA overlays, open-ISA RISC-V extensions (e.g., custom CSRs for SIMEMP metric-budget counters), and dedicated cryptographic receipt/CAM hash-consing IP blocks** ($\$10^5$–$\$10^6$) than for a full-custom processor designed "from zero" ($\$10^7$–$\$10^8$+).

- **Actionable Recommendation (RQ8):** Replace the exogenous "competitive market share" threshold with an endogenous, `SIMEMP`-native **Physical Efficiency Gate (Class P / Roofline Threshold)**: open Phase B (starting with modular FPGA soft-cores, RISC-V extensions, and hardware `CAE-DAG`/receipt offload IP blocks rather than full-custom ASICs from zero) when verified Class P/I execution receipts across lineage-disjoint `Class S` roots prove that `Q(rsi)` software optimizations have saturated the Roofline ceilings of existing Layer-0 hardware, that software emulation of `QnIR` receipting and hash-consing consumes more than a declared fraction θ of total target-envelope energy/memory bandwidth, and that FPGA/ASIC simulations demonstrate a declared order-of-magnitude (e.g., $\geq 10\times$) Energy-Delay Product improvement.

Minor Corrections and Typos

- **Page 7, Section 3.4, lines 4–5 vs. Page 8, Table 1:** Section 3.4 cites `[9–11]` for Mojo's stated design intent of combining dynamic-language ergonomics with systems-language performance, whereas Table 1 (row 5, Mojo) cites `[9, 10]`. Because `[11]` is a 2026 Hacker News thread on open-sourcing Mojo rather than a design-intent statement, aligning the text with Table 1 by citing `[9, 10]` for design intent (and `[11]` separately if referencing open-source status) will improve citation precision.
- **Page 7, Section 4.2, line 5:** `"the traditional intermediate- representation bargain in reverse"` contains an extraneous space after the hyphen; change to `"intermediate-representation bargain"`.
- **Page 10, Section 4.3, line 13:** `"receipt- car-rying"` contains an extraneous space after the first hyphen; change to `"receipt-carrying"`.

[Q(anchor) Compliance, Lineage Sustainability, & Neighbour Witnessing] (Pages: 10-15)

Potential Mistakes and Improvements

1. **Internal Contradiction Between Equation 2, Condition 1, and Figure 3 in Neighbour-Anchor Witnessing (§5.6, Page 14 — Response to RQ6)**

   - **Definitive Inconsistency in Formalization vs. Diagram:** In Section 5.6, Equation 2 defines the witnessing neighbourhood of an unanchored lineage step s strictly over the set of grounded anchors $\mathcal{A}$:

   $$W_\delta(s) = \{a \in \mathcal{A} : d(s, a) \leq \delta\}$$

   Condition 1 ("Grounded witnesses") explicitly reinforces this constraint: *"Every $a \in W_\delta(s)$ carries a declared grounding class; a witness that is itself ungrounded witnesses nothing."* Furthermore, the text immediately following Figure 3 states: *"Witnessing never promotes a step to anchor status, so anchor sparsity — and with it the structural predicate's meaning — is preserved."*
   - However, Figure 3 and its caption state: *"Steps s_2 and s_4 do not land on known anchors; they are witnessed by receipted links to their grounded neighbours (a_1, s_3 and a_2, s_3 respectively)."* In the diagram, a_1 (I) and a_2 (P) are drawn in the lower row as explicit anchor nodes with declared grounding classes (`I` and `P`), whereas s_3 is drawn in the upper sequence ($s_1 \to s_2 \to s_3 \to s_4 \to s_5$) as an ordinary lineage step without an attached anchor node $a_3 \in \mathcal{A}$, without a declared grounding class label (`I`, `P`, or `S`), and without dashed "receipted link" arrows connecting it to s_2 or s_4. Additionally, the vertical dashed "receipted link" arrow from s_4 to a_2 (P) places a_2 directly underneath s_4 (mirroring how a_1 sits underneath s_1), which visually suggests that s_4 itself landed on a_2, conflicting with the top annotation *"no known anchor occurs at s_2, s_4."*
     - If $s_3 \notin \mathcal{A}$, listing s_3 as a witness in $W_\delta(s_2)$ and $W_\delta(s_4)$ directly violates Equation 2 and Condition 1.
     - Conversely, if $s_3 \in \mathcal{A}$ is intended to be an inline step that landed directly on a known anchor, Figure 3 omits its grounding class and visual receipted-link arrows.

- **Potential Concern — Directed vs. Undirected DAG Distance $d(s, a)$ and Temporal Causal Ordering:** Section 5.6 defines $d(s, a)$ as *"a declared distance on the lineage DAG (for instance, minimum path length through causal-index edges)."* This raises two topological ambiguities:
  1. *Causal Directionality and Ungrounded Execution Windows:* Because the lineage is a directed acyclic graph (DAG) where edges advance a causal index ($s_1 \to s_2 \to s_3 \to s_4 \to s_5$), s_3 lies in the causal future of s_2, and no directed causal path leads from s_4 back to s_3. For both $d(s_2, s_3) \leq \delta$ and $d(s_4, s_3) \leq \delta$ to be finite, $d(s, a)$ must be defined on the *underlying undirected graph* (or allow both ancestor and descendant paths). More critically, if s_2 is witnessed by a future descendant s_3, s_2 cannot be witnessed at the moment it is emitted because s_3 does not yet exist. While Condition 3 permits emitting a separate versioned witnessing artefact retroactively after s_3 is created, relying on future steps to witness past steps leaves s_2 ungrounded during the interim execution window and risks circular grounding if s_3's derivation depends on s_2.
  2. *Inclusion of Witnessing Edges in $d(s, a)$:* In Figure 3, a_1 and a_2 are connected to the $s_1 \ldots s_5$ chain *only* via the dashed "receipted link" arrows, with no solid causal-index edges connecting a_1 to s_1 (or a_2 to s_3/s_5) prior to witnessing. If $d(s, a)$ is measured strictly through causal-index edges, $d(s_2, a_1)$ is undefined unless a_1 is attached to s_1 via a causal edge.
- **Actionable Fix:** Update Figure 3 and its caption so that every listed witness explicitly belongs to $\mathcal{A}$ with a declared grounding class (e.g., either attach an explicit anchor $a_3 \in \mathcal{A}$ to s_3 with dashed receipted links and reposition a_2 (P) under s_5 with a diagonal link from s_4, or revise the witness sets to reference only preceding/concurrent anchors in $\mathcal{A}$). Explicitly specify in Equation 2 whether $d(s, a)$ is restricted to directed causal ancestors $d_{\text{causal}}(a \rightsquigarrow s)$ (enabling synchronous witnessing at creation time) or is an undirected graph metric evaluated within a bounded epoch-sealing window.

2. **Analytical Proof of Unbounded Monotone Growth Under Table 3 Mechanisms (§5.4–5.5, Pages 12–13 — Response to RQ4 & RQ5)**
   - **Substantive Assessment of RQ4 & RQ5:** The decomposition of lineage unsustainability in Section 5.4 into *cost bloat*, *anchor drift*, and *receipt-chain rot* (rather than destructive deletion) accurately captures the core non-destructive failure modes of an append-only architecture (though it should also explicitly account for *DAG edge and witness-metadata blowup* and *global hash-consing index dissipation*). In response to RQ5, however, the joint action of `Q(anchor)` and the seven canon mechanisms in Table 3 bounds *per-generation churn* and *local re-verification traversal*, but is **provably insufficient** to bound total lineage state and active operational costs across an unbounded generation count ($x \to \infty$):
     1. *Why `CAE-DAG` Hash-Consing Cannot Collapse Versioned `Q(rsi)` Successors:* By canonical definition (§2.4), an accepted `Q(rsi)` step emits a *versioned successor* v_{g+1} only if it is *empirically more efficient* than v_g under strict semantic invariance and a finite metric budget. Even though input-output semantics are invariant, v_{g+1} must possess a strictly distinct syntactic/operational AST or IR structure (e.g., modified control flow, memory layout, or instruction schedule); otherwise its execution cost on context hardware would be identical to v_g and it could not pay its derivation budget. Because `CAE-DAG` content addressing (Table 3, Row 1) collapses *structurally isomorphic* symbolic substructures rather than undecidable semantic equivalence classes, every accepted generation $g \in \{1, \ldots, x\}$ introduces at least one new unique root commitment hash $h(v_{g+1}) \neq h(v_g)$ and $\mathcal{O}(\text{depth})$ new spine hashes along modified paths (analogous to path-copying in persistent Merkle DAGs). Furthermore, because physical execution work on a fixed workload is lower-bounded by Landauer/Memory-Wall limits ($W_{\text{min}} > 0$), $x \to \infty$ generations can only occur if the system continuously ingests new front inputs or shifts hardware envelopes, injecting unbounded unique content.
     2. *Why Receipt Chains and Negative Receipts Force $\Omega(x)$ Monotone Growth:* Even when program sub-expressions share hash-consed nodes in the `CAE-DAG`, lowering and execution boundaries emit deterministic receipts chained to advancing causal indices, envelope measurements, and parent lineage hashes ($h_g = \text{Hash}(\text{payload}_g \parallel h_{g-1})$, §2.2). By cryptographic collision resistance, distinct causal events produce unique receipt hashes that achieve zero deduplication under hash-consing. Moreover, under the BOC policy and negative-receipt discipline (§4.4, Table 3 Row 5), every sub-optimal candidate $k_g \geq 1$ rejected during generation g is permanently recorded with its negative receipt ("nothing is ever freed", Table 3 Row 3). While per-step finite metric budgets (Table 3, Row 2) bound the *marginal* cost ΔS_g of a single step, summing bounded positive additions over x generations yields a strict linear lower bound:

$$\text{Storage}(x) \geq \text{Storage}(0) + \sum_{g=1}^{x} \left(|\Delta\text{UniqueNodes}_g| + \sum_{j=1}^{k_g} |\text{Receipt}_{g,j}| \right) = \Omega(x)$$

     3. *Missing Failure Mode in §5.4 (Global Hash-Consing Index Dissipation / Memory-Wall Trap):* Beyond passive storage bloat, the decomposition in §5.4 omits an active operational failure mode induced by `CAE-DAG` hash-consing itself. To detect structural isomorphism across $N = \Omega(x)$ unique historical nodes, every lowering pass must query a global content-addressed lookup index of size $\Omega(N)$. As $N \to \infty$, this index exceeds SRAM/L3 cache and eventually DRAM capacity on priority consumer devices (§4.1). Under SIMEMP §4.1 (where memory movement is accounted physical dissipation), the energy and latency cost of querying the deduplication table grows monotonically with lineage depth, eventually exhausting finite compilation budgets on hash-table lookups alone.
     4. *Anchor Drift vs. Anchor Bloat Dilemma and Receipt-Chain Rot:* Nothing in Table 3 enforces a lower bound on anchor emergence density $\rho = |\mathcal{A}|/|V|$. If `Q(rsi)` optimizes internal lowering paths over x generations without minting new emergent anchors, the distance $d(s, a)$ from frontier artefacts to base anchors grows as $\Omega(x)$ (*anchor drift*).

Conversely, if new anchors are minted at least once every δ steps to keep navigation bounded, the anchor population $|\mathcal{A}(x)| = \Omega(x/\delta)$ itself grows without bound. Meanwhile, historical-horizon ceilings (Table 3, Row 7) cap deep-history claims, meaning any Class P or Class S anchor from early generations that is not periodically re-witnessed eventually crosses the horizon ceiling (*receipt-chain rot*), while re-witnessing it under append-only rules emits new versioned artefacts that further grow the ledger.

- **Missing Mechanism (Answer to RQ5):** Table 3 lacks (i) a **succinct cryptographic receipt-folding primitive** (collapsing $\mathcal{O}(\delta)$ or $\mathcal{O}(x)$ transitive receipt chains into $\mathcal{O}(1)$ verifiable commitments) and (ii) an **epoch-tiered cold-demotion / horizon-compaction protocol** that decouples *cryptographic addressability* ($\mathcal{O}(1)$ accumulator roots in hot memory) from *active hot-working-set retention* (demoting superseded intermediate nodes and historical negative receipts to cold archival tiers).

3. **Epistemic Compliance of Base and Emergent Anchors: "Class-I-via-Isomorphism" as Modality Smuggling (§5.2–5.3, Table 2, Pages 11–12 — Response to RQ3)**

   - **Base Anchor Audit (Table 2, §5.2):** Section 2.3 explicitly establishes that under the October 2026 consultation (`qs_history.md` §1–2), *every* `Q(anchor)` must declare **both** a structural predicate (out-cone centrality) *and* an epistemic predicate chosen from the three Anchor Grounding Classes (`I`, `P`, or `S`). However, in Table 2 (Page 11), the fourth and fifth rows (*"Sparse navigation anchors"* in `qlog.md` §4, §8 and *"Structural indirection exemplars"* in `qm.md` §8) list `"structural"` in the epistemic `Class` column. Listing `"structural"` as their epistemic class conflates the two orthogonal predicates and leaves those two base-anchor families without a declared epistemic grounding class. Those rows should either be assigned an explicit epistemic class (e.g., `Class I` if routing and interface contracts are deterministically self-witnessed) or typed as `"polymorphic / inherited (I / P / S)"` from the wrapped domain anchor.
   - **Emergent Anchor Audit & Modality Smuggling (§5.3, Condition 1 — Answer to RQ3):** Condition 1 proposes that *"A lowered COBOL routine is Class I only in the narrow sense that its isomorphism to legacy behaviour is deterministically re-derivable; the legacy behaviour itself was witnessed, at best, by Class S quorums. Collapsing that distinction would be modality smuggling."* Section 5.3 asks reviewers to test whether this Class-I-via-isomorphism reading is honest. Tagging an emergent lowered legacy anchor *node* itself as `Class I`—even in a narrow sense—is **not epistemically honest** and enables **modality smuggling across structural indirection boundaries**:

     1. *Edge Grounding vs. Node Grounding:* In formal compiler verification (translation validation), a deterministic equivalence check witnesses a *relational morphism on the lowering edge* $e : P_{\text{COBOL}} \xrightarrow{\sim} P_{\text{QnLang}}$ (proving $[\![P_{\text{QnLang}}]\!] \cong_{\mathcal{E}} [\![P_{\text{COBOL}}]\!]$ inside a declared SIMEMP envelope). While the compilation step e is deterministically re-derivable (`Class I` for formal proof, or `Class P` for bounded empirical envelope testing), the lowered *artefact node* P_{QnLang} encodes legacy domain logic whose specification was witnessed at best by `Class S` institutional quorums.
     2. *Downstream Bypass via Structural Indirection:* Because structural indirection (`qm.md` §8; Table 3, Row 4) cuts off upstream cone traversal and dependency invalidation at `Q(anchor)` boundaries, downstream workflows inspecting only the emergent anchor's node metadata (`Class I`) will treat socially contingent legacy business rules (or legacy bugs) as self-witnessed mathematical invariants on par with $\partial\partial \equiv \emptyset$ or π, thereby bypassing the modality ceilings (`qs.md` §4.2) and historical-horizon decay rules required for `Class S` artefacts.

   - **Insufficiency of the Four Candidate Conditions (§5.3):** To achieve full SIMEMP compliance, the four conditions in Section 5.3 require three targeted additions:

     1. *Modality Meet-Semilattice Rule / Product Typing (Condition 1 Fix):* Separate the *lowering edge receipt* ($\text{Class}(e) \in \{\text{I}, \text{P}\}$) from the *emergent anchor node's grounding* by typing the node as an explicit composite product tuple $\text{Class}(\text{Anchor}) = \text{Origin}(\mathbf{S}) \times \text{Lowering}(\mathbf{I})$, governed for downstream modality ceilings by a weakest-link meet rule under the epistemic ordering Class S $\prec$ Class P $\prec$ Class I:

$$\text{Class}(P_{\text{QnLang}}) = \min_{\text{epistemic}} \big(\text{Class}(P_{\text{legacy}}),\, \text{Class}(e)\big) = \min(\text{S}, \text{I}) = \text{Class S}$$

     2. *Strengthening Source Termini (Condition 4 Fix):* Condition 4 permits receipts to terminate in "author-identity receipts." For legacy ingestion, Condition 4 should require an explicit provenance receipt for the *original legacy specification* (e.g., a `Class S` quorum attestation of the COBOL repository), preventing an ingestion operator from laundering unverified legacy code simply by signing the frontend invocation with an author-identity receipt.
     3. *Adding a 5th Condition (Anti-Cascade Emergence & Stability Gate):* An emergent anchor is recognized locally only *after* a large downstream out-cone (e.g., *"two hundred lowered programmes"*) already depends on it. Under append-only immutability, fixing a DCC contract *at emergence* (Condition 3) without invalidating and re-emitting all 200 existing dependents requires that lowered routines emit a provisional DCC envelope header at initial ingestion (which can be promoted to a `Q(anchor)` contract via versioned index receipts without mutating the target payload hash), paired with a minimum stability gate so frequently mutated utility routines do not repeatedly trigger out-cone supersession.

4. **Critique of the Snapshot / Check-Node Analogy and Attack on the Three Failure Modes of Neighbour Witnessing (§5.6, Page 14 — Response to RQ6)**

   - **Breakdown of the Error-Correcting Code (Check-Node) Analogy:** Section 5.6 compares neighbour-anchor witnessing to *"snapshots with deltas, or error-correcting codes in which anchors are check nodes and intervening steps are variable nodes."*

  - The **snapshots-with-deltas** analogy holds accurately, provided the receipted links carry constructive transition proofs (or deterministic replay deltas along the path $a \rightsquigarrow s$ of length $\leq \delta$) rather than bare reachability pointers.
  - The **error-correcting code (LDPC / Tanner graph check-node)** analogy is **mathematically misleading**: in linear error-correcting codes, check nodes enforce redundant algebraic parity constraints ($\sum H_{ij} v_j = 0 \pmod 2$) over multiple variable nodes so that an erased or corrupted variable node can be *algebraically reconstructed* from its intact neighbours without re-running the source process. In a `CAE-DAG`, cryptographic receipt links provide *tamper detection (soundness/provenance integrity)*, not *payload recovery*. Furthermore, Equation 2 permits a single witness ($|W_\delta(s)| = 1$), which provides zero parity redundancy; even when $|W_\delta(s)| \geq 2$, upstream anchors a_1, a_2 sealed prior to s contain zero mutual information about the specific delta in s unless the full deterministic derivation trace between a_1 and s is also retained.

- **Adversarial Attack on the Three Named Failure Modes (Plus Versioned Witnessing Bloat):**

  1. *Correlated Witness Falsification:* Requiring lineage-disjoint roots per the polyroot discipline (`chronosa.md` §1.4) prevents single-trunk citation laundering, but fails in three ways:
     - *Missing Cardinality & Disjointness Constraints in Equation 2:* Conditions 1–4 do not enforce a minimum witness degree $|W_\delta(s)| \geq k \geq 2$ or require that witnesses in $W_\delta(s)$ reside in disjoint root trees.
     - *Single-Root Derivation Deadlock:* Within a vertical `Q(rsi)` self-bootstrapping chain or a single lowered legacy codebase prior to any cross-tree merge, all reachable steps $s_1 \ldots s_t$ descend from the *same* root subtree, making lineage-disjoint witnessing structurally impossible without artificially injecting cross-root dependencies every δ steps.
     - *Common-Mode Toolchain / Hardware Faults:* Even when two witnesses originate from lineage-disjoint legacy roots, lowering both through the *same buggy or compromised compiler frontend* or calibrating them on the same Layer-0 host introduces correlated errors that polyroot graph topology cannot detect (requiring *diverse double-lowering* or cross-class witnessing, e.g., pairing `Class S` with `Class I`).
  2. *δ-Gaming (Topological Shortcut, Step-Packing, and Synthetic Anchor Attacks):* Declaring δ as a static, BOC-auditable constant in a DCC contract prevents runtime parameter inflation, but leaves unweighted shortest-path distance $d(s, a)$ vulnerable to three structural exploits:
     - *Hub-and-Spoke Topological Shortcutting:* Inserting a trivial, no-op dependency edge (e.g., importing an unused constant from base anchor a_0 such as π or $\partial\partial \equiv \emptyset$) into every step s_t, which collapses minimum graph path length to $d(s_t, a_0) = 1 \leq \delta$ while the actual computational derivation chain runs unanchored for thousands of steps.
     - *Step-Granularity Packing:* Packing a massive, computationally expensive semantic transformation into a single macro-lowering edge so that topological hop count is $d(s, a) = 1 \leq \delta$ while actual verification work is unbounded.
     - *Synthetic Anchor Inflation:* Periodically minting trivial `Class I` identity wrappers every δ steps and routing dummy dependencies through them to inflate their local out-cone centrality.
     - *Fix:* Define $d(s, a)$ not as unweighted shortest-path hop count, but require $W_\delta(s)$ to form an **active dataflow dominator cut** separating s from unverified ancestors, where path distance bounds the *maximum (longest) unanchored directed ancestral derivation path* weighted by cumulative **SIMEMP work/memory-dissipation budget**:

$$d_{\text{work}}(W_\delta(s), s) = \max_{\text{path } p:\, W_\delta(s) \rightsquigarrow s} \sum_{e \in p} \text{Work}(e) \leq \delta_{\text{work}}$$

  3. *Receipt Transit Cost and Recursive Witnessing Bloat:* Condition 2 requires that the Source-Receipt Invariant hold *transitively* through witnessing edges back to permitted termini. In a DAG with branching factor $b > 1$, traversing δ pointer-linked hops (or chasing upstream anchor chains to depth $L \gg \delta$) incurs $\mathcal{O}(|W_\delta(s)| \cdot \delta)$ or $\mathcal{O}(b^\delta)$ non-contiguous memory dereferences, directly colliding with SIMEMP's Memory-Wall discipline (§2.1) unless Condition 4 caps witness cardinality ($|W_\delta(s)| \leq \kappa$) and folds intermediate receipts into a contiguous, fixed-width accumulator at sealing time. Moreover, Condition 3 ("The witnessing is itself a versioned artefact") risks recursive metadata bloat if witness links are minted as separate lineage nodes that themselves require witnessing, rather than being folded directly into s's immutable birth receipt.

5. **Concrete SIMEMP-Compatible Candidate Designs Mapped to the Five-Row Contract Surface (§5.7, Page 15 — Response to RQ7)**

- To address the open call in Section 5.7 and RQ7, the following four candidate mechanisms directly target the author's wish list (append-only cold demotion, succinct receipt verification, amortized deep-history re-verification, and formal growth bounds via provenance pricing) and map explicitly onto the five-row `qstructures.md` contract surface (`footprint`, `access`, `locality`, `determinism`, `work`):

Candidate Mechanism	Core Operation (Append-Only Compatible)	`footprint` (Storage Bound)	`access` (Addressing & Lookup)	`locality` (Memory Wall Discipline)	`determinism` (Execution Invariant)	`work` (Verification Cost)
1. Incrementally Verifiable Receipt Folding (IVC / Algebraic Rollups at δ-Cuts, e.g., Nova)	Instead of retaining and traversing δ raw intermediate receipts in hot memory, each lowering step s_t folds its predecessor's accumulated proof π_{t-1} and local step receipt r_t into a constant-size cryptographic fold $\pi_t = \text{Fold}(\pi_{t-1}, r_t)$ anchored at `Q(anchor)` cuts. Unfolded intermediate step receipts are demoted to cold archival storage while remaining content-addressed.	$\mathcal{O}(1)$ hot verification footprint per witnessed node or inter-anchor interval regardless of δ or generation depth x; $\mathcal{O}(\lvert\mathcal{A}\rvert)$ active anchor state.	$\mathcal{O}(1)$ single-read verification at step s_t or enclosing `Q(anchor)` boundary; $\mathcal{O}(\log \delta)$ cold Merkle-inclusion lookup only on explicit audit queries.	Contiguous fixed-size witness block co-located in the node header (fits in L1/L2 cache); eliminates $\mathcal{O}(\delta)$ DAG pointer-chasing.	Strictly deterministic arithmetic/hash circuit folding over fixed finite fields declared in the DCC profile (`No-Implicit Rule`).	$\mathcal{O}(1)$ downstream verifier work; $\mathcal{O}(1)$ incremental folding work paid once upfront inside the producer's `Q(rsi)` derivation budget.
2. Epoch-Stratified Cold Demotion with MMR Compaction & Rejection Certificates (cf. Certificate Transparency / EIP-4444)	Partitions the causal index into fixed-budget epochs $E_1, \ldots, E_m$. At epoch closure, only the active frontier and live `Q(anchor)` contracts remain in **Tier-0 (Hot)**; superseded internal ancestor payloads ($s \notin \mathcal{A}$) and full AST payloads of losing BOC negative receipts are sealed into an immutable `Class I` Merkle Mountain Range (MMR) root R_k and demoted to **Tier-1 (Cold)**, retaining in Tier-0 only a compact **Cuckoo/Quotient Rejection Filter** and minimal unsat-core hashes to block duplicate re-attempts (*Missing row to name explicitly:* cold-retrieval latency and physical archival retention pricing when deep historical re-auditing is triggered).	Bounds Tier-0 hot footprint and active `CAE-DAG` hash-table size to $\mathcal{O}(\lvert\text{Active Frontier}\rvert + \lvert\mathcal{A}\rvert + m) + \approx$ 16 bits per rejected candidate, decoupling operational memory from x.	$\mathcal{O}(1)$ Tier-0 hash/filter lookup (at most 2 cache-line ~~r~~eads) for active nodes and rejected candidate screening; $\mathcal{O}(\log \lvert E_k \rvert)$ caller-supplied MMR witness required to fault-in cold history.	Hot `CAE-DAG` frontier compacted into contiguous epoch arena pages within consumer-device DRAM/cache ceilings; cold payloads never cross the memory bus during normal compilation.	Deterministic epoch cutoffs triggered at fixed causal-index strides using canonical lexicographic hash ordering; zero de jure deletion or OS cache heuristics.	Zero ongoing hot-loop lookup or verification overhead for superseded epochs; cold re-hydration work paid exclusively by workflows requesting historical fault-ins.

Candidate Mechanism	Core Operation (Append-Only Compatible)	`footprint` (Storage Bound)	`access` (Addressing & Lookup)	`locality` (Memory Wall Discipline)	`determinism` (Execution Invariant)	`work` (Verification Cost)
3. Causal-Horizon Provenance Leases & Amortized Re-Verification Schedules	Prevents *receipt-chain rot* and bounds deep-history traversal by assigning every `Class P` and `Class S` anchor a mandatory **staleness horizon** τ measured strictly in discrete causal generations (never wall-clock time), paired with **dyadic skip-list fingers** (at distances $2^0, 2^1, \ldots, 2^{\lfloor \log_2 x \rfloor}$). When $\text{gen}_{\text{curr}} - \text{gen}_{\text{witness}} > \tau$, downstream consumers must pay an amortized `Q(rsi)` budget to attach a fresh calibration (P) or quorum (S) receipt, or the branch transitions to a `COLD_HERITAGE` modality ceiling (addressable, but barred from active financial/actuator lowering).	$\mathcal{O}(\log x)$ routing pointers per lineage trunk; active `Class P/S` verification set bounded strictly by current maintenance budget.	$\mathcal{O}(\log x)$ hop count to any historical generation $g < x$; $\mathcal{O}(1)$ local header check ($\text{gen}_{\text{curr}} - \text{gen}_{\text{witness}} \leq \tau$) at structural-indirection boundaries.	Skip-list fingers and integer TTL fields packed inline within the fixed 64/128-byte DCC contract descriptor; zero global DAG scans.	Pure integer arithmetic on discrete generational counters x and τ declared in the DCC contract; zero wall-clock non-determinism.	Amortizes Layer-0 physical re-calibration and social quorum re-witnessing work at $\mathcal{O}(1/\tau)$ per generation across consuming workflows.
4. Pigouvian Lineage-Entropy Pricing (`SIMEMP_RENT`)	Modifies the BOC objective (Eq. 1) and `Q(rsi)` acceptance gate so that every candidate c prices its permanent lineage storage externality into its finite metric budget: $\text{NetGain}(c) = \Delta\text{Work}_{\text{saved}}(c) - \lambda_{\text{store}} \cdot \Delta U(c) \cdot \delta(c) > 0$, where $\Delta U(c)$ is net new `CAE-DAG` bytes (including negative receipts) and $\delta(c)$ is work-weighted distance to the nearest `Class I` anchor.	**Formal Growth Bound:** Cumulative unique lineage expansion $\sum_{g=1}^{x} \Delta U_g \leq \frac{W_0 - W_{\min}}{\lambda_{\text{store}} \cdot \delta_{\min}}$ is strictly bounded on any stationary workload because physical runtime work $W_g \geq W_{\min} > 0$ cannot drop below Landauer/ Memory-Wall limits.	$\mathcal{O}(1)$ byte-delta counter read from the `CAE-DAG` ingestion table during BOC evaluation.	Forces `Q(rsi)` to compact deep delta chains into new anchors when $\delta(c)$ transit cost exceeds anchor-minting cost; zero runtime memory overhead.	Exact byte-count and node-count deltas computed deterministically by the `CAE-DAG` engine at compile/sealing time.	$\mathcal{O}(1)$ integer arithmetic check at the `Q(rsi)` sealing gate; internalizes long-term storage bloat into the finite derivation budget.

Minor Corrections and Typos

- **Page 11, Section 5.2, Paragraph 1, lines 9–12:** Remove the extraneous space after the hyphen in *"its re- derivation is deterministic"* → *"its re-derivation is deterministic"*. In the same sentence, the clause *"and its convergence bounds ($0 \leq |r_m| < 1/(q_m q_{m+1})$ for continued-fraction convergents) are stated in the canon"* is attributed to the exterior-derivative invariant $\partial\partial \equiv \emptyset$ (`qm_calculus.md` §3), whereas continued-fraction convergent bounds actually belong to the second anchor family in Table 2 (*"Transcendental continued-fraction generators ($\pi, e, \tau, \ln 2$)"*, `qexpr.md` §4.3 / `qm_calculus.md` §4). Additionally, add a thin space between the denominator factors in $1/(q_m q_{m+1})$ so `qmqm+1` does not visually run together.
- **Page 11, Section 5.2, Table 2 (Row 2, "Transcendental continued-fraction generators"):** Remove the extra space before the comma after τ in `"(\pi, e, \tau , \ln 2)"` → `"(\pi, e, \tau, \ln 2)"`.
- **Page 14, Section 5.6, Figure 3:** Inside the bottom-left green anchor box, the label `a1 (I)` is flush-left against the box border, whereas the labels in `a2 (P)` and `s1`–`s5` are horizontally centered. Furthermore, the bottom graphic annotation (`"witnesses: neighbours a1, s3 for s2; a2, s3 for s4"`) uses unsubscripted plain text (`a1, s3, s2, a2, s4`) instead of the subscripted notation (a_1, s_3, s_2, a_2, s_4) used inside the node boxes and in the caption.
- **Page 15, Section 6.1, Paragraph 1, last line:** The phrase *"aligned with humanity purpose"* is missing a possessive or adjectival inflection; change to *"aligned with humanity's purpose"* or *"aligned with human purpose"* (or enclose *"humanity purpose"* in quotation marks if it is a verbatim term from the repository's design notes).

[Alignment Motivation, Review Protocol, Seed Finding & Reviewer Deliverables] (Pages: 15-19)

Potential Mistakes and Improvements

1. **Adjudicating the Seed Finding: Formal Divergence Between Deterministic Bounded Execution and Budgeted Termination (§7.2)**

   - **Analysis of the Inconsistency:** Section 7.2 identifies a canonical inconsistency between `simemp.md` (§2, line 105), which requires every program $p \in \text{QnIR}(\mathcal{A})$ to satisfy ExecutesDeterministicallyWithinBounds(p), and `index.md` (v1.12.0), which states TerminatesWithinBudget(p), and invites the review pass to adjudicate how they should be aligned. Leaving this divergence unresolved is a substantive validity risk because `QnIR` cannot verify "equivalence consistency inside declared SIMEMP envelopes" (§3.2) without specifying whether non-terminating service loops or non-deterministic bounded runs are admissible. Formally, even in finite-state or finite-resource systems, neither predicate entails the other:

     1. *Why TerminatesWithinBudget(p) $\not\Rightarrow$ ExecutesDeterministicallyWithinBounds(p):* A program p that branches on unseeded thread-scheduling races, asynchronous message arrival order, hash-table iteration order, or hardware entropy can strictly halt in $O(1)$ steps well inside its cumulative energy/step budget (TerminatesWithinBudget(p) holds), yet produce different output states and receipt hashes across runs—violating the `determinism` row of the five-row contract surface and breaking lowering-receipt isomorphism. Furthermore, a program can satisfy an aggregate work budget while momentarily spiking past an instantaneous spatial/memory-footprint or power-dissipation bound.
     2. *Why ExecutesDeterministicallyWithinBounds(p) $\not\Rightarrow$ TerminatesWithinBudget(p):* Conversely, there are two operational regimes where bounded deterministic execution fails to satisfy budgeted termination:
        - *Infinite Reactive Cycles within Spatial Bounds:* If "Bounds" constrain spatial footprint, cache locality, and per-step power dissipation—or if p is a continuous legacy service such as the bank COBOL settlement system in §3.3 ("translation is an event, execution is a service")—a deterministic transition system over a finite M-bit memory envelope can enter a deterministic state cycle of length $\leq 2^M$ (livelock) or run indefinitely as an event-loop service without ever violating its spatial or per-step bounds, yet never reach a global halting state.
        - *Abnormal Watchdog Abort vs. Valid Semantic Termination:* Even if "Bounds" include a hard monotonic step ceiling $W_{\max}$ that forces a runtime trap when exhausted, an forced watchdog timeout or unhandled hardware exception is an *abnormal abort* (resource exhaustion), not a valid semantic *termination* that emits a well-formed output receipt isomorphic to legacy behavior.

   - **Recommended Canonical Resolution:** Neither predicate should be erased in favor of the other, because the five-row contract surface (`footprint, access, locality, determinism, work`) requires both step-level determinism within spatial envelopes and normal halting within work budgets. To accommodate both batch derivations (`Q(rsi)`) and continuous legacy services (§3.3), both `simemp.md` and `index.md` should be unified into a single conjunctive validity formula scoped to each bounded transaction/lowering epoch over a five-row envelope $E = (B_{\text{space}}, B_{\text{work}})$:

$$\begin{aligned}\text{Valid}(\mathcal{A}) \iff\ & \text{Consistent}(\mathcal{A}) \wedge \forall p \in \text{QnIR}(\mathcal{A}), \\ & \Big(\text{DeterministicTransition}(p) \wedge \text{WithinSpatialBounds}(p, B_{\text{space}}) \\ & \quad \wedge \text{HaltsWithReceipt}(p, B_{\text{work}})\Big)\end{aligned}$$

     where for reactive services (§3.3), HaltsWithReceipt(p, B_{work}) applies to each atomic event-handling transition rather than requiring the service daemon itself to terminate.

2. **Distinguishing Alignment from Surveillance and Testing the "Natural Language as Projection" Premise (§4.3, §6.1–6.2, §8 RQ9)**

   - **Potential Concern:** Section 4.3 asserts as a "testable design premise (RQ9), not a slogan" that "natural language is not the substrate but a projection: a low-dimensional view rendered from the lower levels of the Qn abstraction stack for human consumption," while Section 6.2 candidly draws a tight boundary between *conformity to process* (auditability, deterministic envelopes, refusal via modality ceilings) and *judgement of purpose* (deferred to `Chronosa`'s unimplemented "global section of an intent sheaf"). `RQ9` asks what measurable properties would distinguish real progress toward humanity-purpose alignment from a "well-organised surveillance capability with a friendly narrative," as well as whether "natural language as a projection of the Qn stack" is a coherent and testable position. While replacing unconstrained natural language with a deterministic, schema-governed language for agent-to-agent coordination is well-supported to prevent hallucination cascades and steganographic collusion (cf. Hong et al., *MetaGPT*, ICLR 2024; Ghaemi, 2025), two substantive gaps affect this thesis:

     1. *Structural Isomorphism to a Panopticon & Steganographic Collusion:* Total receipt-chaining, mandatory author-identity provenance, and deterministic execution envelopes are structurally indistinguishable from a centralized compliance and surveillance ledger: an authoritarian controller or harmful cartel can encode coercive rules into DCC contracts and enforce strict obedience with 100% receipt coverage. Conversely, process conformity alone does not prevent *steganographic collusion* among agent swarms, which can encode covert coordination signals inside the remaining degrees of freedom of valid `QnLang` payloads or cross-boundary invocation timing while respecting all SIMEMP envelopes.

2. *Lossiness and Bootstrap Asymmetry of the Natural-Language Projection:* Mathematically, a projection π : QnStack $\twoheadrightarrow$ NL from a high-dimensional formal DAG to a lower-dimensional natural-language rendering is a **many-to-one (lossy/non-injective)** mapping. First, because π has a non-trivial kernel ($\pi(s_1) = \pi(s_2)$ for distinct states $s_1 \neq s_2$), an unaligned agent swarm can construct a formal `QnLang` execution graph whose lossy natural-language summary $\pi(s)$ appears benign ("friendly narrative") to a human supervisor while exploiting subtle boundary conditions projected away in $\pi(s)$. Second, human intent, legal delegations, and social norms (`Class S`) originate *outside* the Qn stack in high-entropy, ambiguous natural language; recovering formal `QnLang` contracts from human directives requires an inverse *intent-lifting* map (π^{-1} : NL $\to$ QnStack) where semantic drift, silent narrowing, and specification gaming enter before a single deterministic receipt is ever emitted.

- **Suggested Improvement (Addressing `RQ9`):**
  - **Falsifiable Metrics Distinguishing Alignment from Surveillance:**
    1. *Bidirectional Audit Symmetry & Revocation Latency (Anti-Panopticon Metric):* Measure the ratio of verification work and latency required for a leaf node (an individual human user on a consumer-device envelope) to audit an upstream orchestrator's receipt chain and execute a capability revocation or modality-ceiling refusal, divided by the cost for an authority to audit the leaf node. In a surveillance apparatus this ratio is $\gg 1$ (opaque root, transparent leaves); in an alignment instrument it is ≈ 1 (enforced by $O(\delta)$ local neighbour verification).
    2. *Sheaf Cohomology / Obstruction Preservation Rate in `Chronosa`:* In sheaf theory, a presheaf of local sections across diverse communities (`Class S` quorums) fails to admit a global section whenever genuine value conflicts exist (non-vanishing first cohomology $H^1 \neq 0$). A testable metric distinguishing pluralistic alignment from hegemonic control is whether `Chronosa` deterministically detects and emits negative receipts flagging local sheaf obstructions (halting automated actuator execution at the modality ceiling and returning conflict resolution to human deliberation) rather than silently overriding minority subtrees to fabricate a consensus.
    3. *Canonicalized Covert-Channel Capacity, Coordination Dissipation, & Pre-Actuator Harm Interception:* Benchmark agent swarms coordinating exclusively in `QnLang` against unconstrained natural-language baselines on standardized multi-agent tasks, measuring (i) bit/memory dissipation reduction and the residual Shannon entropy (in bits per message) available between equivalent `QnLang` expressions after CAE-DAG hash-consing (proving whether canonicalization reduces covert syntactic channel capacity toward zero), and (ii) the **pre-execution refusal rate** at the NL $\to$ QnLang lifting and `qs.md` §4.2 modality-ceiling boundaries (detecting and refusing underspecified, collusive, or harmful swarm trajectories *a priori* before actuator commitment, rather than merely logging them post-hoc).
  - **Testing "Natural Language as a Projection":** Formulate the projection premise as a falsifiable **Round-Trip Semantic Reconstruction & Faithfulness Contract**: given a formal `QnLang` contract or state transition s, project it to natural language $\pi(s)$ for an independent blinded human quorum or translator to lift back into $\hat{s} = \rho(\pi(s)) \cong s$, measuring semantic drift across the five-row contract surface and testing whether human auditors reading only $\pi(s)$ can reliably detect injected adversarial policy violations in s. Specifically, π is non-deceptive only if its kernel is strictly orthogonal to safety invariants—meaning no two states s_1, s_2 with $\pi(s_1) = \pi(s_2)$ may differ in their declared SIMEMP resource budgets, causal dependencies, or modality ceilings.

3. **Epistemic Labelling Hygiene and Factual Accuracy of IRL Citations (§7.1, Table 4, References [1–12])**

- **Potential Concern:** Section 7.1 states that "mislabelling is treated here as a defect of the same severity as a wrong citation" and defines `IRL design premises` as statements carrying verified real-world citations. Auditing the three-layer labels and Reference list (Page 19) reveals three additional factual and bibliographic discrepancies (beyond the `[7, 8]` RubyGems/Hugging Face package-ecosystem mismatch noted under Section 3.4):
  1. *Misattribution of Organization and Double-Counting in References [4, 5]:* Reference [5] cites `"Federation of American Scientists. On recursive self-improvement (Part I). thefai.org, February 2026... URL https://www.thefai.org/posts/on-recursive-self-improvement-part-i."` However, the domain `thefai.org` belongs to **The Foundation for American Innovation (FAI)**, not the Federation of American Scientists (`fas.org`). Furthermore, the post at that URL is a cross-post of Dean W. Ball's February 5, 2026 *Hyperdimensional* essay (already cited as Reference [4], as Ball is a Senior Fellow at FAI), rather than an independent institutional report. Citing `[4, 5]` together in §1.2 double-counts a single essay under two separate reference numbers.
  2. *Author Omission in Reference [6] and Co-Author Order in Reference [2]:* Reference [6] attributes `"Recursive self-improvement in 2026: The evidence"` solely to the company name `"Pelles"`; the actual author of the `pelles.ai` post (published September 16, 2026; updated October 14, 2026) is **Matan Weksler**. Additionally, Reference [2] cites the authors of `arXiv:2606.28277` as `"Rajesh Jayaram, Vincent Cohen-Addad, et al."` While Jayaram and Cohen-Addad are the two corresponding authors (and blog post authors in `[1]`), Cohen-Addad is the seventh/last author on the arXiv preprint (after Drew Tyler, David Woodruff, Corinna Cortes, Yossi Matias, and Vahab Mirrokni), and `[1, 2]` report piloting the tool on "more than 16,000" conference submissions rather than "more than fifteen thousand" (§1.1, p. 2).
  3. *Clarifying Canon vs. CANDIDATE Scope for `Q(anchor)` in Table 4:* Table 4 (Page 16) lists `Q(anchor) definition` purely under the `Canon` row (`d50ee1e`). While the base glossary entry for `Q(anchor)` is canon, Sections 2.3 and 5.2 explicitly note that the two-predicate decomposition (structural out-cone centrality vs. epistemic Grounding Classes

`I, P, S`) used throughout the paper is an unsealed `CANDIDATE` proposition from `qs_history.md`. Noting this split in Table 4 will keep the summary table strictly aligned with Section 5.2.

Minor Corrections and Typos

- **Page 17, Table 5 (`Pinned path` column):** In rows 6, 9, and 10, literal LaTeX escape backslashes appear before underscores in the rendered paths (`docs/qs\_history.md`, `docs/\_data/glossary/architecture.yml`, and `docs/\_data/glossary/primitives.yml`). Because Section 7.4 states that each row's path is appended directly to `https://github.com/cdqn5249/cdqn/blob/d50ee1e/`, appending the literal `\_` strings produces broken 404 URLs. Remove the backslashes so the paths read `docs/qs_history.md`, `docs/_data/glossary/architecture.yml`, and `docs/_data/glossary/primitives.yml`.
- **Page 18, Section 8 (`RQ1`, `RQ8`, and Question Ordering):**
  - In `RQ8` (line 2), remove the extraneous space after the second hyphen in `"software-cheaper-than- hardware premise"` so it reads `"software-cheaper-than-hardware premise"`.
  - In `RQ1` (line 4), `"minimal counterexample programme"` uses British spelling (matching p. 12 `"lowered programmes"`), whereas Section 3.2 (p. 7) and Section 7.2 (p. 16) use American spelling (`"counterexample program"`, `"every program p"`). Standardize the spelling across the draft.
  - The questions in Section 8 are ordered `RQ4, RQ5, RQ6, RQ7, RQ1, RQ2, RQ3, RQ8, RQ9`. While the text notes they are ordered by author priority, adding a brief parenthetical note that the `RQ1`–`RQ9` numbering is retained from earlier drafts for cross-reference stability (or renumbering them sequentially) would improve readability.
- **Page 19, References `[1]`, `[4]`, `[5]`, `[7]`, `[8]`:** Five bibliography entries duplicate the publication month and year at the end of the citation string due to overlapping BibTeX fields:
  - `[1]`: `"launch post of October 14, 2026, October 2026."`
  - `[4]`: `"February 5, 2026, February 2026."`
  - `[5]`: `"February 2026, February 2026."`
  - `[7]`: `"September 11, 2026, September 2026."`
  - `[8]`: `"September 15, 2026, September 2026."` Remove the redundant month/year fields in the bibliography source.